On Ramanujan Challenge Problem 3.2: The Apéry GCD Conjecture

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Abstract

We study $G_n = \gcd(d_n a_n, d_n b_n)$ for the Apéry sequences (a_n), (b_n) with $d_n = \mathrm{lcm}(1,\dots,n)^3$. We prove *unconditionally* that $G_n = e^{o(n)}$ for a set of positive integers of natural density 1, with the quantitative bound $\#\{n \le N : \log G_n > \varepsilon\, n\} = O_\varepsilon\big((\log N)^2\big)$ for every $\varepsilon > 0$; in particular, the exceptional set has upper Banach density zero and finite harmonic weight. We further prove a *Cartier–Frobenius block law* for the Apéry *pair*: for $p \ge 7$, $1 \le q < p$ and $0 \le r < p$,

$$\big(p^3 a_{qp+r},\, b_{qp+r}\big) \equiv b_r\,(a_q, b_q) \pmod{p},$$

whose second coordinate is the classical Apéry–Lucas congruence and whose numerator coordinate appears to be unrecorded; the same law holds for the $\zeta(2)$ pair with p^2. Combining it with Apéry's Casoratian and the no-two-consecutive-zeros theorem gives the exact valuation law $v_p(a_n) = -3 + v_p(b_n)$, i.e. $e_p(n) = v_p(b_n)$, for every $p \ge 7$ in the top window with $v_p(b_n) \le 2$. Consequently the top-window contribution to $\log G_n$ is *exactly* the logarithm of the $(n/2, n]$-part of b_n, and, since all primes below $n/\log n$ contribute $O(n/\log n)$ by Chebyshev alone, the conjecture is equivalent to the statement that this part of b_n is subexponential. The key ingredient of the density-one theorem is a bound $Z(p) = O(p^{2/3})$ on the zero-count function $Z(p) = \#\{j < p : p \mid b_j\}$, derived from the Apéry recurrence via a gap polynomial argument. A companion centered-coefficient analysis of the bordered value-return certificates proves the *uniform* fiber bound $\max_a \#\{j < p : b_j \equiv a\} = O(p^{3/4})$ and the collision-energy bound $\sum_a \#\{j : b_j \equiv a\}^2 = O(p^{7/4})$, both unconditional, with the underlying certificate nonvanishing proved in the sharp range $1 \le h < k < p$. Computation for all 78,496 primes $5 \le p \le 10^6$ gives $Z(p) \le 12$ with mean ≈ 1. The pair count $Z(p)/2$ fits Poisson(1/2). For each canonical gap polynomial N_h whose splitting field has full hyperoctahedral Galois group, we prove via Chebotarev that its root-pair count follows the signed fixed-point distribution, tending to Poisson(1/2) as $h \to \infty$; the corresponding law for the actual coefficient zeros remains conjectural. The Poisson tail predicts $\max_{p \le N} Z(p) \sim 2 \log\log N$, so $Z(p)$ is almost certainly unbounded. Under the weaker Cesàro hypothesis $\sum_{p \le x} Z(p) \ll \pi(x)$ (Hypothesis $\bar{Z}$), which the Poisson model supports, we prove $\log G_n = O(\sqrt{n})$ for density 1, matching the unconditional small-prime floor, with the conditional exceptional set shrinking to $O(N^{1/3})$ and having upper Banach density zero. The proof combines a dyadic divisor-incidence bound with a sub-interval zero-count argument to reduce the leading-digit bottleneck from $O(N^{3/2})$ to $O(N^{10/7}/\log N)$ unconditionally, and to $O(N^{4/3}/\log N)$ under Hypothesis $\bar{Z}$. A prime-wise decomposition of the residual L^2-norm $\|\eta\|^2 = D + O$ (diagonal + off-diagonal) shows that $D = T + O(1/\ln N)$ unconditionally, reducing the all-n conjecture $G_n = e^{o(n)}$ to a two-prime dispersion input: the cross-prime interaction O is $o(T)$. Computation through $N = 262{,}144$ confirms $|O/T| < 2 \times 10^{-3}$. Two structural reductions further localize the gap: prime counting alone disposes of every prime $p \le n/(f(n)\log n)$ for any $f \to \infty$, so only the quotient classes $\lfloor n/p \rfloor < f(n)\log n$ matter; and a single fixed factorial moment $(\mathrm{HM})_k$ with $k > 6$—at the independence scale, compatible with $Z(p) \ll p^{2/3}$, and proved here unconditionally for $k = 2$—would give the full conjecture with the power saving $\log G_n \ll n^{2/3 + 2/k + o(1)}$.

1 Setup

The Apéry sequences satisfy the three-term recurrence

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5)\, u_n - n^3\, u_{n-1}, \quad n \geq 1, \tag{1}$$

with $(a_0, a_1) = (0, 6)$ and $(b_0, b_1) = (1, 5)$. Here $b_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 \in \mathbb{Z}$ is the n-th Apéry number. Set $d_n = \operatorname{lcm}(1, \ldots, n)^3$; Apéry [3] showed $d_n a_n, d_n b_n \in \mathbb{Z}$. Define $G_n = \gcd(d_n a_n, d_n b_n)$. Write $P(n) = 34n^3 + 51n^2 + 27n + 5$ for the middle coefficient.

2 The Wronskian identity

Lemma 1. *For all $n \geq 1$, $W_n := a_n b_{n-1} - a_{n-1} b_n = 6/n^3$.*

Proof. The Casorati determinant satisfies $W_{n+1}/W_n = n^3/(n+1)^3$ from (1). Since $W_1 = a_1 b_0 - a_0 b_1 = 6$, telescoping gives $W_n = 6/n^3$. □

Corollary 2. *For every prime $p \geq 5$,*

$$v_p(G_n) \leq 3\lfloor \log_p n \rfloor. \tag{2}$$

Proof. G_n divides $d_n(a_n b_{n-1} - a_{n-1} b_n) = 6 d_n / n^3$. For $p \geq 5$, $v_p(6) = 0$ and $v_p(d_n/n^3) = 3\lfloor \log_p n \rfloor - 3 v_p(n) \leq 3\lfloor \log_p n \rfloor$. □

3 Small primes: unconditional bound

Proposition 3. $\displaystyle\sum_{p \leq \sqrt{n}} v_p(G_n) \log p = O(\sqrt{n})$.

Proof. By (2), $v_p(G_n) \log p \leq 3 \log n$ for every prime $p \geq 5$. Primes $p = 2, 3$ contribute at most $O(\log n)$ each. There are $O(\sqrt{n}/\log n)$ primes up to $\sqrt{n}$, so the sum is at most $3 \log n \cdot O(\sqrt{n}/\log n) = O(\sqrt{n})$. □

4 The multiplicative Lucas congruence

Theorem 4 (Gessel [18])**.** *For every prime p and $n = \sum_{i=0}^{s} n_i p^i$ in base p,*

$$b_n \equiv \prod_{i=0}^{s} b_{n_i} \pmod{p}. \tag{3}$$

This follows from the combinatorial identity $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ and termwise application of Lucas's congruence to each binomial coefficient. See also Mimura [22] and McIntosh [21].

5 The zero-count function

Definition 5. For a prime $p \geq 5$, define $Z(p) = \#\{j \in \{0, \ldots, p-1\} : b_j \equiv 0 \pmod{p}\}$.

5.1 Unconditional bound

Lemma 6 (No consecutive zeros)**.** *For every prime $p \geq 5$ and every $j \in \{0, \ldots, p-2\}$, b_j and b_{j+1} cannot both vanish modulo p.*

Proof. Suppose $b_m \equiv b_{m+1} \equiv 0 \pmod p$ with $0 \leq m \leq p-2$. The recurrence (1) at $n = m$ gives $(m+1)^3 b_{m+1} = P(m)\, b_m - m^3\, b_{m-1}$, so $m^3 b_{m-1} \equiv 0 \pmod p$. Since $1 \leq m \leq p-2$ implies $p \nmid m$, we get $b_{m-1} \equiv 0$. Iterating gives $b_0 \equiv 0 \pmod p$, contradicting $b_0 = 1$. □

Lemma 7 (Gap polynomial)**.** *Fix $h \geq 2$. Suppose $b_m \equiv 0 \pmod p$ and $m + h \leq p - 1$. Then*

$$b_{m+h} \equiv C_h(m)\, b_{m+1} \pmod p, \tag{4}$$

where $C_h(m)$ is a rational function of m whose numerator (after clearing the denominator $\prod_{j=1}^{h-1}(m+j+1)^3$) is a polynomial of degree exactly $3(h-1)$ with positive leading coefficient.

In particular, if $b_m \equiv b_{m+h} \equiv 0 \pmod p$, then by Lemma 6 $b_{m+1} \not\equiv 0$, so $C_h(m) \equiv 0 \pmod p$, and m is constrained to at most $3(h-1)$ residue classes modulo p, provided C_h is not identically zero over $\mathbb{F}_p$.

Proof. We iterate the recurrence from the initial condition $(b_m, b_{m+1}) = (0, b_{m+1})$. At step $n = m+1$: $b_{m+2} = P(m+1)\, b_{m+1}/(m+2)^3$, since $b_m = 0$ kills the second term. At step $n = m+j$ for $j \geq 2$: b_{m+j+1} is a $\mathbb{Q}(m)$-linear combination of b_{m+j} and b_{m+j-1}, both already rational multiples of b_{m+1}. The leading coefficient of $P(n)$ is 34 and the subtracted term has leading coefficient 1, so at each step the degree in m increases by 3 and the leading coefficient remains positive. After $h-1$ steps the numerator has degree $3(h-1)$.

For $h = 2$: $C_2(m) = P(m+1)/(m+2)^3$, with numerator $P(m+1) = (2m+3)(17m^2+51m+39)$, a cubic with leading coefficient 34. □

Lemma 8 (Nonvanishing over $\mathbb{F}_p$)**.** *For every prime $p \geq 7$ and every $2 \leq h \leq p$, the numerator of C_h is not the zero polynomial modulo p.*

Proof. Write N_h for the numerator of C_h (a polynomial of degree $3(h-1)$), satisfying $N_{h+1}(m) = P(m+h)\, N_h(m) - (m+h)^6\, N_{h-1}(m)$, with $N_1 = 1$ and $N_2 = P(m+1)$.

First evaluation. Equation (4) at $m = -1$ (where $b_{-1} = 0$, $b_0 = 1$) gives $C_h(-1) = b_{h-1}$, hence $N_h(-1) = b_{h-1} \cdot \big((h-1)!\big)^3$.

Second evaluation. Setting $m = -2$ in the recurrence gives $N_{h+1}(-2) = P(h-2)\, N_h(-2) - (h-2)^6\, N_{h-1}(-2)$, with $N_1(-2) = 1$ and $N_2(-2) = P(-1) = -5$. By induction using the Apéry recurrence at $n = h-2$:

$$N_h(-2) \;=\; -5\, b_{h-2}\,\big((h-2)!\big)^3 \qquad (h \geq 2). \tag{5}$$

Conclusion. If $N_h \equiv 0 \pmod p$ as a polynomial ($2 \leq h \leq p$, $p \geq 7$), then $N_h(-1) \equiv N_h(-2) \equiv 0$, forcing $b_{h-1} \equiv b_{h-2} \equiv 0 \pmod p$ (since $p \nmid 5$ and $p \nmid (h-1)!$, $(h-2)!$). These are consecutive zeros, contradicting Lemma 6. □

Proposition 9. *For every prime $p \geq 7$,*

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \tfrac{3}{2}(H-1)(H-2) + 1, \tag{6}$$

where H is any integer with $2 \leq H \leq p$. Optimizing $H = \lfloor (p/3)^{1/3} \rfloor + 1$ gives $Z(p) \leq \frac{3^{4/3}}{2}\, p^{2/3} + O(p^{1/3}) = O(p^{2/3})$ unconditionally for all primes. In particular, $Z(p) = o(p)$.

Proof. Partition $\{0,\dots,p-1\}$ into $\lceil p/H\rceil$ consecutive blocks of length at most H. In each block, mark the first zero; all other zeros have gap $h \in \{2,\dots,H-1\}$ to some earlier zero in the same block (gap 1 is impossible by Lemma 6). By Lemma 7, a gap-h pair constrains m to at most $3(h-1)$ residue classes mod p (the denominator factors $(m+j+1)^3$ are nonzero since $0 \le m+j+1 \le p-1$). By Lemma 8, $C_h \not\equiv 0$ over $\mathbb{F}_p$, so the numerator roots are at most $3(h-1)$. Summing:

$$Z(p) \le \left\lceil \frac{p}{H} \right\rceil + \sum_{h=2}^{H-1} 3(h-1) = \frac{p}{H} + \tfrac{3}{2}(H-1)(H-2) + O(1).$$

The minimum over real $H > 0$ occurs at $H = (p/3)^{1/3}$, giving the bound. For $p = 2, 3, 5$: $Z(p) = 0$ (since b_j for $j < 5$ is coprime to $2, 3, 5$). $\square$

Corollary 10 (Sub-interval zero count)**.** *For every prime $p \ge 7$ and every interval $I \subseteq \{0,\dots,p-1\}$,*

$$\#\{j \in I : p \mid b_j\} = O(|I|^{2/3}).$$

Proof. Apply the proof of Proposition 9 to the sub-interval I in place of $\{0,\dots,p-1\}$: partition I into $\lceil |I|/H \rceil$ blocks of length H, bound the non-first zeros in each block by the same gap polynomial argument (Lemma 7 applies regardless of the interval), and optimize. $\square$

Remark 11. The leading coefficient of the numerator of C_h is $U_{h-1}(17)$, the Chebyshev polynomial of the second kind evaluated at 17, satisfying $\ell_1 = 1$, $\ell_{h+1} = 34\ell_h - \ell_{h-1}$. Let $N_h(m)$ denote the cleared numerator of C_h. The identity $P(-u-1) = -P(u)$ (evident from $P(n) = (2n+1)(17n^2 + 17n + 5)$) propagates through the recurrence to give $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$, verified computationally for $h \le 200$. For even h, N_h is an *odd* polynomial in $(m + (h+1)/2)$, forcing the structural factor $(2m+h+1)$. At $h = 2$: factor $(2m+3)$ in $P(m+1) = (2m+3)(17m^2+51m+39)$. The computational data (Section 10) suggests $Z(p) = O(1)$, far stronger than $O(p^{2/3})$.

Remark 12 (Content restriction)*.* A stronger nonvanishing holds: if $p \ge 7$ divides the content of N_h (the gcd of all its coefficients), then $h \ge 2p$. Indeed, $N_h(-r) = (-1)^{r-1} b_{r-1} b_{h-r} \big((r-1)!\big)^3 \big((h-r)!\big)^3$ for $1 \le r \le h$ (by induction from the Apéry recurrence); choosing $r = 1, 2$ and using $p \nmid (h-1)!$, $(h-2)!$, 5 for $h \le p$ recovers Lemma 8, while the range $p < h < 2p$ is excluded similarly via $r = h - p + 1$. Setting $\beta_n = (n!)^3 b_n$ and $F(z) = \sum_{n\ge0} b_n z^n$, the endpoint factorization packages into the double generating function $\sum_{r,s\ge0} \frac{N_{r+s+1}(-r-1)}{(r!)^3 (s!)^3} u^r v^s = F(-u)\,F(v)$, showing that the boundary array is a rank-one tensor.

Remark 13 (Adjacent resultant formula)*.* The adjacent resultants satisfy the exact formula

$$\big|\mathrm{Res}(N_h, N_{h+1})\big| = \prod_{j=1}^{h-1} \big(j!^3\, b_j\big)^6. \tag{7}$$

This follows from the recursion $|\,\mathrm{Res}(N_h, N_{h+1})| = |N_h(-h)|^6\, |\,\mathrm{Res}(N_{h-1}, N_h)|$ (since $N_{h+1} \equiv -(m+h)^6 N_{h-1} \pmod{N_h}$; with the standard signed convention the recursion carries an additional factor $(-1)^{h-1}$ from the scalar minus sign, which is why the formula is stated in absolute value), combined with $N_h(-h) = (-1)^{h-1} N_h(-1) = (-1)^{h-1}((h-1)!)^3 b_{h-1}$ (the reflection identity and the first evaluation from Lemma 8). Verified for $h = 2, 3, 4, 5$; the cases $h = 2, 3, 4$ give $\mathrm{Res}(N_2, N_3) = -5^6$, $\mathrm{Res}(N_3, N_4) = -2^{18} \cdot 5^6 \cdot 73^6$, $\mathrm{Res}(N_4, N_5) = 2^{36} \cdot 3^{18} \cdot 5^{12} \cdot 17^{12} \cdot 73^6$.

A key consequence: $p \mid \mathrm{Res}(N_h, N_{h+1})$ if and only if $p \mid b_j$ for some $1 \le j \le h-1$ (when $p > h$, so factorials are coprime to p). In particular, N_2 and N_3 have disjoint root sets modulo p for all $p \ge 7$ (since $5^6 \ne 0$).

Remark 14 (Sharpness of the bound). The exponent 2/3 in Proposition 9 is optimal for any argument using only the degree bounds $\deg N_h = 3(h-1)$. The exact extremal problem $\max\{\sum r_h : r_h \le 3(h-1),\ \sum h\, r_h \le p\}$ has value $\frac{3}{2}\, p^{2/3} + O(p^{1/3})$, with leading constant 3/2 achieved by filling gap sizes $h = 2, 3, \ldots, H$ to capacity and using the remaining space at $h = H + 1$, where $H \sim p^{1/3}$. Moreover, degree bounds alone (even using all pairs of zeros, not just consecutive pairs) cannot yield $O(p^{1/2})$: an abstract polynomial model on $\sim p^{2/3}$ points can satisfy $\deg F_h \le 3(h-1)$ for all relevant h. Improving the exponent requires genuinely arithmetic input about the family (N_h).

Remark 15 (Dodgson condensation). The Desnanot–Jacobi identity (Lewis Carroll's condensation) applied to the tridiagonal determinant gives the bilinear relation

$$N_{h+2}(m)\, N_h(m{+}1) - N_{h+1}(m)\, N_{h+1}(m{+}1) = -\prod_{j=2}^{h+1}(m{+}j)^6. \tag{8}$$

This is an identity in $\mathbb{Z}[m]$, verified for $h \le 7$. A consequence: for $m \not\equiv -j \pmod p$ $(2 \le j \le h{+}1)$, the pairs $(m, h{+}2)$ and $(m{+}1, h)$ cannot both lie in the zero relation $\mathcal{Z} = \{(m,h) : N_h(m) \equiv 0\}$, since $N_{h+1}(m)\, N_{h+1}(m{+}1)$ would equal a nonzero product of sixth powers. After a meromorphic gauge $\tau_h(x) = N_h(x)/d_h(x)$ satisfying $d_{h+2}(x)\, d_h(x{+}1) = \prod_{j=2}^{h+1}(x{+}j)^6$, this becomes an exact SL_2-tiling: $\tau_{h+1}(x)\, \tau_{h+1}(x{+}1) - \tau_{h+2}(x)\, \tau_h(x{+}1) = 1$, equivalently an A_1 T-system (discrete Liouville equation). A zero of τ forces its transverse neighbors to be units (*singularity confinement*), but abstract SL_2-tilings can have zeros of density 1/2 (e.g., $q_n = 0, 1, 0, -1, \ldots$ with $q_n^2 - q_{n-1}q_{n+1} = 1$), so the bilinear structure alone does not force a sublinear zero count. The arithmetic content—endpoint factorization, restart identity (Lemma 21), and polynomial degree bounds—is essential.

Remark 16 (Projective orbit reformulation). Let $B_n = n!^3\, b_n$ and D_n be the second solution of the renormalized recurrence $w_{n+1} = P(n)\, w_n - n^6\, w_{n-1}$ with $(D_0, D_1) = (0, 1)$. Their Casoratian is $B_n D_{n+1} - D_n B_{n+1} = (n!)^6$, nonzero for $n < p$. Define the *projective Apéry orbit* $\pi\colon \{0, \ldots, p{-}1\} \to \mathbb{P}^1(\mathbb{F}_p)$ by $\pi_n = [B_n : D_n]$. For $0 \le m < m + h < p$, $N_h(m) \equiv 0 \pmod p$ if and only if $\pi_m = \pi_{m+h}$. Thus the zero-count $Z(p)$ equals the number of distinct collisions in the orbit π. The reflection symmetry $P(-1{-}n) = -P(n)$ implies $b_{p-1-n} \equiv b_n \pmod p$, making the orbit a palindrome: $\pi_n = \pi_{p-1-n}$. In particular, $Z(p)$ is even except at the non-ordinary primes $p \mid a_p(f)$ (at most two primes $p \le 10^5$).

Remark 17 (Galois structure). For each fixed h, the Galois group of the splitting field of N_h over $\mathbb{Q}$ is not the full symmetric group: the reflection $N_h(-m{-}h{-}1) = (-1)^{h-1} N_h(m)$ forces roots into pairs $\{r, -r{-}h{-}1\}$, and $\mathrm{Gal}(N_h) \cong C_2 \wr S_m$ (hyperoctahedral group) with $m = \lfloor 3(h{-}1)/2 \rfloor$ (confirmed exactly for $h \le 5$, $2 \le h \le 9$ by obstruction tests). By the Chebotarev density theorem, the average number of $\mathbb{F}_p$-roots of N_h is 1 for odd h and 2 for even h. This prediction is verified: for 262 primes in $[1000, 3000]$, the measured averages are $0.98, 2.03, 1.11, 2.07, 1.11, 1.88$ for $h = 3, \ldots, 8$.

Proposition 18 (Hyperoctahedral Poisson law)**.** *Assume* $\mathrm{Gal}(N_h) \cong C_2 \wr S_m$ *with* $m = \lfloor 3(h{-}1)/2 \rfloor$. *Let* $Y_h(p)$ *be the number of non-central reflection-pairs of roots of* N_h *that split completely over* $\mathbb{F}_p$. *Then for each fixed* $k \le m$, *the factorial moments in the prime-aspect satisfy*

$$\lim_{x \to \infty} \frac{1}{\pi(x)} \sum_{p \le x} (Y_h(p))_k = \left(\frac{1}{2}\right)^k,$$

where $(Y)_k = Y(Y{-}1)\cdots(Y{-}k{+}1)$. *Since these match the factorial moments of* Poisson(1/2) *for all* k, *the prime-aspect distribution of* $Y_h(p)$ *converges to* Poisson(1/2) *as* $h \to \infty$.

Proof. In $G = C_2 \wr S_m = \{(\sigma,\varepsilon) : \sigma \in S_m,\ \varepsilon \in \{\pm 1\}^m\}$, a *positive fixed point* of $g = (\sigma,\varepsilon)$ is an index i with $\sigma(i) = i$ and $\varepsilon_i = +1$. The Frobenius at p acts on the m reflection-pairs, and $Y_h(p)$ equals the number of positive fixed points. For any k distinct indices $i_1,\dots,i_k$, the fraction of $g \in G$ fixing all k positively is

$$\frac{2^{m-k}\,(m-k)!}{2^m\,m!} = \frac{1}{2^k\,(m)_k}.$$

Summing over all $(m)_k$ ordered k-tuples, $\mathbf{E}_{g\in G}[(Y(g))_k] = (1/2)^k$, which matches the factorial moments of Poisson$(1/2)$. The Chebotarev density theorem then gives the prime-aspect convergence (Remark 20 supplies the effective error term under GRH). Since the factorial moments determine the Poisson distribution and they hold for all $k \le m$, $Y_h(p) \to$ Poisson$(1/2)$ as $m \to \infty$. □

Remark 19 (Prime-aspect vs. orbit-aspect). Proposition 18 is a *prime-aspect* statement: for fixed h and varying p, the root-pair count $Y_h(p)$ converges to Poisson$(1/2)$. The zero-count $Z(p)$ aggregates contributions from all gaps $h = 1,\dots,p-1$ at a *fixed* prime p, so the orbit-aspect independence needed to deduce $Z(p)/2 \sim$ Poisson$(1/2)$ is not implied by Chebotarev alone. The gap between the two aspects parallels the gap between the Sato–Tate distribution (average over primes) and pointwise distribution (a single prime).

Remark 20 (GRH-effective error). For each fixed k, the k-th factorial moment does not require the full splitting field K_h (of degree $2^m m!$, double-exponential in h). Let $E_{h,k}$ be the fixed field of the stabilizer of an ordered signed k-tuple with distinct underlying reflection pairs; then $[E_{h,k} : \mathbb{Q}] = 2^k\,(m)_k = O_k(h^k)$. Under GRH for $\zeta_{E_{h,k}}$,

$$\frac{1}{\pi(x)} \sum_{\substack{p\le x\\ p\text{ good}}} (Y_h(p))_k = 2^{-k} + O_k\!\left(\frac{\log D_{h,k} + 2^k (m)_k \log x}{\sqrt{x}}\right),$$

where $D_{h,k} = |\operatorname{Disc}(E_{h,k})|$. The Apéry recurrence gives the coefficient-height bound $\log \|N_h\|_1 = O(h\log h)$, from which $\log D_{h,k} \ll_k h^{k+2}\log h$. Thus an $O(h^{-C})$ error holds in the polynomial range $x \ge h^{2k+4+2C+\varepsilon}$, for each fixed k.

At $x = 10^6$ with $h = 10$ $(m = 13)$, the degree contributions alone for $k = 1,2,3$ are 0.25, 5.99, and 131.7 respectively: the observed Poisson accuracy is far better than present effective Chebotarev can certify.

5.2 Dual bounds: the restart identity and column estimate

The bound $Z(p) = O(p^{2/3})$ controls zeros along a *fixed row* (varying m for a given gap h). A dual bound controls zeros along a *fixed column* (varying h for a given starting point x_0).

Lemma 21 (Restart identity)**.** *Suppose $N_r(x_0) = 0$ and $N_{r+1}(x_0) \ne 0$ for some $x_0 \in \mathbb{F}_p$ and $r \ge 1$. Then for every $d \ge 0$,*

$$N_{r+d}(x_0) = N_{r+1}(x_0)\cdot N_d(x_0 + r). \tag{9}$$

In particular, if $N_{r+d}(x_0) = 0$ as well, then $N_d(x_0 + r) = 0$.

Proof. The sequence $Y_d = N_{r+d}(x_0)/N_{r+1}(x_0)$ has $Y_0 = 0$, $Y_1 = 1$, and satisfies $Y_{d+1} = P((x_0{+}r){+}d)\,Y_d - ((x_0{+}r){+}d)^6\,Y_{d-1}$, which is the defining recurrence for $N_d(x_0 + r)$. □

Call a column $x_0 \in \mathbb{F}_p$ *polluted* if $x_0 \equiv -m \pmod p$ for some $2 \le m < p$ with $p \mid b_{m-1}$. By the endpoint factorization $N_h(-m) = \pm\, b_{m-1} b_{h-m}((m{-}1)!)^3((h{-}m)!)^3$ (Remark 12), a polluted column satisfies $N_h(x_0) \equiv 0$ for *every* $h \ge m$: the column recurrence in h reaches two consecutive zeros and then vanishes identically. Conversely, two consecutive zero levels $N_{h-1}(x_0) = N_h(x_0) = 0$ propagate backwards through the column recurrence $N_{h+1} = P(x_0{+}h)N_h - (x_0{+}h)^6 N_{h-1}$ until a vanishing coefficient blocks the descent, which forces $x_0 \equiv -m$ with $N_m(-m) = \pm\, b_{m-1}((m{-}1)!)^3 \equiv 0$, i.e. $p \mid b_{m-1}$. Thus pollution is *exactly* the presence of two consecutive zero levels, and the number of polluted columns with $m \le M$ is the zero count $\#\{j < M : p \mid b_j\} = O(M^{2/3})$ (Corollary 10).

Proposition 22 (Column bound, unpolluted case)**.** *Let $x_0 \in \mathbb{F}_p$ be a column whose zero levels in the interval I (of H consecutive h-values, $H \le p$) contain no two consecutive integers—in particular, any unpolluted column. Then*

$$\#\{h \in I : N_h(x_0) \equiv 0 \pmod p\} \le \tfrac{3^{4/3}}{2}\, H^{2/3} + O(H^{1/3} + 1). \tag{10}$$

Proof. Order the zero levels in I. For threshold $L \ge 2$: a gap of length $d \le L$ between consecutive zeros at levels $r < r + d$ gives $N_d(x_0 + r) = 0$ by Lemma 21 (the hypothesis $N_{r+1}(x_0) \ne 0$ holds because the zero levels contain no two consecutive integers). For fixed d, there are at most $3(d{-}1)$ values of $x_0 + r$ satisfying this (by Lemma 8 and $\deg N_d = 3(d{-}1)$). Gaps longer than L contribute at most $H/(L{+}1)$ zeros. Summing:

$$\text{zero count} \le \frac{H}{L+1} + \sum_{d=2}^{L} 3(d{-}1) + O(1) = \frac{H}{L+1} + \tfrac{3}{2}L(L{-}1) + O(1).$$

Optimizing $L \asymp (H/3)^{1/3}$ gives the stated bound. $\square$

Polluted columns genuinely violate the bound (10): at $p = 73$ (where $b_2 = 73$), the column $x_0 \equiv -3$ has $N_h(-3) \equiv 0$ for every $h \ge 3$. Computationally, at mesoscopic scale $H = \lfloor \sqrt{p} \rfloor$ the *unpolluted* column counts are far below the $O(H^{2/3})$ bound: for all primes $p \le 5000$ and all unpolluted x_0, $\#\{h \le H : N_h(x_0) \equiv 0\} \le 4$ (maximum attained at $p = 653$), while the polluted columns in this range reach counts as large as 50 (at $p = 3331$, where $b_7 = 3331 \cdot 175415$, so the column $x_0 \equiv -8$ vanishes for every $h \ge 8$).

Remark 23 (Rectangular incidence estimate). The row bound $|R_h| \le 3(h{-}1)$ and the column bound (10) together give the rectangular incidence estimate

$$\#\{(x,h) : x \in \mathbb{F}_p,\ 2 \le h \le H,\ N_h(x) \equiv 0\} \le \min\{H^2,\ pH^{2/3}\}.$$

The first bound sums row degrees; the second sums the unpolluted column estimates plus the polluted-column contribution $\le Z(p)\cdot H \ll p^{2/3}H \le pH^{2/3}$ (for $H \le p$), which does not change the order. They cross at $H \sim p^{3/4}$. Note that a uniform bound $R_p(H) \le CH$ for all primes is *false*: by the Chebotarev density theorem, for each fixed H, infinitely many primes $p > H^2$ have every N_h $(2 \le h \le H)$ splitting completely over $\mathbb{F}_p$, giving $R_p(H) = \frac{3}{2}H(H-1)$. The correct target is the *mesoscopic* estimate $R_p(\lfloor\sqrt{p}\rfloor) = O(\sqrt{p})$, where the split-prime obstruction does not apply because $H = \sqrt{p}$ grows with p. More generally, if $R_p(H) \ll H^{\beta}$ then $Z(p) \le p/H + R_p(H)$ is optimized at $H \sim p^{1/(\beta+1)}$, giving $Z(p) \ll p^{\beta/(\beta+1)}$: the current $\beta = 2$ gives $2/3$ and $\beta = 1$ yields $1/2$. Reflection forces $R_p(H) \ge H/2$ from central roots, so $\beta = 1$ is the intrinsic limit of this first-moment method.

5.3 The continuant Bezout identity

Write $S_d(x) = \prod_{j=2}^{d}(x + j)$.

Proposition 24 (Bezout identity). *For integers $2 \le d < e$,*

$$N_{e-1}(x{+}1)\, N_d(x) - N_{d-1}(x{+}1)\, N_e(x) = S_d(x)^6\, N_{e-d}(x+d). \tag{11}$$

In particular, after saturating the cut-edge divisor S_d,

$$(N_d, N_e) : S_d^\infty = (N_d,\ N_{e-d}(x+d)) : S_d^\infty.$$

Proof. The continuant addition formula (splitting the interval $[1, e{-}1]$ at position d) gives $N_e = N_d\, N_{e-d+1}(x{+}d{-}1) - (x{+}d)^6\, N_{d-1}\, N_{e-d}(x{+}d)$. The Dodgson identity at index $d{-}2$ gives $N_{d-1}(x{+}1)\, N_{d-1} - N_{d-2}(x{+}1)\, N_d = \prod_{j=2}^{d-1}(x{+}j)^6$. Combining these two identities yields (11). Verified computationally for all $2 \le d < e \le 9$. $\square$

Corollary 25 (Separated-block criterion). *Away from the cut-edge divisor $S_d(x) = 0$,*

$$N_d(\alpha) \equiv N_e(\alpha) \equiv 0 \pmod p \quad\Longleftrightarrow\quad N_d(\alpha) \equiv N_{e-d}(\alpha + d) \equiv 0 \pmod p.$$

In particular, the saturated *gcd satisfies* $\deg\big(\mathrm{gcd}_{\mathbb{F}_p}(N_d, N_e) : S_d^\infty\big) \le 3(\min(d, e{-}d) - 1)$*, which improves the naive $3(d{-}1)$ when $e - d < d$. The saturation is essential: the full gcd may contain cut-edge factors dividing S_d. For example, at $p = 73$ (where $b_2 = 73$),* $\mathrm{gcd}_{\mathbb{F}_{73}}(N_3, N_4) = X + 3$ *even though $N_{e-d} = N_1 = 1$; the factor $X + 3$ divides $S_3 = (x{+}2)(x{+}3)$. The unsaturated degree can exceed the displayed bound by at most* $\deg S_d = d - 1$.

Remark 26 (Separated-block resultant). Define $\mathcal{R}_{d,r} = \mathrm{Res}_x(N_d(x), N_r(x{+}d))$. The Bezout identity gives

$$|\,\mathrm{Res}(N_d, N_e)| = \prod_{j=1}^{d-1} \big((j!)^3 b_j\big)^6 \cdot |\mathcal{R}_{d,e-d}|.$$

For adjacent indices ($e = d + 1$), $N_1 = 1$ gives $\mathcal{R}_{d,1} = 1$, recovering the adjacent resultant formula. For nonadjacent indices, $\mathcal{R}_{d,r}$ is genuinely new. The reflection identity gives $|\mathcal{R}_{d,r}| = |\mathcal{R}_{r,d}|$. The subtraction step does *not iterate*: after one application, $N_d(x)$ and $N_{e-d}(x + d)$ live on disjoint shifted intervals with opposite orientations, so no second nested division is possible. For $d = 2$, writing $P(y) = (2y{+}1)Q(y)$ with $Q(y) = 17y^2{+}17y{+}5$ and θ a root of Q, the norm decomposition $\mathcal{R}_{2,r} = 2^{3(r-1)} G_r(-\tfrac{1}{2}) \cdot 17^{3(r-1)}\, \mathrm{N}_{\mathbb{Q}(\theta)/\mathbb{Q}}(G_r(\theta))$ (where $G_r = N_r(x + 1)$) gives $\mathcal{R}_{2,2} = 2^5 \cdot 5^4 \cdot 11^2 \cdot 17^3 \cdot 71$. At $p = 71$: $N_2(-5/2) \equiv N_4(-5/2) \equiv 0 \pmod{71}$ with $-5/2 \notin \{-1, -2, -3, -4\}$, a nonboundary common root inside the mesoscopic range ($4^3 < 71$). This shows the boundary-only gcd statement is *false*.

Remark 27 (Collision energy and the $H^{1/2}$ barrier). The column exponent $2/3$ is the natural barrier for the consecutive-gap argument: even if each gap of length d has only d (rather than $3(d{-}1)$) admissible starts, the sum $\sum_{d \le L} d \asymp L^2$ still gives $H/L + L^2$, optimized at $H^{2/3}$. To achieve $H^{1/2}$, one needs the aggregate collision energy $\mathcal{E}_p(H) = \sum_{d+r \le H} \#\{x : N_d(x) \equiv N_r(x{+}d) \equiv 0\} \ll H^{3/2}$, which is a sparse-exception theorem for the separated-block resultant family—beyond the scope of universal continuant identities. (The A_1 T-system $\tau_{h+1}(x)\tau_{h+1}(x + 1) - \tau_{h+2}(x)\tau_h(x + 1) = 1$ admits tame solutions with zero density 1/2, e.g. $\tau_{2k} = 0$, $\tau_{2k+1} = (-1)^k$, so singularity confinement alone cannot force sparsity.) In the transfer-matrix formulation (writing $M(n) = \left(\begin{smallmatrix} P(n) & -n^6 \\ 1 & 0 \end{smallmatrix}\right)$, so that $N_h(x) = 0$ iff a fixed matrix coefficient of the interval product $M(x{+}h{-}1) \cdots M(x{+}1)$ vanishes), the conjectured $\mathcal{E}_p(H) \ll H^{3/2}$ is analogous to a Rudnev-type point–plane incidence bound in $\mathrm{PGL}_2(\mathbb{F}_p)$, subject to a nonconcentration hypothesis for the deterministic Apéry cocycle. Computation for selected primes:

p	71	503	2003	5003	7919	20011	50021	10^5+3
$H = \lfloor\sqrt{p}\rfloor$	8	22	44	70	88	141	223	316
$R_p(H)/H$	1.75	1.59	1.36	1.50	1.75	1.63	1.56	1.46
$\mathcal{E}_p(H)$	1	2	2	2	10	0	2	0
$\mathcal{E}_p(H)/H^{3/2}$	.044	.019	.007	.003	.012	0	.001	0

The ratio $R_p(H)/H$ clusters around 3/2 (the Chebotarev prediction from Remark 17), and $\mathcal{E}_p(H)/H^{3/2} \to 0$, consistent with $\mathcal{E}_p(H) \ll H^{3/2}$. The key arithmetic object is the shifted resultant $S_{d,r} = \mathrm{Res}_x(N_d(x), N_r(x+d)) \in \mathbb{Z}$: for $p \nmid S_{d,r}$, the gap polynomials N_d and $N_r(\cdot+d)$ share no root over $\mathbb{F}_p$, so that gap pair contributes zero to $\mathcal{E}_p$. A Hadamard bound gives $\log|S_{d,r}| \ll (d+r)^2 \log(d+r)$, so the total number of prime divisors across all pairs (d,r) with $d+r \le H$ is $O(H^4 \log H)$. Two caveats are essential here (they are developed in the separated-resultant subsection below). First, this counts the *support* $W_p(H) = \#\{(d,r) : \exists x,\ N_d(x) \equiv N_r(x+d) \equiv 0\}$, not the energy: one supported pair can carry up to $3(\min(d,r)-1)$ root witnesses, so the averaging conclusion $W_p(H) \ll H^{1/2}$ for all but $o(X/\log X)$ primes holds for $H \le X^{2/7-\varepsilon}$, while the corresponding energy statement $\mathcal{E}_p(H) \ll H^{1/2}$ only follows in the shorter range $H \le X^{2/9-\varepsilon}$. Second, $p \mid S_{d,r}$ detects a common *projective* root over $\overline{\mathbb{F}}_p$; pairs with $p \mid \ell_d, \ell_r$ (an infinity collision) or with a nonsplit common quadratic factor lie in the resultant support without contributing any $\mathbb{F}_p$-root witness. Either way this is rigorous but far from the global target $H = p-1$.

A sharper route uses the bipartite incidence graph $\Gamma_p(H)$: left vertices are gap lengths d, right vertices are gap lengths r, with an edge when N_d and $N_r(\cdot + d)$ share a root over $\mathbb{F}_p$. Each edge certifies a triple collision, so $\mathcal{E}_p(H) = e(\Gamma_p(H))$. A uniform fiber bound $\deg_L(d) \le C$ (the number of r-partners for each d) and a uniform codegree bound (at most C common right neighbors for any two left vertices) would place $\Gamma_p(H)$ in the regime of the Kővári–Sós–Turán theorem, giving $\mathcal{E}_p(H) \ll H^{3/2}$—the collision-energy scale needed for the mesoscopic target $R_p(\sqrt{p}) = O(\sqrt{p})$. The Dodgson identity (Remark 15) restricts local configurations in Γ_p, but by itself controls $O(H)$ pairs out of $O(H^2)$; upgrading it to a global codegree bound is the key missing step.

5.4 Value fibers: the bordered certificate

The zero-count bound $Z(p) = O(p^{2/3})$ (Proposition 9) concerns the single fiber $a = 0$, where one return condition $b_t \equiv b_{t+h} \equiv 0$ already forces a gap polynomial to vanish. For a *general* fiber $\{t : b_t \equiv a \pmod{p}\}$ with $a \neq 0$, two return conditions are needed to eliminate the two unknowns (b_{t+1}, a), and the resulting certificate is a bordered 2×2 determinant $D_{h,k}$ built from the gap polynomials and the second solution of the cleared recurrence. The determinant factors exactly as $D_{h,k} = \Pi_h \cdot \Delta_{h,k}$, and the analogue of Lemma 8 for the reduced certificate $\Delta_{h,k}$ is subtler: the endpoint evaluations assert equalities of separated Apéry values rather than consecutive zeros, so the no-consecutive-zero mechanism does not apply (an explicit two-point obstruction occurs at $p = 131$, $(h,k) = (12,55)$; see the proof below). Nevertheless, a centered-coefficient analysis—using the Pell–Chebyshev structure of the leading coefficients rather than endpoint values—proves the nonvanishing in the *full* range $1 \le h < k < p$ (Theorem 32 below), and the range $k < p$ is sharp: at $p = 7$ the certificates $\Delta_{h,21}$ vanish identically. Consequently the same block-partition count that proves $Z(p) = O(p^{2/3})$ yields an *unconditional uniform* fiber bound $O(p^{3/4})$. We first state the counting mechanism conditionally, then prove the nonvanishing.

Proposition 28 (Uniform fiber bound conditional on bordered nonvanishing)**.** *Let*

$$P(n) = 34n^3 + 51n^2 + 27n + 5$$

and let $(b_n)_{n\ge 0}$ *be the Apéry numbers, with* $b_0 = 1$*,* $b_1 = 5$*, and*

$$(n+1)^3 b_{n+1} = P(n)b_n - n^3 b_{n-1}.$$

For a prime $p \ge 7$ *and* $a \in \mathbb{F}_p$*, put*

$$N_p(a) = \#\{0 \le t < p : b_t \equiv a \pmod p\}.$$

Define $N_0(x) = 0$*,* $N_1(x) = 1$*, and, for* $m \ge 1$*,*

$$N_{m+1}(x) = P(x+m)N_m(x) - (x+m)^6 N_{m-1}(x),$$

and set

$$\Pi_m(x) = \prod_{j=1}^{m} (x+j)^3, \qquad \begin{aligned} \Delta_{h,k}(x) = N_h(x)\Pi_{k-h}(x+h) - N_k(x) \\ + \Pi_h(x)N_{k-h}(x+h). \end{aligned}$$

Let H *be an integer with* $2 \le H \le (p-1)/2$*, and assume*

$$\Delta_{h,k} \not\equiv 0 \pmod p \quad \text{in } \mathbb{F}_p[x] \qquad (1 \le h < k \le H). \tag{NV$_{p,H}$}$$

Then, for every $a \in \mathbb{F}_p^\times$*,*

$$N_p(a) \le 2\left\lfloor \frac{p-1}{H+1} \right\rfloor + \frac{H(H-1)(2H-1)}{2} + 2. \tag{12}$$

In particular, if $(\mathrm{NV}_{p,H})$ *holds for*

$$H = \left\lceil (2p/3)^{1/4} \right\rceil,$$

then

$$N_p(a) \le 4(2/3)^{3/4} p^{3/4} + \sqrt{6}\, p^{1/2} + 3(2/3)^{1/4} p^{1/4} + 3. \tag{13}$$

Thus the leading constant supplied by this argument is

$$4(2/3)^{3/4} = 2.951151\ldots.$$

Proof. We first establish the exact algebra behind the bordered certificate. For a base point r put

$$Y_m = \left(\frac{(r+m)!}{r!}\right)^3 b_{r+m} = \Pi_m(r) b_{r+m}.$$

As long as $0 \le r$ and $r + m \le p - 1$, clearing the denominators in the Apéry recurrence gives

$$Y_{m+1} = P(r+m)Y_m - (r+m)^6 Y_{m-1}, \tag{14}$$

with $Y_0 = b_r$ and $Y_1 = (r+1)^3 b_{r+1}$. Let $B_0(x) = 1$, $B_1(x) = 0$, and let B_m satisfy the same recurrence in m as N_m. The two fundamental solutions of (14) give

$$Y_m = N_m(r)Y_1 + B_m(r)Y_0. \tag{15}$$

Moreover,

$$B_m(x) = -(x+1)^6 N_{m-1}(x+1) \qquad (m \ge 1). \tag{16}$$

Indeed, this has the correct value at $m = 1$, and its induction step is exactly the recurrence for $N_{m-1}(x+1)$.

Write $R_m(x) = \Pi_m(x) - B_m(x)$. If $b_t = b_{t+m} = a$, equation (15) becomes

$$N_m(t)(t+1)^3 b_{t+1} - R_m(t)a = 0. \tag{17}$$

Consequently, if $b_t = b_{t+h} = b_{t+k} = a$ with $1 \le h < k$ and $t + k \le p - 1$, the nonzero vector $((t+1)^3 b_{t+1}, a)$ is killed by the two rows in (17). Hence

$$D_{h,k}(t) = 0, \qquad D_{h,k}(x) := N_h(x)R_k(x) - N_k(x)R_h(x). \tag{18}$$

The structural factors of this determinant are important. A continuant Casoratian calculation gives

$$N_h B_k - N_k B_h = -\Pi_h^2 N_{k-h}(x+h). \tag{19}$$

For completeness, both sides, as functions of k, satisfy the same second-order recurrence; at $k = h$ they vanish, while at $k = h+1$ they equal $-\Pi_h^2$. Substituting (19) in (18) yields the exact factorization

$$\boxed{D_{h,k}(x) = \Pi_h(x)\Delta_{h,k}(x).} \tag{20}$$

Let $\ell_0 = 0$, $\ell_1 = 1$, and $\ell_{m+1} = 34\ell_m - \ell_{m-1}$. Since ℓ_m is the leading coefficient of N_m, the leading coefficient of $\Delta_{h,k}$ is

$$\ell_h + \ell_{k-h} - \ell_k < 0.$$

The inequality follows, for example, from the strict increase of (ℓ_m) and $\ell_{m+1} > 2\ell_m$. Therefore, over $\mathbb{Z}$,

$$\deg \Delta_{h,k} = 3(k-1), \qquad \deg D_{h,k} = 3(h+k) - 3. \tag{21}$$

The leading coefficient can, however, vanish modulo p, so its characteristic-zero nonvanishing does not prove $(\mathrm{NV}_{p,H})$.

Besides the cubic boundary factor Π_h, interval reflection gives all three immediately forced midpoint factors:

$$\begin{aligned} 2 \mid h &\implies 2x + h + 1 \mid \Delta_{h,k}, \\ 2 \mid k &\implies 2x + k + 1 \mid \Delta_{h,k}, \\ 2 \mid (k-h) &\implies 2x + h + k + 1 \mid \Delta_{h,k}. \end{aligned} \tag{22}$$

These follow by applying $N_m(-x-m-1) = (-1)^{m-1}N_m(x)$ to the first interval, the whole interval, and the translated final interval, respectively. Exactly one of the three lengths $h, k, k-h$ is even unless h and k are both even, in which case all three are even. These are the factors forced by the three reflections; in the exceptional smallest case $\Delta_{1,2} = -4(2x+3)^3$.

We record why the known endpoint facts do not establish the missing nonvanishing. Using Remark 12 in (20) gives, for $1 \le r \le h$,

$$\Delta_{h,k}(-r) = (-1)^{r-1} b_{r-1}\big(b_{h-r} - b_{k-r}\big)\big((r-1)!(k-r)!\big)^3, \tag{23}$$

whereas, for $h < r \le k$,

$$\Delta_{h,k}(-r) = (-1)^{r-1} b_{k-r}\big(b_{r-h-1} - b_{r-1}\big)\big((r-1)!(k-r)!\big)^3. \tag{24}$$

In particular, the raw determinant vanishes structurally at $-1, \ldots, -h$. After removing Π_h, its first two evaluations (when $h \ge 2$) are

$$\Delta_{h,k}(-1) = (k-1)!^3(b_{h-1} - b_{k-1}), \quad \Delta_{h,k}(-2) = -5(k-2)!^3(b_{h-2} - b_{k-2}).$$

Thus their simultaneous vanishing gives two equalities, not two consecutive zeros, and Lemma 6 does not contradict it. For example, modulo 131, the recurrence gives

$$b_{10} = b_{53} = 15, \qquad b_{11} = b_{54} = 15;$$

hence both evaluations vanish for $(h,k) = (12,55)$, while $\Delta_{12,55}(-3) = 2$. Removing the forced midpoint factor $2x+13$ does not change this example, since it is nonzero at -1 and -2. Lemma 8 proves the corresponding assertion for the individual N_m, but it does not imply $(\mathrm{NV}_{p,H})$ for this bordered combination.

We now use the explicitly stated hypothesis $(\mathrm{NV}_{p,H})$. List the a-fiber in increasing order as

$$t_1 < t_2 < \cdots < t_s.$$

For $1 \le i \le s-2$ put

$$h_i = t_{i+1} - t_i, \qquad k_i = t_{i+2} - t_i.$$

If $k_i \le H$, then (20) and $t_i + k_i \le p-1$ show that $\Pi_{h_i}(t_i)$ is a unit modulo p. Thus t_i is a root of the nonzero polynomial Δ_{h_i,k_i}. For a fixed pair (h,k) there are at most $3(k-1)$ such bases, by (21). Summing over all short patterns gives

$$\sum_{k=2}^{H}\sum_{h=1}^{k-1} 3(k-1) = 3\sum_{j=1}^{H-1} j^2 = \frac{H(H-1)(2H-1)}{2}. \tag{25}$$

For the remaining indices, $t_{i+2} - t_i > H$, hence this integer is at least $H+1$. Splitting the indices i according to parity, the corresponding increments telescope and have total at most $p-1$ in each parity class. The number of long indices is therefore at most

$$2\left\lfloor \frac{p-1}{H+1} \right\rfloor.$$

Together with the two final fiber elements, this proves (12).

Finally put $x = (2p/3)^{1/4}$ and $H = \lceil x \rceil$. This H lies in $[2,(p-1)/2]$ for $p \ge 7$. Using $H(H-1)(2H-1)/2 < H^3 \le (x+1)^3$ and $2(p-1)/(H+1) < 2p/x = 3x^3$, the right-hand side of (12) is at most

$$4x^3 + 3x^2 + 3x + 3,$$

which is exactly (13). $\square$

Corollary 29 (Collision energy, conditional form)**.** *Under the bordered nonvanishing hypothesis of Proposition 28, let $U(p)$ denote the right-hand side of* (13)*. Then*

$$E(p) := \sum_{a\in\mathbb{F}_p} N_p(a)^2 \le p\max\{Z(p), U(p)\} = O(p^{7/4}).$$

Proof. By Proposition 9, $N_p(0) = Z(p) = O(p^{2/3})$, while the conditional proposition gives $N_p(a) \le U(p) = O(p^{3/4})$ for $a \ne 0$. Since $\sum_a N_p(a) = p$,

$$E(p) \le \big(\max_a N_p(a)\big)\sum_a N_p(a) \le p\max\{Z(p), U(p)\} = O(p^{7/4}).$$

$\square$

Proposition 30 (An exact degeneration outside the nonsingular range)**.** *Over* $\mathbb{F}_7[x]$ *one has*

$$\Delta_{h,21}(x) = 0 \qquad (1 \le h < 21).$$

In particular, a restriction on the size of k relative to the characteristic is essential.

Proof. Put

$$A_j(x) = \begin{pmatrix} P(x+j) & -(x+j)^6 \\ 1 & 0 \end{pmatrix}, \qquad T = A_7 A_6 \cdots A_1, \qquad q = x^7 - x.$$

All matrices in this proof are over $\mathbb{F}_7[x]$. Expanding the seven factors, equivalently applying the continuant recurrence seven times, gives the exact identities

$$\operatorname{tr} T = -q^3, \qquad \det T = \prod_{j=1}^{7} (x+j)^6 = q^6.$$

The second identity also follows immediately by multiplying the determinants. Cayley–Hamilton now gives

$$T^2 + q^3 T + q^6 I = 0, \qquad \text{hence} \qquad T^3 = q^9 I. \tag{26}$$

Since $A_{j+7} = A_j$ in characteristic 7, the transfer matrix for twenty-one steps is T^3. Applying (26) to the initial vectors $(N_1, N_0)^\mathsf{T} = (1,0)^\mathsf{T}$ and $(B_1, B_0)^\mathsf{T} = (0,1)^\mathsf{T}$ yields

$$N_{21} = 0, \qquad B_{21} = q^9.$$

On the other hand,

$$\Pi_{21} = \left(\prod_{j=1}^{21} (x+j) \right)^3 = q^9.$$

Thus $R_{21} = \Pi_{21} - B_{21} = 0$, and consequently

$$D_{h,21} = N_h R_{21} - N_{21} R_h = 0.$$

The factorization $D_{h,21} = \Pi_h \Delta_{h,21}$ and the fact that $\mathbb{F}_7[x]$ is an integral domain finish the proof. □

We next compute the few leading coefficients that detect every possible degeneration before a singular step is reached. Recall that $\ell_0 = 0$, $\ell_1 = 1$, and $\ell_{m+1} = 34\ell_m - \ell_{m-1}$.

Lemma 31 (Centered continuant coefficients)**.** *For $m \ge 1$, set*

$$\widehat{N}_m(Y) = N_m\left(Y - \frac{m+1}{2} \right).$$

Let u_m and v_m be respectively the coefficients of Y^{3m-5} and Y^{3m-7} in $\widehat{N}_m$; a coefficient with a negative exponent is understood to be zero. Then

$$u_m = \frac{-(m-1)(32m^2 - 64m - 11)\ell_m + 5m\ell_{m-1}}{256}, \tag{27}$$

$$v_m = \frac{(m-1)(5120m^5 - 31744m^4 + 62016m^3 - 19264m^2 - 37024m - 2095)\ell_m}{655360} - \frac{m(320m^3 - 1600m^2 - 320m + 621)\ell_{m-1}}{131072}. \tag{28}$$

Moreover, if

$$\widehat{\Pi}_m(Y) = \Pi_m\left(Y - \frac{m+1}{2}\right),$$

then

$$\widehat{\Pi}_m(Y) = Y^{3m} - \frac{m(m^2-1)}{8}Y^{3m-2} + \textit{terms of degree at most } 3m-4. \tag{29}$$

All these identities hold over $\mathbb{Z}[1/10]$.

Proof. The reflection identity $N_m(-x-m-1) = (-1)^{m-1}N_m(x)$ shows that $\widehat{N}_m$ contains only powers of the same parity as $3m-3$. To compare its first three possible powers, center the recurrence for N_{m+1} by putting $T = x + (m+2)/2$. The natural centered variables for N_m and N_{m-1} are then $T - 1/2$ and $T - 1$, respectively, and

$$P(x+m) = 34\left(T + \frac{m-1}{2}\right)^3 + \frac{3}{2}\left(T + \frac{m-1}{2}\right), \qquad (x+m)^6 = \left(T + \frac{m-2}{2}\right)^6.$$

Comparison of the coefficients two and four places below the leading one gives

$$u_{m+1} = 34u_m - u_{m-1} - \frac{3}{4}(17m^2 - 17m - 2)\ell_m + \frac{3}{4}m(m-2)\ell_{m-1}, \tag{30}$$

$$v_{m+1} = 34v_m - v_{m-1} - \frac{3}{4}(17m^2 - 17m - 36)u_m + \frac{3}{4}(m^2 - 2m - 4)u_{m-1} \tag{31}$$

$$+ \frac{3(m-1)(17m^3 + 51m^2 - 90m - 24)}{64}\ell_m - \frac{3m(m-2)(m^2-6)}{16}\ell_{m-1}. \tag{32}$$

For completeness, the coefficient extraction in the second line uses only

$$e_2 = \frac{S_1^2 - S_2}{2}, \qquad e_4 = \frac{S_1^4 - 6S_1^2S_2 + 3S_2^2 + 8S_1S_3 - 6S_4}{24}$$

for the appropriate repeated shifts. The cases $m = 1, 2, 3$ are checked directly from $N_1 = 1$ and $N_2 = P(x+1)$. Substitution of (27) and (28) into (30)–(32), followed by $\ell_{m+1} = 34\ell_m - \ell_{m-1}$, makes the coefficients of both ℓ_m and ℓ_{m-1} agree identically. This proves the two closed forms by induction.

Finally,

$$\widehat{\Pi}_m(Y) = \prod_{j=1}^{m}\left(Y + j - \frac{m+1}{2}\right)^3.$$

The repeated shifts have sum zero, while the coefficient two below the top is

$$-\frac{3}{2}\sum_{j=1}^{m}\left(j - \frac{m+1}{2}\right)^2 = -\frac{m(m^2-1)}{8}.$$

This proves (29). □

Theorem 32 (Range nonvanishing of the bordered certificate)**.** *Let* $p \geq 7$ *be prime. For every pair of integers*

$$1 \leq h < k < p,$$

the reduced bordered certificate satisfies

$$\Delta_{h,k} \not\equiv 0 \pmod{p} \qquad \text{in } \mathbb{F}_p[x].$$

Consequently $(\mathrm{NV}_{p,H})$ *holds for every* $H \leq p-1$*; in particular it holds throughout the range required in Proposition 28.*

Proof. Work in $\mathbb{F}_p$. Put

$$d = k - h, \qquad n = 3(k-1), \qquad z = x + \frac{k+1}{2}.$$

The two centered variables in the first product defining $\Delta_{h,k}$ are $z - d/2$ and $z + h/2$; in the last product they occur in the opposite order. Write

$$\Delta_{h,k}(x) = Lz^n + C_1 z^{n-1} + C_2 z^{n-2} + \cdots .$$

Multiplying the centered expansions in Lemma 31 gives

$$L = \ell_h + \ell_d - \ell_k, \tag{33}$$

$$C_1 = \frac{3}{2}(d\ell_h - h\ell_d), \tag{34}$$

$$\begin{aligned} C_2 &= u_h + u_d - u_k \\ &\quad + \frac{d(-d^2 + 1 + 12d - 3hk)\ell_h + h(-h^2 + 1 + 12h - 3dk)\ell_d}{8}. \end{aligned} \tag{35}$$

For example, in the first product the degree-two loss from the shifted leading powers is $3d(4d-hk)/8$; adding the second coefficient of $\widehat{\Pi}_d$ gives the coefficient displayed in (35).

Suppose for a contradiction that $\Delta_{h,k} = 0$. Then $L = C_1 = C_2 = 0$. Set

$$A = \ell_h, \quad B = \ell_{h-1}, \quad C = \ell_d, \quad D = \ell_{d-1}.$$

The recurrence for ℓ_m gives, by induction,

$$\ell_{h+d} = 34AC - AD - BC, \tag{36}$$

$$\ell_{h+d-1} = AC - BD, \tag{37}$$

$$B^2 - 34AB + A^2 = 1, \qquad D^2 - 34CD + C^2 = 1. \tag{38}$$

Since h, d, k are nonzero in $\mathbb{F}_p$, equation (34) gives a unique $\rho \in \mathbb{F}_p$ with

$$A = h\rho, \qquad C = d\rho. \tag{39}$$

We first show that $\rho = 0$. Otherwise put $U = h\rho$. From $L = 0$, (39), and (36), after division by ρ,

$$hD = d(34U - B) - k. \tag{40}$$

Substitute (39) and (40) into the second Cassini identity in (38), multiply it by h^2, and subtract d^2 times the first Cassini identity. Direct expansion leaves

$$2dk(B - 17U + 1) = 0.$$

Thus $B = 17U - 1$. The first Cassini identity now reduces to $-288U^2 = 0$, impossible because $p \geq 7$ and $U \neq 0$. Therefore

$$A = C = 0. \tag{41}$$

Equations (36)–(38) now give

$$\ell_k = 0, \qquad \ell_{k-1} = -BD, \qquad B, D \in \{1, -1\}.$$

Using (27) in (35), together with (41), we obtain

$$C_2 = \frac{5}{256}\big(hB + dD + kBD\big). \tag{42}$$

For $(B, D) = (1, 1), (1, -1), (-1, 1), (-1, -1)$, the parenthesis in (42) is respectively $2k, -2d, -2h, 0$. Since $h, d, k < p$, its only possible zero is therefore

$$B = D = -1, \qquad \ell_{k-1} = -1. \tag{43}$$

It remains to inspect one more even coefficient. Under (41) and (43), formulas (27) and (28) become, for $m = h, d, k$,

$$u_m = -\frac{5m}{256}, \qquad v_m = \frac{m(320m^3 - 1600m^2 - 320m + 621)}{131072}. \tag{44}$$

(The dangerous case cannot contain $m = 1$, since $\ell_1 = 1$.) Let C_4 denote the coefficient of z^{n-4} in $\Delta_{h,k}$. In the present case $A = C = 0$, the shifted products and (29) give

$$\begin{aligned} C_4 &= v_h + v_d - v_k && (45) \\ &\quad + \frac{1}{8}u_h d(-d^2 + 1 + 30d - 3hk) + \frac{1}{8}u_d h(-h^2 + 1 + 30h - 3dk). && (46) \end{aligned}$$

Indeed, by the parity structure of $\widehat{N}_m$ and $\widehat{\Pi}_m$, every contribution to the z^{n-4} coefficient of the two shifted products other than those displayed carries the factor ℓ_h (in $\widehat{N}_h \cdot \widehat{\Pi}_d$) or ℓ_d (in $\widehat{\Pi}_h \cdot \widehat{N}_d$)—namely the pairings of the leading coefficient of $\widehat{N}$ with the levels $0, -2, -4$ of $\widehat{\Pi}$ and with the shift binomials—and these vanish here; equation (46) is *not* asserted for general (h, k). For the u_h term the degree-two loss from the shifted leading powers is $3d(10d - hk)/8$, and the contribution from $\widehat{\Pi}_d$ is $-d(d^2 - 1)/8$; the other term is symmetric. Substitution of (44) into (46) gives

$$\begin{aligned} v_h + v_d - v_k &= -\frac{5hd(4d^2 + 6dh - 15d + 4h^2 - 15h - 2)}{2048}, \\ \text{the two } u\text{-terms} &= \frac{10hd(2d^2 + 3dh - 15d + 2h^2 - 15h - 1)}{2048}. \end{aligned}$$

Consequently

$$C_4 = -\frac{75hdk}{2048}.$$

This is nonzero in $\mathbb{F}_p$, because $p \geq 7$ and $0 < h, d, k < p$. This final contradiction proves the theorem. $\square$

Corollary 33 (Unconditional uniform fiber and energy bounds)**.** *For every prime $p \geq 7$, every $a \in \mathbb{F}_p^\times$, and every $2 \leq H \leq (p-1)/2$,*

$$N_p(a) \leq 2\left\lfloor \frac{p-1}{H+1} \right\rfloor + \frac{H(H-1)(2H-1)}{2} + 2.$$

In particular, with $H = \lceil (2p/3)^{1/4} \rceil$,

$$N_p(a) \leq 4(2/3)^{3/4}p^{3/4} + \sqrt{6}\,p^{1/2} + 3(2/3)^{1/4}p^{1/4} + 3.$$

Moreover,

$$\sum_{a \in \mathbb{F}_p} N_p(a)^2 = O(p^{7/4}).$$

Proof. Theorem 32 supplies hypothesis ($\mathrm{NV}_{p,H}$) in Proposition 28. That proposition gives the first two assertions, and Corollary 29 gives the last one. $\square$

Remark 34 (Verification and sharpness). The nonvanishing theorem and its boundary are machine-verified from several independent directions: (i) direct polynomial-level scans find no identically vanishing $\Delta_{h,k}$ for any prime $7 \le p \le 700$ and $1 \le h < k \le 40$, nor in 1000 random cases with $p \le 997$; (ii) the centered coefficient formulas for L, C_1, C_2 were checked against exact expansions over $\mathbb{Q}$, and the danger-case pipeline ($L = C_1 = C_2 = 0$, $B = D = -1$, $C_3 = 0$, $C_4 = -75hdk/2048$) was confirmed on 309 constructed rank-of-apparition cases across all primes $p \le 3000$ and on three genuine drop-four cases at $p \approx 10^7$, in every instance against the actual centered polynomial; (iii) the $p = 7$, $k = 21$ degeneration is exact: $T^3 = (x^7 - x)^9 I$ for the seven-step transfer matrix, so $N_{21} \equiv 0$ and $R_{21} \equiv 0$ identically. The theorem makes the range restriction $k < p$ explicit and sharp; for the fiber count only $k \le H \asymp p^{1/4}$ is used.

Remark 35 (Why the uniform fiber bound matters). The energy bound $E(p) = \sum_a N_p(a)^2 = O(p^{7/4})$ (Corollary 33) is the first *unconditional* power saving over the trivial $E(p) \le p \max_a N_p(a) \le p^2$ scale for the value distribution of Apéry numbers modulo p; the random-map prediction is $E(p) \asymp p$. Any bound $E(p) \ll p^{2-\eta}$ gives $Z(p) \le E(p)^{1/2} \ll p^{1-\eta/2}$, so energy savings are a strictly weaker—hence potentially more accessible—avenue than direct zero-set bounds, and the value-collision structure enters the two-prime dispersion analysis of Section 12 through the equal-value pattern counts. Note that the $p^{3/4}$ fiber bound also complements the zero-fiber bound: $Z(p) = O(p^{2/3})$ remains the stronger statement for $a = 0$, so $\max_a N_p(a) = O(p^{3/4})$ with the maximum over *all* a.

The next block of results develops the arithmetic of the separated-block resultants $\mathcal{R}_{d,r} = \mathrm{Res}_x(N_d(x), N_r(x+d))$ themselves: complex root localization (hence positivity), a centered norm decomposition, half-integer center divisor lattices, an exact square law on the diagonal, the precise projective meaning of $p \mid \mathcal{R}_{d,r}$, and the two counting lemmas that separate the present state of the collision-energy program from the mesoscopic target $Z(p) \ll \sqrt{p}$.

5.5 Separated root strips and centered resultant structure

Retain the notation

$$P(n) = (2n+1)(17n^2 + 17n + 5), \qquad N_{h+1}(x) = P(x+h)N_h(x) - (x+h)^6 N_{h-1}(x),$$

with $N_0 = 0$ and $N_1 = 1$, and, for $d, r \ge 2$, put

$$\mathcal{R}_{d,r} = \mathrm{Res}_x\big(N_d(x), N_r(x+d)\big).$$

We use the standard resultant convention. The first result proves, in particular, that every integer $\mathcal{R}_{d,r}$ is nonzero.

We will use only the absolute adjacent-resultant formula from Remark 13. With the standard signed convention, its one-step recursion for $h \ge 2$ has the correction

$$\mathrm{Res}(N_h, N_{h+1}) = (-1)^{h-1} N_h(-h)^6 \,\mathrm{Res}(N_{h-1}, N_h);$$

the factor $(-1)^{h-1}$ comes from the scalar minus sign in the recurrence.

Proposition 36 (Separated root strips)**.** *For $d \ge 2$, every complex root α of N_d satisfies*

$$-d < \mathrm{Re}\,\alpha < -1. \tag{47}$$

Consequently, for all $d, r \ge 2$,

$$\mathcal{R}_{d,r} > 0.$$

Proof. Let $\mathsf{A}_d(x)$ be the $(d-1)\times(d-1)$ symmetric tridiagonal matrix with

$$(\mathsf{A}_d)_{ii} = P(x+i), \qquad (\mathsf{A}_d)_{i,i+1} = (\mathsf{A}_d)_{i+1,i} = (x+i+1)^3.$$

Expansion at the last row gives the defining continuant recurrence, so $\det \mathsf{A}_d(x) = N_d(x)$.

Suppose $\operatorname{Re} x \geq -1$ and set $z_i = x + i + \frac{1}{2}$. Then $|z_i| \geq \frac{1}{2}$ and

$$P(x+i) = 34z_i^3 + \frac{3}{2}z_i.$$

It follows that

$$|P(x+i)| \geq |z_i|\big(34|z_i|^2 - \tfrac{3}{2}\big) \geq 28|z_i|^3.$$

The sum of the absolute values of the two possible off-diagonal entries in row i is at most

$$|z_i - \tfrac{1}{2}|^3 + |z_i + \tfrac{1}{2}|^3 \leq 2(|z_i| + \tfrac{1}{2})^3 \leq 16|z_i|^3.$$

Thus $\mathsf{A}_d(x)$ is strictly row-diagonally dominant and is invertible. There is no root with $\operatorname{Re} x \geq -1$.

If J is the reversal matrix, the definitions and $P(-z-1) = -P(z)$ give

$$\mathsf{A}_d(-x-d-1) = -J\mathsf{A}_d(x)J.$$

Taking determinants proves the reflection identity

$$N_d(-x-d-1) = (-1)^{d-1}N_d(x)$$

then excludes $\operatorname{Re} x \leq -d$, proving (47). The roots of $N_r(x+d)$ lie in the disjoint strip

$$-d-r < \operatorname{Re} x < -d-1,$$

so the two polynomials have no common complex root and their resultant is nonzero.

Finally, the leading coefficients ℓ_d, ℓ_r are positive. In the root-product formula for the resultant, a difference of two real roots is positive because the first root strip lies strictly to the right of the second. Nonreal roots can be grouped with their conjugates, making the corresponding products positive squared absolute values. Hence the resultant itself is positive. $\square$

It is useful to clear all dyadic denominators in the natural centered coordinate. For $h \geq 1$, if $n_h = 3(h-1)$, define

$$M_h(Y) = 2^{n_h} N_h\left(\frac{Y-h-1}{2}\right) \in \mathbb{Z}[Y]. \tag{48}$$

Then

$$M_h(-Y) = (-1)^{h-1}M_h(Y), \qquad \operatorname{lc} M_h = \ell_h,$$

and a linear change of variable gives the exact normalization

$$\operatorname{Res}_Y\big(M_d(Y), M_r(Y+d+r)\big) = 2^{n_d n_r}\mathcal{R}_{d,r}. \tag{49}$$

Proposition 37 (Centered norm decomposition)**.** *For $d, r \geq 2$, write*

$$\widehat{N}_d(Y) = N_d\left(Y - \frac{d+1}{2}\right) = \begin{cases} A_d(Y^2), & d \text{ odd,} \\ Y A_d(Y^2), & d \text{ even,} \end{cases}$$

where $A_d \in \mathbb{Z}[1/2][U]$, and set

$$G_{d,r}(Y) = N_r\left(Y + \frac{d-1}{2}\right), \qquad \Psi_{d,r}(Y^2) = G_{d,r}(Y)G_{d,r}(-Y).$$

Then

$$|\mathcal{R}_{d,r}| = \begin{cases} |\operatorname{Res}_U(A_d, \Psi_{d,r})|, & d \text{ odd}, \\ \big|N_r\big((d-1)/2\big)\big|\,|\operatorname{Res}_U(A_d, \Psi_{d,r})|, & d \text{ even}. \end{cases} \tag{50}$$

Moreover, the centered-coefficient formulas of Lemma 31 give

$$A_d(U) = \ell_d U^m + u_d U^{m-1} + v_d U^{m-2} + \cdots$$

with the evident adjustment when d is even.

Proof. In the centered variable $Y = x + (d+1)/2$, the second polynomial is $G_{d,r}(Y)$. If d is odd, the roots of $A_d(Y^2)$ occur in pairs $\{\sqrt{u}, -\sqrt{u}\}$, and their two contributions to the root-product formula multiply to $\Psi_{d,r}(u)$. This is exactly the first resultant in (50). If d is even, the additional root $Y = 0$ contributes $G_{d,r}(0) = N_r((d-1)/2)$. The last assertion is the parity statement and the coefficient formulas of Lemma 31. $\square$

Equivalently, in the finite $\mathbb{Q}$-algebra

$$K_d = \mathbb{Q}[X]/(N_d), \qquad \alpha = X \bmod N_d,$$

one has

$$\mathcal{R}_{d,r} = \ell_d^{3(r-1)} \operatorname{Nm}_{K_d/\mathbb{Q}}\big(N_r(\alpha + d)\big). \tag{51}$$

No irreducibility or separability assumption is needed for this determinantal norm identity. For $d = 2$, $A_2(U) = 34U + \frac{3}{2}$; Proposition 37 is precisely the rational linear factor times quadratic norm decomposition recorded in Remark 26.

5.6 Half-integer center factors and their propagation

The even centered factor gives a genuine divisibility lattice, rather than only a numerical repetition pattern.

Proposition 38 (Even-block center recurrence)**.** *Fix an even integer $a \geq 2$ and define*

$$T_0^{(a)} = 0, \qquad T_1^{(a)} = 1, \qquad T_b^{(a)} = 2^{3(b-1)} N_b\left(\frac{a-1}{2}\right) \quad (b \geq 2).$$

If $z = a + 2b - 1$, then

$$T_{b+1}^{(a)} = \big(34z^3 + 102z^2 + 108z + 40\big)T_b^{(a)} - z^6 T_{b-1}^{(a)}. \tag{52}$$

In particular, every $T_b^{(a)}$ is an integer. Furthermore,

$$T_b^{(a)} \mid \mathcal{R}_{a,b}, \tag{53}$$

and the odd part of $T_b^{(a)}$ divides $\mathcal{R}_{b,a+b}$. By symmetry, if b is even, the odd part of $T_a^{(b)}$ divides both $\mathcal{R}_{a,b}$ and $\mathcal{R}_{a,a+b}$.

Proof. Substitute $x = (a-1)/2$ in the recurrence for N_b and multiply by 2^{3b}. Since $2(x+b) = a + 2b - 1 = z$ and

$$8P(z/2) = 34z^3 + 102z^2 + 108z + 40,$$

one obtains (52).

Because a is even, reflection shows that $c = -(a+1)/2$ is a root of N_a. Gauss's lemma gives

$$N_a(X) = (2X + a + 1)Q_a(X), \qquad Q_a \in \mathbb{Z}[X].$$

Resultant multiplicativity and the linear-root formula yield

$$\mathrm{Res}_X(2X + a + 1, N_b(X + a)) = 2^{3(b-1)} N_b(c + a) = T_b^{(a)},$$

which proves (53).

Let p^e be any odd prime power dividing $T_b^{(a)}$. Since 2 is a unit modulo p^e, this says

$$N_b(t) = 0 \pmod{p^e}, \qquad t = (a-1)/2.$$

The restart identity is valid without division in the present situation: the two sides satisfy the same recurrence and hence

$$N_{a+b}(c) = N_{a+1}(c)N_b(c + a) = N_{a+1}(c)N_b(t) = 0 \pmod{p^e}.$$

Put $y = -t - b - 1$. Reflection gives $N_b(y) = (-1)^{b-1} N_b(t) = 0$, while $y + b = c$, so $N_{a+b}(y+b) = 0$. The Sylvester–Bézout identity for the resultant, evaluated at $y \in \mathbb{Z}/p^e\mathbb{Z}$, gives $p^e \mid \mathcal{R}_{b,a+b}$. Taking all odd prime powers proves the claim. The symmetric assertion follows from $\mathcal{R}_{a,b} = \mathcal{R}_{b,a}$. □

There is also a simultaneous center factor for every pair. With $T_b^{(a)}$ as above, define

$$\mathfrak{C}_{d,r} = \begin{cases} 1, & d, r \text{ odd}, \\ |T_r^{(d)}|, & d \text{ even}, \ r \text{ odd}, \\ |T_d^{(r)}|, & d \text{ odd}, \ r \text{ even}, \\ \dfrac{|T_r^{(d)} T_d^{(r)}|}{2(d+r)}, & d, r \text{ even}. \end{cases}$$

Then $\mathfrak{C}_{d,r}$ is an integer and

$$\mathfrak{C}_{d,r} \mid \mathcal{R}_{d,r}. \tag{54}$$

Indeed, factor every available central linear factor before applying resultant multiplicativity. When both indices are even, the two linear factors have resultant $2(d+r)$, which accounts for the denominator and prevents counting their mutual resultant twice.

5.7 The diagonal square law

For the diagonal family, the centered pairing becomes an exact square.

Theorem 39 (Diagonal square law)**.** *For $d \geq 2$, let $n = 3(d-1)$ and put*

$$f_d(T) = N_d\left(T - \frac{2d+1}{2}\right) = E_d(T^2) + T O_d(T^2), \qquad E_d, O_d \in \mathbb{Q}[U].$$

Then

$$\mathcal{R}_{d,d} = (-1)^{d-1} 2^n \ell_d N_d \left(-\frac{2d+1}{2}\right) \operatorname{Res}_U(O_d, E_d)^2. \tag{55}$$

Moreover,

$$D_d = 2^n N_d(-1/2) \in \mathbb{Z}, \qquad Q_d = \operatorname{Res}_U(O_d, E_d) \in \mathbb{Z},$$

and therefore

$$\boxed{\mathcal{R}_{d,d} = \ell_d |D_d| Q_d^2.} \tag{56}$$

For every odd prime p,

$$v_p(\mathcal{R}_{d,d}) = v_p(\ell_d) + v_p(D_d) + 2v_p(Q_d). \tag{57}$$

In particular, if $p \nmid \ell_d D_d$, then $v_p(\mathcal{R}_{d,d})$ is even.

Proof. We first record a general identity. Let

$$f(T) = aT^n + \cdots = E(T^2) + TO(T^2),$$

and suppose the two parity parts have the full degrees forced by nonzero coefficients in degrees n and $n-1$. If α_i are the roots of f, then

$$f(-\alpha_i) = -2\alpha_i O(\alpha_i^2),$$

and the root-product formula gives

$$\operatorname{Res}_T(f(T), f(-T)) = 2^n a^{n-1} f(0) \prod_i O(\alpha_i^2). \tag{58}$$

If $s = \deg O$, $e = \deg E$, and $b = \operatorname{lc} O$, then

$$\begin{aligned} a^{2s} \prod_i O(\alpha_i^2) &= \operatorname{Res}_T(f(T), O(T^2)) \\ &= b^{n-2e} \operatorname{Res}_U(O, E)^2. \end{aligned}$$

For $n = 2m$, one has $(e, s) = (m, m-1)$. For $n = 2m+1$, one has $(e, s) = (m, m)$ and $b = a$. In either case, substitution into (58) gives

$$\operatorname{Res}_T(f(T), f(-T)) = 2^n a f(0) \operatorname{Res}_U(O, E)^2. \tag{59}$$

For $f = f_d$, the natural centered parity of N_d shows that the coefficient of T^{n-1} is $-nd\ell_d/2$, so the stated degree conditions hold. Reflection gives

$$N_d(T - 1/2) = (-1)^{d-1} f_d(-T).$$

After the translation $T = x + (2d+1)/2$, scaling the second argument of the resultant contributes $(-1)^{(d-1)n} = (-1)^{d-1}$. Equation (59) is therefore exactly (55).

It remains only to justify the integral form. Integrality of D_d is immediate from evaluating an integer polynomial of degree n at a half-integer. Put $c = 2d+1$ and $V = X(X+c)$. Reduction by the monic relation $X^2 = V - cX$ gives the free-module decomposition

$$\mathbb{Z}[X] = \mathbb{Z}[V] \oplus X\mathbb{Z}[V].$$

Thus there are $A, B \in \mathbb{Z}[V]$ such that

$$N_d(X) = A(V) + XB(V).$$

Since $T = X + c/2$ and $U = T^2$, one has $V = U - c^2/4$, and comparison with $f_d(T) = E_d(U) + TO_d(U)$ gives

$$O_d(U) = B(U - c^2/4), \qquad E_d(U) = A(U - c^2/4) - \frac{c}{2}B(U - c^2/4).$$

Translation invariance reduces Q_d to $\mathrm{Res}_V(B, A - \frac{c}{2}B)$. In fact $\deg A = \deg E_d$. When $n = 2m$, this follows from $\deg E_d = m$ and $\deg B = m - 1$. When $n = 2m + 1$, the leading coefficients computed above give

$$\operatorname{lc} B = \ell_d, \qquad \operatorname{lc} A = \frac{c - nd}{2}\ell_d,$$

and $c - nd = -3d^2 + 5d + 1 \neq 0$ for every integer $d \geq 2$. The root-product formula therefore gives

$$Q_d = \mathrm{Res}_U(O_d, E_d) = \mathrm{Res}_V(B, A - \tfrac{c}{2}B) = \mathrm{Res}_V(B, A) \in \mathbb{Z}.$$

Finally, reflection gives

$$\left| N_d\left(-\frac{2d+1}{2}\right)\right| = |N_d(-1/2)|.$$

Proposition 36 supplies the sign, and (56) follows. The valuation formula is immediate. □

In odd characteristic the affine diagonal roots can also be read off exactly. In the coordinate used above,

$$f_d(T) = f_d(-T) = 0 \iff E_d(T^2) = 0, \quad TO_d(T^2) = 0.$$

Thus the number of distinct $\mathbb{F}_p$-roots is

$$\mathbf{1}_{p|f_d(0)} + 2\#\{u \in (\mathbb{F}_p^\times)^2 : E_d(u) = O_d(u) = 0\}. \tag{60}$$

This distinguishes an actual split collision from a nonsplit quadratic factor of the resultant.

5.8 What resultant divisibility does and does not mean

Proposition 40 (Projective validity domain)**.** *For integer polynomials of their nominal degrees, $p \mid \mathcal{R}_{d,r}$ if and only if their nominal homogenizations have a common point in $\mathbb{P}^1(\overline{\mathbb{F}}_p)$. If the two leading coefficients do not both vanish modulo p, this is equivalent to an affine common root in $\overline{\mathbb{F}}_p$. It is not, in general, equivalent to a common root in $\mathbb{F}_p$. The universally valid implication is only*

$$\{x \in \mathbb{F}_p : N_d(x) = N_r(x + d) = 0\} \neq \varnothing \implies p \mid \mathcal{R}_{d,r}.$$

Two mesoscopic examples show both failures of the converse. For $d = r = 2$ and $p = 17$,

$$N_2(X) \equiv 5(2X + 3), \qquad N_2(X + 2) \equiv 5(2X + 7) \pmod{17}.$$

They have no affine common root, although $17^3 \mid \mathcal{R}_{2,2}$; both nominal cubics lost their leading terms and meet at infinity. On the other hand, for $d = r = 3$ and $p = 110629$, the leading coefficient is a unit and

$$\gcd\bigl(N_3(X), N_3(X + 3)\bigr) = X^2 + 7X - 15495.$$

Its discriminant is 62029, a nonsquare modulo 110629. Hence there is an affine common root over $\overline{\mathbb{F}}_p$ but none over $\mathbb{F}_p$, even though $110629^2 \mid \mathcal{R}_{3,3}$.

The centered coefficients identify the complete lattice of automatic infinity edges. Recall

$$\ell_0 = 0, \quad \ell_1 = 1, \quad \ell_{n+1} = 34\ell_n - \ell_{n-1}.$$

For $p \geq 5$, choose $\lambda \in \mathbb{F}_{p^2}$ with $\lambda + \lambda^{-1} = 34$. Then

$$\ell_n = \frac{\lambda^n - \lambda^{-n}}{\lambda - \lambda^{-1}}.$$

If $\rho_p = \operatorname{ord}(\lambda^2)$, then

$$p \mid \ell_n \iff \rho_p \mid n, \qquad \rho_p \mid p - \left(\frac{2}{p}\right). \tag{61}$$

Consequently every pair with $\rho_p \mid d, r$ is in the raw resultant support, independently of affine collisions. If $M = \lfloor H/\rho_p \rfloor$, these pairs alone number

$$V_p^{\text{infinity}}(H) = \frac{M(M-1)}{2}. \tag{62}$$

There is a legitimate centered-coefficient transfer here, but it detects only the false edge at infinity. Suppose $p \geq 7$ and $d < p$. If $\ell_d = 0$, Cassini gives $\ell_{d-1}^2 = 1$, while Lemma 31 gives

$$u_d = \frac{5d\ell_{d-1}}{256} \not\equiv 0 \pmod{p}.$$

Because the natural centered polynomial contains only every other power, its degree drops by exactly two. If $\rho_p \mid d, r$, the two nominal homogenizations have a common factor of order two at infinity; equivalently, the nominal Sylvester matrix has nullity at least two modulo p. Its Smith normal form therefore gives

$$p^2 \mid \mathcal{R}_{d,r}. \tag{63}$$

For example, $p = 665857$ has $\rho_p = 8$ and $H = \lfloor \sqrt{p} \rfloor = 816$. Formula (62) already gives 5151 raw resultant pairs, without producing a single affine collision. Thus the unsaturated integer resultant support is not the correct collision graph.

The centered/Pell method from Theorem 32 cannot be used pointwise in the opposite direction. At $(d, r, p) = (2, 2, 71)$ and $x = -5/2$, the two natural centered variables are -1 and 1, and

$$\widehat{N}_2(Y) = 34Y^3 + \frac{3}{2}Y, \qquad \widehat{N}_2(\pm 1) = \pm\frac{71}{2} \equiv 0 \pmod{71},$$

while $\ell_2 = 34 \not\equiv 0 \pmod{71}$. The proof of Theorem 32 compares coefficients only after assuming that an entire certificate polynomial is identically zero. Two point-value equations do not make its coefficients L, C_1, C_2, C_4 vanish. In particular, its C_4 formula is valid only inside the stated danger case and cannot be transferred to this collision.

5.9 The two missing counting lemmas

For an integer $H \geq 2$, define

$$\mathscr{D}_H = \{(d, r) \in \mathbb{Z}_{\geq 2}^2 : d + r \leq H\},$$

$$X_{d,r}(p) = \{x \in \mathbb{F}_p : N_d(x) = N_r(x + d) = 0\}, \qquad m_{d,r}(p) = |X_{d,r}(p)|.$$

The collision energy is the weighted count

$$\mathcal{E}_p(H) = \sum_{(d,r)\in\mathscr{D}_H} m_{d,r}(p).$$

Define also the simple affine support

$$W_p(H) = \#\{(d,r) \in \mathscr{D}_H : m_{d,r}(p) > 0\}.$$

The raw resultant support

$$V_p(H) = \#\{(d,r) \in \mathscr{D}_H : p \mid \mathcal{R}_{d,r}\}$$

is the edge count of a still larger projective-support graph. Thus

$$W_p(H) \le \mathcal{E}_p(H), \qquad W_p(H) \le V_p(H).$$

Remark 27 identified $\mathcal{E}_p(H)$ with the edge count $W_p(H)$ by forgetting that one pair can have several root witnesses. For example, at $p = 3109$ the pair $(d,r) = (8,8)$ has two distinct $\mathbb{F}_p$ common roots, so it contributes one to W_p and two to $\mathcal{E}_p$. The examples above exhibit the separate failure of $W_p(H) = V_p(H)$.

The same distinction changes the averaging statement in that remark. The Hadamard argument controls only

$$\sum_{X<p\le 2X} V_p(H) \ll \frac{H^4 \log H}{\log X}.$$

After inserting the available multiplicity bound, its valid consequence is

$$\sum_{X<p\le 2X} \mathcal{E}_p(H) \ll \frac{H^5 \log H}{\log X}.$$

Since there are $\asymp X/\log X$ primes in the interval, this proves $\mathcal{E}_p(H) \ll H^{1/2}$ for all but $o(X/\log X)$ of them only in the range $H \le X^{2/9-\varepsilon}$, rather than the stated $X^{2/7-\varepsilon}$ range. The latter exponent applies to simple support, not to the collision energy.

The existing gcd estimate gives

$$m_{d,r}(p) \le 3(\min(d,r) - 1).$$

Accordingly, define the degree-weighted support

$$\mathcal{W}_p(H) = \sum_{(d,r)\in\mathscr{D}_H} (\min(d,r) - 1)\mathbf{1}_{p|\mathcal{R}_{d,r}}.$$

Then the rigorous implication is

$$\mathcal{E}_p(H) \le 3\mathcal{W}_p(H). \tag{64}$$

In particular,

$$V_p(H) \ll H \quad\Longrightarrow\quad \mathcal{E}_p(H) \ll H^2, \tag{65}$$

not $H^{3/2}$. This loss is real at the level of the proposed graph argument: take $\asymp H$ exceptional pairs with $d, r \asymp H$, make them a matching (maximum degree one and codegree zero), and assign $\asymp H$ distinct root witnesses to every edge. This obeys the available per-pair degree bound and has energy $\asymp H^2$. The ordinary Kővári–Sós–Turán theorem applies to a simple graph and cannot control those parallel root witnesses.

Thus a sufficient replacement for G1 is the genuinely weighted estimate

$$\boxed{\mathcal{W}_p(H) \ll H^{3/2}.} \tag{66}$$

Alternatively, together with $V_p(H) \ll H$, it would suffice to prove an average actual gcd-degree bound $O(H^{1/2})$ on the exceptional pairs, or a corresponding multiplicity-tail estimate.

There is a second, independent gap between energy and the first moment $R_p(H) = \sum_{h=2}^{H} r_p(h)$. Let

$$k_x = \#\{2 \le h \le H : N_h(x) = 0\}, \qquad \partial_H = \{-2, -3, \dots, -H\} \subseteq \mathbb{F}_p.$$

For $x \notin \partial_H$, Proposition 24 maps every pair $2 \le d < e \le H$ counted by $\binom{k_x}{2}$ injectively to the energy term $(x, d, e-d)$. Indeed $e - d = 1$ cannot occur: in that case the Bézout right-hand side is $S_d(x)^6 N_1 = S_d(x)^6 \neq 0$. Thus every resulting pair has $e - d \geq 2$ and lies in $\mathscr{D}_H$. Hence

$$\sum_{x \notin \partial_H} \binom{k_x}{2} \le \mathcal{E}_p(H). \tag{67}$$

The exclusion of ∂_H is essential. The unsaturated degree claim following Proposition 24 fails at cut-edge roots: for $(d, e, p) = (3, 4, 73)$,

$$\gcd_{\mathbb{F}_{73}[X]}(N_3, N_4) = X + 3,$$

whereas $N_{e-d} = N_1 = 1$. Thus that claim is valid only for the saturated gcd, or as an away-from-boundary root count; the use above has precisely that domain.

The same example invalidates the proof of the unrestricted column bound in Proposition 22. Since $b_2 = 73$, the endpoint formula gives

$$N_h(-3) \equiv 0 \pmod{73} \qquad (h \geq 3).$$

At the singular transition the recurrence has reached two consecutive zeros, so the entire later tail remains zero; it cannot be charged as an $O(1)$ transition cost. Thus that proof does not establish the proposition or the rectangular incidence bound derived from it; a valid version needs an explicit boundary term or an unpolluted nonsingular initial state. The present argument excludes such columns from (67) and retains their full contribution in $R_p^{\partial}(H)$. But columns with $k_x = 1$ contribute nothing to either side of (67). Energy alone cannot bound their total mass.

For a precise sufficient condition, set $K = \lceil \sqrt{H} \rceil$ and

$$R_p^{\partial}(H) = \sum_{x \in \partial_H} k_x, \qquad L_p(H) = \sum_{\substack{x \notin \partial_H \\ 0 < k_x < K}} k_x.$$

Since $k \le 2\binom{k}{2}/(K-1)$ for $k \geq K$, (67) gives the deterministic inequality

$$\boxed{R_p(H) \le R_p^{\partial}(H) + L_p(H) + \frac{2}{\lceil \sqrt{H} \rceil - 1} \mathcal{E}_p(H).} \tag{68}$$

Thus the missing low-fiber amplification is

$$R_p^{\partial}(H) + L_p(H) \ll H, \tag{69}$$

or, in compressed form,

$$R_p(H) \ll H + H^{-1/2}\mathcal{E}_p(H).$$

This is a lower-participation assertion; Kővári–Sós–Turán is an upper-bound theorem and does not supply it. The boundary term cannot be discarded, since the endpoint factorization of $N_h(-j)$ can make a cut-edge column large when $p \mid b_{j-1}$.

Combining the two genuinely missing estimates (66) and (69) would give

$$\mathcal{E}_p(H) \ll H^{3/2}, \qquad R_p(H) \ll H.$$

Only then does the valid last step

$$Z(p) \leq \left\lceil \frac{p}{H} \right\rceil + R_p(H-1)$$

give $Z(p) \ll \sqrt{p}$ at $H = \lfloor \sqrt{p} \rfloor$.

For completeness, suppose only the proposed G2 support estimate

$$V_p(H) \ll H^{2-\delta}$$

is available. The present ingredients give merely

$$\mathcal{E}_p(H) \ll H^{3-\delta}.$$

Partition $\{0, \dots, p-1\}$ into nonwrapping consecutive blocks of length at most H, and let their Apéry zero counts be z_i. Three zeros $m < m+d < m+d+r$ in one block have $d, r \geq 2$ by Lemma 6. All intervening recurrence denominators are units modulo p, so the gap-polynomial identity gives the distinct energy witness (m, d, r). Hence

$$\sum_i \binom{z_i}{3} \leq \mathcal{E}_p(H)$$

and Hölder's inequality yields

$$Z(p) \ll \frac{p}{H} + p^{2/3} H^{(1-\delta)/3}. \tag{70}$$

If the G2 estimate holds uniformly for $H \leq \sqrt{p}$, then, after combining with the existing $p^{2/3}$ bound,

$$Z(p) \ll \begin{cases} p^{2/3}, & 0 < \delta \leq 1, \\ p^{(5-\delta)/6}, & 1 < \delta \leq 2. \end{cases}$$

Thus even $V_p(H) \ll H$ ($\delta = 1$) recovers only the existing exponent from the currently proved inputs. Even a separately proved $\mathcal{E}_p(H) \ll H^{3/2}$ would give only $Z(p) \ll p^{7/12}$ by this triple-counting route unless the low-fiber amplification is also proved.

Finally, the recurrence does not close on the scalar resultants. In the algebra K_d of (51), put $u_r = N_r(\alpha + d)$. Then

$$u_{r+1} = P(\alpha + d + r)u_r - (\alpha + d + r)^6 u_{r-1}.$$

The norm of a sum depends on all polarized mixed norms, so $\mathrm{Nm}(u_r)$ and $\mathrm{Nm}(u_{r-1})$ do not determine $\mathrm{Nm}(u_{r+1})$. Modulo p, different values of r may also vanish in different residue-field components, whereas restart requires the same root witness. This root-witness alignment, the weighted support estimate (66), and the low-fiber amplification (69) are the exact missing lemmas left by the present attack.

5.10 Collision energy is an off-boundary factorial moment

Retain

$$\mathscr{D}_H = \{(d,r) \in \mathbb{Z}^2_{\geq 2} : d + r \leq H\}, \qquad \mathcal{E}_p(H) = \sum_{(d,r)\in\mathscr{D}_H} m_{d,r}(p),$$

and assume throughout that $p \geq 7$ and $H \leq \sqrt{p}$; the omitted small cases are empty or immediate. For $x \in \mathbb{F}_p$ put

$$k_x = k_x(H) = \#\{2 \leq h \leq H : N_h(x) = 0\}, \qquad R_p(H) = \sum_{x\in\mathbb{F}_p} k_x,$$

and put

$$\partial_H = \{-2, -3, \ldots, -H\} \subseteq \mathbb{F}_p.$$

Split the energy as

$$\mathcal{E}_p(H) = \mathcal{E}_p^\circ(H) + \mathcal{E}_p^\partial(H)$$

according as its root witness lies outside or inside ∂_H.

Proposition 41 (Exact off-boundary column identity)**.** *One has*

$$\boxed{\mathcal{E}_p^\circ(H) = \sum_{x\notin\partial_H} \binom{k_x}{2}.} \tag{71}$$

At the boundary only the following direction is universally valid:

$$0 \leq \mathcal{E}_p^\partial(H) \leq \sum_{x\in\partial_H} \binom{k_x}{2}. \tag{72}$$

Consequently

$$\sum_{x\notin\partial_H} \binom{k_x}{2} \leq \mathcal{E}_p(H) \leq \sum_{x\in\mathbb{F}_p} \binom{k_x}{2}. \tag{73}$$

Proof. The continuant addition formula used in the proof of Proposition 24 gives

$$N_{d+r}(x) = N_d(x)N_{r+1}(x+d-1) - (x+d)^6 N_{d-1}(x)N_r(x+d).$$

Thus an energy witness (x,d,r) gives the two zero levels d and $d+r$ in the column x. This map is injective, since its lower level and the difference recover (d,r). It proves the upper bound in (72) and the upper bound in (73).

Conversely, suppose that $x \notin \partial_H$ and that $2 \leq d < e \leq H$ are two zero levels. Then $S_d(x)$ is a unit, so the saturated Bézout identity gives

$$N_{e-d}(x+d) = 0.$$

The difference $e-d$ cannot be one: in that case the right-hand side of (11) would be $S_d(x)^6 N_1 = S_d(x)^6 \neq 0$. Hence $(x, d, e-d)$ is an energy witness. This is inverse to the preceding map and proves (71). □

The equality is stronger than (67), but it also shows why the boundary exclusion cannot be repaired by changing an inequality sign. At $p = 73$, $H = 8$, the polluted column $x = -3$ has six zero levels and hence 15 column pairs, while only two boundary energy witnesses occur in the entire computation.

Corollary 42 (What the present column estimates prove)**.** *Uniformly for $H \le \sqrt{p}$,*

$$\mathcal{E}_p(H) \ll H^{8/3}. \tag{74}$$

Proof. The row-degree bound gives

$$R_p(H) \le \sum_{h=2}^{H} 3(h-1) = \frac{3}{2}H(H-1). \tag{75}$$

Define the window-polluted set

$$\mathcal{P}_H = \{-m : 2 \le m \le H,\ p \mid b_{m-1}\}.$$

The pollution characterization following Lemma 21 and Corollary 10 give

$$P_H := |\mathcal{P}_H| \ll H^{2/3}.$$

Every column outside $\mathcal{P}_H$ has no adjacent zero levels in $[2, H]$. (A globally polluted column whose cut occurs after H has no adjacent pair in this window.) Proposition 22 therefore gives

$$M_H := \max_{x \notin \mathcal{P}_H} k_x \ll H^{2/3}.$$

By Proposition 41,

$$\begin{aligned} \mathcal{E}_p(H) &\le \sum_x \binom{k_x}{2} \\ &\le \frac{M_H - 1}{2} \sum_{x \notin \mathcal{P}_H} k_x + P_H \binom{H-1}{2} \\ &\ll H^{2/3}H^2 + H^{2/3}H^2. \end{aligned}$$

This is (74). □

This saves $H^{1/3}$ over the direct $O(H^3)$ degree bound, but its exponent is still larger than two and hence has no consequence for the record exponent of $Z(p)$.

5.11 The proposed fixed point is not a contraction

For an arbitrary threshold $2 \le K \le H$, define

$$R_p^{\circ}(H) = \sum_{x \notin \partial_H} k_x, \qquad L_{p,K}(H) = \sum_{\substack{x \notin \partial_H \\ 0 < k_x < K}} k_x,$$

and let

$$M_p^{\circ}(H) = \max_{x \notin \partial_H} k_x.$$

Proposition 43 (Exact fixed-point audit)**.** *For every such K,*

$$R_p(H) \le R_p^{\partial}(H) + L_{p,K}(H) + \frac{2}{K-1}\mathcal{E}_p^{\circ}(H), \tag{76}$$

where $R_p^{\partial}(H) = \sum_{x \in \partial_H} k_x$. On the other hand,

$$\mathcal{E}_p^{\circ}(H) \le \frac{M_p^{\circ}(H) - 1}{2} R_p^{\circ}(H). \tag{77}$$

Combining the two estimates never gives an absorptive fixed point.

Proof. For $k \geq K$ one has
$$k \leq \frac{2}{K-1}\binom{k}{2}.$$
Summing this over the high off-boundary columns and using (71) proves (76). The elementary inequality $\binom{k}{2} \leq (M_p^\circ - 1)k/2$ proves (77).

Substitution gives
$$\mathcal{E}_p^\circ \leq \frac{M_p^\circ - 1}{2} L_{p,K} + \frac{M_p^\circ - 1}{K-1}\mathcal{E}_p^\circ. \tag{78}$$
If $K \leq M_p^\circ$, the feedback coefficient is at least one. If $K > M_p^\circ$, every nonzero off-boundary column is already included in $L_{p,K}$, so $L_{p,K} = R_p^\circ$ and (78) is the original tautology. These two cases exhaust all thresholds. □

In particular, the proposed $K = \lceil\sqrt{H}\rceil$ does not interact contractively with the available $M_p^\circ \ll H^{2/3}$ estimate. The amplification inequality is a lower bound on the factorial moment carried by high fibers; it is not a second upper bound for that moment. Singleton columns can make $R_p(H)$ large while contributing nothing to energy, and even an independent $R_p(H) \ll H$ estimate together with only $k_x \ll H^{2/3}$ permits the abstract scale $\mathcal{E}_p(H) \asymp H^{5/3}$.

There is also a notation obstruction in the suggested partition. In `meso_result.tex`, $W_p(H)$ denotes simple affine support and $\mathcal{W}_p(H)$ denotes degree-weighted raw resultant support. A raw resultant-supported pair need have no $\mathbb{F}_p$ witness at all, and therefore cannot be assigned to a witness column. Even after restricting to affine support, the artificial weight $\min(d, r) - 1$ is not bounded by the number of actual witnesses. Thus an inequality of the form "$\mathcal{W}_p(H) \leq \sum_x \binom{k_x}{2}$" is unavailable; the valid column statement is Proposition 41 for the actual energy.

5.12 The exact triple-counting exponent

Proposition 44 (Optimization from an energy exponent)**.** *Suppose, uniformly in a range of H, that*
$$\mathcal{E}_p(H) \ll H^\beta.$$
Then block triple-counting gives
$$Z(p) \ll \frac{p}{H} + p^{2/3}H^{(\beta-2)/3}. \tag{79}$$
If the estimate is uniform for every $H \leq \sqrt{p}$, the optimized triple-counting exponent is
$$\theta_{\text{triple}}(\beta) = \begin{cases} 1/2, & \beta \leq 1, \\ (\beta+2)/6, & 1 \leq \beta \leq 2, \\ \beta/(\beta+1), & \beta \geq 2. \end{cases} \tag{80}$$
After combining with the existing $Z(p) \ll p^{2/3}$ estimate, every $1 < \beta < 2$ gives
$$Z(p) \ll p^{(\beta+2)/6},$$
and in particular $\beta = 3/2$ gives $Z(p) \ll p^{7/12}$.

Proof. Partition the index interval into $B \ll p/H$ nonwrapping blocks and let z_i be their zero counts. As in the proof of (70),
$$\sum_i \binom{z_i}{3} \leq \mathcal{E}_p(H).$$

The elementary bound

$$z \le 2 + \left(6\tbinom{z}{3}\right)^{1/3}$$

and Hölder's inequality give

$$Z(p) \ll B + B^{2/3}\mathcal{E}_p(H)^{1/3},$$

which is (79).

Write $H = p^{\alpha}$, $0 < \alpha \le 1/2$. The two exponents are

$$1-\alpha, \qquad \frac{2}{3} + \frac{\beta-2}{3}\alpha.$$

When $\beta < 2$, both decrease with α, so one takes $\alpha = 1/2$. This gives $\max\{1/2, (\beta+2)/6\}$. At $\beta = 2$ the second exponent is $2/3$. For $\beta > 2$, balancing the two terms at $\alpha = 1/(\beta+1)$ gives $\beta/(\beta+1)$. This proves (80) and the stated comparison. □

Thus the power $(\beta-1)/3$ displayed in the specification is not the output of block triple-counting: the factor $B^{2/3}$ contributes $H^{-2/3}$, not $H^{-1/3}$. Equivalently, the displayed formula is missing one further factor $H^{-1/3}$. More generally, if $\beta < 2$ is available only through $H \le p^{\eta}$, then taking the largest permitted H gives

$$\theta(\beta,\eta) = \max\left\{1-\eta, \frac{2}{3} + \frac{\eta(\beta-2)}{3}\right\}. \tag{81}$$

It beats $2/3$ exactly when $\eta > 1/3$ and $\beta < 2$. No interior balancing is required in this decreasing regime.

5.13 Structured subfamilies

To avoid the clash between simple affine support and weighted resultant support, write, for $\mathscr{S} \subseteq \mathscr{D}_H$,

$$\mathcal{W}_p^{\mathrm{res}}(\mathscr{S}) = \sum_{(d,r)\in\mathscr{S}} (\min(d,r)-1)\mathbf{1}_{p\mid\mathcal{R}_{d,r}}, \qquad \mathcal{E}_p(\mathscr{S}) = \sum_{(d,r)\in\mathscr{S}} m_{d,r}(p).$$

Proposition 45 (Shallow strips and diagonal bands)**.** *Let $2 \le K \le H/2$ and $k = \lfloor K \rfloor$. Then*

$$\mathcal{W}_p^{\mathrm{res}}\{(d,r) : \min(d,r) \le K\} \le B_H(k), \tag{82}$$

$$\mathcal{E}_p\{(d,r) : \min(d,r) \le K\} \le 3B_H(k), \tag{83}$$

where

$$B_H(k) = \sum_{s=2}^{k}(s-1)(2H-4s+1) \le Hk(k-1). \tag{84}$$

In particular, both quantities are $O(H^{3/2})$ in the deterministic range

$$\min(d,r) \le H^{1/4}.$$

The larger suggested strip $\min(d,r) \le \sqrt{H}$ has only the bound $O(H^2)$ after the necessary degree weight is included.

For a fixed $C \ge 0$ and $D \le H/2$,

$$\mathcal{W}_p^{\mathrm{res}}\{(d,r) : |d-r| \le C,\ \min(d,r) \le D\} \ll (C+1)D^2, \tag{85}$$

and the same bound, up to a factor three, holds for the energy. Hence the part $\min(d,r) \le H^{3/4}$ of a fixed-width diagonal band is $O_C(H^{3/2})$, whereas the full band has only the present bound $O_C(H^2)$.

Proof. For a fixed value $s = \min(d, r)$ there are exactly $2H - 4s + 1$ ordered pairs in $\mathscr{D}_H$. Summing the weight $s - 1$ gives (84). Since the reductions of N_h are nonzero in the present range,

$$m_{d,r}(p) \leq 3(\min(d, r) - 1),$$

which proves (83).

In the band $|d - r| \leq C$, each possible minimum occurs in at most $2C + 1$ ordered pairs. Summing its degree weight gives (85). □

The thresholds are not artifacts of a loose summation: exactly

$$B_H(k) = Hk(k - 1) + O(k^3).$$

Thus if $k \to \infty$ and $k = o(H)$, then $B_H(k) = (1 + o(1))Hk^2$. The unweighted support of the $\sqrt{H}$ strip is $O(H^{3/2})$, but this says nothing about its energy multiplicities. Even after discarding shifted columns which are polluted, Proposition 22 gives only

$$\sum_{\substack{2 \leq d \leq K,\ 2 \leq r,\ d+r \leq H \\ x \in X_{d,r}(p),\ x+d \text{ unpolluted}}} 1 \ll K^2 H^{2/3}; \tag{86}$$

The symmetric orientation obeys the same estimate. For $K = \sqrt{H}$ this is $H^{5/3}$, still above the target.

The ambient degree weight satisfies

$$\sum_{(d,r) \in \mathscr{D}_H} (\min(d, r) - 1) = \frac{H^3}{12} + O(H^2). \tag{87}$$

Removing the whole $\sqrt{H}$ strip and any fixed-width diagonal band changes (87) by only $O_C(H^2)$. Thus the ambient dominant region is the balanced off-diagonal bulk

$$\min(d, r) > \sqrt{H}, \qquad |d - r| > C.$$

This does not mean that the actual worst contribution has been localized: the shallow annulus $H^{1/4} < \min(d, r) \leq \sqrt{H}$ and the high part of the diagonal band are themselves unresolved at the $H^{3/2}$ scale.

5.13.1 Infinity and center lattices

Proposition 46 (Exact weight of the infinity lattice)**.** *Let $\rho = \rho_p$ and $M = \lfloor H/\rho \rfloor$. The automatic projective-infinity pairs $\rho \mid d, r$ have raw degree weight*

$$\mathcal{I}_p(H) = \rho \sum_{s=2}^{M} \left\lfloor \frac{s^2}{4} \right\rfloor - \binom{M}{2} = \frac{H^3}{12\rho^2} + O\left(\frac{H^2}{\rho}\right). \tag{88}$$

They contribute no automatic affine witness and must be saturated before using resultant support as a collision proxy.

Proof. Write $d = \rho a$, $r = \rho b$. For $s = a + b$,

$$\sum_{a=1}^{s-1} \min(a, s - a) = \left\lfloor \frac{s^2}{4} \right\rfloor.$$

There are $\binom{M}{2}$ ordered pairs, and subtracting the 1 in each weight gives (88).

Let $\overline{N}_d$ be the reduction of N_d modulo p, let $n_d = 3(d-1)$ be its nominal degree, and define its nominal homogenization by

$$\overline{F}_d(X,Z) = Z^{n_d}\overline{N}_d(X/Z).$$

When $\rho \mid d$ and $d < p$, the centered-coefficient calculation after (61) shows that the nominal homogenization has the exact form

$$\overline{F}_d(X,Z) = Z^2 F_d^{\mathrm{sat}}(X,Z), \qquad F_d^{\mathrm{sat}}(1,0) \neq 0.$$

Modulo p, dividing Z^2 from each nominal form removes the automatic infinity intersection exactly. One must not delete the whole pair: its saturated forms may still have a separate affine common root. □

For example, the already recorded values $p = 665857$, $\rho_p = 8$, and $H = 816$ give $\mathcal{I}_p(H) = 712657$, whereas $H^{3/2} = 23309.6\ldots$. This is why the raw weighted resultant support is not the statement to which a witness-column proof can apply.

The center factors have the opposite status: they are genuine affine witnesses and cannot be saturated away.

Proposition 47 (Present bound for distinguished center witnesses)**.** *For $(d,r) \in \mathscr{D}_H$, define*

$$c_{d\to r}(p) = \begin{cases} \mathbf{1}_{p\mid T_r^{(d)}}, & d \text{ even},\\ 0, & d \text{ odd}, \end{cases} \qquad c_{d,r}(p) = c_{d\to r}(p) + c_{r\to d}(p).$$

Then $c_{d,r}(p)$ is the number of distinguished central $\mathbb{F}_p$-witnesses supplied by the two even blocks, and

$$\sum_{(d,r)\in\mathscr{D}_H} c_{d,r}(p) \ll H^{5/3}. \tag{89}$$

Their degree-weighted support is consequently at most $O(H^{8/3})$ with the present inputs.

Proof. If d is even, $x = -(d+1)/2$ is a root of N_d, and

$$N_r(x+d) = N_r((d-1)/2) = 0 \iff p \mid T_r^{(d)},$$

because p is odd. The symmetric root is $x = -d-(r+1)/2$. When both occur, their difference is $(d+r)/2 \neq 0$ modulo p, so they are distinct.

For fixed even d, the column $y = (d-1)/2$ is nonsingular throughout $2 \le r \le H-d$: indeed

$$0 < 2(y+r) = d - 1 + 2r < 2H < p.$$

Proposition 22 bounds its zero levels by $O(H^{2/3})$. Summing over the $O(H)$ even values of d and adding the symmetric orientation proves (89). Multiplication by the largest possible degree weight gives $O(H^{8/3})$. □

Thus the center structure explains repeated genuine factors, but the proved column exponent leaves a gap of $H^{1/6}$ even for their unweighted witness count.

5.13.2 What the diagonal square does and does not buy

Proposition 48 (Diagonal midpoint and split off-center roots)**.** *Put*

$$M_p(H) = \#\{2 \le d \le H/2 : p \mid D_d\}.$$

Then $M_p(H) \ll H^{2/3}$. If

$$q_d(p) = \#\{u \in (\mathbb{F}_p^\times)^2 : E_d(u) = O_d(u) = 0\},$$

then the diagonal energy has the exact decomposition

$$\mathcal{E}_p^\Delta(H) := \sum_{2d \le H} m_{d,d}(p) = \sum_{2 \le d \le H/2} \big(\mathbf{1}_{p \mid D_d} + 2q_d(p)\big) = O(H^{2/3}) + 2 \sum_{2 \le d \le H/2} q_d(p). \tag{90}$$

The corresponding degree-weighted support caused by the D_d factors is $O(H^{5/3})$. The unresolved diagonal contribution is the sum of split off-center counts $q_d(p)$. Each such root forces $p \mid Q_d$, but the converse need not hold.

Proof. Since p is odd,

$$p \mid D_d \iff N_d(-1/2) = 0.$$

The column $x = -1/2$ is nonsingular, so Proposition 22 gives $M_p(H) \ll H^{2/3}$. Formula (60) is exactly (90). Weighting the $O(H^{2/3})$ midpoint indices by at most H gives the last support bound. □

In the factorization $\mathcal{R}_{d,d} = \ell_d |D_d| Q_d^2$, the explicit factor ℓ_d comes from the leading coefficient and D_d detects the midpoint $T = 0$; Q_d is the remaining parity resultant. Its square reflects the formal parity pairing $T, -T$, which may coincide or degenerate, not two independent supported indices. The three factors need not be coprime, and $p \mid Q_d$ does not imply a split affine root: it may instead come from a nonsplit common u, or, when $p \mid \ell_d$, from a degree degeneration at U-infinity. For example, $p = 577$, $d = 4$ has $p \mid \ell_4 Q_4$ but no affine common root.

Proposition 49 (The valuation gain is only a constant in averaging)**.** *Let $H \le \sqrt{X}$ and define*

$$\mathscr{G}_p(H) = \{2 \le d \le H/2 : p \mid \mathcal{R}_{d,d},\ p \nmid \ell_d D_d\}.$$

Then

$$\sum_{X < p \le 2X} \#\mathscr{G}_p(H) \ll \frac{H^3 \log H}{\log X}, \tag{91}$$

$$\sum_{X < p \le 2X} \sum_{d \in \mathscr{G}_p(H)} (d-1) \ll \frac{H^4 \log H}{\log X}. \tag{92}$$

Proof. The diagonal valuation law gives $p^2 \mid \mathcal{R}_{d,d}$ for every indicated pair. Therefore

$$2 \log X \sum_{X < p \le 2X} \#\mathscr{G}_p(H) \le \sum_{2 \le d \le H/2} \log \mathcal{R}_{d,d}.$$

The Hadamard estimate $\log \mathcal{R}_{d,d} \ll d^2 \log(2d)$ proves (91). Applying the same argument to $\prod_{2 \le d \le H/2} \mathcal{R}_{d,d}^{d-1}$ proves (92). □

Without the square valuation, the same argument merely omits the factor 2 on the left. Thus it improves a prime-averaged constant, not an H-exponent. At the mesoscopic endpoint $H \asymp \sqrt{X}$, both resulting estimates are weaker than the trivial $O(H)$ count and $O(H^2)$ degree weight. For much shorter ranges they can exclude an initial polylogarithmic segment, but they give no uniform H-power gain in the target range. Hence the diagonal square law does not prove the requested diagonal-band case.

5.14 The exact remaining lemma and stall report

For $(d, r) \in \mathscr{D}_H$, define the split affine gcd degree

$$g^{\mathrm{sp}}_{d,r}(p) = \deg \gcd_{\mathbb{F}_p[X]} \big(\overline{N}_d(X), \overline{N}_r(X + d), X^p - X\big).$$

The polynomial $X^p - X$ is squarefree, and N_d, N_r are nonzero in the present range, so

$$g^{\mathrm{sp}}_{d,r}(p) = m_{d,r}(p). \tag{93}$$

If

$$A_p(H; j) = \#\{(d, r) \in \mathscr{D}_H : g^{\mathrm{sp}}_{d,r}(p) \geq j\},$$

then the layer-cake identity is

$$\boxed{\mathcal{E}_p(H) = \sum_{j\geq 1} A_p(H; j).} \tag{94}$$

Accordingly, the minimal missing arithmetic statement can be isolated as the following split affine gcd-tail lemma:

$$\boxed{\sum_{j\geq 1} \# \big\{(d, r) \in \mathscr{D}_H : \deg\gcd\big(\overline{N}_d, \overline{N}_r(\,\cdot + d), X^p - X\big) \geq j\big\} \ll H^{3/2}.} \tag{95}$$

It is exactly equivalent to the energy target, automatically discards the infinity lattice and nonsplit common factors, and charges the actual multiplicity instead of the worst possible degree. A slightly stronger purely algebraic sufficient statement replaces the three-polynomial gcd by the degree of the radical of the affine gcd $\gcd(\overline{N}_d, \overline{N}_r(\,\cdot + d))$.

The degree-weighted raw resultant estimate is a convenient sufficient condition, but it is not minimal: it both inserts the worst-case degree $3(\min(d, r) - 1)$ and counts projective and nonsplit support. Likewise, the fixed-point inequalities contain no mechanism capable of proving (95); a new arithmetic incidence or split-gcd tail estimate is required.

Finite verification. The script `energy_verify.py` evaluates the continuant recurrence pointwise for all 669 primes $p \leq 5000$, always with $H = \lfloor\sqrt{p}\rfloor$. It computes $R_p(H)$, total and boundary energy, both factorial column moments, $R^{\partial}_p(H)$, low-fiber mass, and polluted-column counts. It verifies (71), (73), and the integer form of (76) for $K = \lceil\sqrt{H}\rceil$ whenever $H \geq 2$ (the 667 primes $p \geq 5$). The column identities are checked for all 669 primes. Selected records are:

p	H	R	$\mathcal{E}$	$\mathcal{E}^{\circ}$	$\sum_x \binom{k_x}{2}$	R^{∂}	P_H	$\sum_{x\in\mathcal{P}_H} k_x$
73	8	19	2	0	16	11	1	6
131	11	17	6	6	6	0	0	0
653	25	38	16	16	16	0	0	0
3331	57	187	0	0	1225	98	1	50
4283	65	203	4	2	1084	93	1	47

The canonical comma-separated 669-record digest printed by the script is recorded in the verifier itself. Across the scan,

$$\max \mathcal{E}_p(H) = 16 \quad (p = 653), \qquad \max \frac{\mathcal{E}_p(H)}{H^{3/2}} = \frac{6}{11^{3/2}} \quad (p = 131),$$

and the largest unpolluted column has four returns. These data are only tests of the deterministic identities, not evidence used in their proofs. In fact every off-boundary column in the 667 records with $K \geq 2$ has $k_x < K$, so $R_p(H) = R_p^{\partial}(H) + L_{p,K}(H)$ there. The tested amplification inequality is therefore completely vacuous on every record in its stated domain, exactly as the fixed-point audit predicts.

The dynamic-height scan also corrects one reconnaissance datum in the existing text. Since

$$b_7 = 584307365 = 3331 \cdot 175415,$$

the column $x = -8$ at $p = 3331$ vanishes at every level $8 \leq h \leq 57$ and has $k_{-8} = 50$. Thus the statement that the largest polluted count through $p = 5000$ is 19 at $p = 541$ is false for $H = \lfloor \sqrt{p} \rfloor$; 19 is the value at that particular prime.

STALL REPORT. The target $\mathcal{E}_p(H) \ll H^{3/2}$ is not proved. The rigorous outputs are:

1. the exact off-boundary identity (71) and the sharp boundary directions;
2. the proof that the proposed $\mathcal{E} \leftrightarrow R$ fixed point cannot contract, together with the unconditional $\mathcal{E}_p(H) \ll H^{8/3}$ consequence of current inputs;
3. the corrected optimization $Z(p) \ll p/H + p^{2/3}H^{(\beta-2)/3}$, under which every $\beta < 2$ available past $H = p^{1/3}$ improves 2/3;
4. $H^{3/2}$ bounds for the deterministic shallow strip $\min(d, r) \leq H^{1/4}$ and for the low part $\min(d, r) \leq H^{3/4}$ of a fixed-width diagonal band;
5. exact removal of the infinity lattice, a bound for genuine center witnesses, and the isolation of the actual split off-center counts $q_d(p)$, each of which forces $p \mid Q_d$;
6. the exact missing split affine gcd-tail statement (95).

5.15 Computational evidence

Proposition 50. *For all* 78,496 *primes* $5 \leq p \leq 10^6$*:*

1. *$Z(p) \in \{0, 2, 4, 6, 8, 10, 12\}$ for all but two primes; the two exceptions ($p = 11$ and $p = 3137$) have $Z(p) = 1$ (non-ordinary primes, where $p \mid a_p(f)$).* $\max Z(p) = 12$ *at* $p = 159{,}977$.
2. *The exact histogram is:*

$Z(p)$	*0*	*1*	*2*	*4*	*6*	*8*	*10*	*12*
count	*47632*	*2*	*23730*	*6045*	*951*	*123*	*12*	*1*
fraction	*.607*	*.000*	*.302*	*.077*	*.012*	*.002*	*.000*	*.000*
Poi$(\frac{1}{2})$	*.607*	—	*.303*	*.076*	*.013*	*.002*	*.000*	*.000*

The mean $Z(p) = 1.000$, stable across all ranges ($\chi^2 = 3.42$ on 4 d.f. against Poisson(1/2) *for the pair count). The empirical factorial moments of $K(p) = Z(p)/2$ over the* 78,494 *ordinary primes are:*

r	*1*	*2*	*3*	*4*	*5*	*6*
$\frac{1}{\pi_{\mathrm{ord}}}\sum(K)_r$	.4998	.2490	.1210	.0605	.0275	.0092
2^{-r}	.5000	.2500	.1250	.0625	.0312	.0156

The first four moments match Poisson(1/2) to better than 3%; the higher-moment deviation is consistent with sampling variance ($K \geq 5$ has only 13 primes in this range).

3. *The symmetry $b_j \equiv b_{p-1-j} \pmod p$ holds for all tested primes: zeros come in palindromic pairs $\{j, p-1-j\}$, so $Z(p)$ is even (except at non-ordinary primes). The pair count $Z(p)/2$ fits* Poisson(1/2) *to within sampling noise (table above). Proposition 18 proves* Poisson(1/2) *for the root-pair count $Y_h(p)$ of each fixed gap polynomial in the prime-aspect; the transfer to the actual coefficient zero count $K(p) = Z(p)/2$ (Remark 19) remains conjectural.*

Remark 51 (Mesoscopic root count). At the scale $H = \lfloor\sqrt{p}\rfloor$, the total root count $R_p(H) = \sum_{h=2}^{H} r_p(h)$ satisfies $R_p(H)/H \approx 3/2$ for all tested primes, consistent with the Chebotarev prediction from Remark 17:

p	$H = \lfloor\sqrt{p}\rfloor$	$R_p(H)$	$R_p(H)/H$
503	22	35	1.59
2003	44	60	1.36
5003	70	105	1.50
10007	100	154	1.54
20011	141	230	1.63
50021	223	347	1.56
100003	316	460	1.46

The ratio $R_p(H)/H$ stabilizes near 3/2 (range 1.32–1.75 for $p > 500$), supporting the mesoscopic conjecture in Remark 23.

5.16 Hypothesis Z

Hypothesis 52 (Hypothesis Z)**.** *There exists an absolute constant C such that $Z(p) \leq C$ for all primes p.*

Remark 53. The analogous problem for the $\zeta(2)$ Apéry numbers $A_n = \sum_k \binom{n}{k}^2\binom{n+k}{k}$ has a fundamentally different structure. Stienstra–Beukers [25] showed that A_p is related to a *CM* modular form, yielding $A_{(p-1)/2} \equiv 0 \pmod p$ for all $p \equiv 3 \pmod 4$ — a *positive density* of primes with guaranteed zeros. In contrast, the $\zeta(3)$ Apéry numbers are governed by the non-CM newform $f(z) = \eta(2z)^4\eta(4z)^4 \in S_4(\Gamma_0(8))$ (LMFDB label 8.4.a.a), for which no such systematic zero-forcing occurs. This is a structural reason why Hypothesis $\bar{Z}$ should hold for the $\zeta(3)$ sequence, even though the pointwise bound $Z(p) \leq C$ (Hypothesis Z) likely fails.

Remark 54. Even the single-row slice $Z^*(p) := \mathbf{1}[p \mid b_{(p-1)/2}]$ is connected to the non-ordinary primes of f via the Beukers congruence $b_{(p-1)/2} \equiv a_p(f) \pmod p$ [5, 1]. The density-zero conjecture for non-ordinary primes of non-CM weight-4 forms is itself open [19]. Thus Hypothesis Z contains a known open problem as a special case. The heuristic density of non-ordinary primes is $\sim C_f/p$ (the Sato–Tate distribution on the normalized coefficients $a_p/p^{3/2}$ admits $\sim 4p^{1/2}$ multiples of p in $[-2p^{3/2}, 2p^{3/2}]$, each contributing mass $\sim p^{-3/2}$). The expected count up to x is thus $C_f \log\log x$ with $C_f \approx 1$. Finding 2 non-ordinary primes ($p = 11, 3137$) below 10^6 is consistent with $\log\log(10^6) \approx 2.9$. This is not a standard Lang–Trotter problem: the target set $\{a : p \mid a\}$ varies with p, and no analogue of Elkies' theorem (infinitely many supersingular primes) is known for weight ≥ 4.

Remark 55 (Evidence against Hypothesis Z). The $2 \cdot \mathrm{Poisson}(1/2)$ distribution (Remark 10 below) has unbounded support: extreme-value theory predicts $\max_{p \le N} Z(p) \sim 2 \log\log N$. The current record $Z(p) = 12$ at $p = 159{,}977$ (with record progression $2, 4, 6, 8, 10, 12$) is consistent with this prediction. We therefore expect Hypothesis Z to fail: the first prime with $Z(p) = 14$ should appear near $p \approx 2 \times 10^7$ (the Gumbel prediction for the Poisson(1/2) pair maximum). A computation to $p = 5 \times 10^7$ would either refute Hypothesis Z or provide strong evidence for a structural cutoff.

Hypothesis 56 (Hypothesis $\bar{Z}$)**.** *$\sum_{p \le x} Z(p) \ll \pi(x)$, equivalently, the average of $Z(p)$ over primes $p \le x$ is bounded.*

Remark 57. Hypothesis $\bar{Z}$ is strictly weaker than Hypothesis Z and consistent with $Z(p) \sim 2 \cdot \mathrm{Poisson}(1/2)$ (which gives $\mathbb{E}[Z(p)] = 1$, independent of p). It suffices for the conditional bound $\log G_n = O(\sqrt{n})$ (Theorem 80).

We state the full distributional prediction of the Poisson model (cf. Proposition 18):

Hypothesis 58 (Poisson conjecture)**.** *For every good prime $p \ge 5$, put*

$$\nu_p := \mathbf{1}_{\{p \mid a_p(f)\}}, \qquad K_*(p) := \frac{Z(p) - \nu_p}{2},$$

where f is the weight-4 newform associated to the Apéry differential operator. Thus $K_*(p)$ counts noncentral reflected pairs of zeros of $(b_j \bmod p)_{0 \le j < p}$. For every fixed $r \ge 1$,

$$\frac{1}{\pi(X)} \sum_{p \le X} \binom{K_*(p)}{r} \longrightarrow \frac{1}{2^r\, r!}.$$

On ordinary primes ($\nu_p = 0$), this reduces to $K(p) = Z(p)/2 \to \mathrm{Poisson}(1/2)$, consistent with the data in Proposition 50.

Remark 59. The Poisson conjecture implies Hypothesis $\bar{Z}$. Taking $r = 1$ gives $\sum_{p \le X} K_*(p) = (\frac{1}{2} + o(1))\, \pi(X)$. Since $Z(p) = 2K_*(p) + \nu_p$ and $0 \le \nu_p \le 1$,

$$\sum_{p \le X} Z(p) = 2 \sum_{p \le X} K_*(p) + \sum_{p \le X} \nu_p \le (2 + o(1))\, \pi(X),$$

which is Hypothesis $\bar{Z}$. A proof would require the fixed-order joint independence of reflection-pair zero events—the correlation theorem underlying the factorial-moment computation in Proposition 18—for the deterministic Apéry orbit, not just the hyperoctahedral random model.

Remark 60 (Computational evidence for $\bar{Z}$). Over dyadic blocks of primes, $(\sum_{P < p \le 2P} Z(p))/|\{P < p \le 2P\}|$ ranges from 0.86 to 1.07 for $P = 2^k$, $3 \le k \le 12$ (666 primes up to 4999), with overall average $\sum Z(p)/\pi = 0.91$. The Poisson prediction is 1.

6 The denominator connection lemma

Lemma 61. *Let $p \ge 7$ be prime with $n/2 < p \le n$, and set $r = n - p$. Then $v_p(G_n) = 0$ if and only if $p \nmid b_r$.*

Proof. Since $p > n/2$, the only multiple of p in $\{1, \ldots, n\}$ is p itself, so $v_p(d_n) = 3$.

Step 1. $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$, so $v_p(G_n) = 0$ requires $v_p(d_n a_n) = 0$, i.e., $v_p(a_n) = -3$.

Step 2 (variation of parameters). From Lemma 1, the Casorati identity telescopes to

$$\frac{a_n}{b_n} = \sum_{m=1}^{n} \frac{6}{m^3\, b_m\, b_{m-1}}. \tag{96}$$

For $p > n/2$, the only $m \in \{1, \ldots, n\}$ with $p \mid m$ is $m = p$.

Step 3. By Theorem 4, $b_p \equiv b_0\, b_1 = 5 \pmod p$ (since p has base-p digits $(0, 1)$), and by the symmetry $b_j \equiv b_{p-1-j}$, $b_{p-1} \equiv b_0 = 1 \pmod p$. Since $p \geq 7$, both are nonzero mod p.

The $m = p$ term in (96) has $v_p = -3$; all other terms have $v_p \geq 0$. Therefore

$$p^3 \cdot \frac{a_n}{b_n} \equiv \frac{6}{b_p\, b_{p-1}} \equiv \frac{6}{5} \pmod p.$$

Using $b_n \equiv 5\, b_r \pmod p$ (Theorem 4, since $n = 1 \cdot p + r$):

$$p^3\, a_n \equiv \tfrac{6}{5} \cdot 5\, b_r = 6\, b_r \pmod p.$$

Thus $v_p(a_n) = -3$ iff $p \nmid b_r$. $\square$

Verification. For all $n \leq 300$ and primes $p \in (n/2, n]$ with $p \geq 7$: $v_p(G_n) = 0 \iff p \nmid b_{n-p}$ confirmed in 2101/2101 cases (100%).

Corollary 62 (Lucas reduction)**.** *For $p \geq 7$ and $p \in (n/2, n]$: $v_p(G_n) \geq 1$ iff $p \mid b_{n-p}$. By the Lucas congruence $b_n \equiv b_1\, b_{n-p} = 5\, b_{n-p} \pmod p$, this is equivalent (for $p \neq 5$) to $p \mid b_n$. Thus $B(n) := \#\{p \in (n/2, n], p \text{ prime} : v_p(G_n) \geq 1\}$ equals $\omega_{(n/2,n]}(b_n) + O(1)$, the number of* distinct prime factors of b_n lying between $n/2$ and n*, up to an absolute constant.*

The full conjecture $G_n = e^{o(n)}$ for all n is therefore equivalent to the *top-half radical estimate*:

$$\log \operatorname{rad}_{(n/2,n]}(b_n) = o(n). \tag{97}$$

Since $\log b_n \sim n \log(17 + 12\sqrt{2}) \approx 3.52\, n$, the height bound gives $\omega_{(n/2,n]}(b_n) \leq \log b_n / \log(n/2) = O(n/\log n)$, precisely one $o(1)$ factor short of the target. Computation for $n \leq$ 2,000,000 gives $B(n) \leq 3$ (with exactly 53 maximizers and no value $B(n) \geq 4$) and $B(n) = 0$ for 94.8% of values (see Remark 63 for the Poisson analysis).

Remark 63 (Random-model prediction). Under a random-integer model, $\Pr(p \mid b_n) \approx 1/p$ for each prime p, so by Mertens' theorem

$$\mathbb{E}[B(n)] = \sum_{n/2 < p \leq n} \frac{1}{p} = \log\Big(\frac{\log n}{\log(n/2)}\Big) + O(1/\log n) \approx \frac{\log 2}{\log n} \ \to\ 0.$$

Computation for $n \leq$ 200,000 confirms: $\mathbb{E}[B] \approx 0.065$, $\max B(n) = 3$, and the histogram matches Poisson$(\log 2/\log n)$ to high accuracy: $B(n) = 0$ for 93.7%, $B(n) = 1$ for 6.0%, $B(n) = 2$ for 0.20%, $B(n) = 3$ for 0.006% (the Poisson predictions with $\lambda = 0.065$ are 93.7%, 6.1%, 0.20%, 0.004%). The dispersion ratio $\operatorname{Var}(B)/\mathbb{E}[B] = 1.003$ is the exact Poisson signature. The extended scan to $n \leq$ 2,000,000 (with all 148,930 primes $7 \leq p \leq 2 \times 10^6$) confirms the picture at ten times the range: $\max B(n) = 3$ still, overall $\operatorname{Var}(B)/\mathbb{E}[B] = 0.9991$, per-shell $\operatorname{Var}/\mathbb{E}$ between 0.95 and 1.00, and windowed sums over 64 windows per dyadic shell show no localized pile-up (largest window sum 481 against a window mean of 429.5 at $N = 2^{19}$). By CRT-based approximate independence across primes, $B(n)$ should follow Poisson$(\log 2/\log n)$ with $\lambda \to 0$. Extreme-value theory then predicts $\max_{n\le N} B(n) = O(\log N/\log\log N)$—trivially $o(n/\log n)$—so the random model *strongly* predicts the full conjecture. The obstruction is proving that b_n behaves "randomly" with respect to its prime factors in $(n/2, n]$, despite its highly structured factorization (supercongruences, Lucas decomposition, palindromic symmetry).

7 The unconditional density-one theorem

Theorem 64. *For a set of positive integers n of natural density 1, $G_n = e^{o(n)}$.*

Proof. Fix N large and consider $n \in (N, 2N]$. Write $B(n) = \#\{\text{primes } p > \sqrt{n} \text{ with } v_p(G_n) \ge 1\}$. By Proposition 3, the small-prime contribution is $O(\sqrt{n})$ for every n. Each such prime contributes $v_p(G_n)\log p \le 3\log n$ (since $v_p(d_n) = 3$ for $p > \sqrt{n}$), so $\log G_n \le O(\sqrt{n}) + 3B(n)\log n$. It suffices to show $B(n) = o(n/\log n)$ for all but $o(N)$ values of $n \in (N, 2N]$.

For $p > \sqrt{n}$, write $n = qp + r$ with $1 \le q \le \sqrt{2N}$ and $0 \le r < p$. By Lemma 69(iii), $v_p(G_n) \ge 1$ requires $D_q b_r \equiv 0 \pmod{p}$. By Lemma 70, leading-digit bad primes ($p \mid b_q$, $p \nmid b_r$) have $v_p(G_n) = 0$ and do not contribute to $B(n)$. The remaining contributions come from *lower-digit bad primes* ($r \in \mathcal{Z}_p$) and *companion-block primes* ($D_q \equiv 0 \pmod{p}$, $p \nmid b_r$).

Lower-digit incidence. For fixed $p \in (\sqrt{N}, 2N]$, an interval of N consecutive integers contains at most $N \cdot Z(p)/p + Z(p)$ values of n whose residue $r = n \bmod p$ lies in $\mathcal{Z}_p$. Set $\rho_p = Z(p)/p$. The total lower-digit incidence is

$$R_N \le \sum_{\sqrt{N}<p\le 2N} \big(N\rho_p + Z(p)\big).$$

By Proposition 9, $\rho_p \to 0$. Since $\pi(2N) = O(N/\log N)$:

$$\sum_p \rho_p = o(\pi(2N)) = o(N/\log N), \quad \sum_p Z(p) = \sum_p p\rho_p \le 2N\sum_p \rho_p = o(N^2/\log N).$$

Hence $R_N = o(N^2/\log N)$.

Companion-block incidence. By Proposition 73, the companion-block contribution to $\log G_n$ is $O(n^{2/3}) = o(n)$ at every index. Hence companion-block primes do not affect the density-one statement, and we set $P_N = 0$ in the counting argument.

Conclusion. Write $\sum_{n\in(N,2N]} B(n) \le R_N + P_N = o(N^2/\log N)$. For any fixed $\delta > 0$, Markov's inequality gives

$$\#\big\{n \in (N, 2N] : B(n) > \delta n/\log n\big\} \le \frac{o(N^2/\log N)}{\delta N/\log N} = o(N).$$

A dyadic decomposition shows the exceptional set has density zero, so $B(n) = o(n/\log n)$ for density 1. For all such n, $\log G_n = O(\sqrt{n}) + 3B(n)\log n = O(\sqrt{n}) + o(n) = o(n)$. ☐

Corollary 65 (Quantitative exceptional set)**.** *For every $\varepsilon > 0$,*

$$\#\{n \le N : \log G_n > \varepsilon\, n\} = O_\varepsilon(N^{2/3}).$$

Proof. We make the density-one argument quantitative using the explicit bound $Z(p) = O(p^{2/3})$ from Proposition 9. For $n \in (N, 2N]$, $\log G_n \le O(\sqrt{n}) + 3B(n)\log n$, so $\log G_n > \varepsilon n$ requires $B(n) > \varepsilon N/(7\log N)$ for N large.

Partial summation with $Z(p) \le Cp^{2/3}$ gives

$$\sum_{\sqrt{N}<p\le 2N} \frac{Z(p)}{p} \le C \sum_{p\le 2N} p^{-1/3} = \tfrac{3}{2}\, C\, \frac{(2N)^{2/3}}{\log(2N)} + O(N^{1/3}) = O\left(\frac{N^{2/3}}{\log N}\right).$$

The lower-digit incidence satisfies $R_N = N\sum Z(p)/p + \sum Z(p) = O(N^{5/3}/\log N)$ (the second sum, $\sum Z(p) \le C\sum p^{2/3}$, contributes the same order). The companion-block incidence $P_N = O(N^{3/2})$ is dominated by R_N. Hence $\sum_{n\in(N,2N]} B(n) = O(N^{5/3}/\log N)$.

By Markov's inequality,

$$\#\{n \in (N, 2N] : B(n) > \varepsilon N/(7\log N)\} \le \frac{O(N^{5/3}/\log N)}{\varepsilon N/(7\log N)} = O(N^{2/3}/\varepsilon).$$

A dyadic decomposition over intervals $(N/2^{j+1}, N/2^j]$ gives the global bound, since $\sum_{j\ge 0}(N/2^j)^{2/3} = O(N^{2/3})$. □

In particular, the exceptional set has finite harmonic weight:

Corollary 66 (Finite harmonic weight)**.** *For every $\varepsilon > 0$,*

$$\sum_{\substack{n\ge 1\\ \log G_n > \varepsilon n}} \frac{1}{n} < \infty.$$

Proof. The contribution from $(2^k, 2^{k+1}]$ is $O(2^{-k/3})$ by Corollary 65, and the geometric series converges. □

Remark 67. The exponent $2/3$ in the exceptional set is inherited from the lower-digit incidence R_N, which uses $Z(p) = O(p^{2/3})$. More generally, if $Z_p(I) \le C|I|^\alpha$ uniformly for all primes and sub-intervals (as Corollary 10 provides with $\alpha = 2/3$), then $R_N = O(N^{1+\alpha}/\log N)$ and $L_N = O(N^{(4-\alpha)/(3-\alpha)}/\log N)$. The exceptional set is therefore $O(\max(N^\alpha,\ N^{1/(3-\alpha)})/\varepsilon)$, with the crossover at $\alpha = (3-\sqrt{5})/2 \approx 0.382$. Under Hypothesis $\bar{Z}$, the dyadic bound $D(P, Q) \le \sum Z(p) = O(P/\log P)$ is equivalent to $\alpha = 0$ in this formula, giving the exceptional set $O(N^{1/3}/\varepsilon)$ (Corollary 82). Theorem 80 shows the matching density-one bound $\log G_n = O(\sqrt{n})$, where $O(\sqrt{n})$ is now the small-prime floor, not the leading-digit contribution.

8 Codegree amplification

We begin with a structural rigidity lemma: consecutive integers cannot share a top-half bad prime.

Lemma 68 (Consecutive coprimality)**.** *Let $p > 5$ be prime and $n \ge p$. If $p \mid b_{n \bmod p}$ and $p \mid b_{(n+1) \bmod p}$, then $p \le n/2$. Equivalently, no prime $p > n/2$ with $p > 5$ can be lower-digit bad for both n and $n+1$.*

Proof. Since $p > n/2$, both n and $n+1$ have base-p representations $n = p+r$, $n+1 = p+(r+1)$ with $0 \le r < r+1 < p$ (provided $r < p-1$; the case $r = p-1$ gives $n+1 = 2p$ and $b_{n+1} \equiv b_2 b_0 = 73 \not\equiv 0$ for $p > 73$, a finite check).

Suppose $p \mid b_r$ and $p \mid b_{r+1}$ with $r, r+1 < p$. The Apéry recurrence gives

$$(k+1)^3 b_{k+1} = P(k) b_k - k^3 b_{k-1},$$

and since $k+1 < p$ implies $(k+1)^3$ is invertible mod p, the conditions $p \mid b_k$ and $p \mid b_{k+1}$ force $p \mid b_{k-1}$. Iterating downward from $k = r$ gives $p \mid b_j$ for all $0 \le j \le r+1$, contradicting $b_0 = 1$. □

A prime that divides b_n only through a leading base-p digit contributes nothing to G_n.

Lemma 69 (Block system for $p^3 a_n \bmod p$)**.** *Let $p \geq 7$, $p \leq n < p^2$, and write $n = qp + r$ with $1 \leq q < p$ and $0 \leq r < p$. Define $\Delta_{q,r} \equiv p^3 a_{qp+r} \pmod p$ and $D_q = \Delta_{q,0}$. Then:*

(i) $\Delta_{q,r} \equiv D_q b_r \pmod p$ for every $0 \leq r < p$;

(ii) $D_q b_{q-1} - D_{q-1} b_q \equiv 6/q^3 \pmod p$;

(iii) $p \mid G_n$ if and only if $D_q b_r \equiv 0 \pmod p$.

Proof. For $p > \sqrt{n}$, $v_p(\mathrm{lcm}(1, \ldots, n)) = 1$, so $v_p(d_n) = 3$. Since $d_n a_n \in \mathbf{Z}$, $p^3 a_n \in \mathbf{Z}_p$, and $v_p(d_n a_n) = v_p(p^3 a_n)$. Meanwhile $v_p(d_n b_n) = 3 + v_p(b_n) \geq 3$, so $p \mid G_n \Leftrightarrow p \mid p^3 a_n$.

(i) Multiply the Apéry recurrence at index $qp + r$ by p^3 and reduce modulo p. Since $qp + r \equiv r \pmod p$, the sequence $r \mapsto \Delta_{q,r}$ satisfies the same recurrence $(r+1)^3 \Delta_{q,r+1} = P(r)\Delta_{q,r} - r^3 \Delta_{q,r-1}$ as $r \mapsto D_q b_r$, with matching initial values $\Delta_{q,0} = D_q$ and $\Delta_{q,1} = 5D_q = D_q b_1$. Since $(r + 1)^3$ is invertible mod p for $0 \leq r \leq p - 2$, uniqueness gives the identity.

(ii) Multiply the Wronskian $a_n b_{n-1} - a_{n-1} b_n = 6/n^3$ at $n = qp$ by p^3 and reduce mod p, using $b_{qp} \equiv b_q$ and $b_{qp-1} \equiv b_{q-1} b_{p-1} \equiv b_{q-1}$ (Lucas), and $p^3 a_{qp-1} \equiv D_{q-1} b_{p-1} \equiv D_{q-1}$ (part (i)).

(iii) Immediate from (i) and the observation above. □

Lemma 70 (Leading-digit vanishing)**.** *Let $p \geq 7$, $p \leq n < p^2$, $n = qp + r$ with $q \geq 1$. If $p \mid b_q$ and $p \nmid b_r$, then $v_p(G_n) = 0$.*

Proof. Since consecutive Apéry numbers below the p-th cannot both vanish mod p (the recurrence propagates to $b_0 = 1$), $p \nmid b_{q-1}$. Part (ii) of Lemma 69 gives $D_q b_{q-1} \equiv 6/q^3 \not\equiv 0 \pmod p$, so $D_q \not\equiv 0$. By part (iii), $p \mid G_n$ iff $D_q b_r \equiv 0$, but $D_q \not\equiv 0$ and $p \nmid b_r$, so $D_q b_r \not\equiv 0$ and $v_p(G_n) = 0$. □

Remark 71 (Block constants are Apéry numbers)*.* The block constant D_q of Lemma 69 equals $a_q \bmod p$, where a_q is the Apéry rational companion. Indeed, a_q satisfies the same Wronskian recurrence $a_q b_{q-1} - a_{q-1} b_q = 6/q^3$ with $a_0 = 0$, $a_1 = 6$, so $D_q = a_q$ by uniqueness. Consequently, $D_q \equiv 0 \pmod p$ iff p divides the numerator of a_q (in lowest terms) and p does not divide its denominator.

Since $|a_q| \leq C(1 + \sqrt{2})^{4q}$ and $\mathrm{denom}(a_q) \mid \mathrm{lcm}(1, \ldots, q)^3$, the numerator of a_q has $O(q)$ prime divisors. In particular, for fixed q, at most $O(q)$ primes p can be companion-block bad at quotient q. Computation through $n = 2000$ confirms that the companion-block count $\#\{p > \sqrt{n} : p \mid \mathrm{num}(a_{\lfloor n/p \rfloor})\}$ is at most 3 for every n in this range.

Proposition 72 (Reflection of the companion solution)**.** *Let $p \geq 5$ be prime. Every p-integral solution (u_n) of the Apéry recurrence* (1) *on $\{0, \ldots, p-1\}$ is palindromic modulo p:*

$$u_{p-1-j} \equiv u_j \pmod p \qquad (0 \leq j \leq p-1).$$

In particular, for the companion solution $a_0 = 0$, $a_1 = 6$,

$$a_{p-1-j} \equiv a_j \pmod p, \qquad a_{p-1} \equiv 0 \pmod p.$$

Proof. Let R denote the reversal operator, $(Ru)_j = u_{p-1-j}$. The coefficient polynomial $P(n) = 34n^3 + 51n^2 + 27n + 5$ satisfies $P(-1-x) = -P(x)$, so substituting $n \mapsto p - 2 - n$ in the recurrence shows that R maps p-integral solutions to p-integral solutions modulo p.

The integer Apéry solution (b_n) satisfies $b_{p-1} \equiv 1$ and $b_{p-2} \equiv 5 \pmod p$: in the binomial sum $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$, only the $k = 0$ term survives modulo p for $n = p - 1$, while for $n = p - 2$ only

$k = 0, 1$ survive, contributing $1+4 = 5$. Since Rb and b have the same initial values $(b_0, b_1) = (1, 5)$, uniqueness gives $Rb = b$.

For two solutions x, y, the discrete Wronskian $W_n(x, y) = x_n y_{n-1} - x_{n-1} y_n$ satisfies $(n+1)^3 W_{n+1} = n^3 W_n$, so $W_n(x, y) = C(x, y)/n^3$. Using $Rb = b$:

$$W_n(Ru, b) = -W_{p-n}(u, b) = W_n(u, b).$$

Hence $W_n(Ru - u, b) = 0$, so $Ru - u = cb$ for some $c \in \mathbf{F}_p$. Applying R: on one hand $R(Ru - u) = -(Ru - u)$, on the other $R(cb) = cb$. Since p is odd, $c = 0$ and $Ru = u$. □

Proposition 73 (Companion-block height bound)**.** *Assume $D_q \equiv a_q \pmod p$. For every $n \geq 1$, the total logarithmic weight of companion-block primes is at most $O(n^{2/3})$:*

$$\sum_{\substack{p > \sqrt{n} \\ p \mid \mathrm{num}(a_{\lfloor n/p \rfloor})}} \log p \;\leq\; C\, n^{2/3}$$

for an absolute constant C. In particular, companion-block primes contribute $o(n)$ to $\log G_n$ at every index.

Proof. Write $a_q = A_q/C_q$ in lowest terms. From the telescoping identity $a_q = b_q \sum_{k=1}^{q} 6/(k^3 b_k b_{k-1})$ and $b_k \geq 1$, we obtain $|A_q| \leq |a_q| \cdot C_q \leq C_0(1+\sqrt{2})^{4q}$, hence each numerator A_q has at most $O(q)$ prime divisors.

For a fixed n, set $Q = \lfloor n^{1/3} \rfloor$. A companion-block prime $p > \sqrt{n}$ has quotient $q = \lfloor n/p \rfloor < \sqrt{n}$.

Small quotients ($q \leq Q$): for each q, the primes dividing A_q contribute $\sum \log p \leq \log |A_q| = O(q)$. Summing over $q \leq Q$: $\sum_{q=1}^{Q} O(q) = O(Q^2) = O(n^{2/3})$.

Large quotients ($q > Q$): each such prime satisfies $p < n/Q = n^{2/3}$. Since each prime determines its quotient uniquely, the total log-weight from this range is bounded by the Chebyshev sum $\sum_{p < n^{2/3}} \log p = \theta(n^{2/3}) = O(n^{2/3})$. □

Proposition 74 (Quotient reduction)**.** *Let f be any function with $f(n) \to \infty$ and $f(n) \leq \log n$. Then*

$$\sum_{\substack{\sqrt{n} < p \leq n/(f(n)\log n) \\ p \mid G_n}} v_p(G_n) \log p \;=\; O\!\left(\frac{n}{f(n)}\right) = o(n).$$

Consequently,

$$\log G_n = O(\sqrt{n}) + O(n^{2/3}) + O\!\left(\frac{n}{f(n)}\right) + \sum_{\substack{n/(f(n)\log n) < p \leq n \\ p \mid b_{n \bmod p}}} v_p(G_n) \log p,$$

and the full conjecture $G_n = e^{o(n)}$ is equivalent to the last sum being $o(n)$ for some (equivalently, every) such f. Every prime in that sum has quotient $q = \lfloor n/p \rfloor < f(n) \log n$.

Proof. For $p > \sqrt{n}$ we have $p^2 > n$, so $\lfloor \log_p n \rfloor = 1$ and $v_p(G_n) \leq 3$ by (2); each bad prime therefore contributes at most $3 \log p \leq 3 \log n$. The number of primes up to $Y = n/(f(n)\log n)$ is $\pi(Y) \ll Y/\log Y \ll n/(f(n)\log^2 n)$, using $\log Y \geq \frac{1}{2}\log n$ for large n (as $f(n) \leq \log n$). Multiplying by $3\log n$ gives the first display. For the second display, split $\log G_n$ into: primes $p \leq \sqrt{n}$ (Proposition 3: $O(\sqrt{n})$); companion-block primes $p > \sqrt{n}$ with $p \mid \mathrm{num}(a_{\lfloor n/p \rfloor})$ (Proposition 73: $O(n^{2/3})$); the remaining bad primes $p > \sqrt{n}$, which by Lemma 69(iii) and Lemma 70 satisfy the lower-digit condition $p \mid b_{n \bmod p}$, split at $n/(f(n)\log n)$ by the first display. For the equivalence: one direction is the display; the other is immediate since any sub-sum of $\sum_p v_p(G_n)\log p$ is at most $\log G_n$. $\square$

Remark 75 (The conjecture lives at bounded quotients). Proposition 74 localizes the entire remaining difficulty to the top $O(\log\log n)$ dyadic blocks of primes: only $p \in (n/(f(n)\log n), n]$ matters, i.e., the quotient classes $q = \lfloor n/p \rfloor < f(n)\log n$. For each fixed q, the bad primes with quotient q form the shifted-divisibility set

$$\{\, p = (n-r)/q \text{ prime} : \ 0 \le r < p, \quad p \mid b_r \,\},$$

in which both the modulus and the index move with r. The case $q = 1$ is precisely the top-half radical estimate (97); the bounded-quotient cases are its mild deformations. Thus any method that resolves the top-half channel uniformly in $q < f(n)\log n$ for a single slowly growing f proves the full conjecture. Conversely, prime counting alone is already sufficient below this range, so no further structural information about small bad primes is needed anywhere in the argument.

The first-moment bound in Corollary 65 can be sharpened to a polylogarithmic exceptional set by exploiting the gap polynomials a second time, now as a *codegree* bound between nearby integers.

Lemma 76 (Common bad-prime codegree). *Let $m, n \in (N, 2N]$ with $h = n - m \ge 1$, and let $P_0 \ge 1$. The number of primes $p > P_0$ that are simultaneously lower-digit bad for m and for n (i.e., $p \mid b_{m \bmod p}$ and $p \mid b_{n \bmod p}$) is at most $O(h \log N/\log P_0)$. In particular, if $P_0 \ge N^\delta$ for any fixed $\delta > 0$, the codegree is $O(h)$.*

Proof. We split the common bad primes into three classes.

Small primes ($P_0 < p \le h$): counting all of them trivially, $\pi(h) \ll h \ll h\log N/\log P_0$ (as $P_0 \le 2N$ in any nonvacuous application).

No wrap ($p > h$ and $(m \bmod p) + h < p$): here $n \bmod p = (m \bmod p) + h$, so the two zeros of b modulo p are at distance exactly h, and the gap implication (Lemma 7) gives $p \mid N_h(m \bmod p)$; polynomial periodicity $N_h(m \bmod p) \equiv N_h(m) \pmod p$ then gives $p \mid N_h(m)$. The gap polynomial has degree $3(h-1)$ in m (the tridiagonal determinant representation gives exactly this), so for $m \in (N, 2N]$, $|N_h(m)| \le (CN^3)^h$. Since $N_h(m) > 0$ (the associated tridiagonal matrix is positive definite at positive integers), the number of prime factors exceeding P_0 is at most $3h\log(CN)/\log P_0$.

Wrap ($p > h$ and $(m \bmod p) + h \ge p$): writing $r = m \bmod p$, the wrap condition says $r = p - j$ for some $1 \le j \le h$, i.e. $p \mid m + j$. Hence every wrapped prime divides the nonzero integer $\prod_{j=1}^h (m+j) \le (3N)^h$, so there are at most $h\log(3N)/\log P_0$ of them—no zero-set information is needed in this case. (If $r + h = p$ exactly, then $n \equiv 0$ and $b_0 = 1$, so p is not lower-digit bad for n; this boundary case is vacuous but in any event covered by $j = h$.)

Summing the three classes gives $O(h\log N/\log P_0)$. $\square$

Remark 77 (Wrap is essential). The no-wrap case alone does not exhaust the lemma: at $p = 37$ one has $\mathcal{Z}_{37} = \{17, 19\}$, and $m = 93$, $n = 128$ (so $h = 35$) give residues 19 and 17—not differing by h. Here $p \mid m + 18$, and the wrap class accounts for it. In the top-half application ($p \in (N, 2N]$, quotient 1) wrap cannot occur, since $n \le 2N < 2p$.

Theorem 78 (Polylogarithmic exceptional set). *For every $\varepsilon > 0$,*

$$\#\{n \le N : \log G_n > \varepsilon\, n\} = O_\varepsilon\big((\log N)^2\big).$$

Proof. Fix $\varepsilon > 0$ and consider $n \in (N, 2N]$ with $\log G_n > \varepsilon n$. By Proposition 3, the small-prime contribution is $O(\sqrt{n})$. Each prime $p > \sqrt{n}$ satisfies $v_p(G_n) \leq 3$ (since $v_p(d_n) = 3$). By Lemma 70, leading-digit bad primes ($p \mid b_{\lfloor n/p \rfloor}$ with $n \bmod p \notin \mathcal{Z}_p$) satisfy $v_p(G_n) = 0$. The remaining primes with $v_p(G_n) \geq 1$ are the lower-digit bad primes ($n \bmod p \in \mathcal{Z}_p$) and the *companion-block primes* ($D_q \equiv 0 \pmod{p}$ with $p \nmid b_r$, cf. Lemma 69). By Proposition 73, the total logarithmic weight of companion-block primes is $O(n^{2/3})$ at every index, hence $o(n)$. Hence $\log G_n \leq O(\sqrt{n}) + 3\, t(n) \log n$, and $\log G_n > \varepsilon n$ forces

$$t(n) := \#\{p > \sqrt{n} : n \bmod p \in \mathcal{Z}_p\} \geq \frac{\varepsilon n}{4 \log n} \tag{98}$$

for N sufficiently large.

Partition $(N, 2N]$ into intervals J of length $Y = c_1 \varepsilon^2 N / \log N$, where $c_1 > 0$ is a small constant to be chosen. Write $\mathcal{E}_J = \{n \in J : \log G_n > \varepsilon n\}$ and $M = |\mathcal{E}_J|$.

For each prime $p \in (\sqrt{N}, 2N]$, let d_p count the elements of $\mathcal{E}_J$ for which p is lower-digit bad. By (98),

$$I := \sum_p d_p \geq \frac{\varepsilon N M}{4 \log N}.$$

The prime universe has size $L = \pi(2N) \leq 4N / \log N$.

Upper bound on pair incidence. For $m, n \in \mathcal{E}_J$ with $m < n$, Lemma 76 gives at most $C(n - m)$ common lower-digit bad primes (taking $P_0 = N^\delta$ in the codegree lemma). Hence

$$\sum_p \binom{d_p}{2} = \sum_{\substack{m<n \\ m,n \in \mathcal{E}_J}} |\mathcal{P}_m \cap \mathcal{P}_n| \leq C \sum_{\substack{m<n \\ m,n \in \mathcal{E}_J}} (n - m) \leq C Y M^2.$$

Lower bound on pair incidence. By Cauchy–Schwarz applied to the d_p,

$$\sum_p \binom{d_p}{2} \geq \frac{1}{2}\left(\frac{I^2}{L} - I\right) \geq \frac{1}{2}\left(\frac{\varepsilon^2 N M^2}{64 \log N} - I\right).$$

Combining gives $CYM^2 \geq \varepsilon^2 N M^2 / (128 \log N) - I/2$. Since $I \leq LM \leq 4NM/\log N$ and $Y = c_1 \varepsilon^2 N / \log N$, for c_1 chosen smaller than $1/(128C)$, the YM^2 term cannot absorb the quadratic lower bound once $M > C'/\varepsilon^2$. Hence $M = O(1/\varepsilon^2)$.

There are $N/Y = O(\log N / \varepsilon^2)$ intervals. The total on $(N, 2N]$ is therefore $O(\log N / \varepsilon^4)$. Summing over $O(\log N)$ dyadic blocks $(N/2^{j+1}, N/2^j]$ gives the result. $\square$

Theorem 78 supersedes Corollary 65. The proof's critical-window estimate—every interval of length $\asymp \varepsilon^2 N / \log N$ in $(N, 2N]$ contains $O_\varepsilon(1)$ exceptional integers—directly gives the uniform local bound

$$\sup_{M \geq 0} \#\{M < n \leq M + L : \log G_n > \varepsilon n\} \ll_\varepsilon L^{2/3} + 1,$$

by applying the critical-window argument to any interval $[M, M + L]$ with $N = M$ and $Y = c\varepsilon^2 M / \log M$. Dividing by L and letting $L \to \infty$ yields *upper Banach density zero* unconditionally, strengthening the harmonic-weight corollary 66.

9 Counting bad primes: the conditional result

We first sharpen the leading-digit incidence from $O(N^{3/2})$ to $O(N^{10/7}/\log N)$ via a dyadic argument combining divisor-counting with the sub-interval zero bound; this is unconditional and also strengthens the conditional theorem below.

Proposition 79 (Dyadic leading-digit bound)**.** *The leading-digit incidence L_N (counting pairs (n,p) with $n \in (N, 2N]$, $p > \sqrt{n}$ prime, and $p \mid b_{\lfloor n/p \rfloor}$) satisfies*

$$L_N \;=\; O\!\left(\frac{N^{10/7}}{\log N}\right).$$

Proof. For dyadic blocks $P = 2^j$, $Q = 2^k$ with $Q < P$ and $PQ \in [N/4,\, 4N]$, define

$$D(P,Q) = \#\{(p,q) : P < p \le 2P,\ Q < q \le 2Q,\ p \mid b_q\}.$$

Two bounds hold unconditionally:

- *Sub-interval zero counting:* By Corollary 10, each prime $p \in (P, 2P]$ has at most $O(Q^{2/3})$ zeros of b_j in $(Q, 2Q]$. Summing over $\pi(2P)$ primes gives $D(P,Q) = O(P\,Q^{2/3}/\log P)$.
- *Divisor counting:* Since $\log b_q \le (q{+}1)\log 64$, the number of prime factors of b_q exceeding P is at most $\log b_q/\log P = O(q/\log P)$. Summing over $q \in (Q, 2Q]$ gives $D(P,Q) \le \sum_q O(q/\log P) = O(Q^2/\log P)$.

Hence $D(P,Q) \le \min(P\,Q^{2/3},\, Q^2)/\log P$.

Each pair (p,q) in $D(P,Q)$ contributes to at most $p \le 2P$ values of n with $\lfloor n/p \rfloor = q$, so the block's contribution to L_N is at most $D(P,Q)\cdot 2P \le 2\min(P^{4/3}\,N^{2/3},\ N^2/P)/\log P$ (using $Q = \Theta(N/P)$). The crossover is at $P_0 = N^{4/7}$. Summing $O(\log N)$ dyadic blocks $P \in [\sqrt{N},\, 2N]$:

$$L_N \;\le\; \sum_{P \le N^{4/7}} \frac{2P^{4/3}\,N^{2/3}}{\log P} \;+\; \sum_{P > N^{4/7}} \frac{2N^2}{P\log P} \;=\; O\!\left(\frac{N^{10/7}}{\log N}\right),$$

since both sums are geometric series dominated by the $P = N^{4/7}$ boundary term. ☐

Theorem 80. *Assume Hypothesis $\bar{Z}$. Then for a set of positive integers n of natural density* 1*,*

$$\log G_n = O(\sqrt{n}).$$

Proof. Fix N and $\omega(N) \to \infty$ arbitrarily slowly. As in Theorem 64, decompose $\log G_n = \sum_p v_p(G_n)\log p$. The small-prime contribution is $O(\sqrt{n})$. Split $B(n) = B_{\text{low}}(n) + B_{\text{lead}}(n)$, where B_{low} counts bad primes $p > \sqrt{n}$ with $n \bmod p \in \mathcal{Z}_p$ and B_{lead} counts those with $\lfloor n/p \rfloor \in \mathcal{Z}_p$.

Lower digit. Under Hypothesis $\bar{Z}$, set $M(t) = \sum_{p \le t} Z(p) \ll t/\log t$. Abel summation gives $\sum_{\sqrt{N} < p \le 2N} Z(p)/p = O(1)$, so $R_N = \sum B_{\text{low}}(n) = O(N)$. By Markov, $B_{\text{low}}(n) \le \omega(N)$ for all but $O(N/\omega(N)) = o(N)$ values.

Leading digit. As in the proof of Corollary 82, Hypothesis $\bar{Z}$ gives $L_N = O(N^{4/3}/\log N)$. By Markov with threshold $\omega(N)\,N^{1/3}/\log N$:

$$\#\big\{n \in (N, 2N] : B_{\text{lead}}(n) > \omega(N)\,N^{1/3}/\log N\big\} \le \frac{O(N^{4/3}/\log N)}{\omega(N)\,N^{1/3}/\log N} = O(N/\omega(N)) = o(N).$$

Conclusion. For all remaining n:

$$\log G_n \le O(\sqrt{n}) + 3\big(\omega(N) + \omega(N)\, N^{1/3}/\log N\big) \log n.$$

Since $\omega(N) \cdot N^{1/3} \ll N^{1/2}$ for any $\omega(N) \to \infty$ slowly enough ($\omega \le N^{1/6}$ suffices), the second term is $O(N^{1/2}) = O(\sqrt{n})$. □

Remark 81 (The leading-digit bottleneck). The gap between the conditional bound $O(\sqrt{n})$ (Theorem 80) and the conjectured $O(\log n)$ comes from the small-prime contribution (Proposition 3), which gives an unconditional floor of $O(\sqrt{n})$. Under Hypothesis $\bar{Z}$, the big-prime contribution is now absorbed into this floor, thanks to Proposition 79.

Heuristically, $B_{\text{lead}}(n) = O(1)$ because the leading digit $\lfloor n/p \rfloor$ lands in $\mathcal{Z}_p$ with probability $\le C/p$ for each prime, giving expected $B_{\text{lead}} = \sum C/p = O(1)$. Making this rigorous would require $D(P,Q) \ll Q/\log P$ (the random-divisibility prediction), improving the proven $D(P,Q) \le \min(P\,Q^{2/3},\, Q^2)/\log P$. The small-prime bound $O(\sqrt{n})$ itself would need a finer analysis of $v_p(G_n)$ for small primes.

Corollary 82 (Conditional exceptional set)**.** *Assume Hypothesis $\bar{Z}$. For every $\varepsilon > 0$,*

$$\#\{n \le N : \log G_n > \varepsilon\, n\} = O_\varepsilon(N^{1/3}).$$

Proof. Under Hypothesis $\bar{Z}$, $R_N = O(N)$. For the leading-digit incidence, Hypothesis $\bar{Z}$ also strengthens the dyadic bound: since each prime $p \in (P, 2P]$ contributes at most $Z(p)$ zeros to any sub-interval, $D(P,Q) \le \sum_{P<p\le 2P} Z(p) = O(P/\log P)$ (using $\sum_{p\le x} Z(p) = O(\pi(x))$). The crossover with the divisor bound $D(P,Q) \le Q^2/\log P$ shifts from $P_0 = N^{4/7}$ to $P_0 = N^{2/3}$, giving $L_N = O(N^{4/3}/\log N)$. The threshold $B(n) > \varepsilon N/(7\log N)$ is exceeded by at most

$$\frac{O(N) + O(N^{4/3}/\log N)}{\varepsilon N/(7\log N)} = O(N^{1/3}/\varepsilon).$$

Dyadic summation over $(N/2^{j+1}, N/2^j]$ gives the global bound. □

Under Hypothesis $\bar{Z}$, the exceptional set is not only sparse globally but cannot cluster in any interval:

Corollary 83 (Upper Banach density zero)**.** *Assume Hypothesis $\bar{Z}$. For every $\varepsilon > 0$, the set $E_\varepsilon = \{n : \log G_n > \varepsilon n\}$ has upper Banach density zero:*

$$\lim_{L\to\infty} \sup_{M\ge 0} \frac{|E_\varepsilon \cap (M, M+L]|}{L} = 0.$$

Proof. For $I = (M, M+L]$ with $M > 2L$, every prime $p > n/2$ with $n \in I$ lies in $(M/2,\, M+L]$. Under Hypothesis $\bar{Z}$, each such prime contributes at most $Z(p)$ candidate bad integers to I, giving $\sum_{n\in I} B(n) \le \sum_{M/2<p\le M+L} Z(p) \ll \pi(M+L) = O(M/\log M)$. For $n \in I \cap E_\varepsilon$, the bound $\log G_n \le O(\sqrt{n}) + 3B(n)\log n$ and $\log G_n > \varepsilon n$ force $B(n) \gg \varepsilon M/\log M$ for M large. Therefore $|I \cap E_\varepsilon| \ll_\varepsilon 1 = o(L)$. The case $M \le 2L$ follows similarly by covering $(0, 3L]$ with $O(\log L)$ dyadic intervals. □

Corollary 84. *The condition $Z(p) = o(p)$ is* necessary *for the conjecture: if $Z(p) \ge \varepsilon p$ for a positive proportion of primes, double-counting gives* $\log G_n \ge c\varepsilon n$ *for a positive proportion of n. Conversely, $Z(p) = o(p)$ implies $G_n = e^{o(n)}$ for density* 1 *(Theorem 64).*

Proof. For each prime $p \in (n/2, n]$ with $Z(p) \geq \varepsilon p$, Lemma 61 gives $v_p(G_n) \geq 1$ whenever $n - p \in \mathcal{Z}_p$. Suppose a proportion $\delta > 0$ of primes satisfy $Z(p) \geq \varepsilon p$. Summing over $n \in (N, 2N]$ and bad primes $p \in (n/2, n]$, the total incidence count is $\geq \delta\, \pi(2N) \cdot \varepsilon N/2 = \Omega(N^2/\log N)$. Dividing by N and applying Markov's inequality: a proportion $\geq \varepsilon\delta/4$ of integers n have $B(n) \geq \varepsilon\delta N/(4 \log N)$, giving $\log G_n \geq 3B(n) \log(N/2) \geq c\varepsilon\delta\, n$. □

Remark 85 (The pointwise gap). The full conjecture $G_n = e^{o(n)}$ for *every* n is strictly stronger than the density-one statement. By Corollary 62, it is equivalent to the top-half radical estimate (97). The height bound gives $B(n) = O(n/\log n)$, but for an *adversarial* family (Z_p) with $|Z_p| \leq 1$, one may choose $Z_p = \{N - p\}$ for primes in $(N/2, N]$, producing $B(N) \geq \pi(N) - \pi(N/2) \sim N/(2 \log N)$. Thus the needed pointwise bound $B(n) = o(n/\log n)$ requires arithmetic information specific to the Apéry sequence, not just the cardinalities $Z(p)$.

By Corollary 62, every bad prime $p > n/2$ satisfies $p \mid b_n$. Hence the conjecture implies $\log \operatorname{rad}_{(n/2,n]}(b_n) = o(n)$: the large prime radical of b_n must be subexponential, even though $|b_n| = e^{\Theta(n)}$.

The Poisson model predicts that $B(n) \geq 1$ occurs infinitely often: $\mathbb{P}(B(n) \geq 1) \sim \log 2/\log n$, and $\sum (\log n)^{-1}$ diverges. This is compatible with the conjecture, since one bad prime contributes only $O(\log n)$ to $\log G_n$. To violate $G_n = e^{o(n)}$, one needs $B(n) \geq \varepsilon n/(6 \log n)$; the Poisson tail gives $\mathbb{P}(B(n) \geq k_n) \leq e^{-cn}$ for $k_n = \lceil \varepsilon n/(6 \log n) \rceil$, and the sum converges. Thus the model predicts: bad-prime occurrences are inevitable, but catastrophic pile-ups are finitely many.

10 Computational verification

We computed G_n exactly for $n = 1, \ldots, 1000$ using rational arithmetic.

n range	avg $\log(G_n)/n$	max $\log(G_n)/n$
1–50	0.276	0.93
51–100	0.108	0.26
101–200	0.086	0.18
201–300	0.048	0.10
301–400	0.038	0.08
401–500	0.024	0.05
501–600	0.029	0.06
601–700	0.017	0.04
701–800	0.014	0.04
801–900	0.016	0.04
901–1000	0.013	0.03

A power-law fit gives $\log(G_n)/n \sim 5\, n^{-0.87}$, i.e., $\log G_n \sim 5\, n^{0.13}$, far stronger than the density-one bound $o(n)$ of Theorem 64.

The exponent 0.13 should be interpreted with caution: if $\log G_n \sim \kappa \log n$, the effective log-log slope is $1/\log n = 0.145$ at $n = 1000$, close to the fitted 0.13. A random-gcd model—assigning each prime $p \leq n$ an independent probability $\sim 1/p$ of dividing G_n—predicts $\mathbb{E}[\log G_n] \sim \log n$ by the weighted Mertens theorem, with standard deviation $\sim (\log n)/\sqrt{2}$ of the same order. The data thus cannot yet distinguish $\kappa \log n$ from $n^{0.13}$; the logarithmic prediction is arithmetically more natural.

10.1 Extended modular verification

For each n, define $B(n) = \#\{p > \sqrt{n} : n \bmod p \in \mathcal{Z}_p\}$, so that $\log G_n \le O(\sqrt{n}) + 3\,B(n)\log n$ (Section 7). The ratio $B(n) \cdot \log n/n$ governs $\log G_n/n$: whenever it tends to zero, $G_n = e^{o(n)}$. We computed $B(n)$ for all $n \le 100{,}000$ by modular arithmetic (iterating the Apéry recurrence mod p for each prime $p \le 200{,}001$ and checking whether $n \bmod p \in \mathcal{Z}_p$).

n range	max $B(n)$	avg $B/T(n)$	max $B/T(n)$
1–100	3	0.038	0.260
101–500	4	0.013	0.124
501–1,000	4	0.006	0.036
1,001–2,000	4	0.003	0.027
2,001–5,000	4	0.001	0.015
5,001–10,000	5	0.0008	0.007
10,001–20,000	5	0.0004	0.004
20,001–50,000	6	0.0002	0.002
50,001–100,000	6	0.0001	0.001

Here $T(n) = n/\log n$, the density-one threshold. The ratio $\max B/T$ decreases monotonically. The overall maximum is $B(n) = 6$ at $n = 20{,}546$, giving the explicit bound $\log G_n/n \le 3 \cdot 6 \cdot \log(20{,}546)/20{,}546 + O(1/\sqrt{20{,}546}) = 0.009$, which confirms $G_n = e^{o(n)}$ at every index $n \le 100{,}000$. The Poisson prediction for the maximum is $\max_{n \le N} B(n) \approx 2\log\log N \approx 5.6$ at $N = 10^5$, consistent with the observed 6.

11 Remarks

1. The Beukers supercongruence $b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{3r}}$ [4] provides tower rigidity: for each fixed prime, $v_p(b_n)$ is controlled by the base-p digits. Quantitatively, since $b_n \equiv \prod b_{d_i} \pmod{p}$ (Theorem 4), we have $p \nmid b_n$ iff every base-p digit of n avoids $\mathcal{Z}_p$. Hence

$$\#\{n < p^r : p \mid b_n\} = p^r - (p - Z(p))^r.$$

2. For the gap-2 polynomial $C_2(m) = P(m+1)$, the factorization $P(x) = (2x+1)(17x^2+17x+5)$ shows that gap-2 pairs can only occur at $m = (p-3)/2$ (when $2(m+1)+1 = 2m+3 \equiv 0$) or at roots of the quadratic $17x^2 + 17x + 5$ (discriminant -51; exists over $\mathbb{F}_p$ iff $\left(\frac{-51}{p}\right) = 1$). Empirically, all observed gap-2 pairs arise from the linear factor at $m = (p-3)/2$.

3. For $1 \le j \le p-2$, the Apéry number $b_j \bmod p$ admits a character-sum representation as a Greene [14] Gaussian hypergeometric ${}_4F_3$ over $\mathbb{F}_p$ with parameter $\chi = \omega^j$ (Teichmüller character). However, the Weil bound controls the archimedean size $|b_j|_\infty = O(p^{3/2})$ but does *not* bound $Z(p)$, since zeros of b_j modulo p are a p-adic event. The gap polynomial argument above is specifically adapted to this p-adic zero-count problem.

4. Malik–Straub [20] studied Lucas congruences and zero-divisibility for all sporadic Apéry-like sequences; Rowland–Yassawi [23] proved automaticity of the Lucas property for diagonals of rational functions, which includes the Apéry sequence. Our $Z(p)$ data for the $\zeta(3)$ sequence agrees with their empirical findings.

5. In the standard toric coordinate, the truncated polynomial $H_p(t) = \sum_{j=0}^{p-1} b_j\, t^j \in \mathbb{F}_p[t]$ is the scalar Hasse–Witt invariant of the Apéry K3 pencil. Its *roots* in $\mathbb{F}_p$ are the nonordinary fibers; because the family has generic Picard rank 19, nonordinary implies supersingular at good primes. Our statistic $Z(p)$ counts vanishing *coefficients* of H_p, equivalently the vanishing finite Mellin coefficients $b_j = -\sum_{t\in\mathbb{F}_p^\times} H_p(t)\, t^{-j}$ for $1 \le j \le p-2$. Thus $Z(p)$ is a spectral sparsity statistic for the global Hasse/point-count function, *not* a count of supersingular fibers. Standard Hasse-stratification, Newton-polygon, and overconvergent F-crystal theory do not by themselves bound $Z(p)$; they control the root locus, not the Fourier support. A bounded-complexity geometric lift *does* exist: the lisse sheaf $\mathcal{V}_\ell = R^2 f_*\mathbb{Q}_\ell$ on the smooth locus of the K3 pencil (rank 22, conductor independent of p) satisfies $\mathrm{Tr}(\mathrm{Frob}_t \mid \mathcal{V}_\ell) \equiv A_p(t) \pmod p$ by the Hasse–Witt congruence. The Mellin sum $\mathscr{M}_{p,j} = \sum_t \tau_p(t)\, \omega_p(t)^{-j}$ is then realized on $R\Gamma_c(U_{\overline{\mathbb{F}}_p}, \mathcal{V}_\ell \otimes \mathcal{L}_{\omega_p^{-j}})$ with bounded rank and conductor. However, the event $b_j \equiv 0 \pmod p$ is the *same-characteristic* divisibility $\mathfrak{p}_p \mid \mathscr{M}_{p,j}$, a crystalline condition outside the scope of ℓ-adic equidistribution (Forey–Fresán–Kowalski [13]). The missing theorem is a *crystalline Mellin anti-concentration* estimate controlling $\#\{j : p \mid \mathscr{M}_{p,j}\}$.

6. **Jacobi counterexample: bounded complexity is not enough.** Fix an integer $d \ge 2$. For every prime $p \equiv 1 \pmod d$, the rank-1 Kummer sheaf $\mathcal{L}_\psi$ on $\mathbb{P}^1 \setminus \{0, 1, \infty\}$ attached to $\psi = \omega_p^{-(p-1)/d}$ (order d) has Mellin transforms $M_j = J(\omega_p^{-j}, \psi)$. The Gross–Koblitz/Stickelberger valuation $v_p(g(\omega^{-k})) = k/(p-1)$ and the Jacobi factorization $J(\alpha, \beta) = g(\alpha)g(\beta)/g(\alpha\beta)$ give $v_p(M_j) = 1$ whenever $j \ge (p-1)(d-1)/d$, yielding at least $(p-1)/d - 1$ characters with $p \mid M_j$. Thus a sheaf of *fixed* rank, conductor, and Weil bound $O(\sqrt{p})$ can have $\gg p$ bad tame characters. This shows that bounded cohomological complexity alone cannot control $Z(p)$; any proof of Hypothesis Z must use Apéry-specific arithmetic.

7. **Sufficient criteria for Hypothesis Z.** Two clean algebraic conditions, either of which would imply $Z(p) = O(1)$:

   (a) Bounded tame-Hasse polynomial. Suppose there exist an absolute integer D and, for each sufficiently large p, a nonzero Laurent polynomial $Q_p \in \mathbb{F}_p[X, X^{-1}]$ of degree at most D and units $u_{p,j} \in \mathbb{F}_p^\times$ such that $\mathscr{M}_{p,j} \equiv u_{p,j}\, Q_p(\omega_p^j) \pmod p$ for every nontrivial tame character. Then $Z(p) \le 2D$, since Q_p (after clearing negative exponents) has at most $2D$ roots.

   (b) Bounded nonordinary twists. Suppose for each p the set E_p of tame characters χ for which the twisted crystal $H^1_{\mathrm{rig},c}(U, \mathcal{V} \otimes \mathcal{K}_\chi)$ fails to have a unique unit-root eigenvalue satisfies $|E_p| \le C$. Then $Z(p) \le C + O(1)$, since a unique unit root forces $v_p(\mathscr{M}_{p,j}) = 0$.

   Criterion (a) is the algebraic version of Hypothesis Z; criterion (b) is its p-adic-slopes formulation. The Jacobi counterexample above shows that neither criterion holds for general bounded-complexity sheaves.

8. **Linear-complexity obstruction.** The Sym^2 factorization $H_p(t) = \Delta(t)^{\varepsilon_p} \cdot B_p(t)^2$ with B_p squarefree of degree $\lfloor (p-1-2\varepsilon_p)/2 \rfloor$ implies that H_p has at most $\lfloor (p-1)/2 \rfloor + 2$ distinct zeros in $\mathbb{F}_p^\times$. Choosing a primitive root g and forming the canonical character interpolation $Q(X) = -\sum_{r=1}^{p-2} H_p(g^r)\, X^r$, whose values $Q(g^j) = b_j$ for $1 \le j \le p-2$, we find that Q has at least $(p-5)/2$ nonzero coefficients. Any Laurent polynomial of degree at most D representing the same function on the character group must have $2D + 1 \ge (p-5)/2$, giving $D \ge (p-7)/4$. Thus criterion (a) cannot be satisfied in its direct (unnormalized) form: a nontrivial Gauss/Jacobi normalization of the Mellin coefficients is necessary before bounded tame degree could emerge.

9. **Palindromic symmetry.** For $0 \le j \le p-1$ and $0 \le k \le p-1-j$, the identity $\binom{p-1-j}{k} \equiv (-1)^k \binom{j+k}{k} \pmod p$ holds (each of the k falling factors $p{-}1{-}j{-}i \equiv -(j{+}1{+}i)$). When $k \le j$, $\binom{p-1-j+k}{k} \equiv (-1)^k \binom{j}{k}$; when $k > j$, Lucas's theorem gives $\binom{p-1-j+k}{k} \equiv 0$ (since $p{-}1{-}j{+}k \ge p$ and $\binom{k-j-1}{k} = 0$). Therefore each term of $b_{p-1-j} = \sum_k \binom{p-1-j}{k}^2 \binom{p-1-j+k}{k}^2$ with $k > j$ vanishes, and for $k \le j$ the $(-1)^{2k}$ signs cancel, giving $b_{p-1-j} \equiv b_j \pmod p$. The extra terms of b_j when $j > (p{-}1)/2$ (i.e., $k > p{-}1{-}j$) likewise vanish: $\binom{j+k}{k} \equiv \binom{j+k-p}{k} \equiv 0$ since $j{+}k{-}p < k$. Equivalently, $H_p(t)$ is palindromic: $t^{p-1} H_p(1/t) \equiv H_p(t)$. This explains the observed pairing: whenever $b_j \equiv 0$, also $b_{p-1-j} \equiv 0$, so zeros of b_j come in pairs $\{j,\, p-1-j\}$ (except at the self-paired center $j = (p-1)/2$, which vanishes only at non-ordinary primes). Proposition 72 proves the stronger statement that *every* solution of the Apéry recurrence is palindromic modulo p, in particular the companion solution a. The "pair count" $K(p) = Z(p)/2$ (or $(Z(p)-1)/2$ at non-ordinary primes) follows Poisson(1/2) to the precision of the data (Proposition 50).

10. **Sym^2 squareness (theorem).** The Apéry differential operator is the symmetric square of the Picard–Fuchs operator of an underlying elliptic family (Stienstra–Beukers [25]). Caruso, Fürnsinn, Vargas-Montoya, and Zudilin [7] proved the mod-p squareness rigorously: for every prime $p \ge 5$, setting $\varepsilon_p = \frac{1}{2}(1 - \left(\frac{-6}{p}\right))$, there exists a unique (up to sign) polynomial $B_p \in \mathbb{F}_p[t]$ such that

$$H_p(t) \;=\; \Delta(t)^{\varepsilon_p} \cdot B_p(t)^2,$$

where $\Delta(t) = t^2{-}34t{+}1$ is the conifold discriminant and B_p is squarefree and coprime to Δ. (A caveat on attribution: Theorem 1.2 of [7] establishes the quadratic-times-square *factorization*; the squarefreeness and coprimality-to-Δ assertions used here are verified computationally for all $p \le 20{,}000$ (Subsection 12.20) but are not formulated in [7] as a Hasse-invariant, Picard-group, or elliptic-pencil statement, and a separate reference or proof is needed for their use in full generality. Crucially, $\mathcal{Z}_p$ is the set of vanishing *coefficients* of H_p (the finite Mellin transform), *not* the evaluation roots of H_p or of B_p; the two loci are genuinely different—e.g. at $p = 7$, $H_7 = (t{-}1)^2(t^2{+}1)^2$ has the evaluation root $t = 1$ yet no vanishing coefficient, so $\mathcal{Z}_7 = \varnothing$ (Proposition 123). A Weil bound for the curve $B_p(t) = 0$ therefore does *not* bound the Fourier sum $F_p(a) = \sum_{j \in \mathcal{Z}_p} e_p(aj)$.) The proof uses the explicit identity $F(t(x)) = (1+x)\,h(x)^2$ between the Apéry and Franel generating functions, the p-Lucas property of both sequences, and a quadratic-descent argument through the covering $\mathbb{F}_p(x)/\mathbb{F}_p(t)$ where $t = x(1-8x)/(1+x)$. The deck-transformation character $(-6/p)$ determines whether $\varepsilon_p = 0$ or 1:

$$\varepsilon_p = 0 \iff \left(\frac{-6}{p}\right) = 1, \qquad \varepsilon_p = 1 \iff \left(\frac{-6}{p}\right) = -1.$$

Equivalently, $\varepsilon_p = 0$ iff $p \equiv 1, 5, 7, 11 \pmod{24}$. These two cases occur with equal density 1/2 by Dirichlet's theorem.

Splitting the mean $Z(p)$ by residue class modulo 8 (and modulo 24) yields no significant difference (mean ≈ 1.00 in each class, $n \ge 2000$ primes per class up to 10^5); the squareness structure does not directly influence $Z(p)$, consistent with $Z(p)$ counting coefficient zeros rather than root zeros.

The factorization imposes a convolution structure: b_j is (up to the Δ^{ε_p} correction) the j-th coefficient of B_p^2, i.e., $b_j \equiv \sum_k \beta_k\, \beta_{j-k} \pmod p$ where β_k denotes the coefficients of B_p. The vanishing $b_j \equiv 0$ thus requires cancellation across $\sim p/2$ convolution terms. A Monte Carlo simulation (1000 random monic polynomials of degree $(p-1)/2$ over $\mathbb{F}_p$, for $p \in$

$\{101, 503, 1009, 5003\}$) shows that the zero-count of the *squared* polynomial follows Poisson(1) (mean ≈ 1, variance ≈ 1, total variation distance to Poisson(1) equal to 0.018). However, the observed $Z(p)$ distribution over 667 primes $p \leq 5000$ is *not* Poisson(1) (TV = 0.43); instead it matches $2 \cdot$ Poisson(1/2) with TV = 0.023. The explanation is the palindromic pairing: since $b_{p-1-j} \equiv b_j$, zeros come in pairs $\{j, p-1-j\}$ and $Z(p)$ is almost always even. Denoting by $Z_{\text{pair}}(p)$ the number of zero pairs, the data show $Z_{\text{pair}} \sim \text{Poisson}(1/2)$, so $Z(p) = 2Z_{\text{pair}}(p) \sim 2 \cdot \text{Poisson}(1/2)$. Thus squareness explains the $O(1)$ mean, while palindromic symmetry supplies the parity constraint. The Poisson tail predicts $\max_{p \leq N} Z(p) \sim 2 \log\log N$ via extreme-value theory, strongly suggesting that $Z(p)$ is unbounded (i.e., Hypothesis Z is false). Whether the algebraic provenance of B_p imposes a bounded *average* (Hypothesis $\bar{Z}$) remains the central open question.

The factorization does *not* improve the divisor-counting bound $D(P, Q) \leq Q^2/\log P$. In the regime $P \geq Q^{4/3}$ where the divisor estimate is active, one has $q < p/2$, so the coefficient $b_q \bmod p$ lies in the formally free triangular range of the square root: the map $(\beta_1, \ldots, \beta_q) \mapsto (b_1, \ldots, b_q) \bmod p$ is a triangular automorphism with Jacobian 2^q, hence the single equation $b_q \equiv 0$ imposes no algebraic constraint beyond fixing one coefficient of B_p. Thus the Sym^2 structure gives the correct *random model* (uniform square root $\Rightarrow$ Poisson(1/2)) but not the missing *distribution theorem* for the deterministic Apéry square root.

11. **Toric framework.** The Laurent polynomial $f = w(x + x^{-1} + y + y^{-1} + z + z^{-1})$ on $(\mathbb{F}_p^\times)^4$ has Newton polytope $\Delta = \text{conv}(\{0\} \cup \text{Supp}(f))$, a 4-dimensional pyramid over the regular octahedron in $\mathbb{R}^3$. Its f-vector is $(7, 18, 20, 9)$, its normalized volume is $4! \cdot \text{vol}(\Delta) = 8$, and it is Kouchnirenko nondegenerate for every prime $p \geq 5$: for each of the 27 nonempty faces σ not containing the origin, the toric critical system $\{x_i \partial f_\sigma / \partial x_i = 0\}$ has no solution in $(\mathbb{F}_p^\times)^4$. For the Adolphson–Sperber twisted Hodge polygon with w-Kummer twist $\chi_w = \omega^{-j}$ (character parameter $a = j/(p-1)$), the slopes are a, $a+1$, $a+2$, $a+3$ with multiplicities $1, 3, 3, 1$, confirming that the first slope equals a for every $a \in \{k/100 : 0 \leq k \leq 99\}$. By Newton-above-Hodge (Adolphson–Sperber [2]), the p-adic Newton polygon lies on or above this Hodge polygon, and ordinarity (Newton = Hodge) implies the existence of a unit root. The $Z(p)$ bound thus reduces to proving ordinarity for all but $O(1)$ twist parameters j.

12. **Branch obstruction.** The $p-1$ Kummer twists χ^{-j} ($j = 0, \ldots, p-2$) are distinct characters of conductor p. In Iwasawa-theoretic coordinates $\chi^j = \omega^a \langle \cdot \rangle^s$ with $a = j \bmod (p-1)$, the horizontal family $\{j : 0 \leq j < p-1\}$ hits *each* of the $p-1$ branches ω^a exactly once. Weierstrass preparation and bounded-λ rigidity control zeros *within one branch* (the vertical direction j, $j + (p-1)$, $j + (p-1)p, \ldots$, where Dwork congruences live) but not *across* branches. Thus there is no p-adic analytic function of one variable whose zero count majorizes $Z(p)$. Any "rigidity" proof of Hypothesis $\bar{Z}$ would need a mechanism not supplied by existing Iwasawa–Dwork theory.

13. Any estimate $\log G_n = o(n)$—whether $O(\log n)$, $O(\sqrt{n})$, or $O(n^{0.13})$—yields $\gamma = 0$ in the exponent $G_n = e^{\gamma n + o(n)}$ and therefore does not improve the Rhin–Viola irrationality measure $\mu(\zeta(3)) \leq 5.513$ [24], which depends only on the exponential growth rate $|b_n \zeta(3) - a_n| = e^{-(\log \alpha + o(1))n}$. An improvement would require an exponentially large common divisor, which is not predicted by any current model.

14. **Comparison with previous work.** Classical refinements of Apéry's method (Beukers [4], Rhin–Viola [24], Zudilin [27]) exhibit explicit common factors of the hypergeometric coefficients to improve denominator estimates and irrationality measures, but always uniformly for

every index. They do not provide exceptional-set estimates for the *actual* reduced gcd G_n. A separate body of work, based on the Subspace Theorem, proves $\gcd(a^n-1, b^n-1) = e^{o(n)}$ pointwise for multiplicatively independent a, b (Bugeaud–Corvaja–Zannier [6]); these results apply to constant-coefficient linear recurrences via Binet representations, but the Apéry recurrence's polynomially varying coefficients preclude the multiplicative-independence hypothesis. To our knowledge, no previous work gives a quantitative power-saving exceptional-set estimate for gcd cancellation in a variable-coefficient holonomic recurrence.

15. **The remaining open problem.** The companion-block height bound (Proposition 73) shows that primes $p > \sqrt{n}$ with $D_q \equiv 0 \pmod{p}$ contribute $O(n^{2/3})$ to $\log G_n$ at *every* index. After removing this channel, the conjecture $G_n = e^{o(n)}$ for all n is equivalent to the *pointwise radical estimate*
$$\sum_{\substack{p>\sqrt{n} \\ p \mid b_{n \bmod p}}} \log p \;=\; o(n).$$
Theorem 78 proves this for all but $O_\varepsilon((\log N)^2)$ integers in $[1, N]$. However, the codegree method (Lemma 76) controls primes *shared* between pairs of integers, and does not bound the total weight of "private" primes at a single index. Closing the gap requires a one-point rainbow estimate: an argument that one fixed integer cannot simultaneously be bad at too many distinct primes. By the quotient reduction (Proposition 74), the estimate is only needed for the primes $p > n/(f(n)\log n)$ with f growing arbitrarily slowly—equivalently, uniformly over the quotient classes $q = \lfloor n/p \rfloor < f(n)\log n$—since prime counting alone handles the range $\sqrt{n} < p \le n/(f(n)\log n)$.

12 The second-moment approach

The codegree method of Section 8 exploits primes shared between pairs of integers to prove a density-one theorem. To close the remaining gap—proving $W(n) = o(n)$ for *every* n—we study the *second moment* of the bad-prime weight function.

12.1 The pointwise reduction and why it is deep

We first record the cleanest form of the all-index target and a structural obstruction that any proof must overcome. For a dyadic shell $I_N = (N, 2N]$ and $m \in I_N$, the Lucas collapse (Corollary 62) gives, for every prime $N < p \le m$, the congruence $b_m \equiv 5\, b_{m-p} \pmod{p}$; hence the detector $X_p(m) = \mathbf{1}[p \mid b_{m-p}]$ equals $\mathbf{1}[p \mid b_m]$, and the incidence count collapses to a single integer:
$$L_N(m) \;=\; \#\{p \in (N, m] : p \mid b_m\} \;=\; \omega_{(N,m]}(b_m). \tag{99}$$

Proposition 86 (Pointwise reduction)**.** *The all-index conjecture $G_n = e^{o(n)}$ follows from the pointwise radical anti-concentration*
$$(\diamondsuit) \qquad \max_{m \in (N,2N]} \#\{p \in (N, m] : p \mid b_m\} \;\ll\; N^{o(1)}. \tag{100}$$

Proof. By (99) and the trivial bound $\sum_{m \in I_N}(L_N(m) - \mu)^2 \le (\max_m L_N(m)) \sum_m L_N(m) = (\max_m L_N(m))\, T_N$, the hypothesis $(\diamondsuit)$ gives the centered second moment $V_w^\circ(N) \ll N^{o(1)} T_N$. The L^2-to-uniform upgrade (Lemma 100) and Corollary 101 then give $W(n) = o(n)$ for every n, hence $G_n = e^{o(n)}$. The cross-prime coupling of the second moment is thus a *detector artifact*: both conditions $p \mid b_{m-p}$ and $q \mid b_{m-q}$ are statements about the single integer b_m. □

Remark 87 (Size slack and the sieve-impossibility barrier). The estimate ($\diamondsuit$) is deep for a precise reason. Since $\log b_{2N} = 2N\log(17+12\sqrt{2}) + O(\log N) \approx 7.05\,N$, while $\sum_{N<p\le 2N}\log p \sim N$ by the prime number theorem, the integer b_{2N} has room to contain the *entire* product of primes in $(N, 2N]$ about seven times over. The trivial height bound gives only $\omega_{(N,m]}(b_m) \le \log b_m/\log N = O(N/\log N)$, one $o(1)$ factor short. Consequently the adversarial "anchored star" of Section 12.5—the hypothetical scenario in which b_{2N} is divisible by every prime in $(N, 2N]$—is not excluded by size. It satisfies every structural axiom available to a sieve: the vertical bound $Z(p) = O(1)$, reflection, no consecutive zeros, the sub-interval count $O(|I|^{2/3})$, and generic prime spacing, yet it violates ($\diamondsuit$) with $L_N(2N) \gg N/\log N$. Therefore *no* sieve-theoretic or counting argument blind to the specific non-archimedean arithmetic of the Apéry numbers can prove ($\diamondsuit$): a genuine input about the prime factorization of the individual integer b_m is mandatory. This is the precise sense in which the all-index conjecture, though numerically overwhelming (Remark 63), lies beyond the reach of the vertical and dispersion machinery developed above.

12.2 The criterion

For a dyadic shell $I_N = (N, 2N]$, define

$$B(m) \;=\; \sum_{\substack{p>\sqrt{m}\\ p\mid b_{m \bmod p}}} \log p, \qquad M_2(N) \;=\; \sum_{m\in I_N} B(m)^2.$$

Proposition 88 (Second-moment criterion)**.** *If $M_2(N) \le C\cdot N^{2-\delta}$ for some $C>0$ and $\delta>0$, then $W(n) = o(n)$ for all n. More precisely, for every $n \in I_N$,*

$$W(n) \;\le\; B(n) \;\le\; \sqrt{M_2(N)} \;\le\; C^{1/2}\cdot N^{1-\delta/2}.$$

Proof. $B(n)^2$ is one nonneg. term of the sum $M_2(N)$, hence $B(n) \le \sqrt{M_2(N)}$. ☐

In particular, $M_2(N) = O(N\cdot(\log N)^A)$ for any fixed A gives $B(n) = O(\sqrt{N}\cdot(\log N)^{A/2}) = o(N)$.

Remark 89 (Shifted-gcd identity). The bad-prime weight admits a combinatorial reformulation. By the injectivity of $p \mapsto n \bmod p$ on primes $p > \sqrt{n}$ (two such primes giving the same residue would force their product $\le n$),

$$W(n) \;=\; \sum_{0<r<n} \log \operatorname{rad}_{>\max(\sqrt{n},\,r)} \gcd(n-r,\, b_r).$$

Each inner radical is 1 or a single prime $p = n-r$. For the top-half block $p > n/2$, the condition becomes "$n-r$ is prime and $(n-r)\mid b_r$."

Proposition 90 (Hard-core reduction)**.** *For each bad prime $p > \sqrt{n}$, write $n = q_p p + r_p$, $s_p = \min(r_p, p-1-r_p)$, $h_p = |p-1-2r_p|$. Let $1 \le Q, R, H < n/2$. Then the total weight of primes satisfying at least one of $q_p > Q$, $s_p \le R$, $h_p \le H$ is $O(n/Q + (R+H)\log n)$. In particular, with $Q = R = H = \sqrt{n/\log n}$, all "easy channels" contribute $O(\sqrt{n\log n})$; the remaining hard core consists of primes $p \gg \sqrt{n\log n}$ with zero positions far from both endpoints and the reflection centre.*

Proof. Large quotient ($q_p > Q$): then $p < n/Q$, so $\sum \log p \le \vartheta(n/Q) \ll n/Q$. Near endpoint ($s_p \le R$): by the injectivity of $p \mapsto n \bmod p$, at most two primes $> \sqrt{n}$ contribute for each value of s_p. Near centre ($h_p \le H$): the identity $(2q_p+1)p = 2n+1\pm h_p$ forces p to divide a fixed integer of size $O(n)$, giving $O(1)$ primes per value of h_p. ☐

Remark 91 (The vertical main term is harmless). Write $W(n) = M(n) + \mathcal{E}(n)$ where $M(n) = \sum_{p>\sqrt{n}} Z(p)\,(\log p)/p$. The unconditional bound $Z(p) \le Cp^{2/3}$ gives $M(n) \ll \sum_{p\le n} p^{-1/3}\log p \ll n^{2/3}$. The entire problem is the oscillatory error $\mathcal{E}(n)$, which at each prime is $(\log p)\cdot(1_{n \bmod p\in\mathcal{Z}_p} - Z(p)/p)$.

12.3 Empirical verification

The following table, computed for all dyadic shells with $64 \le N \le 4096$ using the full zero-set data for all primes $p \le 20{,}000$, gives:

N	$M_2(N)$	M_2/N	$M_2/(N\log^2 N)$	$\max B(m)$
64	672	10.5	0.605	10.4
128	1,684	13.2	0.560	15.9
256	4,376	17.1	0.555	15.8
512	11,896	23.2	0.598	22.3
1024	28,632	28.0	0.583	21.7
2048	69,226	33.8	0.582	27.1
4096	166,671	40.7	0.587	32.1
8192	421,868	51.5	0.634	37.2

The ratio $M_2(N)/(N\log^2 N)$ stabilizes near 0.59, strongly suggesting

$$M_2(N) \;\sim\; C_2\cdot N\cdot(\log N)^2, \qquad C_2 \approx 0.59. \tag{101}$$

This gives $B(n) \le \sqrt{0.59\,N}\,\log N \approx 0.77\sqrt{N}\,\log N = o(N)$.

12.4 CRT expansion

Expand M_2 dyadically. For distinct primes $p, q \in (P, 2P]$ with $P > \sqrt{N}$, write $\Omega_p = \{m \in I_N : m \bmod p \in \mathcal{Z}_p\}$. Then

$$M_2(N) \;=\; \underbrace{\sum_p (\log p)^2\,|\Omega_p|}_{\text{diagonal}} \;+\; \underbrace{\sum_{p\ne q} \log p\,\log q\,|\Omega_p\cap\Omega_q|}_{\text{off-diagonal}}.$$

Diagonal. $|\Omega_p| = Z(p)\cdot N/p + O(Z(p))$, so

$$\text{diag} \;=\; N\sum_p \frac{Z(p)\,(\log p)^2}{p} \;+\; O\Big(\sum_p Z(p)\,(\log p)^2\Big).$$

Under the Poisson model $\mathbb{E}[Z(p)] = 1$, the main term is $\sim N\cdot\sum_{P<p\le 2P}(\log p)^2/p \sim N\cdot(\log P)^2\cdot\log 2$. Summing dyadically $P = \sqrt{N}, 2\sqrt{N}, \ldots, N$ gives diagonal $\sim \frac{1}{2}\,N\,(\log N)^2$, matching $\approx 63\%$ of the empirical (101).

Off-diagonal. Since $p, q > \sqrt{N}$ we have $pq > N$, so each CRT class $r_p \pmod p$, $r_q \pmod q$ contains at most one element of I_N. The CRT main term of the off-diagonal is $\sim N\big(\sum Z(p)\log p/p\big)^2 \sim \frac{1}{4}\,N\,(\log N)^2$.

The predicted constant. Under the independent random model—each prime p hits a uniform m with probability $\sim 1/p$, independently across primes—the second moment of $B(m)$ at one integer is $\mathbb{E}[B(m)^2] = (\mathbb{E}[B])^2 + \mathrm{Var}(B)$. By weighted Mertens, $\mathbb{E}[B] = \sum_{\sqrt{m}<p\le m}(\log p)/p = \frac{1}{2}\log m + O(1)$, and

$$\mathrm{Var}(B) \;=\; \sum_{\sqrt{m}<p\le m} \frac{(\log p)^2}{p}\left(1-\frac{1}{p}\right) \;=\; \tfrac{3}{8}\,(\log m)^2 + O(\log m).$$

Therefore

$$\mathbb{E}[B(m)^2] \;=\; \left(\tfrac{1}{4}+\tfrac{3}{8}\right)(\log m)^2 + O(\log m) \;=\; \tfrac{5}{8}\,(\log m)^2 + O(\log m), \tag{102}$$

predicting $M_2(N) \sim \frac{5}{8}\,N\,(\log N)^2$. The diagonal constant is $3/8$ (matching the observed $\mathrm{Diag}/N\log^2 N \approx 0.37$) and the off-diagonal contributes $1/4$. The empirical ratio $M_2/(N\log^2 N) \approx 0.59$ is consistent with convergence to $5/8 = 0.625$, with lower-order boundary effects.

Centered variance and cross-prime covariance. The centered dispersion $V^\circ = M_2 - M_1^2/N$ decomposes as

$$V^\circ \;=\; \underbrace{\sum_p (\log p)^2\,|\Omega_p|\big(1-|\Omega_p|/N\big)}_{\le\,\mathrm{Diag}} \;+\; \underbrace{\sum_{p\ne q} \log p\,\log q\,\big(|\Omega_p\cap\Omega_q| - |\Omega_p|\,|\Omega_q|/N\big)}_{E_N^\circ}. \tag{103}$$

The first sum is at most the diagonal S (since $|\Omega_p|/N \ge 0$). The term E_N° is the centered cross-prime covariance: it vanishes exactly when the events $\{m \in \Omega_p\}$ are pairwise uncorrelated. The computed values are:

N	$M_2/N\log^2 N$	$\mathrm{Diag}/N\log^2 N$	V°/S	E_N°
256	0.556	0.359	1.014	+110
512	0.597	0.376	1.010	+215
1024	0.582	0.375	0.981	−152
2048	0.581	0.370	0.992	+7
4096	0.588	0.371	0.995	+41
8192	0.634	0.387	0.995	−188
16384	0.669	0.400	0.995	−922
32768	0.646	0.394	0.988	−14065

The ratio V°/S converges to 1 from below for $N \ge 1024$, with $V^\circ/S < 1$ at every scale tested beyond 512. The negative sign of E_N° for large N reflects genuine *anti-correlation* among the CRT-constrained incidence events: at $N = 1024$, over 96% of cross-prime covariance pairs $C_{p,q}$ are negative. This is consistent with a weak form of negative dependence arising from the partial-period CRT constraint. The magnitude $|E_N^\circ|$ is $O(\sqrt{N}\,(\log N)^2)$, so $|E_N^\circ|/S \to 0$ at rate $\sim 1/\sqrt{N}$.

Three-model comparison. To test whether the observed $V^\circ/S \approx 1$ is an Apéry-specific phenomenon or a generic consequence of the cardinality and reflection structure, we compare three models on each dyadic block I_N: (1) the actual Apéry zero-sets; (2) uniformly random reflected subsets of $\mathbb{F}_p^*$ with the same cardinalities $|Z_p|$; (3) random reflected subsets with cardinalities drawn from $2\cdot\mathrm{Poisson}(1/2)$. Over 20 independent trials of each random model and $N \in \{256, 512, 1024, 2048\}$, all three models yield $V^\circ/S = 1.00 \pm 0.05$, with no statistically significant difference between the Apéry data and either random model. This rules out any detectable cross-prime alignment or repulsion specific to the Apéry sequence.

Remark 92 (Boundary correction). The naïve CRT main term $Z(p)Z(q)N/(pq)$ overestimates the random prediction for primes near $\sqrt{N}$ or N, because the admissible interval $J_{p,q} = J_p \cap J_q$ may be shorter than N. After replacing N by $L_{p,q} = |J_{p,q}|$ in the main term, the boundary-corrected CRT residual is positive and $O(\sqrt{N}\,(\log N)^2)$—the apparent negative sign in the uncorrected decomposition is entirely a boundary artifact.

12.5 The no-go model

Define the *adversarial zero-sets*: for one fixed $m_0 \in I_N$, set $\mathcal{Z}_p^* = \{m_0 \bmod p,\ p-1-(m_0 \bmod p)\}$. Then $|\mathcal{Z}_p^*| = 2$ (satisfying reflection), yet $m_0 \in \Omega_p^*$ for *every* prime $p > \sqrt{N}$, giving $B^*(m_0) = \sum_{p>\sqrt{N}} \log p = n - o(n)$. In particular:

- The adversarial model satisfies $Z^*(p) = 2$, $Z^*_{\text{pair}}(p) = 1$, and all proved vertical bounds.
- Yet $M_2^*(N) \geq B^*(m_0)^2 \geq c\,n^2$, which is $\Omega(N^2)$, not $O(N)$.

Conclusion: *No proof using only vertical information (the sizes $Z(p)$, the reflection symmetry, nearest-neighbor exclusion, and gap-polynomial bounds) can establish $W(n) = o(n)$ for all n.* The missing theorem must use cross-prime *horizontal* information: the locations of the zeros $\mathcal{Z}_p$ across different primes p.

12.6 The Fourier form of the CRT error

Define the modular Fourier transform of $\mathcal{Z}_p$:

$$S_p(a) \;=\; \frac{1}{p}\sum_{r\in\mathcal{Z}_p} e\!\left(\frac{ar}{p}\right).$$

Then $S_p(0) = Z(p)/p$ and, by the palindromic pairing, $e(a(p-1)/(2p))\,S_p(a)$ is real-valued for every a. Fourier inversion gives

$$E_N(p,q) \;=\; \sum_{\substack{(a,b)\neq(0,0)\\ 0\leq a<p,\ 0\leq b<q}} S_p(a)\,S_q(b)\sum_{m\in I_N} e\!\left(\frac{am}{p}+\frac{bm}{q}\right).$$

The inner sum is a standard exponential sum $|K_N(\alpha)| \leq \min(N, 1/\|\alpha\|)$ where $\alpha = a/p + b/q$.

Applying the classical additive large sieve to bound $\sum_{p\neq q} |E_N(p,q)|$ gives $O(N^2/\log N)$ at the top prime scale $p \sim N$—one full logarithm larger than the threshold $o(N^2/\log^2 N)$ required for $M_2 = O(N(\log N)^A)$. Thus the classical large sieve does not close the gap.

12.7 The dispersion target

Define the centered dispersion

$$V^\circ(P,N) \;=\; \sum_{m\in I_N}\left|B_P(m) - \frac{S(P,N)}{N}\right|^2,$$

where $B_P(m) = \sum_{P<p\leq 2P} 1_{\Omega_p}(m)$ and $S(P,N) = \sum_p |\Omega_p|$.

Hypothesis 93 (AP–BDH dispersion). $V^\circ(P,N) \ll N^{o(1)}\cdot S(P,N)$.

Proposition 94. *Hypothesis 93 together with the unconditional bound $Z(p) = O(p^{2/3})$ implies $W(n) = o(n)$ for all n.*

Proof. By hypothesis, for every $m \in I_N$,

$$B_P(m) \;\leq\; \frac{S(P,N)}{N} + \sqrt{V^\circ(P,N)} \;\leq\; \frac{S(P,N)}{N} + N^{o(1)}\sqrt{S(P,N)}.$$

By the zero-count bound, $S(P,N) \le (N/P+1)\cdot\sum_{P<p\le 2P} Z(p) \ll NP^{2/3}/\log P$. Therefore $B_P(m) \ll N^{o(1)}\cdot(NP^{2/3}/\log P)^{1/2}$. The weighted sum satisfies $W_P(m) \le B_P(m)\cdot\log(2P)$. Summing over $O(\log N)$ dyadic blocks gives

$$W(n) \ll N^{o(1)}\cdot N^{1/2}\cdot N^{1/3} = N^{5/6+o(1)} = o(N). \qquad \square$$

The dispersion hypothesis is a cross-prime equidistribution theorem stating that the Apéry zero-sets $\mathcal{Z}_p$ do not systematically align at one index across different primes. The no-go model of Section 12.5 shows that this hypothesis genuinely requires arithmetic input about the Apéry family, beyond what any vertical bound can provide.

Remark 95 (Covariance sign distribution). At $N = 1024$, more than 93% of the $\binom{113}{2} = 6{,}328$ cross-prime covariance terms $C_{p,q}$ are negative. This negativity has a clean structural explanation: for primes $p,q > \sqrt{N}$, the Poisson parameter $\lambda_{p,q} = A_pA_q/N < 1$, so $C_{p,q} < 0$ precisely when the joint count $J_{p,q}$ vanishes. In the sparse regime, $J_{p,q} = 0$ for $e^{-\lambda} \approx 96\%$ of pairs, while the rare pairs with $J_{p,q} = 1$ have $C_{p,q} = 1 - \lambda_{p,q} \approx 1$. Over a complete period $\pmod{pq}$, the average of $C_{p,q}$ is exactly zero: many small negatives are balanced by few large positives. Consequently, the high percentage of negative signs does *not* imply negative dependence and cannot be used directly. The aggregate off-diagonal $E^\circ = \sum_{p\ne q} w_pw_qC_{p,q}$ oscillates in sign: positive at $N = 1024, 4096, 8192$ and negative at $N = 2048$ (Table 1). The relevant theorem is the aggregate *sub-Poisson collision* estimate $|E^\circ| = o(S)$, which requires cancellation in the signed sum rather than a universal sign or a factored negative-dependence property. Neither the Dubhashi–Ranjan negative-lattice theorem (designed for balls-into-bins, not partial-period CRT) nor Janson's inequality (which requires nonnegativity of intersection parameters) applies to this partial-period setting.

Remark 96 (The dispersion route bypasses $\bar{Z}$). Hypothesis 93 does *not* require Hypothesis $\bar{Z}$. The adversarial singleton model $\mathcal{Z}_p^* = \{m_0 \bmod p, p-1-(m_0 \bmod p)\}$ satisfies $\bar{Z}$ in its strongest form ($Z(p) = 2$ for all p), yet violates the dispersion hypothesis. Conversely, the dispersion hypothesis with $Z(p) = O(p^{2/3})$ already gives $W(n) = o(n)$, even without any average bound on $\sum Z(p)$. This makes the dispersion route *strictly stronger* than the $\bar{Z}$ route: it is both sufficient on its own and robust against arbitrary zero-count fluctuations. The three-model comparison confirms that $V^\circ/S \approx 1$ empirically, matching the prediction of the independent random model.

Remark 97 (Martingale decomposition). Order the active primes as $p_1 < \cdots < p_K$ and define the Doob martingale $M_k = \mathbb{E}[B \mid X_{p_1},\ldots,X_{p_k}]$. The increments $D_k = M_k - M_{k-1}$ are orthogonal: $\mathrm{Var}(B) = \sum_k \mathbb{E}[D_k^2]$. However, $D_k = (X_{p_k} - \pi_k)\Delta_k$ where the "predictive cascade" $\Delta_k = w_{p_k} + \sum_{j>k} w_{p_j}(\mathbb{P}(X_{p_j}{=}1 \mid \mathcal{F}_{k-1}, X_{p_k}{=}1) - \mathbb{P}(X_{p_j}{=}1 \mid \mathcal{F}_{k-1}, X_{p_k}{=}0))$ contains all future primes. Under independence $\Delta_k = w_{p_k}$ and the variance equals the diagonal. For the Apéry measure on a partial CRT period, revealing that $m \bmod p_k$ falls in a zero can nearly identify m (when $A_{p_k} \approx 1$), making most subsequent indicators nearly deterministic. Controlling the cascade $\sum \mathbb{E}[\pi_k(1-\pi_k)\Delta_k^2] \le (1+o(1))\sum w_{p_k}^2\alpha_{p_k}$ is equivalent to the dispersion inequality itself.

Remark 98 (Concentration and the all-n gap). The L^2-to-uniform upgrade (Lemma 100) gives $\max|B(m)-\mu| \le \sqrt{V^\circ}$. If $V^\circ \le C\,S$ for some constant C, then $\sqrt{V^\circ} = O(\sqrt{N}\,(\log N)^{3/2}) = o(N/\log N)$, so the all-$n$ theorem $W(n) = o(n)$ follows without any higher-moment, sub-Gaussian, or negative-association input (Corollary 101). Empirically, the actual concentration is much stronger: $\max B(m)/(N/\log N) \approx 0.07$ at $N = 4096$ (with $\max B \approx 36$ vs. $N/\log N \approx 492$), and the maximum $|B(m)-\mu|/\sigma$ stabilizes near 6–7 standard deviations, consistent with the Poisson prediction for the extremal order statistic.

Remark 99 (Block decomposition of V°). Decompose the prime sum $B(m) = \sum_j B_{P_j}(m)$ by dyadic prime blocks $P_j < p \le 2P_j$. Then $V^\circ = \sum_j V_{P_j}^\circ + 2\sum_{j<k}\mathrm{cross}(P_j,P_k)$. At $N = 4096$, the cross-block covariance is 0.03% of the full V°. This is because cross-block pairs (p,q) with $p \in (P,2P]$,

$q \in (Q, 2Q]$, $Q \ge 2P$ satisfy $pq \ge 2P^2 > 2N$, so the CRT approximation is even tighter for cross-block than within-block terms. Consequently, the dispersion hypothesis reduces to $V_P^\circ \ll N^{o(1)}\, S_P$ for each dyadic block *separately*—a within-scale CRT independence statement.

12.8 The covariance structure and the L^2-to-uniform upgrade

N	K	V°/S	E°	$\lvert E^\circ\rvert/S$	$J{=}0$	$J{=}1$
512	67	0.998	+130.6	0.015	95.6%	4.1%
1024	113	0.998	+215.6	0.010	96.6%	2.9%
2048	205	0.994	+96.2	0.002	98.1%	1.8%
4096	380	0.996	+120.0	0.001	98.9%	1.1%
8192	742	1.001	+1344	0.005	99.3%	0.6%

Table 1: Covariance structure of the off-diagonal sum $E^\circ = V^\circ - S_c$, computed over all $\binom{K}{2}$ pairs of active primes $p > \sqrt{N}$. $J{=}0$ and $J{=}1$ are the fractions of pairs (p, q) with joint occupancy $J_{p,q} = 0$ or 1. The Cauchy–Schwarz bound on $|E^\circ|$ exceeds the actual value by a factor of 3,000–6,000.

The following deterministic lemma is the bridge from the dispersion hypothesis to the all-index conclusion.

Lemma 100 (L^2-to-uniform upgrade)**.** *Let $F_N\colon I_N \to \mathbb{R}$ with $\mu_N = \frac{1}{N}\sum_{m\in I_N} F_N(m)$ and $\mathcal{V}_N = \sum_{m\in I_N}(F_N(m) - \mu_N)^2$. Then*

$$\max_{m\in I_N} |F_N(m) - \mu_N| \;\le\; \sqrt{\mathcal{V}_N}. \tag{104}$$

Proof. Each squared deviation $(F_N(m_0) - \mu_N)^2$ is a single summand of $\mathcal{V}_N$, hence at most $\mathcal{V}_N$. □

Corollary 101 (Dispersion implies the all-index Apéry bound)**.** *If there exists a constant C such that $V_w^\circ(N) \le C\, S_w(N)$ for all sufficiently large dyadic N, where*

$$V_w^\circ(N) = \sum_{m\in I_N} (B(m) - \mu)^2, \qquad S_w(N) = \sum_{p>\sqrt{N}} (\log p)^2 A_p,$$

then $W(n) = o(n)$ for all n, and consequently $G_n = e^{o(n)}$ for every positive integer n.

Proof. By Lemma 100 with $F_N = B$,

$$\max_{m\in I_N} |B(m) - \mu| \;\le\; \sqrt{V_w^\circ} \;\le\; \sqrt{C\, S_w}.$$

Since $S_w \le (\log 2N)^2 \sum A_p$ and $\sum A_p = O(N\log N)$ (from the bound $Z(p) \le Cp^{2/3}$ or even $\sum Z(p) \ll \pi(x)$), we have $S_w = O(N(\log N)^3)$. Thus $\sqrt{V_w^\circ} = O(\sqrt{N}\,(\log N)^{3/2})$. The mean satisfies $\mu = O((\log N)^2)$. Therefore

$$W(n) \;\le\; B(n) \;\le\; \mu + \sqrt{V_w^\circ} = O(\sqrt{N}\,(\log N)^{3/2}) = o(N/\log N),$$

which gives $W(n) = o(n)$ upon taking $N = n$. □

Table 1 and the extended computation of V°/S through $N = 32{,}768$ (Section 10) provide strong evidence for the bound $V^\circ \le C\cdot S$ with $C \le 1.01$. The ratio $|E^\circ|/S$ decreases from 0.015 at $N = 512$ to 0.001 at $N = 4096$; the brief increase at $N = 8192$ is consistent with fluctuations of $O(1/\sqrt{N})$. No concentration inequality, negative-association theorem, or fourth-moment bound is needed: the L^2-to-L^∞ trivial inequality together with $V^\circ = O(S)$ completes the proof.

12.9 The exact random CRT model

The independent fixed-margin CRT model gives a rigorous null prediction for the covariance structure. For each active prime p, retain the observed column size A_p and choose the incidence set Ω_p uniformly among all A_p-subsets of I_N, independently across primes. Put $D_p = A_p(1 - A_p/N)$ and $D_w = \sum_p w_p^2 D_p = S_{\text{diag}}$.

Proposition 102 (Random CRT second moment)**.** *Under the independent fixed-margin model,* $\mathbf{E}\, C_{p,q} = 0$ *and* $\operatorname{Var}(C_{p,q}) = D_p D_q/(N-1)$. *Moreover,* $\mathbf{E}(C_{p,q}C_{p,r}) = 0$ *for all triples of distinct primes. Consequently,*

$$\mathbf{E}\big[(E^\circ)^2\big] \;=\; \frac{2}{N-1}\Big(D_w^2 - \sum_p w_p^4 D_p^2\Big). \tag{105}$$

If no single prime column carries a positive proportion of D_w*, then*

$$\frac{\operatorname{sd}(E^\circ)}{S} \;\sim\; \sqrt{\frac{2}{N}}, \tag{106}$$

matching the empirical ratio $|E^\circ|/S$ *in Table 1 to within the expected fluctuations.*

Proof. Given Ω_p, the intersection count $J_{p,q}$ is hypergeometric with parameters (N, A_p, A_q), giving $\mathbf{E}\, C_{p,q} = 0$ and $\operatorname{Var}(C_{p,q}) = D_p D_q/(N-1)$. For triples, conditioning on Ω_p makes $C_{p,q}$ and $C_{p,r}$ conditionally independent with zero mean, so $\mathbf{E}(C_{p,q}C_{p,r}) = 0$. Expanding $(E^\circ)^2$ and using orthogonality gives (105). □

12.10 The operator norm and eigenvector localization

Define the normalized off-diagonal covariance matrix H with entries

$$H_{p,q} = \frac{C_{p,q}}{\sqrt{D_p D_q}} \quad (p \neq q), \qquad H_{p,p} = 0.$$

Then $V^\circ = S_{\text{diag}} + v^T H v$ where $v_p = w_p\sqrt{D_p}$ and $|v|^2 = D_w$. Consequently, $V^\circ \le (1 + \|H\|_{\text{op}})\, S_{\text{diag}}$.

N	K	$\|H\|_{\text{op}}$	$\lambda_{\max}$	$\lambda_{\min}$	Rayleigh(v)
256	39	0.824	+0.824	−0.706	0.022
512	67	0.987	+0.987	−0.576	0.031
1024	113	1.088	+1.088	−0.707	0.021
2048	205	0.896	+0.896	−0.735	0.004
4096	380	0.904	+0.904	−0.734	0.002
8192	742	1.210	+1.210	−1.000	0.009

Table 2: Spectral data for the normalized covariance matrix H. The operator norm $\|H\|_{\text{op}}$ appears bounded (fluctuating between 0.8 and 1.2, not growing with N). The Rayleigh quotient $v^T Hv/|v|^2 = E^\circ/D_w$ for the logarithmic weight vector v is much smaller and non-monotone with mean ≈ 0.01, explaining why $V^\circ \approx S$. At $N = 8192$, two eigenvalues equal -1 exactly, arising from a three-prime collision cluster with $A_p \in \{1, 2, 3\}$.

The striking gap between $\|H\|_{\text{op}} \approx 1$ and the Rayleigh quotient ≈ 0.01 is explained by *eigenvector localization.* At every computed N, the top eigenvector concentrates on two or three primes with very sparse columns ($A_p \le 4$): these are large primes that share a common marked integer by

CRT coincidence. At $N = 8192$ ($\lambda_{\max} = 1.21$), the cluster consists of $p = 14851, 7937, 7559$ with $A_p = 1, 2, 3$ respectively. The logarithmic weight vector $v_p = (\log p)\sqrt{D_p}$ assigns negligible mass to these sparse primes (combined contribution $< 0.5\%$ of $|v|^2$ at every N), so it is nearly orthogonal to the localized extreme eigenvectors. At $N = 4096$, only 3% of $|v|^2$ lies in the top 5 eigenvectors; the largest projection of v (10%) falls on an eigenvalue of magnitude 0.14, deep in the spectral bulk. At $N = 8192$, the projection drops to 0.5%.

This localization has a combinatorial origin: the extreme eigenvalues of H arise from k-fold CRT collisions—integers m marked by $k \geq 3$ distinct large primes. The corresponding eigenvectors span a $(k-1)$-dimensional space of eigenvalue -1 and one direction of eigenvalue $k-1$. The logarithmic weight vector, being smoothly spread across all K active primes, is structurally immune to these localized collision artifacts.

12.11 Palm decorrelation and the cancellation mechanism

The identity (103) decomposes $V^\circ = S_{\text{diag}} + E^\circ$ where the off-diagonal $E^\circ = v^T H v$. The Rayleigh quotient $R_v = v^T H v/|v|^2 = E^\circ/D_w$ admits a transparent conditional-expectation interpretation. Define the *load excess*

$$\eta_p \;=\; \frac{(Hv)_p}{v_p} \;=\; \frac{\dfrac{1}{A_p}\displaystyle\sum_{m\in\Omega_p}\bigl(B_{-p}(m) - \mu_{-p}\bigr)}{w_p\,(1 - A_p/N)}, \tag{107}$$

where $B_{-p}(m) = B(m) - w_p X_p(m)$ is the total load excluding prime p, and $\mu_{-p} = \mu - w_p A_p/N$. The quantity η_p measures the average excess load from all *other* primes, conditional on $m \in \Omega_p$, normalized by the centered self-weight. Under exact independence, $\eta_p = 0$ for every p. The Rayleigh quotient is the weighted average

$$R_v \;=\; \sum_p \frac{v_p^2}{D_w}\,\eta_p. \tag{108}$$

Thus $V^\circ \approx S$ says: averaged with the natural variance weights, conditioning on one prime hitting does not bias the load from all other primes. This is the arithmetic content of the observed approximate orthogonality.

Remark 103 (Cancellation, not near-null)*.* The small Rayleigh quotient does *not* mean that v is a near-null vector of H. Computation gives $\|Hv\|/|v| \approx 0.31$ at all tested sizes $N = 512, \ldots, 4096$, while $R_v = v^T H v/|v|^2$ decreases from 0.031 to 0.002. The cancellation indicator $\|Hv\|/(|v|\,|R_v|)$ grows from 12 to 154: positive and negative spectral contributions to R_v individually exceed 0.13 but nearly cancel. This is the standard trace-zero mechanism: since $\operatorname{tr} H = \sum_j \lambda_j = 0$ and the projections $|\langle u, e_j\rangle|^2$ are approximately uniform ($u = v/|v|$ is delocalized), the identity $R_v = \sum_j \lambda_j(|\langle u, e_j\rangle|^2 - 1/K)$ produces cancellation to order $K^{-1/2}$. The extreme eigenvalues are localized (IPR of top eigenvector ≈ 0.25, effective dimension ≈ 4), so their projections deviate from $1/K$ but contribute little to the weighted sum.

12.12 Dyadic block compression

Partition the active primes into $L = O(\log N)$ dyadic blocks $\mathcal{B}_1, \ldots, \mathcal{B}_L$ according to $\lfloor \log_2 p \rfloor$. For each block, define the restricted weight $s_j = \|v|_{\mathcal{B}_j}\|$ and the unit share $a_j = s_j/|v|$. The *compressed covariance matrix* $\tilde{H} \in \mathbb{R}^{L\times L}$ is

$$\tilde{H}_{i,j} \;=\; \frac{(v^{(i)})^T H\, v^{(j)}}{s_i\, s_j}, \tag{109}$$

where $v^{(j)}$ is v restricted to $\mathcal{B}_j$. Then the Rayleigh quotient factors exactly as

$$R_v \;=\; a^T \tilde{H}\, a. \tag{110}$$

This reduces the $K \times K$ problem to an $L \times L$ one ($L \leq 8$) in the single direction of interest.

N	K	L	$\|H\|_{\text{op}}$	$\|\tilde{H}\|_{\text{op}}$	R_v	$\|Hv\|/\|v\|$
512	67	6	0.987	0.113	0.031	0.386
1024	113	6	1.088	0.111	0.021	0.302
2048	205	7	0.896	0.097	0.004	0.311
4096	380	7	0.904	0.040	0.002	0.313
8192	742	8	1.210	0.052	0.009	0.291

Table 3: Full operator norm $\|H\|_{\text{op}}$ versus compressed operator norm $\|\tilde{H}\|_{\text{op}}$. The full norm is bounded near 1 with a spike to 1.21 at $N = 8192$ (from the first singleton collision); the compressed norm is much smaller, ranging from 0.113 at $N = 512$ to 0.040–0.052 at $N = 4096$–8192. The column $\|Hv\|/|v|$ shows that v is not a near-null vector of H ($\|Hv\|/|v| \approx 0.3$ at all sizes); the small Rayleigh quotient arises from spectral cancellation.

The compressed norm $\|\tilde{H}\|_{\text{op}}$ controls the Rayleigh quotient via $|R_v| \leq \|\tilde{H}\|_{\text{op}}$, and is both numerically accessible (an $L \times L$ eigenvalue problem) and analytically tractable (each entry is a doubly-weighted average of covariances over a dyadic block pair).

The Marchenko–Pastur null model for K independent isotropic columns in $\mathbb{R}^N$ predicts $\|H\|_{\text{op}} \approx 2\sqrt{K/(N-1)} + K/(N-1)$; with $K \sim N/\log N$, this gives $2/\sqrt{\log N} + 1/\log N$, which matches the observed range 0.8–1.2 to within 20–40% (the excess is consistent with sparse localized spikes from low-A_p columns). For the compressed matrix, the same model predicts $\|\tilde{H}\|_{\text{op}} \sim \sqrt{L/N} \sim \sqrt{\log N/N} \to 0$, consistent with the observed decrease.

Remark 104 (Row leverage). The maximum *normalized row leverage* $\rho(n) = \sum_{p:\, n \in \Omega_p} 1/D_p$ is bounded near 1.5 at all tested sizes, dominated by one singleton column ($1/D_p \approx 1$) and one or two doublet columns ($1/D_p \approx 0.5$). No singleton collisions (two singletons at the same m) occur up to $N = 4096$. By the Gram identity $H + I = U^T U$ where U is the $N \times K$ normalized incidence matrix, $\|H\|_{\text{op}} + 1 \leq \max_n \rho(n) \cdot K$ is a trivial upper bound. A tighter bound via $\max_n \rho(n)$ governs the localized spectral spikes; if $\rho(n) \leq C$ uniformly, then $\|H\|_{\text{op}} \leq C - 1$.

12.13 Gram identity and block aggregates

The compression admits an exact Gram-matrix interpretation. Define the *normalized block aggregate*

$$Y_i(m) \;=\; \frac{1}{s_i} \sum_{p \in \mathcal{B}_i} (\log p) \left(X_p(m) - \frac{A_p}{N} \right), \qquad 1 \leq i \leq L, \tag{111}$$

where each Y_i is a centered function on I_N with $\sum_m Y_i(m) = 0$. Let Y be the $N \times L$ matrix with columns Y_i. Then

$$\boxed{I_L + \tilde{H} \;=\; Y^T Y \;\succeq\; 0,} \tag{112}$$

or entrywise, $\tilde{H}_{i,j} = \langle Y_i, Y_j \rangle - \delta_{ij}$. This is the key structural identity: $I + \tilde{H}$ is the Gram matrix of L vectors in $\mathbb{R}^N$, so every eigenvalue of $\tilde{H}$ is at least -1. Only the positive edge requires an arithmetic bound.

The Rayleigh identity (110) now reads

$$R_v \;=\; a^T \tilde{H}\, a \;=\; \left\| \sum_i a_i Y_i \right\|_2^2 - 1.$$

Thus $R_v = O(1)$ is equivalent to approximate orthonormality of the L aggregate functions Y_i.

Remark 105 (Low-to-high transfer). Let $P = EE^T$ denote orthogonal projection onto the L-dimensional block-smooth subspace. Then $PHu = E\tilde{H}a$ and

$$\|Hu\|^2 \;=\; \|\tilde{H}a\|^2 + \|(I-P)HEa\|^2. \tag{113}$$

At $N = 4096$, the data give $\|Hu\| \approx 0.31$ and $\|PHu\| \le \|\tilde{H}\|_{\mathrm{op}} \approx 0.04$, so $\|(I-P)Hu\| \ge 0.307$. Almost all of Hu lies in the within-block oscillatory subspace, not the block-smooth subspace. The covariance operator converts a smooth prime-scale profile into oscillations inside the blocks. This is a strictly sharper diagnosis than spectral cancellation alone.

12.14 Exact random CRT prediction

The independent *fixed-margin* CRT model (Section 12.9) yields exact second-moment formulas for the compressed matrix. Each normalized column satisfies the isotropy identity $\mathbf{E}[\phi_p \phi_p^T] = \frac{1}{N-1} P_0$, where $P_0 = I - \frac{1}{N}\mathbf{1}\mathbf{1}^T$. For distinct primes, $\mathbf{E}[H_{p,q}^2] = \frac{1}{N-1}$.

Proposition 106 (Compressed second moments)**.** *In the independent fixed-margin model,* $\mathbf{E}[\tilde{H}_{i,j}] = 0$ *for all* i, j*, and*

$$\mathbf{E}[\tilde{H}_{i,j}^2] \;=\; \frac{1}{N-1} \qquad (i \neq j), \tag{114}$$

$$\mathbf{E}[\tilde{H}_{i,i}^2] \;=\; \frac{2}{N-1}(1-\kappa_i), \qquad \kappa_i = \sum_{p \in \mathcal{B}_i} e_i(p)^4. \tag{115}$$

Consequently, the Frobenius norm satisfies

$$\mathbf{E}\|\tilde{H}\|_F^2 \;=\; \frac{L^2 + L - 2\sum_i \kappa_i}{N-1} \;\asymp\; \frac{L^2}{N}. \tag{116}$$

Proof. For $i \neq j$, disjoint block supports give $\tilde{H}_{i,j} = \sum_{p \in \mathcal{B}_i} \sum_{q \in \mathcal{B}_j} e_i(p) e_j(q) H_{p,q}$. Since different unordered prime pairs have uncorrelated entries (including those sharing one prime, by conditioning), only the diagonal terms survive: $\mathbf{E}[\tilde{H}_{i,j}^2] = \frac{1}{N-1}\sum_p e_i(p)^2 \sum_q e_j(q)^2 = \frac{1}{N-1}$. The diagonal entries involve $\sum_{p<q \in \mathcal{B}_i} e_i(p)^2 e_i(q)^2$, giving the κ_i correction. □

This proves $\|\tilde{H}\|_{\mathrm{op}} \to 0$ *in the random model* whenever $L = o(\sqrt{N})$. The GOE refinement predicts $\|\tilde{H}\|_{\mathrm{op}} \approx 2\sqrt{L/N}$, with the factor 2 being the asymptotic edge constant.

Remark 107 (Random surrogate comparison). For each N, we generate 20 random palindromic zero sets of the same sizes $|Z_p|$, build the corresponding $\tilde{H}$, and compare $\|\tilde{H}\|_{\mathrm{op}}$. At $N = 2048$: actual $= 0.097$, random mean $= 0.097$, z-score $= -0.01$. At $N = 1024$: actual $= 0.111$, random mean $= 0.130$, z-score $= -0.91$. The actual Apéry matrix is statistically indistinguishable from its random palindromic surrogate.

N	L	$\sqrt{L/N}$	$\|\tilde{H}\|_{\mathrm{op}}$	$\|\tilde{H}\|_{\mathrm{op}}/\sqrt{L/N}$
512	6	0.108	0.113	1.04
1024	6	0.077	0.111	1.45
2048	7	0.058	0.097	1.66
4096	7	0.041	0.040	0.97
8192	8	0.031	0.052	1.67

Table 4: The compressed operator norm tracks the random CRT prediction $\sqrt{L/N}$ with ratio ≈ 1–1.7.

12.15 AMTD: the precise remaining conjecture

The dispersion target (Hypothesis 93) is *not* a consequence of the Elliott–Halberstam conjecture, the generalized Elliott–Halberstam conjecture, or Barban–Davenport–Halberstam for fixed sequences. Those conjectures concern the discrepancy of a *fixed* arithmetic function in residue classes; here the tested residue function $1_{\mathcal{Z}_p}$ varies with the modulus. This is a fundamental structural distinction, not a technical gap.

We therefore propose:

Hypothesis 108 (Apéry Moving-Target Dispersion (AMTD))**.** *There exists an absolute constant C such that, for every dyadic interval $I_N = (N, 2N]$ and every dyadic prime block $P < p \le 2P$ with $P > \sqrt{N}$,*

$$V^\circ(P,N) \;\le\; C\, N^{o(1)}\, S(P,N).$$

The strongest structural sufficient condition is a *short-arc Fourier orthogonality* conjecture:

$$\int_{\mathbb{R}/\mathbb{Z}} \Bigg| \sum_{\substack{P<p\le 2P \\ 1\le a<p \\ x<a/p\le x+1/N}} \frac{(\log p)\, F_p(a)}{p} \Bigg|^2 dx \;\ll\; \frac{N^{o(1)}}{N} \sum_{P<p\le 2P} \sum_{a=1}^{p-1} \left| \frac{(\log p)\, F_p(a)}{p} \right|^2, \tag{117}$$

where $F_p(a) = \sum_{r\in\mathcal{Z}_p} e_p(ar)$. By Gallagher's lemma, (117) implies the weighted cross-prime dispersion bound of Proposition 116 (Subsection 12.16); a dilation to scale $M = 4N$ is needed because the Fejér weight $\gamma_N(N)$ vanishes at the raw scale, and the resulting majorant is the log-weighted energy Q_P of (123) rather than $S(P,N)$ literally. That weighted bound already suffices for the all-index conclusion of Proposition 94, so no direct comparison with $S(P,N)$ is required (Remark 117).

The classical additive large sieve gives the right-hand side of (117) multiplied by $(1 + P^2/N)$. For $P > \sqrt{N}$, this factor exceeds 2 and grows to $O(N)$ at the top block $P \sim N$. The P^2 barrier is intrinsic to the generic large-sieve proof and cannot be removed by average-spacing refinements alone. A proof of (117) must therefore exploit the special structure of the Apéry Fourier coefficients $F_p(a)$—their palindromic phase constraint ($e_p(a/2)\, F_p(a) \in \mathbb{R}$), their connection to the Sym^2 K3 Hasse polynomial, or the arithmetic of the zero sets $\mathcal{Z}_p$ themselves.

Remark 109 (Partial large-sieve result for small primes)*.* For dyadic blocks with $P \le \sqrt{N}$, the classical additive large sieve gives $V^\circ(P,N) \le (N + 4P^2) \sum_{P<p\le 2P} w_p^2 Z(p)/p \le 5\, S(P,N)$. Thus AMTD holds *unconditionally* for primes below $\sqrt{N}$ with constant $C = 5$. The open case is $P > \sqrt{N}$, where the large-sieve factor $1 + 4P^2/N$ exceeds any fixed constant. At the top block $P \sim N$, this factor is $O(N)$, and the generic large sieve provides no useful bound.

Remark 110 (Linnik dispersion route). The most plausible analytic pathway is Linnik's dispersion method: expand the L^2-discrepancy, isolate the diagonal, and seek cancellation in the off-diagonal via reciprocity and bilinear trace-function estimates in the framework of Fouvry–Kowalski–Michel [11], extended by Kowalski–Michel–Sawin [12] and Drappeau [9]. This reduces the problem to a *bounded-complexity representation theorem*: either the Fourier coefficient $a \mapsto \hat{\mathcal{Z}}_p(a) = \sum_{r\in\mathcal{Z}_p} e_p(ar)$ or the phase $m \mapsto e_p(u\, b_m)$ must belong to a uniformly controlled trace-function family. The obstacle is that b_m mod p is computed from a nonautonomous recurrence whose complexity grows with p, and no fixed bounded-conductor ℓ-adic sheaf with trace function $A_p(t)$ uniformly in p is currently known. The closest available framework is Forey–Fresán–Kowalski [13], which studies arithmetic Fourier transforms of a fixed perverse sheaf over a fixed finite field; four mismatches remain (growing complexity, horizontal vs. extension-degree aspect, crystalline vs. ℓ-adic, and one- vs. two-characteristic dispersion). A polylogarithmic loss in (117) already suffices for the competition target.

Remark 111 (Palindromic Fourier factoring). For doublet primes in the top block, the exponential sum over hit positions is

$$S(\theta) = \sum_{p\in\mathcal{D}} \big[e(\theta\, m_1(p)) + e(\theta\, m_2(p))\big] = 2\sum_{p\in\mathcal{D}} e\Big(\theta\, \tfrac{3p-1}{2}\Big) \cos(\pi\theta\, h_p)\,,$$

where $h_p = m_2(p) - m_1(p) = p - 1 - 2r_p$ is the doublet gap. By Parseval, $\int_0^1 |S(\theta)|^2\, d\theta = T + F$. The diagonal AMTD requires this integral to be $O(T)$, equivalently that $S(\theta)$ has approximately square-root cancellation on the minor arcs.

The palindromic structure converts each doublet edge into a prime-indexed phase $e(3p\theta/2)$ modulated by the cosine $\cos(\pi\theta h_p)$. The gaps h_p are determined by the first zero r_p of b_n mod p. A proof of diagonal AMTD via this route would establish cancellation in $S(\theta)$ by exploiting the "pseudorandom" distribution of h_p as p varies.

The compact two-term factorization above holds only on the clipped subfamily where both lifts $m_1 = p + r_p$ and $m_2 = 2p - 1 - r_p$ actually fall in $I_N = (N, 2N]$, i.e. $2p - 1 - 2N \le r_p \le 2N - p$ (Lemma 118 and the clipping analysis of Subsection 12.16); outside it, one lift is clipped and the reflected residue contributes a pair centred at $(2p-1)/2$ rather than $(3p-1)/2$, and in the block $N/2 < p \le N$ there are additional quotient-2 and quotient-3 lifts. Computationally only a small fraction of top-block doublets (5 of 36 at the tested scale) have the displayed centre $(3p-1)/2$, so the honest cancellation problem is for the full clipped hit set, not the unqualified two-term sum.

Remark 112 (Hierarchy of targets). The compression produces a strict hierarchy of increasingly weak—and therefore increasingly tractable—conjectures:

(A) **Full operator bound:** $\|H\|_{\mathrm{op}} = O(1)$. This implies a Bessel inequality for all prime columns and is far stronger than needed.

(B) **Compressed Bessel bound:** $\|\tilde{H}\|_{\mathrm{op}} = O(1)$. This asks for approximate orthonormality of only $L = O(\log N)$ block-aggregate functions, and suffices for the all-index GCD estimate.

(C) **Scalar Rayleigh bound:** $a^T \tilde{H} a = O(1)$. This is the weakest statement that closes the all-index theorem, equivalent to $V^\circ \ll S$. It preserves all useful signs and remains a two-prime problem.

Each implication (A)⇒(B)⇒(C) is strict. The random CRT model predicts $\|\tilde{H}\|_{\mathrm{op}} \asymp \sqrt{L/N} \to 0$ and $|a^T \tilde{H} a| \asymp N^{-1/2}$, both of which are far stronger than (B) and (C).

Remark 113 (Diagonal AMTD via Cauchy–Schwarz). The all-index GCD theorem admits a simpler closing route that avoids cross-block estimates entirely. Define the uncentered block load $F_i(m) = \sum_{p\in\mathcal{B}_i}(\log p)\, g_p(m) = s_i\, Y_i(m)$, so that $V^\circ = \|\sum_i F_i\|_2^2$. If each dyadic block satisfies

$$\|F_i\|_2^2 \;\le\; C\, s_i^2 \qquad \text{(Diagonal AMTD)}, \tag{118}$$

equivalently $1 + \tilde{H}_{ii} \le C$, then by Cauchy–Schwarz,

$$V^\circ \;\le\; \Big(\sum_i \|F_i\|_2\Big)^2 \;\le\; L \sum_i \|F_i\|_2^2 \;\le\; C L\, S.$$

Since $L = O(\log N) = N^{o(1)}$, this gives $V^\circ \le N^{o(1)}\, S$, which suffices for $\max_m |B(m) - \mu| \le \sqrt{V^\circ} = O\big(\sqrt{N}\,(\log N)^{3/2}\big) = o(N/\log N)$. The within-block estimate (118) is a *two-prime* signed dispersion problem—it preserves useful signs, does not require four-prime correlations, and avoids the full operator norm or Frobenius arguments. At the top block $P \sim N$, it reduces to a sparse translated-set collision problem: bounding the additive energy of $\{p + r : r \in \mathcal{Z}_p\}$ for primes p in a dyadic range.

Block-level data at $N = 4096$. The ratio $\|F_i\|_2^2/s_i^2$ for each dyadic block at $N = 4096$:

Block	K_i	$\|F_i\|_2^2/s_i^2$	$\tilde{H}_{ii}$
2^6	2	1.018	$+0.018$
2^7	13	0.998	-0.002
2^8	17	0.987	-0.013
2^9	30	0.988	-0.012
2^{10}	47	1.015	$+0.015$
2^{11}	94	1.016	$+0.016$
2^{12}	177	0.989	-0.011

Maximum ratio: 1.018 (well below any nontrivial constant). The Cauchy loss is $L \cdot \sum_i \|F_i\|_2^2/S = 7.00 = L$, confirming $V^\circ/S = 1.002$ while the Cauchy bound gives $V^\circ \le 7S$. At $N = 8192$ the maximum block ratio remains 1.029 (block 2^{10}) with $L = 8$ blocks. At $N = 32{,}768$ the collision count for top-block doublets is $F = 38$ with $T = 925$, giving $F/T = 0.041$ (decreasing from 0.055 at $N = 8192$), confirming the conjecture across all tested sizes.

Remark 114 (Translation variance). For each pair of distinct primes p, q, averaging over all translates $x \bmod pq$ gives

$$\frac{1}{pq} \sum_{x \bmod pq} |T_{p,q}(x; N)|^2 \;\le\; \frac{N\, Z(p)^2 Z(q)^2}{pq},$$

where $T_{p,q}(x; L) = \sum_{j=1}^{L} f_p(x + j)\, f_q(x + j)$. For bounded zero counts $Z(p) = O(1)$, this is already at the random scale for a *typical* translate. However, the GCD problem samples the single translate $x = N$ simultaneously for all prime pairs. Summing individual pair bounds absolutely costs $O(N^{3/2})$, which exceeds $S \asymp N(\log N)^2$. The P^2 large-sieve barrier arises because the Fourier support of f_p has $\sim p$ nonzero frequencies (not $Z(p)$), and palindromic symmetry improves only absolute constants.

12.16 The exact short–arc transform and the remaining bilinear form

Write $e(t) = e^{2\pi i t}$ and $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. For a prime block $P < p \le 2P$ put

$$c_{p,a} = \frac{(\log p) F_p(a)}{p}, \qquad Q_P = \sum_{P<p\le 2P} \sum_{a=1}^{p-1} |c_{p,a}|^2,$$

and, for an integer $M \ge 2$, define the cyclic short–arc sum

$$A_{P,M}(x) = \sum_{\substack{P<p\le 2P,\ 1\le a<p \\ a/p\in(x,x+1/M]_{\mathbb{T}}}} c_{p,a}.$$

Endpoints are immaterial. Finally set

$$D_P(k) = \sum_{P<p\le 2P} \sum_{a=1}^{p-1} c_{p,a} e(-ka/p), \qquad k \in \mathbb{Z}.$$

Proposition 115 (Exact triangular-kernel identity)**.** *With $\|t\|$ denoting distance to the nearest integer, one has*

$$\int_{\mathbb{T}} |A_{P,M}(x)|^2 \, dx = \sum_{p,a} \sum_{q,b} c_{p,a} \overline{c_{q,b}} \left(\frac{1}{M} - \left\| \frac{a}{p} - \frac{b}{q} \right\| \right)_+ \tag{119}$$

$$= \sum_{k\in\mathbb{Z}} \gamma_M(k) |D_P(k)|^2, \tag{120}$$

where

$$\gamma_M(0) = \frac{1}{M^2}, \qquad \gamma_M(k) = \left(\frac{\sin(\pi k/M)}{\pi k} \right)^2 \quad (k \ne 0).$$

Moreover Fourier inversion on each $\mathbb{F}_p$ gives the exact identity

$$D_P(k) = \sum_{P<p\le 2P} (\log p) \left(\mathbf{1}_{\mathcal{Z}_p}(k \bmod p) - \frac{Z(p)}{p} \right). \tag{121}$$

Proof. For a point $\lambda \in \mathbb{T}$, the set of x for which $\lambda \in (x, x+1/M]_{\mathbb{T}}$ is a cyclic interval of length $1/M$. The intersection of the two intervals belonging to λ and μ has measure $(M^{-1} - \|\lambda - \mu\|)_+$, proving (119). The k-th Fourier coefficient of an interval of length $1/M$ has squared modulus $\gamma_M(k)$; Parseval proves (120). Finally,

$$\frac{1}{p} \sum_{a=0}^{p-1} F_p(a) e_p(-ak) = \mathbf{1}_{\mathcal{Z}_p}(k),$$

and removing $a = 0$, for which $F_p(0) = Z(p)$, proves (121). □

The two notions of zero frequency are both visible here. The modular frequency $a = 0$ has been removed and produces the vertical centre $\sum_p (\log p) Z(p)/p$. The integer Fourier mode $k = 0$ remains in (120); since $b_0 = 1$, it equals $D_P(0) = -\sum_p (\log p) Z(p)/p$ and is weighted by M^{-2}.

Thus it has neither been silently discarded nor identified with the empirical mean on I_N. If $\mu_{P,I}$ is that empirical mean, then orthogonal projection gives

$$\sum_{k\in I_N} |B_P^{(w)}(k) - \mu_{P,I}|^2 \le \sum_{k\in I_N} |D_P(k)|^2,$$

where $B_P^{(w)}(k) = \sum_{P<p\le 2P}(\log p)\mathbf{1}_{\mathcal{Z}_p}(k \bmod p)$.

There is a small but consequential point in applying this identity. At the scale printed in (117), $\gamma_N(N) = \gamma_N(2N) = 0$, so that estimate at scale N does not by itself control the shell $I_N = (N, 2N]$. Uniformity in the scale repairs the problem without a phase twist.

Proposition 116 (Correct Gallagher reduction by dilation)**.** *Assume that* (117)*, with a factor $M^{o(1)}$, holds for every pair (P, M) satisfying $\sqrt{M} < P \le M$. Then, uniformly for $\sqrt{N} < P \le N$,*

$$\sum_{N<k\le 2N} |D_P(k)|^2 \ll N^{1+o(1)} Q_P. \tag{122}$$

Consequently, together with the unconditional $Z(p) \ll p^{2/3}$, the short–arc target implies $W(n) = o(n)$ for every n and hence $G_n = e^{o(n)}$.

Proof. First suppose $P > 2\sqrt{N}$. Apply (117) with the scale $M = 4N$. This is legitimate because $\sqrt{4N} < P \le 4N$. For $N < k \le 2N$,

$$\gamma_{4N}(k) = \left(\frac{\sin(\pi k/(4N))}{\pi k}\right)^2 \ge \frac{1}{8\pi^2 N^2}.$$

Equations (120) and (117) therefore give

$$\sum_{N<k\le 2N} |D_P(k)|^2 \le 8\pi^2 N^2 \int_{\mathbb{T}} |A_{P,4N}(x)|^2\, dx \ll N^{1+o(1)} Q_P.$$

For $\sqrt{N} < P \le 2\sqrt{N}$, the ordinary additive large sieve, applied on any interval of N consecutive integers, gives

$$\sum_{N<k\le 2N} |D_P(k)|^2 \le (N + 4P^2) Q_P \ll N Q_P.$$

This proves (122).

Finite Fourier Parseval gives

$$Q_P = \sum_{P<p\le 2P} (\log p)^2 \left(\frac{Z(p)}{p} - \frac{Z(p)^2}{p^2}\right) \le \sum_{P<p\le 2P} \frac{(\log p)^2 Z(p)}{p}. \tag{123}$$

Thus $Q_P \ll P^{2/3}\log P$, while

$$\mu_P := \sum_{P<p\le 2P} \frac{(\log p) Z(p)}{p} \ll P^{2/3}.$$

By (121), the logarithmically weighted bad-prime sum in this block is at most

$$\mu_P + |D_P(k)| \ll P^{2/3} + N^{1/2+o(1)} P^{1/3} \ll N^{5/6+o(1)}$$

for every $k \in I_N$. Summing the $O(\log N)$ dyadic blocks and using the small-prime and companion-block estimates already invoked in Proposition 94 proves the claimed all-index bound. □

Remark 117 (What this does and does not imply). The direct assertion in Subsection 12.15 that (117) "implies AMTD block by block" is not the conclusion of the displayed Fourier calculation. Its right-hand side is (123); it is not $S(P,N) = \sum_p |\Omega_p|$, and its coefficients carry the weights $\log p$. In particular, when $p > N$, a residue in $\mathcal{Z}_p$ need not have a representative in I_N. Proposition 116 is the valid conclusion: a weighted dispersion estimate with the vertical majorant NQ_P, which is already strong enough for the conclusion of Proposition 94. No unproved comparison with $S(P,N)$ is needed.

12.17 Palindromic phases and top-block clipping

Lemma 118 (Exact doublet Fourier phase)**.** *Suppose $\mathcal{Z}_p = \{r, p-1-r\}$, choose $0 \le r \le (p-1)/2$, and put $h = p-1-2r$. Then, for every $a \bmod p$,*

$$F_p(a) = 2e_p\left(\frac{a(p-1)}{2}\right)\cos\left(\frac{\pi a h}{p}\right). \tag{124}$$

Here $e_p(a/2)$ in the palindromic reality statement means multiplication by the inverse of 2 in $\mathbb{F}_p$; consequently $e_p(a/2)F_p(a) = 2\cos(\pi a h/p) \in \mathbb{R}$.

Proof. The two integers r and $p-1-r$ have average $(p-1)/2$ and difference h. Factoring their two exponentials proves (124). Since $(p+1)/2$ represents 2^{-1} (mod p), multiplying the phase in (124) by $e_p(a(p+1)/2)$ cancels it. □

In ordinary real arguments this real rotation is $e(-a(p-1)/(2p))F_p(a)$, because $(p+1)/2 \equiv -(p-1)/2 \pmod p$. The expression with the opposite ordinary sign is not generally real; the distinction is checked on the actual doublet $\mathcal{Z}_{17} = \{3, 13\}$ by `oracleA_verify.py`.

The phase $e(\theta(3p-1)/2)$ in Remark 111 is a phase of two particular *integer lifts*; it is not the ordinary real phase used to make $F_p(a)$ real. At the discrete value $\theta = a/p$ it does coincide modulo 1 with the phase $e_p(a(p-1)/2)$ in (124), since the two centres differ by p. For continuous θ, however, it is the centre of the chosen integer lifts and therefore changes when those lifts change. It also has a restricted validity domain. In general the contribution of a residue $u \bmod p$ to the shell is the finite set

$$H_{N,p}(u) = \{u + \ell p : N < u + \ell p \le 2N,\ \ell \in \mathbb{Z}\}. \tag{125}$$

The shell Fourier sum must use $H_{N,p}(r) \cup H_{N,p}(p-1-r)$, including every admissible lift. For example, when $p \in (N, 2N]$, the two displayed lifts

$$m_1 = p + r, \qquad m_2 = 2p - 1 - r$$

both belong to I_N precisely when

$$2p - 1 - 2N \le r \le 2N - p. \tag{126}$$

Only on this subfamily does

$$e(\theta m_1) + e(\theta m_2) = 2e\left(\theta\frac{3p-1}{2}\right)\cos(\pi\theta h)$$

hold for the contribution of the actual hit set $\Omega_p \cap I_N$. Outside (126), one or both lifts are clipped; when the reflected residue $p-1-r$ itself lies in I_N, the relevant pair has centre $(2p-1)/2$, not $(3p-1)/2$. Parseval still gives

$$\int_{\mathbb{T}} \left| \sum_p \sum_{m \in \Omega_p \cap I_N} e(\theta m) \right|^2 d\theta = T + F,$$

but only when the sum uses the actual, clipped hit positions. Thus the unqualified formula in Remark 111 cannot be the starting point for all top-block doublets. In the computational block $N/2 < p \le N$, there can additionally be quotient-2 or quotient-3 lifts in (125); these are also absent from the complete two-term sum in that remark.

12.18 The exact cross-prime lemma and a sharp obstruction

Put $K_M(t) = (M^{-1} - \|t\|)_+$. Splitting (119) into equal and unequal primes gives

$$\int_{\mathbb{T}} |A_{P,M}|^2 = \frac{Q_P}{M} + R_=(P,M) + R_{\neq}(P,M),$$

where

$$R_{\neq}(P,M) = \sum_{\substack{P<p,q\le 2P\\ p\neq q}} \sum_{a=1}^{p-1}\sum_{b=1}^{q-1} K_M\left(\frac{a}{p}-\frac{b}{q}\right)\frac{(\log p)(\log q)}{pq} F_p(a)\overline{F_q(b)}. \tag{127}$$

Since $p \le 2P \le 2M$, each a/p has at most two same-prime neighbours in an arc of length $1/M$, and $|R_=(P,M)| \le 2Q_P/M$ by $2|uv| \le |u|^2 + |v|^2$. Hence the precise missing analytic lemma is the one-sided scalar estimate

$$\operatorname{Re} R_{\neq}(P,M) \le CM^{-1+o(1)}Q_P. \tag{SDC}$$

Equivalently, this asks for cancellation among the near-Farey solutions

$$\left\|\frac{a}{p}-\frac{b}{q}\right\| < \frac{1}{M}, \qquad \min\{|aq-bp|,\, pq-|aq-bp|\} < \frac{pq}{M}.$$

This is a scalar Rayleigh bound for the actual Apéry vector; it neither asks for a full operator norm nor discards the signs as Cauchy–Schwarz does. It is the exact bilinear input still missing from the Linnik route of Remark 110.

There is also an exact reciprocal-fraction parametrization. Let $\langle d\rangle_{pq}$ be the representative of $d \bmod pq$ of least absolute value, and let $\bar q$ and $\bar p$ denote inverses modulo p and q, respectively. For $p \neq q$, putting $k = \langle aq - bp\rangle_{pq}$ is a bijection between the pairs occurring in (127) and

$$0 < |k| < \frac{pq}{M}, \qquad (k,pq) = 1, \qquad a \equiv k\bar q \pmod p, \qquad b \equiv -k\bar p \pmod q.$$

Since $F_q(-u) = \overline{F_q(u)}$, one obtains

$$R_{\neq}(P,M) = \frac{1}{M}\,\mathcal{B}_{P,M}, \tag{128}$$

$$\mathcal{B}_{P,M} = \sum_{\substack{P<p,q\le 2P\\ p\neq q}} \frac{(\log p)(\log q)}{pq} \sum_{\substack{0<|k|<pq/M\\ (k,pq)=1}} \left(1-\frac{M|k|}{pq}\right) F_p(k\bar q)F_q(k\bar p). \tag{129}$$

Thus (SDC) is precisely the one-sided moving reciprocal-fraction estimate $\operatorname{Re}\mathcal{B}_{P,M} \le CM^{o(1)}Q_P$. The numerators $k\bar q$, $k\bar p$ and both zero sets move jointly with p, q; palindromy only replaces each doublet transform by a real cosine after a moving phase rotation. It does not turn (129) into a fixed-numerator bilinear form.

Proposition 119 (One trace-function mismatch removed on doublets)**.** *On the doublet sector, $a \mapsto F_p(a)$ is the trace function of a rank-2, uniformly bounded-conductor sheaf: explicitly it is the direct sum of the two rank-1 Artin–Schreier sheaves with traces $a \mapsto e_p(ar_p)$ and $a \mapsto e_p(-a(1+r_p))$. Thus the growing-complexity obstruction for representing the* local Fourier coefficient *disappears completely on this sector.*

Proof. For $\mathcal{Z}_p = \{r_p, p-1-r_p\}$,

$$F_p(a) = e_p(ar_p) + e_p(-a(1+r_p)).$$

Each summand is a linear additive-character trace of rank and conductor bounded independently of both p and the slope. Direct sum proves the claim. $\square$

This does not make the frameworks discussed in Remark 110 applicable as stated. The slope r_p is selected by a same-characteristic Apéry zero condition and moves with p; there is no fixed sheaf over one fixed finite field whose extension-degree traces give this horizontal family. Moreover (129) couples two varying characteristics at once. The anchored family below has exactly the same rank-2 conductor bound and still violates the target. Hence bounded local conductor removes one of the four mismatches listed in Remark 110, but horizontal control of the moving slopes remains the load-bearing arithmetic input.

Palindromic reality and the bound $Z(p) = O(1)$ cannot prove (SDC). The following construction quantifies the obstruction at exactly the required short-arc scale.

Proposition 120 (Reflected anchored obstruction)**.** *Let*

$$\mathcal{P}_N = \{p \text{ prime} : N < p \le 2N,\ p \nmid (4N+1)(2N+1)\}$$

and, for $p \in \mathcal{P}_N$, set

$$\mathcal{Z}_p^* = \{\, 2N \bmod p,\ -2N-1 \bmod p\,\}.$$

Then every $\mathcal{Z}_p^*$ is a two-element set invariant under $r \mapsto p-1-r$; it contains neither 0 nor adjacent residues. Yet the short–arc estimate at $(P, M) = (N, 4N)$ fails by a factor $N^{1-o(1)}$. Simultaneously, all columns contain the point $2N \in I_N$, so their centered block dispersion is $\gg N^2$ in the logarithmically weighted normalization.

Proof. The divisors of $4N+1$ are exactly those for which the two displayed residues coincide; excluding divisors of $2N+1$ also keeps 0 out of the sets and rules out one of the two possible adjacency congruences. The other would force $p \mid 2N$, impossible for a prime $p \in (N, 2N)$. These exclusions remove only $O(1)$ primes from this dyadic range. For the corresponding Fourier coefficients, let D_N^* denote (121). Since $2N \bmod p \in \mathcal{Z}_p^*$ for every $p \in \mathcal{P}_N$, the prime number theorem gives

$$D_N^*(2N) = \sum_{p\in\mathcal{P}_N} (\log p)(1 - 2/p) \asymp N.$$

At $M = 4N$ the Fejér weight of this mode is

$$\gamma_{4N}(2N) = \frac{1}{4\pi^2 N^2}.$$

Therefore (120) gives

$$\int_{\mathbb{T}} |A^*_{N,4N}(x)|^2 dx \gg 1.$$

On the other hand, finite Fourier Parseval gives

$$Q_N^* = \sum_{p\in\mathcal{P}_N} (\log p)^2 \left(\frac{2}{p} - \frac{4}{p^2}\right) \asymp \log N.$$

Thus the proposed right-hand side is $(4N)^{-1+o(1)} Q_N^* = N^{-1+o(1)} \log N = o(1)$. The same common hit at $2N$ gives a centered deviation $\asymp N$ and hence weighted dispersion $\gg N^2$: indeed every column has at most two hits in the interval of length N, so the empirical weighted mean is $O(N^{-1} \sum_{p\in\mathcal{P}_N} \log p) = O(1)$. $\square$

The obstruction is not an assertion about the Apéry zeros. It proves the required impossibility content: reflection, the exact palindromic phase, doublet cardinality, vertical Parseval, and generic prime spacing do not imply the short–arc bound. A successful proof must insert an Apéry-specific horizontal fact into (127), exactly as anticipated in Remark 110; the separately running arithmetic oracle is responsible for such an input.

12.19 Computational reconnaissance

The reproducible computation in `oracleA_explore.py` and `oracleA_exploration.md` uses the exact zero-set bank and the blocks $N/2 < p \le N$ for $N = 8192, 16384, 32768, 65536$. For the complete two-position doublet sum, the exact Parseval ratios $(T+F)/T$ are between 1.0224 and 1.0319. After declaring the major arcs $|\theta - a/q| \le 16/(qN)$ with $q \le 16$, the largest sampled minor-arc peaks are between $3.38\sqrt{T}$ and $3.69\sqrt{T}$. The largest nondegenerate connected correlation between $e(3p\theta/2)$ and $\cos(\pi\theta h_p)$ at those small-denominator rationals falls from 0.2204 to 0.0657 across the four scales. An independent-uniform-gap collision model predicts the observed complete-pair collision count within ratios 0.756 to 1.128.

The shell computation separately enumerates every lift in (125). With logarithmic weights its true off-diagonal ratios E°/S_c are respectively $0.00497, 0.00185, -0.00689, 0.00340$. Direct counting and an independent finite-group Fourier reconstruction agree within $4.4 \cdot 10^{-10}$. The clipping distinction is numerically substantial: at $N = 8192$ the complete two-position sum retains only 162 of its displayed positions in I_N, whereas the actual doublet columns have 374 shell lifts. These observations support square-root-scale cancellation, but none is used to infer (117).

STALL REPORT. The target (117) is not proved, either for the full Apéry sum or for its doublet sub-sum. The rigorous analytic outputs are:

1. the exact triangular-kernel/Fejér identity (119)–(121);
2. the corrected dilation argument proving that the stated uniform short–arc target really would imply the fully unconditional all-index conclusion, without the unsupported identification of its Fourier energy with $S(P, N)$;
3. the exact doublet phase and the clipping domain (126), correcting the unrestricted use of Remark 111;
4. the one-sided cross-prime short-determinant estimate (SDC), isolated as the precise missing bilinear lemma after the harmless same-prime terms are removed;
5. Proposition 119, which removes the growing-conductor trace-function mismatch on the doublet sector while locating the surviving obstruction in the horizontal distribution of the moving slopes r_p;
6. the reflected anchored family of Proposition 120, which proves that none of palindromy, doublet dominance, vertical bounds, or generic large sieve geometry can supply that lemma.

12.20 Arithmetic oracle B: a fixed-anchor theorem and the sharp horizontal obstruction

Put

$$\mathcal{Z}_p = \{0 \le j < p : b_j \equiv 0 \pmod p\}, \qquad H_p(t) = \sum_{j=0}^{p-1} b_j t^j.$$

The requested AMTD estimate is not proved here. There is nevertheless an unconditional answer to the literal fixed-target question in goal G2. It is stronger than density zero, but it also exposes the decisive quantifier reversal: a target fixed before the prime varies is elementary, whereas the no-go target is chosen after the dyadic scale. Thus the status of the ranked goals is: G1 is open; literal G2 is proved below but is insufficient for the moving target; G3 is not achieved because the required marked crystalline coordinate has not been constructed; and G4, the sharp obstruction with the exact missing input, is achieved.

Proposition 121 (Fixed Apéry anchors are finite)**.** *For $c \in \mathbb{Z}$, let $\bar{c}_p \in \{0, \dots, p-1\}$ be its residue modulo p, and set*

$$c^\flat = \begin{cases} c, & c \geq 0, \\ -c-1, & c < 0. \end{cases}$$

Then

$$\{p \geq 5 : \bar{c}_p \in \mathcal{Z}_p\} \subseteq \{p \geq 5 : p \mid b_{c^\flat}\}. \tag{130}$$

Consequently, for every fixed c,

$$\#\{p \leq X : \bar{c}_p \in \mathcal{Z}_p\} \leq \omega(b_{c^\flat}) = O_c(1) = o(\pi(X)). \tag{131}$$

For $p > |c|$, the inclusion in (130) *is an equivalence.*

Proof. First suppose that $c \geq 0$, and write $c = \sum_i c_i p^i$ in base p. The lowest digit is $c_0 = \bar{c}_p$. If $\bar{c}_p \in \mathcal{Z}_p$, then $p \mid b_{c_0}$, and Gessel's Lucas congruence (Theorem 4) gives

$$b_c \equiv \prod_i b_{c_i} \equiv 0 \pmod{p}.$$

Now let $c = -d < 0$ and write $d = qp + s$ with $0 \leq s < p$. If $s = 0$, then $\bar{c}_p = 0$ and there is no hit because $b_0 = 1$. If $s > 0$, then $\bar{c}_p = p - s$. Reflection (Remark 16, or Proposition 72) gives

$$b_{p-s} \equiv b_{s-1} \pmod{p}.$$

Thus a hit implies $p \mid b_{s-1}$. The lowest base-p digit of $d-1$ is $s-1$, so another application of Lucas gives $p \mid b_{d-1} = b_{c^\flat}$. This proves the inclusion.

The binomial formula for b_n has positive integral summands and its $k = 0$ summand is 1; hence b_n is a fixed nonzero integer for fixed n. It has only finitely many prime divisors, proving (131). Finally, if $p > |c|$, then for $c \geq 0$ one has $\bar{c}_p = c$, while for $c = -d < 0$ reflection gives $b_{\bar{c}_p} = b_{p-d} \equiv b_{d-1}$; both implications are then reversible. □

Remark 122 (Why this does not kill the anchored star)*.* Proposition 121 has the quantifier order

$$\forall c \in \mathbb{Z}\ \exists C_c\ \forall X : \qquad \#\{p \leq X : \bar{c}_p \in \mathcal{Z}_p\} \leq C_c. \tag{132}$$

In Section 12.5, however, the anchor is $m_0 = m_0(N) \in I_N$ and is allowed to move with N. To exclude just one full star in a dyadic prime block, one would already need a uniform statement such as

$$\sup_{m \in I_N} \#\{P < p \leq 2P : m \bmod p \in \mathcal{Z}_p\} = o(P/\log P), \tag{133}$$

in the relevant range $P > \sqrt{N}$. AMTD asks for a centered second moment, which is stronger still.

Lucas merely turns the expression inside the supremum into a count of large prime divisors of the moving integer b_m. The elementary height bound $\log b_m = O(m)$ gives at best $O(N/\log N)$ prime divisors of size $\asymp N$ when $m \asymp N$, exactly the size of a full prime block. Thus this Lucas/height argument supplies no way to move the constants C_c outside the supremum. This is the same distinction exhibited by the anchored-star/no-go construction in Section 12.

12.20.1 The two zero loci are in dual spaces

The proposed B1 and B2 routes make an object-level identification that is false. The following elementary calculation makes the distinction exact.

Proposition 123 (Coefficient zeros are not Hasse-polynomial roots)**.** *For* $1 \le j \le p-2$*, the following identity holds in* $\mathbb{F}_p$*:*

$$b_j = -\sum_{t \in \mathbb{F}_p^\times} H_p(t) t^{-j}. \tag{134}$$

Consequently,

$$\mathcal{Z}_p = \{j : [t^j] H_p(t) = 0\}$$

is a coefficient, or finite-Mellin, zero-set. It is not the evaluation root locus of H_p *or of the square root in Remark 10.*

Proof. After inserting $H_p(t) = \sum_{k=0}^{p-1} b_k t^k$, multiplicative orthogonality gives

$$\sum_{t \in \mathbb{F}_p^\times} H_p(t) t^{-j} = \sum_{k=0}^{p-1} b_k \sum_{t \in \mathbb{F}_p^\times} t^{k-j} = -b_j,$$

because $1 \le j \le p-2$ and the inner sum vanishes unless $k = j$.

For a concrete separation, exact reduction at $p = 7$ gives

$$\begin{aligned} H_7(t) &= 1 + 5t + 3t^2 + 3t^3 + 3t^4 + 5t^5 + t^6 \\ &= (t-1)^2 (t^2+1)^2 \pmod 7. \end{aligned}$$

No coefficient is zero, so $\mathcal{Z}_7 = \varnothing$, whereas $t = 1$ is an evaluation root of both H_7 and its square root. □

In particular, if $\rho_p = \min \mathcal{Z}_p$, then ρ_p need not be a root of $B_p(t)$, and no theorem identifies the two notions; they can coincide accidentally at individual primes. A Weil bound for $V(B_p)$ therefore does not bound

$$F_p(a) = \sum_{j \in \mathcal{Z}_p} e_p(aj).$$

Nor is ρ_p equivalent to the whole zero-set: it is undefined when $\mathcal{Z}_p$ is empty, and for example

$$\mathcal{Z}_{181} = \{19, 47, 133, 161\}.$$

For a center singleton the first zero trivially determines the set; only on the doublet subfamily $Z(p) = 2$ does it determine a noncentral reflected pair and its gap. Once there are at least two noncentral pairs, it does not determine the whole zero-set.

12.20.2 Why the square root is not a bounded-complexity family

Let $\Delta(t) = t^2 - 34t + 1$. The factorization stated in Remark 10 and proved, at the level of factorization, by Caruso–Fürnsinn–Vargas-Montoya–Zudilin [7] is

$$H_p(t) = \Delta(t)^{\varepsilon_p} B_p(t)^2, \qquad \deg B_p = \frac{p - 1 - 2\varepsilon_p}{2}. \tag{135}$$

Thus $\deg B_p = \Theta(p)$. This yields a second, independent obstruction to B2 even if one temporarily replaces coefficient zeros by evaluation roots.

Proposition 124 (Fixed family versus moving Hasse section)**.** *The polynomials B_p cannot be the reductions of one fixed bounded-degree divisor on the t-line. If B_p is squarefree, then $V(B_p)_{\overline{\mathbb{F}}_p}$ has* $\deg B_p = \Theta(p)$ *geometric points. The equation $B_p(t) = 0$ is zero-dimensional and has no genus; if it is replaced by the hyperelliptic curve $y^2 = B_p(t)$, then*

$$g_p = \left\lfloor \frac{\deg B_p - 1}{2} \right\rfloor = \Theta(p).$$

Hence the standard degree/genus form of the Weil–Bombieri estimate supplies no constant uniform in p for either version.

Proof. A fixed relative divisor of degree D has fibers of degree D, whereas (135) has unbounded degree. The remaining assertions are the point count of a squarefree zero-dimensional divisor and the standard genus formula for a squarefree hyperelliptic equation. □

There is no contradiction between this proposition and the existence of a fixed Picard–Fuchs family. A fixed family can produce, separately in each characteristic p, a new Frobenius/Hasse section whose weight and degree grow with p. "The geometric family is fixed" does not imply "the derived Hasse polynomial has bounded complexity." This is precisely the moving-polynomial versus fixed-family quantifier distinction.

There are two source cautions. First, Theorem 1.2 of [7] states the square/quadratic-times-square factorization. Its proof uses the Franel relation

$$f_\alpha(t(x)) = (1 + x)h(x)^2, \qquad t = \frac{x(1 - 8x)}{1 + x}, \tag{136}$$

and the quadratic extension with discriminant Δ. It does not state that B_p is squarefree, and it does not formulate the theorem in terms of a Hasse invariant, a Picard group, or an elliptic pencil. The squarefree and coprimality assertions in Remark 10 therefore require a separate proof or reference; the finite verification below is not such a proof.

Second, the explicit pencil proposed in the oracle specification,

$$E_t : \quad y^2 = x(x - 1)\big(x - t(1 - t)\big), \tag{137}$$

is not identified with the Apéry truncation in the cited material. In the displayed coordinate its discriminant is

$$16t^2(1 - t)^2(1 - t + t^2)^2,$$

whose support differs from $\Delta(t) = 0$. More concretely, at $p = 7$ its raw Deuring polynomial is obtained by substituting $\lambda = t(1 - t)$ in $1 + 2\lambda + 2\lambda^2 + \lambda^3$; its unique $\mathbb{F}_7$ root is $t = 4$, while the Apéry polynomial above has root $t = 1$. This does not exclude an additional base change and gauge. It proves that (137), as printed, cannot simply be declared to produce B_p, and in any event evaluation Frobenius cannot produce the coefficient-first-zero ρ_p without the Mellin step (134).

12.20.3 The moving-target quantifier trap

Remark 16 identifies $b_m \equiv 0 \pmod p$ with collisions in the projective Apéry orbit and supplies reflection. For every fixed gap h, the corresponding gap polynomial $N_h(m) \in \mathbb{Z}[m]$ is fixed, so a prime-aspect Chebotarev theorem may be applied after h is fixed. The first actual gap, however, may range up to $O(p)$, no uniform $o(p)$ bound is known, and it varies with the same prime being counted. The available quantifiers are

$$\text{for every fixed } h, \quad X \longrightarrow \infty;$$

AMTD would require estimates uniform for moving $h \ll p$ inside one dyadic block. Degrees, splitting fields, discriminants, and effective Chebotarev constants all move with h. Interchanging these limits is exactly the forbidden moving-polynomial argument.

The center gives an even sharper warning. Beukers' congruence in Remark 54 says

$$b_{(p-1)/2} \equiv a_p(f) \pmod p, \qquad f = \eta(2z)^4\eta(4z)^4.$$

The index $(p-1)/2$ moves with p, and the density-zero conjecture for the nonordinary primes $p \mid a_p(f)$ is open for this non-CM weight-four form. Ordinary Sato–Tate controls fixed intervals for normalized traces; it does not prove this shrinking, same-characteristic divisibility event. It therefore cannot be invoked to prove B1 or B3.

Finally, equidistribution of ρ_p modulo each fixed integer q would still be only a marginal statement. It has neither the uniformity in growing frequencies needed for the minor arcs nor the two-prime joint information needed for collision hyperplanes. In particular it cannot, by itself, exclude the scale-dependent anchored star.

12.20.4 The correct fixed-family reduction

The minimum horizontal statement can be written without referring to a spurious root of B_p. For $N < p \le 2N$ and $m \in I_N = (N, 2N]$, put

$$X_p(m) = \mathbf{1}_{\{p \le m\}} \mathbf{1}_{\{b_{m-p} \equiv 0 \pmod p\}}, \qquad T_N = \sum_{N<p\le 2N} \sum_{m\in I_N} X_p(m).$$

Then the centered marked two-characteristic Hasse dispersion statement is

$$\left| \sum_{\substack{N<p,q\le 2N \\ p\ne q}} \sum_{m\in I_N} X_p(m)X_q(m) - \frac{T_N^2}{N} \right| \ll N^{o(1)} T_N. \tag{138}$$

Writing

$$F_N := \sum_{m\in I_N} L_N(m)(L_N(m)-1), \qquad L_N(m) = \sum_{N<p\le 2N} X_p(m),$$

for the ordered collision count, the left side of (138) is $|F_N - T_N^2/N|$. The exact identity

$$\sum_{m\in I_N} \left(L_N(m) - \frac{T_N}{N} \right)^2 = T_N + F_N - \frac{T_N^2}{N}$$

shows directly that (138) gives the top-block AMTD bound. It also kills a full top-block anchored star (for example the anchor $m_0 = 2N$), for which $L_N(m_0)$ has order $N/\log N$, and it is the top-block form of the centered second-factorial-moment input used by AMTD. The corresponding statement for all $P > \sqrt{N}$, with the centering of Hypothesis 108, is the precise input needed by the analytic oracle.

A genuine fixed-family route would have to construct, from one integral transcendental crystal (morally the symmetric square of the underlying rank-two Picard–Fuchs connection), the Kummer-twisted objects

$$\mathcal{M}_{p,j} = R\Gamma_{c,\mathrm{rig}}\big(U_p, \mathcal{V}_{\mathrm{tr},p} \otimes \mathcal{K}_{\omega_p^{-j}}\big) \tag{139}$$

and a marked Hasse coordinate $c_{p,j}$ satisfying

$$c_{p,j} = 0 \quad \Longleftrightarrow \quad b_j \equiv 0 \pmod p.$$

It would then have to prove (138) for these marked zeros, uniformly in the two characteristics and in the moving characters j. This is a $1/p$-scale crystalline anti-concentration and horizontal large sieve statement, not ordinary Frobenius equidistribution. Fixed rank/conductor of an ℓ-adic sheaf does not supply it: the Jacobi-sum counterexample discussed with the Sym^2 remarks has fixed complexity but linearly many bad tame characters.

Thus a construction of (139), its marked coordinate, and the centered two-characteristic estimate (138) would be a quantifier-correct reduction of AMTD to a fixed arithmetic family. This is a specification of the missing package, not an achieved G3 reduction to an existing Frobenius/Sato–Tate theorem: neither the marked coordinate with all its integral control nor its horizontal dispersion theorem is presently available.

12.20.5 Finite verification

The reproducible computation in `oracleB_explore.py` recomputes all $\mathcal{Z}_p$ for the 2260 primes $5 \le p \le 20000$ and compares them entrywise with the independent binary zero-set bank. The exact histogram is

$Z(p)$	0	1	2	4	6	8
$\#p$	1356	2	695	176	27	4.

Among the 695 doublets, the KS distance of $2\rho_p/(p-1)$ from the reflection-constrained uniform limit is 0.03894. Small-modulus permutation tests for the outer gap $h_p \bmod q$ against $p \bmod q$, using all 904 active primes and $q = 3, 5, 7, 11, 13$, have p-values between 0.257 and 0.675.

An exact Sage/FLINT worker verified (135), the predicted degree, squarefreeness, and coprimality to Δ at every one of the 2260 primes. It also verified $B_p \mid t^{p^2} - t$ and used $\deg \gcd(B_p, t^p - t)$ to extract the exact linear/quadratic factorization type at every prime in the range. Complete factorization into explicit factor polynomials was restricted to a sparse cross-check sample. At $p = 101, 503, 1009, 5003, 10007, 19997$, the degrees of B_p are respectively $50, 250, 504, 2501, 5002, 9998$, and the numbers of irreducible factors are $30, 148, 262, 1284, 2595, 5058$. These calculations demonstrate growing complexity and catch transcription errors; they prove no assertion beyond the finite range. Exact identity checks used in the propositions above are repeated independently by `oracleB_verify.py`.

STALL REPORT. The actual cross-prime decorrelation of the Apéry coefficient zero-sets, and hence G1/AMTD, is not proved. G3 is not reached for the reason stated above; literal G2 is proved but has insufficient uniformity; the obstruction and exact missing input constitute G4. The rigorous outputs are:

1. Proposition 121, which proves the literal G2 count is $O_{m_0}(1)$ for every fixed anchor;
2. Remark 122, which proves that this has the wrong quantifier order for the moving anchored-star family;
3. Proposition 123 and the exact $p = 7$ counterexample, which show that B1/B2 as stated confuse Mellin coefficient zeros with evaluation roots;
4. Proposition 124, which isolates the linear-degree obstruction even on the evaluation-root side;

5. the fixed-gap/fixed-prime-family audit, including the open nonordinary center slice from Remark 54;
6. the quantifier-correct remaining arithmetic input (138), together with the crystalline Mellin package (139) that a fixed-family proof would have to supply.

12.21 A fixed toric marked coordinate

Put

$$\Lambda(x, y, z) = \frac{(1 + x)(1 + y)(1 + z)\big((1 + y)(1 + z) + xyz\big)}{xyz} \in \mathbb{Z}[x^{\pm 1}, y^{\pm 1}, z^{\pm 1}]. \tag{140}$$

This one Laurent polynomial gives the required marked scalar. In particular, it avoids both loci ruled out by Propositions 123 and 124: it is neither an evaluation of H_p nor an evaluation of its factor B_p.

Lemma 125 (Fixed constant-term realization)**.** *For every integer $n \geq 0$,*

$$\mathrm{CT}_{x,y,z}\, \Lambda(x, y, z)^n = \sum_{r=0}^{n} \binom{n}{r}^2 \binom{n+r}{r}^2 = b_n. \tag{141}$$

Moreover, every exponent of each of x, y, z occurring in Λ lies between -1 and 1.

Proof. Write

$$A = (1 + x)(1 + y)(1 + z), \qquad E = (1 + y)(1 + z) + xyz.$$

Then the constant term in question is

$$[x^n y^n z^n] A^n E^n.$$

If exactly k of the n factors selected in E^n contribute xyz, the x-coefficient contributes a second factor $\binom{n}{k}$, while the y- and z-coefficients both contribute $\binom{2n-k}{n-k}$. Hence

$$\mathrm{CT}\,\Lambda^n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{2n-k}{n-k}^2.$$

The substitution $r = n-k$ gives (141). The exponent assertion follows directly from the numerator of (140), whose degree in each variable is between 0 and 2, followed by division by xyz. ☐

Theorem 126 (The marked coordinate)**.** *Let $p \geq 5$ be prime. For $0 \leq j \leq p-2$, define the scalar*

$$c^{\mathrm{tor}}_{p,j} := -\sum_{(x,y,z)\in(\mathbb{F}_p^\times)^3} \Lambda(x, y, z)^j \in \mathbb{F}_p, \tag{142}$$

where the zeroth power is the constant Laurent polynomial 1, including at the zeros of Λ. Define separately

$$c^{\mathrm{tor}}_{p,p-1} := 1.$$

Then, for every $0 \leq j \leq p-1$,

$$c^{\mathrm{tor}}_{p,j} = b_j \pmod{p}. \tag{143}$$

Consequently

$$c^{\mathrm{tor}}_{p,j} = 0 \iff b_j \equiv 0 \pmod{p}. \tag{144}$$

Proof. For an integer e,

$$\sum_{a\in\mathbb{F}_p^\times} a^e = \begin{cases} -1, & p-1 \mid e, \\ 0, & p-1 \nmid e, \end{cases} \qquad \text{in } \mathbb{F}_p.$$

By Lemma 125, every exponent in Λ^j belongs to $[-j,j]$ in each coordinate. If $0 \le j \le p-2$, the only multiple of $p-1$ in this interval is zero. Three applications of torus orthogonality therefore give

$$\sum_{(\mathbb{F}_p^\times)^3} \Lambda^j = (-1)^3\,\mathrm{CT}\,\Lambda^j = -b_j,$$

which proves (143) on this range. At the remaining endpoint, the reflection in Remark 16 gives $b_{p-1} = b_0 = 1$ modulo p, exactly the separately defined value. □

Remark 127 (The endpoint is not a cosmetic convention). The raw power sum at $j = p-1$ cannot replace the separate definition. If

$$T = (\mathbb{G}_m)^3, \qquad U = T \setminus V(\Lambda),$$

then direct inclusion–exclusion gives

$$\#U(\mathbb{F}_p) = p^3 - 7p^2 + 17p - 14. \tag{145}$$

Indeed, the union of $x = -1$, $y = -1$, and $z = -1$ has $(p-1)^3 - (p-2)^3$ points. Off $y = -1$ and $z = -1$, the fourth zero equation determines a unique nonzero x and has $(p-2)^2$ points; its intersection with the first union has $p-2$ points, given by $x = -1$ and $1 + y + z = 0$. Subtracting this zero locus from $(p-1)^3$ gives (145). Thus $-\sum_{T(\mathbb{F}_p)} \Lambda^{p-1} = -\#U(\mathbb{F}_p) \equiv 14 \pmod{p}$, not $b_{p-1} = 1$ in general. At $p = 7$ the raw endpoint is even zero. For $j = 0$, by contrast, the convention in (142) gives $-(p-1)^3 = 1$ in $\mathbb{F}_p$.

12.21.1 Uniform geometric complexity

For $1 \le j \le p-2$, let $E_p = \mathbb{Q}(\mu_{p-1})$. Choose a prime $\mathfrak{p} \mid p$ and a Teichmüller character $\omega_p \colon \mathbb{F}_p^\times \to \mu_{p-1}$ such that $\omega_p(a) \equiv a \pmod{\mathfrak{p}}$, and put $\chi_{p,j} = \omega_p^j$. Fix $\ell = 2$ (recall that $p \ge 5$), choose an embedding $E_p \hookrightarrow \overline{\mathbb{Q}}_2$, and let $\mathcal{K}_{\chi_{p,j}}$ be the associated rank-one Kummer sheaf, with the Frobenius convention chosen to have trace $\chi_{p,j}$. On the fixed threefold U put

$$\mathcal{L}_{p,j} = \Lambda^* \mathcal{K}_{\chi_{p,j}}. \tag{146}$$

The Grothendieck trace formula gives the cyclotomic integer

$$\widetilde{c}^{\,\mathrm{tor}}_{p,j} = -\operatorname{Tr}_{\mathrm{alt}}\Big(\mathrm{Frob}_p \mid R\Gamma_c(U_{\overline{\mathbb{F}}_p}, \mathcal{L}_{p,j})\Big) = -\sum_{u\in U(\mathbb{F}_p)} \chi_{p,j}(\Lambda(u)). \tag{147}$$

Its reduction at the chosen prime above p is $c^{\mathrm{tor}}_{p,j} = b_j$. Both endpoints may instead be represented by the constant sheaf on T: its negative Frobenius supertrace is $-(p-1)^3$, whose reduction is 1. Thus their cohomological complexity is fixed as well. Using the trivial Kummer sheaf extended by zero from U would give the wrong value in Remark 127.

Proposition 128 (Uniform complexity of the toric object)**.** *The ambient variety and rational map in* (146) *are independent of p and j. The coefficient sheaf has rank one. There is an absolute constant B_Λ such that*

$$\sum_i \dim H_c^i\big(U_{\overline{\mathbb{F}}_p}, \mathcal{L}_{p,j}\big) \le B_\Lambda \tag{148}$$

for all $p \ge 5$ and $1 \le j \le p-2$. In particular, both the total degree of the corresponding cohomological L-factor and the ramification complexity are $O(1)$ uniformly in (p,j). This is geometric complexity: as for the Kummer twists in (139)*, it does not count the degree of the varying cyclotomic coefficient field E_p.*

Proof. In $(\mathbb{P}^1)^3$, the principal divisor of Λ has the four simple zero components

$$x = -1, \qquad y = -1, \qquad z = -1, \qquad \overline{Q}:\ X_0(Y_0 + Y_1)(Z_0 + Z_1) + X_1Y_1Z_1 = 0$$

and the six simple pole components

$$x = 0,\ x = \infty,\ y = 0,\ y = \infty,\ z = 0,\ z = \infty.$$

The fourth zero component is smooth and geometrically irreducible. Indeed, its two coefficients as a linear form in (X_0, X_1) are coprime, and a direct check of the six displayed partial derivatives has no projective common zero. On the torus, smoothness is already immediate from $\partial((1+y)(1+z) + xyz)/\partial x = yz$. Thus the input divisor has ten fixed components of fixed multidegrees. The full divisor is not asserted to be simple normal crossings.

More explicitly, the change of coordinates

$$u = \frac{1+y}{y}, \qquad v = \frac{1+z}{z}, \qquad w = -\frac{x}{uv}$$

identifies U with the fixed toric-arrangement complement

$$(\mathbb{G}_m)^3 \setminus \{u = 1,\ v = 1,\ w = 1,\ uvw = 1\}.$$

In particular, a finite stratification depends only on this four-member arrangement, not on the character.

Every $\chi_{p,j}$ has order dividing $p-1$. Its pullback is therefore tame along every divisorial valuation and has Swan conductor zero. Take one log resolution of this fixed integral divisor and spread it out away from finitely many characteristics. The resulting stratification has a fixed finite number of strata. At every good prime, tame specialization for this log-smooth pair applies because the local monodromy has order prime to p; it compares the relevant rank-one representation with one on the characteristic-zero fiber. Fix a finite CW model of that complex fiber. Its cellular cochain groups have dimension at most the number of cells for every rank-one representation, independently of its monodromy, and hence give (148). The finitely many excluded characteristics contribute only finitely many additional (p,j) and may be absorbed into B_Λ. This proves a uniform bound without claiming that the original ten-component divisor is normal crossings or that the cohomology is concentrated in one degree. ∎

Remark 129 (What vanishes). Equation (144) concerns the reduction of the cyclotomic trace (147) modulo a prime above p. It does not assert that the characteristic-zero Frobenius trace itself vanishes. For example, at $(p,j) = (11,5)$ the quadratic character sum gives $\widetilde{c}^{\,\mathrm{tor}}_{11,5} = -33 \neq 0$, while its reduction and b_5 both vanish modulo 11. This integral distinction is essential in any later attempt to prove (138).

12.21.2 Audits of the two originally suggested coordinates

The toric construction above is not a disguised bounded-length transfer word. With $B_n = (n!)^3 b_n$ and the matrix $M(n)$ from Remark 16, put $T_0 = I$ and $T_j = M(j-1)\cdots M(0)$. The recurrence gives the exact identity

$$T_j = \begin{pmatrix} B_j & 0 \\ B_{j-1} & 0 \end{pmatrix} \qquad (j \ge 1). \tag{149}$$

Thus the marked entry is $(1,1)$, not $(1,2)$, and it detects $b_j = 0$ for $j < p$. Its matrix size is fixed, but its word length is j. Exact symbolic multiplication of $R_\ell(x) = M(x+\ell-1)\cdots M(x)$ gives, for $\ell \ge 2$,

$$\deg R_\ell = \begin{pmatrix} 3\ell & 3\ell+3 \\ 3\ell-3 & 3\ell \end{pmatrix}.$$

For completeness, if $A_\ell = (R_\ell)_{11}$ and λ_ℓ is its leading coefficient, then

$$A_{\ell+1} = P(x+\ell)A_\ell - (x+\ell)^6 A_{\ell-1}, \qquad \lambda_{\ell+1} = 34\lambda_\ell - \lambda_{\ell-1}, \quad (\lambda_0, \lambda_1) = (1, 34).$$

The λ_ℓ are positive and strictly increasing, so the two leading terms never cancel. The remaining entries satisfy

$$(R_\ell)_{21} = A_{\ell-1}(x), \qquad (R_\ell)_{12} = -x^6 A_{\ell-1}(x+1), \qquad (R_\ell)_{22} = -x^6 A_{\ell-2}(x+1),$$

for $\ell \ge 2$. This proves the displayed degree matrix rather than extrapolating it from finitely many products. Likewise, the standard factorial gauge has marked coefficient

$$\frac{N_h(m)}{\prod_{s=2}^{h}(m+s)^3}.$$

The endpoint identity used in Remark 12,

$$N_h(-s) = (-1)^{s-1} B_{s-1} B_{h-s} \ne 0 \qquad (1 \le s \le h),$$

shows that it has exactly $h-1$ uncancelled finite poles, each of order three. These facts rule out the direct word and this direct gauge; they are not a no-go theorem for all constructions, as Theorem 126 demonstrates.

The raw Mellin coordinate from Proposition 123 has a different restricted obstruction. If g is a primitive root, its corrected group-algebra element is

$$Q_p(X) = -\sum_{r=0}^{p-2} H_p(g^r) X^{-r} \quad \text{in } \mathbb{F}_p[X^{\pm 1}]/(X^{p-1}-1). \tag{150}$$

For $1 \le j \le p-2$, $Q_p(g^j) = b_j$. The factorization in Remark 10 implies, without any squarefreeness or coprimality assumption,

$$\#\{t \in \mathbb{F}_p^\times : H_p(t) = 0\} \le \deg B_p + 2\varepsilon_p = \frac{p-1}{2} + \varepsilon_p,$$

and hence

$$\#\operatorname{supp} Q_p = \#\{t \in \mathbb{F}_p^\times : H_p(t) \ne 0\} \ge \frac{p-1}{2} - \varepsilon_p \ge \frac{p-3}{2}.$$

Hence the raw periodic coordinate has growing group-algebra support and growing minimal cyclic constant-coefficient recurrence order: the distinct characters in (150) are linearly independent because the $(p-1)$-point Fourier/Vandermonde matrix is invertible in $\mathbb{F}_p$. This is a no-go only for exact raw-value realizations of that form, not for the zero-equivalent toric coordinate.

Finally, bounded complexity of a Kummer factor on the original t-line does not by itself repair (139). At $p = 31$, the two roots of the conifold discriminant are 14 and 20, with $H_{31}(14) = H_{31}(20) = 7$ and $b_8 = b_{22} = 0$. For $j = 8$ the omitted bad-fiber contribution is

$$7(14^{-8} + 20^{-8}) = 11 \pmod{31},$$

so the Mellin sum restricted to the smooth locus is 20, not zero; the same failure occurs for $j = 22$. Boundary terms require integral control.

12.21.3 Quantifier boundary

The family (146) is a uniformly bounded family of cohomological spaces indexed by the tame character j. It is not a single sheaf on a "j-line" whose trace function is $j \mapsto b_j$. Therefore a number of j-singularities or a conductor of such a j-line sheaf is not defined. In particular, Proposition 128 does not turn the moving-character problem into an ordinary FKM bilinear estimate. It is also an alternative toric Kummer realization of the marked scalar, not a construction of, or a comparison with, the transcendental crystal $\mathcal{V}_{\mathrm{tr},p}$ in (139). Thus it supplies exactly the algebraic coordinate requested here, not the rest of that crystalline Mellin package.

STALL REPORT. The literal algebraic goal G1 is achieved by (142): the marked scalar is exactly b_j modulo p, and its ambient dimension, coefficient rank, divisor data, Swan conductors, and total cohomological dimension are uniformly bounded in (p, j). G2 is false in its universal form and is not claimed; the transfer and raw-Mellin statements above are restricted no-go results. G3 is deliberately not attempted. No estimate for (SDC), no short-arc bound, no two-characteristic dispersion estimate (138), and no reduction of any of them to an existing theorem is proved here. The remaining obstruction is precisely horizontal control of the moving characters and their reductions, not the existence of an algebraic marked coordinate.

12.22 Jacobi–Stickelberger analysis of the marked coordinate

We use the fixed Laurent polynomial (140) and the marked-coordinate identity of Theorem 126. Put

$$h = p - 1, \qquad E_p = \mathbb{Q}(\mu_h), \qquad \chi = \omega^j,$$

where $\omega\colon \mathbb{F}_p^\times \to \mu_h$ is the Teichmüller character associated with the chosen prime $\mathfrak{p} \mid p$ of E_p. Every multiplicative character, including the trivial character ε, is extended by zero at 0. We write

$$J(A, B) = \sum_{t \in \mathbb{F}_p} A(t) B(1-t).$$

All character exponents below are read modulo h.

12.22.1 A1: an exact four-Jacobi skeleton

Theorem 130 (Jacobi skeleton)**.** *For $1 \le j \le p-2$, the cyclotomic lift in* (147) *has the exact expression*

$$\widetilde{c}^{\mathrm{tor}}_{p,j} = -\frac{1}{h} \sum_{k=0}^{h-1} J(\omega^{-k}, \omega^j)^2 J(\omega^{-k}, \omega^{j+k})^2. \tag{151}$$

Thus the skeleton has one character variable and four Jacobi factors, all with arguments affine in (j, k).

At $j = 0$, let the right side of (151), with $\chi = \varepsilon$, be denoted by $\mathscr{S}_{p,0}$. The exact endpoint formula is

$$\widetilde{c}^{\,\mathrm{tor}}_{p,0} = \mathscr{S}_{p,0} - 4p^2 + 14p - 13 = -(p-1)^3. \tag{152}$$

In particular, reduction modulo $\mathfrak{p}$ gives b_j for every $0 \le j \le p-2$.

Proof. For a character A and $t \in \mathbb{F}_p^\times$, finite Mellin inversion gives

$$A(1+t) = \frac{1}{h} \sum_{\rho} \rho(-1) J(\overline{\rho}, A) \rho(t), \tag{153}$$

where ρ runs over all multiplicative characters of $\mathbb{F}_p^\times$. Indeed, the Fourier coefficient at ρ is

$$\sum_{t \neq 0} A(1+t)\overline{\rho}(t) = \rho(-1) J(\overline{\rho}, A).$$

For $x, y, z \neq 0$, put

$$s = \frac{(1+y)(1+z)}{xyz}.$$

The marked coordinate factors as

$$\Lambda(x, y, z) = (1+x)(1+y)(1+z)(1+s). \tag{154}$$

If $j > 0$, the summand is already zero when $y = -1$ or $z = -1$. We may therefore apply (153) to s and then restore those zero-contribution values. With $\rho = \omega^k$, the three variable sums separate:

$$\begin{aligned} \sum_{x \neq 0} \chi(1+x)\overline{\rho}(x) &= \rho(-1) J(\overline{\rho}, \chi), \\ \sum_{y \neq 0} (\chi\rho)(1+y)\overline{\rho}(y) &= \rho(-1) J(\overline{\rho}, \chi\rho), \\ \sum_{z \neq 0} (\chi\rho)(1+z)\overline{\rho}(z) &= \rho(-1) J(\overline{\rho}, \chi\rho). \end{aligned}$$

Together with the factor $\rho(-1)$ in (153), the four signs multiply to 1. The outer minus sign in the definition of $\widetilde{c}^{\,\mathrm{tor}}_{p,j}$ now gives (151).

This argument includes all degenerate values of k: the identities

$$J(\varepsilon, A) = J(A, \varepsilon) = -1, \quad J(A, A^{-1}) = -A(-1), \quad J(\varepsilon, \varepsilon) = p - 2$$

are built into the unnormalized Jacobi sum. In particular, no Gauss-sum quotient is being used illegally when a product character is trivial.

At $j = 0$, zero-extension of ε computes the open-set sum $\mathscr{S}_{p,0} = -\#U(\mathbb{F}_p)$ rather than the prescribed raw zeroth power. Equation (145) gives

$$\#U(\mathbb{F}_p) - (p-1)^3 = -4p^2 + 14p - 13,$$

which proves (152). This is the necessary $0^0 = 1$ correction; it is not a choice of character convention. $\square$

For completeness, the reduction of every Jacobi factor has an $O(1)$ formula. If $a, b \in \{0, \ldots, h-1\}$ and $\overline{J}_p(a,b)$ denotes the reduction of $J(\omega^a, \omega^b)$ modulo $\mathfrak{p}$, then

$$\overline{J}_p(a,b) = \begin{cases} -2, & a = b = 0, \\ -1, & \text{exactly one of } a, b \text{ is zero}, \\ 0, & a, b > 0 \text{ and } h - a > b, \\ -(-1)^{h-a}\binom{b}{h-a}, & a, b > 0 \text{ and } h - a \leq b. \end{cases} \tag{155}$$

This follows by expanding $(1-t)^b$: among the powers t^{a+r}, only $a + r = h$ can survive torus orthogonality. Since $-h^{-1} \equiv 1 \pmod{p}$, equations (151) and (155) are an entirely explicit finite-field formula for b_j.

For $1 \leq j \leq p - 2$, if

$$m = \min(j, h - j),$$

then only $0 \leq k \leq m$ survive modulo $\mathfrak{p}$, and on that range

$$\begin{aligned} \overline{J}_p(-k, j) &= (-1)^{k+1}\binom{j}{k}, \\ \overline{J}_p(-k, j+k) &= (-1)^{k+1}\binom{j+k}{k}. \end{aligned} \tag{156}$$

Consequently the surviving k-th skeleton term is exactly

$$\binom{j}{k}^2\binom{j+k}{k}^2. \tag{157}$$

When $j > h/2$, the omitted terms of the usual Apéry sum have $j + k \geq p$ and hence $p \mid \binom{j+k}{k}$. This also proves directly, without numerical input, that the reduction of the skeleton is b_j.

The script `lemmaA_verify.py` independently compares the skeleton with the Apéry recurrence for every prime $5 \leq p \leq 200$ and every $0 \leq j \leq p - 2$ (4178 pairs). It also compares both with the direct three-dimensional torus sum through $p = 31$. At $j = h/2$, it expands

$$\eta(2z)^4\eta(4z)^4 = q\prod_{n\geq 1}(1-q^{2n})^4(1-q^{4n})^4 = \sum_{r\geq 1} a_r q^r$$

and verifies

$$\widetilde{c}^{\,\mathrm{tor}}_{p,h/2} \equiv b_{h/2} \equiv a_p \pmod{p} \tag{158}$$

for every prime $5 \leq p \leq 200$, reproducing the modular-form anchor in Remark 54.

12.22.2 A2: the explicit Stickelberger container

Choose a prime $\mathfrak{P} \mid \mathfrak{p}$ of $E_p(\mu_p)$ and normalize $v_{\mathfrak{P}}(p) = 1$. With

$$g(A) = \sum_{t\in\mathbb{F}_p} A(t)\zeta_p^t,$$

Gross–Koblitz computes this valuation; its restriction to E_p is the $\mathfrak{p}$-adic valuation used below. For an exponent a, write

$$d_h(a) = \frac{[-a]_h}{h} = \left\langle -\frac{a}{h} \right\rangle,$$

where $[-a]_h \in \{0, \ldots, h-1\}$ and the fractional part of an integer is 0. Gross–Koblitz gives $v_{\mathfrak{P}}(g(\omega^a)) = d_h(a)$. Including the inverse-character degeneracy, the resulting formula valid for every a, b is

$$\nu_h(a,b) := v_{\mathfrak{p}}\Big(J(\omega^a, \omega^b)\Big) = d_h(a) + d_h(b) - d_h(a+b) - \mathbf{1}_{\substack{a\not\equiv 0,\ b\not\equiv 0\ (h) \\ a+b\equiv 0\ (h)}}. \tag{159}$$

Indeed, away from a trivial product character this is the valuation of $g(\omega^a)g(\omega^b)/g(\omega^{a+b})$; when the two nontrivial characters are inverse, $J(A, A^{-1}) = -A(-1)$ is a unit and supplies the last correction. Formula (159) also gives zero in the cases involving a trivial factor. Its values are always 0 or 1.

For the remainder of A2 assume $1 \le j \le p-2$, and let

$$T_{j,k} = J(\omega^{-k}, \omega^j)^2 J(\omega^{-k}, \omega^{j+k})^2.$$

Equations (151) and (159) give the requested digit/fractional-part valuation

$$\begin{aligned} v(k;j) &= 2\nu_h(-k,j) + 2\nu_h(-k, j+k) \\ &= 2\mathbf{1}_{\{k>j\}} + 2\mathbf{1}_{\{k>h-j\}}, \qquad 0 \le k \le h-1. \end{aligned} \tag{160}$$

Thus its minimum is zero and

$$\{k : v(k;j) = 0\} = \{0, 1, \ldots, \min(j, h-j)\}, \qquad U_p(j) = 1 + \min(j, h-j). \tag{161}$$

This is computable in $O(1)$ operations per (p, j).

Theorem 131 (Unique-unit container)**.** *For $1 \le j \le p-2$, if $U_p(j) = 1$, then $b_j \not\equiv 0 \pmod p$. Hence*

$$\mathcal{Z}_p \subseteq \{1 \le j \le p-2 : U_p(j) \ne 1\}. \tag{162}$$

The endpoints $j = 0, p-1$ are absent from $\mathcal{Z}_p$ because their marked values are 1.

Proof. Jacobi sums are algebraic integers, $-1/h$ is a unit at $\mathfrak{p}$, and the minimum in (160) is 0. If exactly one $T_{j,k}$ has valuation zero, reduction of (151) kills every other term and leaves one nonzero residue. Therefore the reduction of $\widetilde{c}^{\mathrm{tor}}_{p,j}$, which is b_j by (143), cannot vanish. $\square$

There are two normalization points in this argument. First, the Gauss sums live in $E_p(\mu_p)$, but the Jacobi sums lie in E_p; because $p \equiv 1 \pmod h$, the residue field at $\mathfrak{p}$ is $\mathbb{F}_p$. Second, if the valuation is instead normalized by $v(1-\zeta_p) = 1$, every value in (159)–(160) must be multiplied by h. No Greene $p^{-1}J$ normalization is used here.

The explicit answer (161) now exposes the decision gate:

$$U_p(j) = 1 + \min(j, p-1-j) \ge 2 \qquad (1 \le j \le p-2). \tag{163}$$

Thus the implication in Theorem 131 is correct but never triggers on a nontrivial character. Moreover, (157) shows that the simultaneous unit terms reconstruct the original Apéry sum modulo p: cancellation among unit terms is the original vanishing problem, not a lower-complexity replacement for it.

It is important here that the h Jacobi summands are a Mellin expansion of the trace, not the at most B_Λ Frobenius eigenvalues in Proposition 128. Their unit count depends on the chosen integral character expansion and is not a crystalline slope multiplicity. The fixed-dimensional cohomological trace occurs only after the length-h Mellin sum has cancelled. Thus (163) does not contradict the uniform Betti bound; it identifies the precise point at which this Jacobi skeleton fails to produce a unique unit summand and hence gives no unique-unit-root certificate.

12.22.3 A3: empirical computation and verdict

The zero sets were computed from the Apéry recurrence, independently of the skeleton. A zero at $j = h/2$ was classified first as *non-ordinary*; every other zero was classified as a collision of at least two unit skeleton terms. The results are:

bound	primes	$\sum_p Z(p)$	zero-free	$U = 0$	$U \geq 2$ collision	non-ordinary
200	44	41	25	0	40	1
500	93	95	53	0	94	1
1000	166	165	96	0	164	1
2000	301	283	184	0	282	1
5000	667	608	418	0	606	2

For $p \leq 2000$, the distribution of $Z(p)$ is

$$\#\{Z = 0, 1, 2, 4, 6\} = \{184, 1, 95, 17, 4\}.$$

For the stretch range $p \leq 5000$, it is

$$\#\{Z = 0, 1, 2, 4, 6, 8\} = \{418, 2, 199, 41, 6, 1\}.$$

The non-ordinary points in the stretch range are

$$(p, j) = (11, 5),\ (3137, 1568); \qquad a_{11} = -44, \quad a_{3137} = 207042 = 66 \cdot 3137.$$

The container size is not merely empirical: by (163), for every $p \geq 5$,

$$\#\{1 \leq j \leq p-2 : U_p(j) \neq 1\} = p - 2. \tag{164}$$

Therefore the decision-gate verdict is **S2**. Generic j has multiple unit Jacobi terms (in fact every nontrivial j does), and all observed zeros are either non-ordinary midpoint zeros or cancellations among these unit terms. This presentation of the crystalline/Jacobi route alone gives no estimate $|\mathcal{Z}_p| \ll p^\theta$ with $\theta < 1$. The unconditional $Z(p) \ll p^{2/3}$ bound in Proposition 9 remains available, but it comes from the gap-polynomial argument, not from (162).

Finally, the Lemma–0 collapse is fully compatible with this negative gate. Corollary 62 gives, for $m \in (N, 2N]$ and $N < p \leq m$,

$$b_m \equiv 5b_{m-p} \pmod{p}, \qquad L_N(m) = \#\{p \in (N, m] : p \mid b_m\}.$$

Since (164) excludes no nontrivial affine target $j = m - p$, it supplies neither the pointwise anti-concentration required after the collapse nor the centered estimate (138). The exact skeleton and its Stickelberger valuation are therefore complete, but their decision-gate output is the valid S2 obstruction specified in the task.

12.23 The eigenvalue Newton polygon of the marked coordinate

We retain the marked Laurent polynomial of Theorem 126 and the exact Jacobi skeleton of Theorem 130. For $q = p^r$ and $1 \leq j \leq p-2$, put

$$A = \chi_{p,j} \circ N_{\mathbb{F}_q/\mathbb{F}_p}, \qquad S^{(r)}_{p,j} = -\sum_{x \in U(\mathbb{F}_q)} A(\Lambda(x)).$$

For an auxiliary prime $\ell \neq p$, write E_ℓ for a coefficient field containing the values of A. Thus $S^{(1)}_{p,j} = \widetilde{c}^{\,\mathrm{tor}}_{p,j}$ and $S^{(1)}_{p,j} \equiv b_j \pmod{\mathfrak{p}}$. We use the effective sign convention in which $S^{(r)}_{p,j}$ is a Frobenius power trace. The signed trace on $R\Gamma_c(U, \mathcal{L}_{p,j})$ is its negative. This sign will be restored in the slope-zero inclusion below.

12.23.1 The corrected Adolphson–Sperber object

The coordinate change already used in the proof of Proposition 128,

$$u = \frac{1+y}{y}, \qquad v = \frac{1+z}{z}, \qquad w = -\frac{x}{uv},$$

gives

$$\Lambda = \frac{uv(w-1)(1-uvw)}{w(u-1)(v-1)}, \qquad U = (\mathbb{G}_m)^3 \setminus \{u=1, v=1, w=1, uvw=1\}. \tag{165}$$

There is an important trap here. Converting the Kummer sum to an additive sum with the phase $t\Lambda$ does *not* permit direct application of the nondegenerate Newton-polytope theorem. At $x = y = -1$ two factors of Λ vanish, so $t\Lambda$ and all four toric logarithmic derivatives vanish simultaneously. In fact, if $P = \mathrm{Newt}(\Lambda)$, then

$$P = \{(X, Y, Z) : -1 \le X, Y, Z \le 1,\ X - Y \le 1,\ X - Z \le 1\}.$$

Its cross-sectional area is 4 for $-1 \le X \le 0$ and $(2-X)^2$ for $0 \le X \le 1$, whence

$$3!\,\mathrm{vol}(P) = 6\left(4 + \int_0^1 (2-X)^2\,dX\right) = 38. \tag{166}$$

The same number is the normalized volume of $\mathrm{conv}(0, P \times \{1\})$. A naive use of this polytope would therefore manufacture 38 slots. It is incompatible with the cancelled rank below (and with $\chi_c(U) = 1 - 7 + 17 - 14 = -3$), precisely because the phase is degenerate.

The correct Adolphson–Sperber/GKZ object follows by first performing the Jacobi convolution. Base change of Theorem 130 gives the exact identity

$$S_{p,j}^{(r)} = -\frac{1}{q-1}\sum_{\rho} J_q(\bar\rho, A)^2 J_q(\bar\rho, A\rho)^2, \tag{167}$$

where *all* $q-1$ characters of $\mathbb{F}_q^\times$ occur. With $g_q(C) = \sum_t C(t)\psi(t)$ and $g_q(\varepsilon) = -1$, including the inverse-character degeneracies, one has

$$J_q(\bar\rho, A)^2 J_q(\bar\rho, A\rho)^2 = q^{-2} g_q(\bar\rho)^4 g_q(A\rho)^2 g_q(A^{-1}\rho)^2. \tag{168}$$

For example, when $\rho = A$ the Jacobi factor $J_q(A^{-1}, A)^2$ and the product $g_q(A^{-1})^2 g_q(A)^2 = q^2$ give the same formula; no exceptional summand has been dropped. To fix all signs, inversions, and Tate normalizations, define

$$\mathcal{H}_{q,A}(t) := -\frac{1}{q^2(q-1)}\sum_{\rho} g_q(\bar\rho)^4 g_q(A\rho)^2 g_q(A^{-1}\rho)^2 \rho(t).$$

This is our trace convention for the Tate-normalized balanced hypergeometric object

$$\mathrm{Hyp}(\varepsilon, \varepsilon, \varepsilon, \varepsilon; A, A, A^{-1}, A^{-1}). \tag{169}$$

Equations (167) and (168) say exactly that $\mathcal{H}_{q,A}(1) = S_{p,j}^{(r)}$. The opposite Katz ordering exchanges the two parameter lists and replaces t by t^{-1}, hence gives the same stalk at $t = 1$. Introduce the usual hypergeometric parameter t. Expanding the eight Gauss factors gives the associated GKZ circuit, which is nondegenerate for generic $t \neq 1$; this, rather than the degenerate phase $t\Lambda$, is the object to which the twisted-height recipe of Adolphson–Sperber [2] applies. The value $t = 1$ is the unique balanced-circuit singularity and must subsequently be treated by middle extension and nearby cycles.

Here is the requested closed form. Put

$$a = \left\langle \frac{j}{p-1} \right\rangle \in (0,1), \qquad \boldsymbol{\alpha} = (0,0,0,0),$$

and sort the fractional-part data

$$\boldsymbol{\beta}(a) = \operatorname{sort}\big(\langle a\rangle, \langle a\rangle, \langle -a\rangle, \langle -a\rangle\big) = \operatorname{sort}(a, a, 1-a, 1-a). \tag{170}$$

For $k = 1, \ldots, 4$, the twisted lattice-height statistic is

$$\rho_a(k) = \#\{i : \alpha_i < \beta_k(a)\} - k = 4 - k. \tag{171}$$

Indeed, every entry of (170) is strictly positive, including at $a = 1/2$. Thus the generic circuit has Hodge polynomial

$$H_a^{\mathrm{gen}}(T) = \sum_{k=1}^{4} T^{\rho_a(k)} = 1 + T + T^2 + T^3. \tag{172}$$

The two character multisets in (169) are disjoint, so the generic hypergeometric object is irreducible of rank four. The balanced hypergeometric local-monodromy theorem (in the convolution normalization of Katz [15]) says that the monodromy at $t = 1$ is a nontrivial pseudoreflection. Its special eigenvalue is

$$\exp\left(2\pi i\left(\sum_k \beta_k - \sum_i \alpha_i\right)\right) = \exp(4\pi i) = 1.$$

Thus $T_1 = \exp N$ with $N \neq 0$, $N^2 = 0$, and $\operatorname{rank} N = 1$. The standard rank-one-unipotent hypergeometric nearby-cycle calculation now applies. The parameter multiset is invariant under complex conjugation, so the variation has a real form. The polarized monodromy filtration $W(N)[3]$ has rank-one graded pieces in weights 2 and 4; the real form forces them to have types $(1,1)$ and $(2,2)$, respectively. Together with the four generic Hodge degrees in (172), this gives, after one complex coefficient embedding,

$$\mathrm{Gr}_2^W \text{ of type } (1,1), \qquad \mathrm{Gr}_3^W \text{ of types } (3,0)+(0,3), \qquad \mathrm{Gr}_4^W \text{ of type } (2,2),$$

where $N : \mathrm{Gr}_4^W \xrightarrow{\sim} \mathrm{Gr}_2^W(-1)$. The middle extension has stalk $\ker(T_1 - 1) = \ker N = W_3$, and strictness computes its induced Hodge filtration from the graded types above. It therefore removes the slope-2 slot, not the slope-0 slot. The toric Hodge polynomial and the requested multiplicity are

$$H_{a,t=1}(T) = 1 + T + T^3, \qquad \boxed{h^0(j) = [T^0]H_{a,t=1}(T) = 1} \quad (1 \le j \le p-2). \tag{173}$$

This is a closed fractional-part formula; it uses no Apéry value and no numerical input.

For completeness, the three slots can also be seen on the arrangement. The only non-simple-normal-crossings point is $(u,v,w) = (1,1,1)$. On its exceptional plane, with local coordinates denoted by a, b, c, the leading Kummer function is

$$\Lambda_E = -\frac{c(a+b+c)}{ab}.$$

For every nontrivial character A over $\mathbb{F}_q$, setting $b = 1$ gives

$$\sum_{E^\circ(\mathbb{F}_q)} A(\Lambda_E) = A(-1) \sum_{a,c\neq 0} A(c)A(a+1+c)A^{-1}(a) = q.$$

For $c \neq -1$ the inner a-sum is -1, while at $c = -1$ it is $q-1$; the last equality follows after summing $A(c)$. In perverse normalization, the Mellin identity identifies the $t = 1$ stalk with the effective class $-[R\Gamma_c]$. The blow-up weight spectral sequence identifies the exceptional-plane contribution with its unique rank-one weight-2 piece, $\mathrm{Gr}_2^W(\ker N) \simeq E_\ell(-1)$ above, and the displayed sum says that geometric Frobenius acts on it by $q = p^r$. Hence its eigenvalue in the effective polynomial is exactly $+p$; the outer minus sign has already been absorbed in $-[R\Gamma_c]$.

The involution

$$\tau(u, v, w) = \big(w, (uvw)^{-1}, u\big), \qquad \Lambda \circ \tau = \Lambda^{-1}, \tag{174}$$

is defined over $\mathbb{F}_p$ and carries $\mathcal{L}_A$ to $\mathcal{L}_{A^{-1}}$. Composing this identification with Poincaré duality gives a Frobenius-equivariant alternating perfect pairing

$$V_A \otimes_{E_\ell} V_A \longrightarrow E_\ell(-3)$$

on the remaining rank-two quotient, with no residual constant Kummer twist. In rank two the determinant equals the similitude multiplier, hence it is p^3. Therefore the effective characteristic factor is the theorem-level identity

$$\boxed{P_{p,j}(X) = (X-p)\big(X^2 - a_{p,j}X + p^3\big), \qquad a_{p,j} = S_{p,j}^{(1)} - p.} \tag{175}$$

The equality is an equality of the semisimplified effective virtual Frobenius class $-[R\Gamma_c(U, \mathcal{L}_{p,j})]$ with the rank-three middle-extension stalk. Thus it does not confuse the $q-1$ Mellin summands in (167) with eigenvalues.

12.23.2 Newton polygons and the inclusion theorem

Since $a_{p,j} \equiv S_{p,j}^{(1)} \equiv b_j \pmod{\mathfrak{p}}$, put $m = v_{\mathfrak{p}}(a_{p,j})$. The two roots of the quadratic factor in (175) have valuations

$$\begin{cases} (0,3), & m = 0, \\ (1,2), & m = 1, \\ (3/2, 3/2), & m \geq 2. \end{cases} \tag{176}$$

Adding the Tate slope 1 gives

$$m_0(p,j) = \begin{cases} 1, & b_j \not\equiv 0 \pmod{p}, \\ 0, & b_j \equiv 0 \pmod{p}. \end{cases} \tag{177}$$

In particular, $m_0(p,j)$ is never at least two. At an ordinary point the Newton slopes are $(0,1,3)$ and equal the Hodge slopes in (173); at a Hasse zero the polygon rises strictly above the Hodge polygon.

We record explicitly the theorem required in the specification. In the effective convention, the unique unit root reduces to b_j. Restoring the sign of $R\Gamma_c$, its signed slope-zero sum is therefore $-b_j \pmod{\mathfrak{p}}$. Consequently

Theorem 132 (Slope-zero inclusion)**.** *For every prime $p \geq 5$,*

$$\mathcal{Z}_p \subseteq \{1 \leq j \leq p-2 : m_0(p,j) \neq 1\}. \tag{178}$$

For this marked coordinate equality holds in (178)*: both sides are exactly the Hasse-zero set.*

Thus the inclusion is a theorem, but it is also a tautology: the moving unit root exists exactly when its Hasse invariant b_j is nonzero.

Extension-field computation. To check (175) without making the circular Jacobi-unit count, the script evaluates the full base-changed skeleton (167). If γ generates $\mathbb{F}_q^\times$, set

$$f_n = A(1-\gamma^n), \qquad g_n = A\left(\frac{\gamma^n}{1+\gamma^n}\right).$$

Multiplicative Fourier inversion turns the complete $q-1$ character sum into the exact cyclic convolution

$$S_{p,j}^{(r)} = -\sum_{c \in \mathbb{Z}/(q-1)} (f * f)_c (g * g)_c. \tag{179}$$

This is the promised $\mathbb{F}_{p^r}$ use of Theorem 130. Only the characters with index $K = (q-1)k/(p-1)$ are Hasse–Davenport lifts of base-field characters; the other extension characters cannot be omitted. At $(p,j,r) = (5,1,2)$ the correct trace is -125, whereas the invalid termwise lifting shortcut gives -153.

Newton's identities recover the cubic coefficients from S_1, S_2, S_3:

$$e_1 = S_1, \qquad e_2 = \frac{S_1^2 - S_2}{2}, \qquad e_3 = \frac{S_1^3 - 3S_1S_2 + 2S_3}{6}.$$

On six rational probe pairs, the portable direct-count and full-$q-1$ A1 engines independently compute all three traces and give $e_2 = pS_1 + p^3 - p^2$ and $e_3 = p^4$, as predicted by (175). The $p = 13$ row below uses an independently computed S_1 and the theorem-level factor. Representative rows are

p	j	S_1	e_2	e_3	Newton slopes
5	1	-5	75	625	$(1,1,2)$
5	2	3	115	625	$(0,1,3)$
7	1	-2	280	2401	$(0,1,3)$
7	3	31	511	2401	$(0,1,3)$
11	5	-33	847	14641	$(1,1,2)$
13	1	5	2093	28561	$(0,1,3)$

For example, the complete traces at $(p,j) = (7,1)$ are

$$(S_1, S_2, S_3, S_4) = (-2, -556, 8875, 133128),$$

and at $(11,5)$ they are $(-33, -605, 91839)$. A genuinely cyclotomic check at $p = 11, j = 1$, with $\zeta = \zeta_{10}$, gives

$$S_1 = 8\zeta^3 - 8\zeta^2 - 27, \qquad e_2 = 88\zeta^3 - 88\zeta^2 + 913, \qquad e_3 = 14641.$$

For every prime $p \leq 60$ and every nontrivial j (405 pairs), an independent production scan computed the complete traces S_1, S_2, S_3 modulo p^6 from (179). Thus every one of the $q-1$

extension characters was retained for $q = p, p^2, p^3$; neither the Apéry recurrence nor (175) was used to construct the traces. The optional `--full-small-scan` mode of `lemmaBprime_explore.py`, run under `sage -python`, is the reproduction path; the standard portable verifier uses the exact anchor probes instead of repeating this seven-minute scan. Newton's identities then gave

range	$(0,1,3) = \mathrm{NP} = \mathrm{HP}$	$(1,1,2)$	$m_0 \geq 2$
$p \leq 60$, all j	390	15	0

The 15 nonordinary points are

$$\begin{gathered}(5;1,3),\ (11;5),\ (17;3,13),\ (19;8,10),\\ (31;8,22),\ (37;17,19),\ (41;10,30),\ (59;9,49).\end{gathered}$$

They are exactly the 15 pairs with $b_j = 0$ in this range.

For every prime $p \leq 300$, the second range run used the already proved factor (175): it evaluated the base-field A1 skeleton p-adically at every $j \in \mathcal{Z}_p$, and classified 20 deterministic random coordinates per prime from the factor and their base trace modulo p (all available coordinates for the smallest primes). It contained 63 zero coordinates and 1140 sampled coordinates. All 63 zeros had $v_{\mathfrak{p}}(a_{p,j}) = 1$ and slopes $(1,1,2)$. The random sample split as 1127 ordinary coordinates with slopes $(0,1,3)$ and 13 hits on the already enumerated zero set, again with slopes $(1,1,2)$. No pair had two slope-zero roots. Thus the $p \leq 60$ table is a full extension-field check of the factor, while the $p \leq 300$ table is a theorem-driven enumeration plus independent base-A1 valuation checks at all Hasse zeros. Neither table is used to prove the non-numerical Hodge formula (173).

12.23.3 Exact eigenvalue tests and the discriminant

The polynomial

$$\Delta(t) = t^2 - 34t + 1$$

does occur naturally, but not as a Frobenius eigenline. From (165), the logarithmic critical equations are

$$u^2vw - 2uvw + 1 = 0, \qquad uv^2w - 2uvw + 1 = 0, \qquad uvw^2 - 1 = 0.$$

On U they give

$$u = v = 2 - w, \qquad w^2 - 2w - 1 = 0,$$

and substitution yields the two critical values

$$\Lambda_{\mathrm{crit}} = 17 \pm 12\sqrt{2} = (1 \pm \sqrt{2})^4. \tag{180}$$

Thus the roots of Δ are conifold critical values of the map Λ. The involution (174) exchanges them. The actual rank-two quotient remains self-dual; neither critical value thereby becomes a rank-one Teichmüller eigenline.

The exact tests make this distinction visible. At the split prime $p = 7$, the roots of Δ are 2 and 4. For $j = 1, 2, 3$ the quadratic factors are

$$X^2 + 9X + 343, \qquad X^2 + 25X + 343, \qquad X^2 - 24X + 343.$$

Their unit-root residues are respectively $5, 3, 3$, whereas the proposed Teichmüller targets are respectively $\{2,4\}$, $\{2,4\}$, and $\{1\}$. Hence the candidate already fails modulo 7. More strongly, for $j = 1, 2$ reduction modulo $\Phi_3(X) = X^2 + X + 1$ leaves $8X + 342$ and $24X + 342$, and for $j = 3$ the last quadratic evaluates to 320 at $X = 1$. There is no exact cyclotomic match.

The three quadratic eigenvalue fields in this test are

$$\mathbb{Q}(\sqrt{-1291}), \qquad \mathbb{Q}(\sqrt{-83}), \qquad \mathbb{Q}(\sqrt{-199}),$$

none of which is the base Jacobi field $E_7 = \mathbb{Q}(\zeta_6) = \mathbb{Q}(\sqrt{-3})$. Thus no base-field Jacobi-sum monomial, affine in j or otherwise, equals the unit root in these exact cases. In $\mathbb{Q}_7$, the unit residues $5, 3, 3$ are all nonsquares; the other two slopes are 1 and 3, both odd. Hence none of the three actual eigenvalues is a square in the natural coefficient completion. This rules out the nonvacuous local Sym^2 square test.

At the inert prime $p = 11$, the roots of Δ lie in $\mathbb{F}_{11^2}$ and have order 6. For $j = 1, 2, 4$ their powers do not even lie in $\mathbb{F}_{11}$; for $j = 3$ the target is -1 but $b_3 \equiv 4 \pmod{11}$; and $j = 5$ has no unit root. Residue coincidences can occur: at $j = 6$ both the target and the unit-root residue are 1.

Fix an embedding $\iota_p : \overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_p$. A Teichmüller lift of either reduction of $17 \pm 12\sqrt{2}$ is $\iota_p(\xi)$ for a prime-to-p root of unity ξ; in the inert case it lies in the unramified quadratic extension. Every complex conjugate of ξ has absolute value 1. By contrast, each root of the pure weight-3 quotient is a Weil p-number whose complex conjugates have absolute value $p^{3/2}$. Injectivity of ι_p therefore forbids exact equality in $\overline{\mathbb{Q}}_p$ with $\omega(17 \pm 12\sqrt{2})^{\pm j}$, although isolated congruences modulo p are possible.

The $p = 7$ field and square computations are additional exact sample checks: they exclude every base-field Jacobi monomial in those cases and every square in the natural coefficient completion $E_{7,\mathfrak{p}} = \mathbb{Q}_7$. We do not claim a universal classification of affine Jacobi monomials, or a square obstruction after arbitrary scalar extension. The sole universally identified rank-one eigenvalue is p, of slope 1, and is irrelevant to a slope-zero collision.

Binary verdict. Equation (173) gives $h^0(j) = 1$ for every nontrivial twist, so Newton-above-Hodge allows at most one slope-zero eigenvalue. Equation (177) says that the Newton polygon equals the Hodge polygon precisely away from the original Hasse zeros. Thus the two-unit-root split required by G2 is structurally impossible, independently of whether the single moving root has an isolated alternative description. The requested verdict is therefore

$$\boxed{\textbf{G1 (KILL).}}$$

The crystalline criterion is exactly the Hasse-invariant tautology $p \mid b_j \iff m_0(p, j) = 0$. There is no second unit root with which a moving root could collide. In particular, the critical values $17 \pm 12\sqrt{2}$ do not open a unit-root anti-concentration problem.

Fail-closed computational consistency verification. Running `python3 problems/3.2/lemmaBprime_verif` ends with

```
PASS full A1 probes: 6 pairs r<=3; p13 r=1; cyclotomic p11,j1 S1
PASS cubic factor: 6 probed cubics; 5 r4 recurrences; p13 predicted
PASS p<=59 classification: 405 factor classifications; base-A1 v(a)=1 at 15/15 zeros
PASS p<=300 sample: base-A1 v(a)=1 at 63/63 zeros; sample 1140, zero hits 13
PASS candidate matches: critical values; p7 exact Teich/Jacobi/Q7-square rejection
PASS binary gate: G1 (h0=1; the only fixed branch is Tate slope 1)
SUMMARY: 11 PASS, 0 FAIL
```

12.24 Poisson variance formulation

The diagonal AMTD (118) admits a clean probabilistic reformulation. For a dyadic block $\mathcal{B}_i$, define the *column multiplicity function*

$$\lambda_i(m) = \#\{p \in \mathcal{B}_i : m \in \Omega_p\} = \sum_{p\in\mathcal{B}_i} X_p(m), \qquad m \in I_N.$$

Its empirical mean and variance over I_N are

$$\mu_i = \frac{1}{N}\sum_m \lambda_i(m) = \frac{1}{N}\sum_{p\in\mathcal{B}_i} A_p, \qquad V_i = \frac{1}{N}\sum_m (\lambda_i(m) - \mu_i)^2.$$

Expanding the square gives

$$N\,V_i = \sum_{p\in\mathcal{B}_i} D_p + \sum_{\substack{p,q\in\mathcal{B}_i\\ p\neq q}} C_{p,q}.$$

Since all primes in a dyadic block satisfy $\log p \asymp \log P$, one has $\|F_i\|_2^2 \asymp (\log P)^2\, N\, V_i$ and $s_i^2 \asymp (\log P)^2 \sum_p D_p$, so the diagonal AMTD is equivalent (up to asymptotically negligible corrections from the non-constant weight $\log p$) to:

$$V_i \;\leq\; C\,\mu_i \qquad \text{(sub-Poisson variance).} \tag{181}$$

If the indicators $X_p(m)$ were independent Bernoulli variables (with $\Pr[X_p = 1] = A_p/N$), each $\lambda_i(m)$ would be a sum of independent sparse Bernoullis and the variance-to-mean ratio would be ≤ 1. Condition (181) therefore says that the deterministic Apéry indicators behave as if they were *approximately independent* at the block level.

Table 5 records the empirical variance-to-mean ratio for the top dyadic block.

N	K_{top}	μ	V/μ	collisions	Poisson pred.
1024	47	0.086	1.073	14	7.6
2048	94	0.079	0.983	10	12.8
4096	177	0.079	0.989	22	25.6
8192	363	0.080	1.011	60	52.9

Table 5: Column multiplicity statistics for the top dyadic block. $\mu = \frac{1}{N}\sum_p A_p$ is the mean, V/μ is the variance-to-mean ratio (Poisson prediction: 1), "collisions" is $\sum_m \lambda(\lambda-1)$, and "Poisson pred." is T^2/N where $T = \sum_p A_p$.

Remark 133 (Collision count). The off-diagonal covariance sum has the exact identity

$$\sum_{p\neq q\in\mathcal{B}_i} C_{p,q} \;=\; \sum_m \lambda_i(m)\big(\lambda_i(m)-1\big) \;-\; \frac{(\sum_p A_p)^2 - \sum_p A_p^2}{N}.$$

The first term counts pairwise collisions (integers hit by at least two primes in the block). For the top block ($P \sim N$), each prime has $A_p \leq Z(p) \leq 3$ hits, so $\lambda_i(m) \leq 3$ for all m at $N = 8192$. The collision count fluctuates around the Poisson prediction T^2/N, confirming the near-independence of the indicators.

Remark 134 (Doublet dominance). Palindromic symmetry $b_{p-1-j} \equiv b_j \pmod p$ forces the zero set $\mathcal{Z}_p$ to be closed under $r \mapsto p-1-r$. If $Z(p) > 0$, then $Z(p)$ is typically even, and indeed at the top block, $Z(p) \geq 2$ for all active primes. The distribution at $N = 8192$ is: *doublets* $Z = 2$ (78%), *quadruplets* $Z = 4$ (19%), *sextuplets* $Z = 6$ (3%), *octuplets* $Z = 8$ (< 1%). No singleton primes appear ($Z = 1$ requires $b_{(p-1)/2} \equiv 0 \pmod p$, which is extremely rare for large p).

For a doublet prime with $\mathcal{Z}_p = \{r_p, p-1-r_p\}$, the two hit positions are

$$m_1(p) = p + r_p, \qquad m_2(p) = 2p - 1 - r_p,$$

satisfying $m_1(p) + m_2(p) = 3p - 1$. A collision between two doublet primes p, q requires one of four types:

$$\begin{aligned}
&\text{(A) } m_1(p) = m_1(q) &&\iff && r_q = r_p + (p-q),\\
&\text{(B) } m_1(p) = m_2(q) &&\iff && r_p + r_q = 2q - 1 - p,\\
&\text{(C) } m_2(p) = m_1(q) &&\iff && r_p + r_q = 2p - 1 - q,\\
&\text{(D) } m_2(p) = m_2(q) &&\iff && r_q = r_p + 2(q-p).
\end{aligned}$$

For $q > p$, type (B) is impossible: $m_1(p) = p + r_p < 3p/2 < 3q/2 < 2q - 1 - r_q = m_2(q)$. The collision energy therefore decomposes into three, not four, pieces:

$$F = F_A + F_C + F_D, \tag{182}$$

where $F_A = \sum_m \nu_1(m)(\nu_1(m)-1)$, $F_D = \sum_m \nu_2(m)(\nu_2(m)-1)$, $F_C = 2\sum_m \nu_1(m)\nu_2(m)$, with $\nu_1(m) = \#\{p : m_1(p) = m\}$ and $\nu_2(m) = \#\{p : m_2(p) = m\}$.

Two distinct doublet columns $\{m_1(p), m_2(p)\}$ and $\{m_1(q), m_2(q)\}$ can share at most one element: if both elements coincided, then $m_1(p)+m_2(p) = m_1(q)+m_2(q)$, i.e., $3p-1 = 3q-1$, forcing $p = q$. Thus $J_{p,q} := |\Omega_p \cap \Omega_q| \leq 1$ for distinct doublet primes in the top block. The doublet incidence system is a *simple graph* (no multi-edges), with the ordered collision count $F = \sum_m \lambda(m)(\lambda(m)-1)$. The exact variance identity is

$$V = T + F - T^2/N.$$

Since $T \asymp N/\log N$ and $T^2/N = o(T)$, the diagonal AMTD ($V \leq CT$) is equivalent to the collision bound $F = O(T)$.

Collision decomposition.

N	K	T	T^2/N	F	F_A	F_C	F_D	F/T
1024	47	39	1.5	2	2	0	0	0.051
2048	93	70	2.4	4	2	2	0	0.057
4096	177	140	4.8	0	0	0	0	0.000
8192	363	293	10.5	16	10	4	2	0.055
16384	669	528	17.0	22	16	6	0	0.042
32768	1145	925	26.1	38	18	20	0	0.041

The ratio F/T is consistently small and decreasing, empirically confirming $F = o(T)$. Under the FPI prediction $F \approx 3K^2/N$, one expects $F/T \approx 3/(6.6 \log N)$, giving 0.044 at $N = 32{,}768$, in good agreement with the observed 0.041. At all sizes $\max_m \lambda(m) \leq 3$.

Remark 135 (Two-prime divisor formulation). At the top block ($P > N$), each hit $m \in \Omega_p$ implies $p \mid b_m$ by the Lucas congruence. Therefore

$$\lambda_{\mathcal{B}}(m) = \omega_{(N,2N]}(b_m) := \#\{p \text{ prime} : N < p \le 2N,\ p \mid b_m\},$$

and the collision count becomes

$$F = \sum_{m \in I_N} \omega_{(N,2N]}(b_m)\big(\omega_{(N,2N]}(b_m) - 1\big).$$

The diagonal AMTD is therefore equivalent to the following *two-large-prime divisor theorem*: on average over m, the Apéry number b_m has at most $O(1)$ prime divisors in the dyadic range $(N, 2N]$.

This formulation reveals the arithmetic core: the collision bound $F = O(T)$ is an Apéry-specific Barban–Davenport–Halberstam theorem, asking for a level-of-distribution estimate at the second prime moment. A single-prime level-of-distribution (equivalently, $T = O(N/\log N)$) follows from the Poisson conjecture and is not in doubt; the missing step is the *bilinear* independence: the events $p \mid b_m$ and $q \mid b_m$ are approximately uncorrelated as p, q range over a dyadic block.

Vertical vs. horizontal projections. The incidence graph has primes on one side and integers on the other; an edge (p, n) records $p \mid b_n$. Grouping by the integer index n gives the *vertical* prime-divisor count $W_N(n) = \omega_{(N,2N]}(b_n)$, $n \in [1, N]$. Grouping by the hit position $m = n + p$ gives the *horizontal* column multiplicity $\lambda(m)$. These are different projections. By the Lucas congruence $b_{p+r} \equiv 5\, b_r \pmod p$ (for $r < p$), the horizontal multiplicity $\lambda(m) = \omega_{(N,m]}(b_m)$—it counts large prime divisors of b_m itself. Crucially, even the pointwise bound $W_N(n) \le 1$ for *every* $n \le N$ does not control $\lambda(m)$: many distinct (n_i, p_i) pairs can satisfy $n_i + p_i = m$ with $p_i \mid b_{n_i}$, all contributing *different* prime divisors of the single integer b_m. The collision bound $F = O(T)$ is therefore a genuinely horizontal, two-prime-divisor assertion about b_m.

Remark 136 (Why classical sieves do not close the argument). The collision bound $F = O(T)$ has the form of a two-dimensional upper-bound sieve (Selberg, Rosser–Iwaniec, Brun, GPY), but no classical sieve theorem deduces it from the one-prime data alone. After weight expansion, every such sieve introduces the counts $A_{pq}(N) = \#\{m : pq \mid b_m\}$ for products $pq \asymp N^2$. Since the observation interval has length only N, each individual expected count is $O(1/N)$ and only an *averaged* large-modulus discrepancy theorem can hold. The difficulty is intrinsic: even the strongest conceivable one-prime bound $A_p(N) \le 1$ for every p does not control cross-prime alignment. An adversarial configuration with all zero sets concentrating at a single index m_0 satisfies $A_p = 1$ while $F \asymp (N/\log N)^2$. Thus the sieve machinery can *package* a two-prime distribution theorem as a collision bound, but it cannot *manufacture* that theorem from one-prime information. The minimum new input is the bilinear estimate

$$\sum_{\substack{p \ne q \\ p,q \in (N,2N]}} A_{pq}(N) \ \ll\ T + \frac{T^2}{N}, \tag{183}$$

which is precisely the block second-factorial-moment conjecture (diagonal AMTD, Hypothesis 108).

Proposition 137 (Conditional collision bound)**.** *Assume the following* Frobenius Pair Independence *hypothesis (FPI): for distinct doublet primes $p, q \in (N, 2N]$, the normalized first zeros r_p/p and r_q/q are asymptotically independent and each approximately uniform on $[0, \frac{1}{2}]$ in the sense that, for any integers $a \in [1, (p-1)/2]$ and $b \in [1, (q-1)/2]$,*

$$\frac{1}{K(K-1)} \#\big\{(p, q) \text{ distinct doublet primes in } (N, 2N] : r_p = a,\ r_q = b\big\} \ =\ \big(1 + o(1)\big)\, \frac{4}{pq}, \tag{184}$$

where K is the number of doublet primes. Then $F = o(T)$ and the all-index GCD theorem $G_n = e^{o(n)}$ holds for all n.

Proof. We bound each piece of the decomposition (182) separately.

Type A. The type-A collision count is

$$F_A = \#\big\{(p,q) \text{ distinct doublet primes} : r_q = r_p + (p-q)\big\}.$$

For a fixed ordered pair (p,q), the collision requires $r_q = r_p + (p-q)$. Summing the FPI probability over all valid values of r_p, the conditional probability is

$$\Pr[\text{type A} \mid p,q] \;=\; \sum_{r=1}^{(p-1)/2} \frac{2}{p}\cdot\frac{2}{q}\cdot\mathbf{1}\big[r+(p{-}q)\in[1,(q{-}1)/2]\big] \;\le\; \frac{4\,\min\big((p{-}1)/2,\,(3q{-}1)/2-p\big)}{pq}.$$

For primes $p,q \in (N,2N]$, this is at most $4/N$ when $|p-q| \le N/2$, and vanishes when $|p-q| > N$. Summing over all $K(K-1)$ ordered pairs:

$$F_A \;\le\; K(K{-}1)\cdot\frac{4}{N}\cdot(1+o(1)) \;\le\; \frac{4K^2}{N}\cdot(1+o(1)).$$

Since $K \asymp N/\log N$ and $T = 2K + O(K^{1/2})$, we obtain $F_A \le 4T^2/(4N)\cdot(1+o(1)) = T^2/N\cdot(1+o(1)) = o(T)$.

Type C. The collision condition $r_p + r_q = 2p-1-q$ constrains a sum of independent uniforms. Under FPI, $r_p + r_q$ has a trapezoidal distribution on $[2,(p+q-2)/2]$ with maximum density $2/\min(p,q) \le 2/N$. The target $2p-1-q$ lies in the support when $q < 2p-1$, which holds for $p > (q+1)/2$. Thus $\Pr[\text{type C} \mid p,q] \le 2/N$, and $F_C \le 2K^2/N\cdot(1+o(1)) = o(T)$.

Type D. The condition $r_q = r_p + 2(q-p)$ has the same structure as type A with gap $2(q-p)$ in place of $p-q$. By the same argument, $F_D \le 4K^2/N\cdot(1+o(1)) = o(T)$.

Combining all three, $F = F_A + F_C + F_D = o(T)$. The diagonal AMTD $\|F_i\|_2^2 \le C\,s_i^2$ follows with $C = 1+o(1)$ at each dyadic block, giving $V^\circ \le (1+o(1))\,L\,S = O(\log N)\,S = N^{o(1)}\,S$. ☐

Remark 138 (Status of the FPI hypothesis). The hypothesis (184) is a pair decorrelation statement for the zero positions across different primes. The marginal equidistribution of r_p/p is supported by the Kummer structure of the reduced generating series (Caruso–Fürnsinn–Vargas-Montoya–Zudilin [7]), which shows that the Galois group of f_α mod p is cyclic of order $(p-1)/\gcd(2,p-1)$. The rank-three Apéry Picard–Fuchs connection is the symmetric square of a rank-two connection, so its connected geometric monodromy is of $\mathrm{Sym}^2\mathrm{SL}_2 \simeq \mathrm{SO}_3$ type.

However, neither the cyclic Kummer group nor the SO_3 monodromy controls the zero *positions* r_p. To see why, observe that the zero-set $\mathcal{Z}_p = \{j < p : p \mid b_j\}$ is a *spectral* zero-set: writing $A_p(t) = \sum_{k=0}^{p-1} b_k t^k \in \mathbb{F}_p[t]$ and $f_p(u) = A_p(g^u)$ for a primitive root g, multiplicative orthogonality gives

$$\sum_{u=0}^{p-2} f_p(u)\,g^{-ju} = -b_j,$$

so $j \in \mathcal{Z}_p$ iff the j-th discrete Mellin coefficient of the Hasse-polynomial value vector vanishes. By contrast, the *roots* of $A_p(t)$ in $\mathbb{F}_p$ are evaluation zeros, controlled by the Kummer factorization $A_p(t) = \Delta(t)^{\varepsilon_p} B_p(t)^2$. The Mellin coefficient index j and the discrete logarithm $\mathrm{ind}_g(x)$ of a root x live in dual spaces; no theorem identifies a coefficient-zero position with an evaluation-zero position.

The pair independence (184) is therefore a genuinely new hypothesis: it concerns the spectral zero process $\mathcal{A}_p(j) = \mathbf{1}_{\{b_j\equiv 0\}}$, not a Frobenius class function or a discrete logarithm. Neither Chebotarev, Sato–Tate, nor Katz equidistribution (which control traces and evaluation points) applies to this spectral-zero process.

A clean unconditional formulation replaces FPI by the *marked two-characteristic Hasse dispersion* conjecture:

$$\sum_{\substack{N<p,q\le 2N\\ p\ne q}} \sum_{m\in I_N} X_p(m)\, X_q(m) \ll T_N + \frac{T_N^2}{N}, \tag{185}$$

where $X_p(m) = \mathbf{1}_{\{p\le m,\ b_{m-p}\equiv 0 \pmod p\}}$. The left side equals the ordered collision count F_N^{ord}. Via the exponential-sum detection $\mathcal{A}_p(j) = p^{-1}\sum_{a \bmod p} e_p(a\, b_j)$, the conjecture (185) reduces to cancellation in the mixed-modulus sum

$$\sum_{p\ne q} \frac{1}{pq} \sideset{}{'}\sum_{\substack{a \bmod p\\ c \bmod q}} \sum_{m\in I_N} e_p(a\, b_{m-p})\, e_q(c\, b_{m-q}), \tag{186}$$

where $\sum'$ omits the term $a = c = 0$ (which produces the main term). The inner sum is over a nonautonomous recurrence orbit: $m \mapsto (b_{m-p}, b_{m-q})$. Standard Weil bounds do not apply.

A proof of (185) would require (1) a uniform crystalline Mellin object whose first Hasse coordinate tracks $b_j \bmod p$; (2) nondegeneracy of this coordinate on the monodromy torsor; (3) product monodromy for distinct Kummer twists; (4) rare-event anti-concentration at $1/p$ scale; and (5) a *horizontal* large sieve across different characteristics—a five-stage program beyond current technology but strongly supported by computation ($F/T \le 0.041$ at $N = 32{,}768$).

Proposition 139 (Profile decomposition)**.** *For each prime $p \in (N, 2N]$, let $A_p = |\Omega_p \cap I_N|$ and $L_p = 2N - p$. Define the smooth tail profile*

$$\rho(m) = \sum_{\substack{N<p\le 2N\\ p<m}} \frac{A_p}{L_p}, \qquad m \in I_N.$$

Then:

(i) $\sum_{m\in I_N} \rho(m) = T_N$*;*

(ii) $\|\rho\|_2^2 := \sum_{m\in I_N} \rho(m)^2 \le \rho_{\max} T_N$*, where* $\rho_{\max} := \max_{m\in I_N} \rho(m) \le \sum_p A_p/L_p$*; under the PNT profile approximation,* $\|\rho\|_2^2 = (c_2 + o(1))\, T_N^2/N$ *with* c_2 *as in* (188) *(note* $c_2 > 1$*: by Cauchy–Schwarz* $\|\rho\|_2^2 \ge T_N^2/N$ *always);*

(iii) $F_N \le 2\|\rho\|_2^2 + 2\|\lambda - \rho\|_2^2 - T_N$*.*

In particular, if $\|\rho\|_2^2 = O(T_N^2/N)$ *(as the profile approximation and the data indicate), the residual dispersion estimate* $\|\lambda - \rho\|_2^2 \ll T_N$ *would imply* $F_N = O(T_N)$*, since* $T_N^2/N = O(T_N)$*.*

Proof. Part (i): $\sum_m \rho(m) = \sum_p (A_p/L_p)\cdot L_p = \sum_p A_p = T_N$. Part (ii): $\|\rho\|_2^2 \le \rho_{\max}\sum_m \rho(m) = \rho_{\max} T_N$, and $\rho(m)$ is a partial sum of the positive terms A_p/L_p. For the profile asymptotics: by the PNT approximation $\rho(m) \approx (2/\log N)\log(N/(2N-m))$ and the integral $\int_0^1 (\log 1/t)^2\, dt = 2$, one obtains $\|\rho\|_2^2 \sim c_2\, T_N^2/N$. Numerically, $\|\rho\|_2^2/(T_N^2/N)$ stabilizes near 1.25 for $N = 512, \ldots, 32{,}768$ (Table 6), matching $c_2 = 1.262\ldots$ from (188). Part (iii) follows from $\|a+b\|^2 \le 2\|a\|^2 + 2\|b\|^2$ applied to $\lambda = \rho + \eta$. □

Remark 140 (Orthogonal decomposition). Setting $\eta = \lambda - \rho$, the identity $T + F = \|\lambda\|_2^2 = \|\rho\|_2^2 + 2\langle\rho,\eta\rangle + \|\eta\|_2^2$ gives

$$F = \|\rho\|_2^2 + 2\langle\rho,\eta\rangle + (\|\eta\|_2^2 - T). \tag{187}$$

Under the independent profile model (each $X_p(m)$ is an independent Bernoulli with mean $\pi_p(m) = A_p/L_p$ when $p < m$), the three terms in (187) have the following expected behavior:

- $\|\rho\|_2^2 = \Theta(T^2/N)$: the deterministic profile energy, tending to 0 relative to T.
- $\|\eta\|_2^2 \approx T$: the shot-noise variance of T independent Bernoulli hits.
- $\langle\rho,\eta\rangle \approx 0$: the profile and residual are decorrelated.

Hence $F \approx \|\rho\|_2^2 \approx T^2/N = o(T)$. Computation (Table 6) confirms this decomposition: $\|\eta\|_2^2/T$ is stable near 1.0 (matching palindromic random surrogates), while $F \approx \|\rho\|_2^2$ to within 1% at $N = 16{,}384$ and within 8% at $N = 32{,}768$. The cross-term $\langle\rho,\eta\rangle$ is $O(1)$ at every tested scale (e.g. -0.03 at $N = 16{,}384$ and 0.40 at $N = 32{,}768$, versus $T = 664$ and 1182). Moreover, the cross-term has a *deterministic* bound: $|\langle\rho,\eta\rangle| \le \rho_{\max}\cdot T$ where $\rho_{\max} = \max_m \rho(m) = \sum_p A_p/(2N-p)$. Since $\rho_{\max}/(T/N)$ stabilizes near 1.8 (Table 6), $|\langle\rho,\eta\rangle| \le 1.8\,T^2/N = o(T)$. The unconditional collision theorem $F = O(T)$ is therefore equivalent to the shot-noise estimate $\|\eta\|_2^2 = O(T)$, which is the residualized AMTD (Hypothesis 108).

Table 6: Profile decomposition diagnostics for the full hit measure μ_N. All norms over $I_N = (N, 2N]$. "Surr" denotes the mean of $\|\eta\|_2^2/T$ over 10 palindromic random surrogates.

N	T	F	$\|\rho\|_2^2$	$\|\rho\|_2^2/(T^2/N)$	$\|\eta\|_2^2/T$	Surr
512	21	2	1.1	1.28	1.05	0.98
1024	50	2	3.0	1.22	0.98	0.99
2048	86	4	4.5	1.26	0.99	1.01
4096	172	2	9.4	1.31	0.96	1.00
8192	354	20	19.6	1.28	1.00	1.00
16384	664	34	33.7	1.25	1.00	1.01
32768	1182	58	53.8	1.26	1.00	—
65536	2237	94	92.9	1.22	1.00	—
131072	4317	174	178.5	1.26	1.00	—
262144	7940	312	301.7	1.25	1.00	—

Remark 141 (Exact profile constant). The column $\|\rho\|_2^2/(T^2/N)$ in Table 6 converges to a constant c_2 as $N \to \infty$, with exact closed form

$$c_2 \;=\; \frac{2(\ln 2)^2 - 4\ln 2 + 2}{(2\ln 2 - 1)^2} \;=\; 1.261982\ldots \tag{188}$$

The derivation is direct. The triangular profile $\rho(m) = \sum_{N<p\le m} Z(p)/p$ for $m \in I_N$ has the continuous approximation $\rho(Nt) \approx (\ln t)/(\ln N)$ for $t \in [1,2]$, by Mertens' theorem. Then

$$T = \sum_m \rho(m) \;\approx\; \frac{N}{\ln N}\int_1^2 \ln t\,dt = \frac{N}{\ln N}\,(2\ln 2 - 1),$$

$$\|\rho\|_2^2 = \sum_m \rho(m)^2 \;\approx\; \frac{N}{(\ln N)^2}\int_1^2 (\ln t)^2\,dt = \frac{N}{(\ln N)^2}\,\big(2(\ln 2)^2 - 4\ln 2 + 2\big).$$

The ratio gives (188). The exact palindromic null expectation confirms the match: at $N = 1024$ the finite-N ratio is 1.209, at $N = 4096$ it is 1.281, at $N = 16{,}384$ it is 1.254, oscillating around c_2 with $O(1/\ln N)$ corrections from the prime number theorem. At all tested sizes, $E_{\text{null}}[F]$ agrees with the observed F to within the expected stochastic fluctuation.

Remark 142 (Concentration of F). Under the palindromic null (zero counts $Z(p)$ fixed, positions uniform among palindromic configurations), F concentrates around its mean. The dependency graph of collision indicators is the line graph of the complete graph on the active primes: two edge indicators share a dependency exactly when the corresponding prime pairs share one prime. This sparse structure gives

$$\operatorname{Var}(F_N) = \big(2 + O(1/\ln N)\big)\, E[F_N], \tag{189}$$

a rigorous Poisson-scale variance. *Proof sketch.* Write $F = 2U$ where $U = \sum_{p<q} I_{p,q}$ counts collision pairs. Edge indicators for disjoint prime pairs $\{p, q\} \cap \{r, s\} = \varnothing$ are independent. For a shared prime p, the conditional bound $\Pr(I_{p,q} = 1 \mid R_p = r) \le 2/h_q$ gives $\sum_{e<f} |\operatorname{Cov}(I_e, I_f)| \le 2\Delta\,\Lambda$ where $\Delta = 2\sum_p h_p^{-1} = O(1/\ln N)$ and $\Lambda = E[U]$. Combined with the Bernoulli variance $\sum \operatorname{Var}(I_e) = \Lambda(1 - O(1/N))$, this yields (189).

Since $E[F] = c_2\, T^2/N\,(1 + o(1)) \asymp N/(\ln N)^2 \to \infty$, Chebyshev gives $F/E[F] \to 1$ in L^2 and in probability, and $\Pr(F \ge C\,T) = O_C(\ln N/N)$ for every fixed $C > 0$.

The profile decomposition $F = \|\rho\|^2 + 2\langle\rho, \eta\rangle + \|\eta\|^2 - T$ separates deterministic and stochastic contributions. The deterministic cross-term satisfies $|\langle\rho, \eta\rangle| \le \rho_{\max}\, T$, with

$$\frac{\rho_{\max}}{T/N} \;\longrightarrow\; \frac{\ln 2}{2\ln 2 - 1} = 1.7943\ldots;$$

this bound is $O(T^2/N)$, the *same* order as $\|\rho\|^2$. The computationally observed $\langle\rho, \eta\rangle \approx 0$ is therefore genuine decorrelation evidence, not a consequence of the profile geometry.

If $\|\eta\|^2 = O(T)$, then all three terms are controlled: $\|\rho\|^2 = O(T^2/N)$, $|\langle\rho, \eta\rangle| = O(T^2/N)$, and $\|\eta\|^2 - T = O(T)$. Since $T^2/N = O(T)$ (because $T = o(N)$), this gives $F = O(T)$, exactly the BFM2 bound. The estimate $\|\eta\|^2 = O(T)$—that is, shot-noise-scale fluctuations of the actual zero positions around the triangular profile—is the sole gap.

Remark 143 (Prime-wise decomposition of $\|\eta\|^2$). For each prime $p \in (N, 2N]$, define the centred indicator

$$X_{m,p} \;=\; \mathbf{1}_{(m-p)\in Z_p} \;-\; \frac{Z(p)}{p}, \qquad m \in [p{+}1,\, 2N].$$

Then $\eta(m) = \sum_{N<p\le m} X_{m,p}$. Note that while distinct primes $p \ne q$ provide separate residue fields (and $b_j \bmod p$ is determined independently of $b_j \bmod q$ by CRT), this does not automatically make $X_{m,p}$ and $X_{m,q}$ independent for deterministic zero-sets—the zero positions within each Z_p are algebraically constrained. Under the independent fixed-cardinality residue model (Section 12.26), however, the columns are independent by construction.

Expanding the square:

$$\|\eta\|^2 \;=\; \underbrace{\sum_p \sum_m X_{m,p}^2}_{D} \;+\; \underbrace{\sum_{p\ne q} \sum_m X_{m,p}\, X_{m,q}}_{O}\,. \tag{190}$$

The diagonal $D = \sum_p D_p$ where $D_p = \sum_m X_{m,p}^2 = Z_p^{\text{act}}(1{-}q_p)^2 + (L_p - Z_p^{\text{act}})\, q_p^2$ with $q_p = Z(p)/p$, $Z_p^{\text{act}} = \#\{j \in [1, L_p] \cap Z_p\}$, $L_p = 2N - p$. Expanding:

$$D \;=\; T \;-\; 2\sum_p Z_p^{\text{act}}\, q_p \;+\; \sum_p L_p\, q_p^2. \tag{191}$$

Both correction terms are $O(1/\ln N)$: since $Z_p^{\mathrm{act}} \le Z(p)$ and $q_p = Z(p)/p \le Z(p)/N$, the second term is $\le 2\sum_p Z(p)^2/p \le (2/N)\sum_p Z(p)^2$, and the third is $\le N\sum_p Z(p)^2/p^2 \le (1/N)\sum_p Z(p)^2$. Under $\sum_p Z(p) = O(K)$ where $K \sim N/\ln N$, the Cauchy–Schwarz bound $\sum_p Z(p)^2 \le (\max Z(p)) \cdot \sum_p Z(p)$ and $\max Z(p) = O(1)$ (empirically) give both corrections $O(K/N) = O(1/\ln N)$. Thus $D = T + O(1/\ln N)$ unconditionally under $\sum_p Z(p)^2 = O(K)$. Under the null model, $E[O] = 0$.

Computation confirms this structure for the actual Apéry zero-sets: D matches T to $< 0.1\%$ for all tested N, and O/T fluctuates around zero at the level 10^{-3}–10^{-2} (see Table 7). Under the null model, $\mathrm{Var}(D) = O(T)$ (fluctuation of the truncated active mass Z_p^{act}) and $\mathrm{Var}(O) = O(T^2/N)$ (CRT averaging: each of the $\binom{K}{2}$ pair terms has overlap $\sim N$ within period $pq \sim N^2$). Together,

$$\frac{\mathrm{Var}(\|\eta\|^2)}{T^2} = O\left(\frac{1}{T} + \frac{1}{N}\right) \to 0,$$

proving $\|\eta\|^2/T \to 1$ in L^2 and in probability within the null model. The unconditional statement $\|\eta\|^2 = O(T)$ for the actual Apéry zero-sets is equivalent to a *two-prime dispersion* input: the zero positions of distinct primes are equidistributed modulo pq in the short interval $[1, N]$.

Table 7: Prime-wise decomposition $\|\eta\|^2 = D + O$.

N	T	D	O	O/T
512	21	20.94	1.14	0.054
1024	50	49.86	−0.83	−0.017
2048	86	85.90	−0.43	−0.005
4096	172	171.90	−6.76	−0.039
8192	354	353.90	1.22	0.003
16384	664	663.90	0.46	0.001
32768	1182	1181.91	3.52	0.003
65536	2237	2236.92	−4.16	−0.002
131072	4317	4316.92	−1.76	−0.0004
262144	7940	7939.93	11.92	0.0015

Note the structural identity $O = F - \|\rho\|^2 - 2\langle\rho, \eta\rangle$, so $O = o(T)$ is equivalent to $F = o(T)$ (since $\|\rho\|^2 = O(T^2/N) = o(T)$), and the prime-wise decomposition reduces the BFM2 bound to the assertion that the cross-prime interaction is asymptotically negligible.

Remark 144 (Higher factorial moments). The third factorial moment $F_3 = \sum_m \lambda(m)(\lambda(m)-1)(\lambda(m)-2)$ equals 6 at both $N = 16{,}384$ and $N = 32{,}768$ (a single row with $\lambda = 3$) but vanishes at $N = 65{,}536$ and $N = 131{,}072$ ($\max_m \lambda(m) = 2$ with 47 and 87 collision positions respectively), then returns to $F_3 = 6$ at $N = 262{,}144$ (a single row with $\lambda = 3$ among 154 collision positions). The palindromic model predicts $E[F_3] \approx c_3 T^3/N^2$ where c_3 is a geometry-of-truncation constant. At $N = 262{,}144$ with $T = 7940$, $T^3/N^2 \approx 7.3$, so sporadic triple collisions at the single-digit level are consistent with the model. The fourth factorial moment $F_4 = 0$ at all tested sizes (no row has $\lambda \ge 4$).

12.25 Gap polynomial sparsity

For the Apéry recurrence, the gap polynomial N_h (whose roots in $\mathbb{F}_p$ are necessary for a simultaneous zero pair $b_x \equiv b_{x+h} \equiv 0$) has $r_p(h) = \#\{x \in \mathbb{F}_p : N_h(x) = 0\} \le 3(h-1)$, giving $\sum_{h<H} r_p(h) \le 3H^2/2$ and hence $Z(p) \le Cp^{2/3}$. However, the actual root counts are drastically smaller: for all primes $p \le 10{,}000$ with $Z(p) \ge 2$ and all gaps $h \le 100$, $\sum_{h\le 100} r_p(h) \le 1$, with most primes having $\sum = 0$ (the zero pairs are spaced more than 100 apart).

Hypothesis 145 (Linear root sum)**.** *$\sum_{1\le h<H} r_p(h) \ll H$ for every prime p and every $H \geq 1$.*

This would improve the zero-count to $Z(p) \ll p^{1/2}$, strengthening the exponent in Proposition 94 from $N^{5/6}$ to $N^{3/4}$. The heuristic justification is that N_h has degree $m_h = 3(h-1)/2$ in the primitive centred variable, and (conjecturally) $\mathrm{Gal}(N_h^{\mathrm{prim}}/\mathbb{Q}) = S_{m_h}$, so that by the Chebotarev density theorem $\mathbb{E}[r_p(h)] = 1$ for each h. For $h = 2,3,4,5$, modular factorization certificates confirm $\mathrm{Gal} = S_{m_h}$ with $(m_2, m_3, m_4, m_5) = (1,3,4,6)$, and the full hyperoctahedral groups $C_2 \wr S_{m_h}$ are verified via the single-flip lemma (the signed-group kernel is detected by a single pair flip in Frobenius factorization).

12.26 The exact random model

The palindromic identity $b_{p-1-j} \equiv b_j \pmod p$ forces the zero-set to consist of reflected pairs. Let V_p^+ denote the space of inversion-invariant functions $f\colon \mathbb{F}_p^\times \to \mathbb{F}_p$, i.e. $f(t) = f(t^{-1})$. The algebraic discrete Fourier transform $\hat f(j) = \sum_{t\in\mathbb{F}_p^\times} f(t)\, t^{-j}$ maps V_p^+ isomorphically onto the subspace of sequences satisfying $\hat f(j) = \hat f(-j)$.

Proposition 146 (Random inversion-invariant model)**.** *Choose f uniformly from V_p^+. Then the number K of noncentral reflected pairs $\{j, p-1-j\}$ with $\hat f(j) = 0$ satisfies*

$$K \;\sim\; \mathrm{Binomial}\!\left(\frac{p-3}{2},\, \frac{1}{p}\right) \xrightarrow{p\to\infty} \mathrm{Poisson}(1/2).$$

Proof. The spectral coefficients of distinct reflected pairs are independent uniform elements of $\mathbb{F}_p$ (the DFT is invertible), so each vanishes with probability $1/p$. □

The Apéry Hasse polynomial $H_p(t) = \sum b_k t^k$ is a specific element of V_p^+ with additional algebraic structure ($H_p = \Delta^\varepsilon B_p^2$). The conjecture that $Z(p) \sim 2\,\mathrm{Poisson}(1/2)$ is equivalent to asserting that H_p behaves, for the spectral-zero statistic, like a generic element of V_p^+.

The square structure can be incorporated directly. Let $V_p^- = \{g\colon \mathbb{F}_p^\times \to \mathbb{F}_p : g(t^{-1}) = -g(t)\}$ denote the anti-invariant subspace, with $\dim V_p^- = (p-3)/2$. For uniform $g \in V_p^-$, the function $f = g^2$ lies in V_p^+ and vanishes at the two fixed points $t = \pm 1$.

Proposition 147 (Random anti-invariant square model)**.** *Choose g uniformly from V_p^- and put $f = g^2$. For any fixed distinct noncentral reflected pairs represented by $j_1, \ldots, j_r$,*

$$\Pr\bigl(\hat f(j_1) = \cdots = \hat f(j_r) = 0\bigr) \;=\; p^{-r} + O_r(p^{-c_r p})$$

for some $c_r > 0$. In particular, the pair-zero count $K^{\mathrm{sq}} \Rightarrow \mathrm{Poisson}(1/2)$.

Proof. Write $u_i = g(x_i)$ for orbit representatives. The spectral coefficient $\hat f(j) = \sum_i u_i^2 (x_i^j + x_i^{-j})$ is a diagonal quadratic form $Q_\lambda(u)$ in the independent uniform variables u_i. For $\lambda \neq 0$, the physical function $h_\lambda(t) = \sum_a \lambda_a (t^{j_a} + t^{-j_a})$ has Mellin support $\le 2r$, so by the finite-field Mattson–Solomon uncertainty inequality, at least $(p-1)/(4r) - 1$ orbit values are nonzero, each contributing a quadratic Gauss sum of absolute value $p^{-1/2}$. The factorial-moment method then gives the result. □

Multiplication by $\Delta(t)^\varepsilon$ affects only $O(1)$ orbits and does not change the limiting law. Thus the Sym^2 squareness of the Apéry Hasse polynomial is fully compatible with Poisson(1/2) in any random model where the square root B_p behaves generically.

Proposition 148 (Quadratic rank and both halves of the spectrum)**.** *Let $B(t)$ be a uniformly random normalized reciprocal polynomial of degree $d = (p-1-2\varepsilon)/2$ over $\mathbb{F}_p$, with free variables $x_1, \ldots, x_m$ where $m = \lfloor d/2 \rfloor$. Put $H = \Delta^\varepsilon B^2$ and split the noncentral reflected-pair representatives into a* triangular *range $\mathcal{J}_{\mathrm{tri}}$ ($|\mathcal{J}_{\mathrm{tri}}| = m$) and a* nonlinear *range $\mathcal{J}_{\mathrm{nl}}$ ($|\mathcal{J}_{\mathrm{nl}}| = p/4 + O(1)$).*

(i) *In the triangular range, H_j is an affine-linear function of $(x_1, \ldots, x_m)$ with a unitriangular Jacobian. Hence $K_{\mathrm{tri}} = \#\{j \in \mathcal{J}_{\mathrm{tri}} : H_j = 0\} \Rightarrow \mathrm{Poisson}(1/4)$.*

(ii) *In the nonlinear range, each H_j is an affine quadratic form in $(x_1, \ldots, x_m)$ whose quadratic part has rank $\geq m/2 - O(1) \gg p$. By the Gauss-sum bound for affine quadratic forms of rank R: $\Pr(F(x) = 0) = 1/p + O(p^{-R/2})$. The pencil-rank estimate from Proposition 147 gives $K_{\mathrm{nl}} \Rightarrow \mathrm{Poisson}(1/4)$.*

(iii) *Without requiring independence between the two halves,*

$$\mathbf{E}\, Z(p) \;=\; 2\,\mathbf{E}\, K \;=\; 2\big(\mathbf{E}\, K_{\mathrm{tri}} + \mathbf{E}\, K_{\mathrm{nl}}\big) \;=\; 2\left(\frac{1}{4} + \frac{1}{4}\right) + o(1) \;=\; 1 + o(1).$$

If the mixed pencil-rank condition holds—i.e., every nonzero $\mathbb{F}_p$-linear combination of $H_{j_1}, \ldots, H_{j_s}$ with at least one $j_\nu \in \mathcal{J}_{\mathrm{nl}}$ has quadratic rank $\gg p/s$—then the full pair-zero count $K = K_{\mathrm{tri}} + K_{\mathrm{nl}} \Rightarrow$ Poisson$(1/2)$.

Remark 149 (Separation of random model from deterministic problem). Proposition 148 applies to the random reciprocal-polynomial model, not to the actual Apéry sequence. The actual B_p supplies one deterministic coefficient vector for each prime; converting the random-model statement into a theorem about $\sum_{p \leq x} Z(p)$ requires an arithmetic anti-concentration theorem asserting that these specific vectors avoid the coefficient quadrics with approximately random frequency. No such theorem is currently known.

12.27 Shift-correlation computation

Define the shift-correlation exponential sum

$$C_p(a) \;=\; \sum_{r=0}^{p-2} e_p(a\,(b_{r+1} - b_r))\,, \qquad a \neq 0.$$

A nontrivial bound $|C_p(a)| \ll p^{1-\eta}$ is the missing input for a Bombieri–Fouvry-type dispersion argument (cf. Section 12.5).

For all 666 primes $p \leq 5000$, we computed $\max_{a \neq 0} |C_p(a)|/\sqrt{p}$. The results:

- $\max_a |C_p(a)|/\sqrt{p}$ has mean 3.35 and maximum 5.07;
- $\max_a |C_p(a)|/\sqrt{p \log p}$ has mean 1.23 and maximum 1.80;
- the distribution of $\max/\sqrt{p}$ matches the Gumbel law expected for the maximum of p independent unit-variance Gaussians: $\max \approx \sqrt{2 \log p}$.

For shifts $s = 2, 3, 5$, the same behavior holds. The histogram of the differences $b_{r+1} - b_r \pmod p$ has std/mean ≈ 1 and the fraction of zero-bins is $\approx 1/e$, precisely the Poisson(1) model.

Conclusion: the Apéry sequence $b_r \bmod p$ is computationally indistinguishable from a random function $\{0, \ldots, p-1\} \to \mathbb{F}_p$ in its additive exponential-sum statistics. This provides strong computational evidence for the shift-correlation bound required by the dispersion approach.

12.28 Value-collision energy

A second pseudorandomness diagnostic is the *value-collision energy*

$$E(p) \;=\; \sum_{a\in\mathbb{F}_p} N_p(a)^2, \qquad N_p(a) = \#\{j \in \{0,\dots,p-1\} : b_j \equiv a \pmod p\}.$$

By Cauchy–Schwarz, $Z(p)^2 = N_p(0)^2 \le E(p)$, so $E(p) \ll p$ would imply $Z(p) \ll \sqrt{p}$.

For a uniformly random function $\{0,\dots,p-1\} \to \mathbb{F}_p$, each $N_p(a) \sim \text{Poisson}(1)$ and $E(p)/p \to 2$. The palindromic symmetry $b_{p-1-j} \equiv b_j$ forces the half-sequence values to be doubled, giving $E(p)/p \to 3$: each half-position contributes independently to $N_p(a)$, and the doubling multiplies the second moment by 4 while the mean remains 1, yielding $\mathbb{E}[N_p(a)^2] = 4\big(\frac{1}{2}+\frac{1}{4}\big) = 3$.

We computed $E(p)/p$ for all 9590 primes $p \le 100{,}000$. The results confirm the palindromic prediction: $\mathrm{mean}(E(p)/p) = 2.9995$, with 99% of primes in $[2.9, 3.1]$ and no dependence on $Z(p)$. The value-collision energy thus confirms that the Apéry sequence modulo p is anti-concentrated at the scale predicted by the palindromic random model.

Remark 150 (Second Fourier layer). By additive orthogonality,

$$E(p) = \frac{1}{p}\sum_{u\in\mathbb{F}_p} |S_p(u)|^2, \qquad S_p(u) = \sum_{j=0}^{p-2} e_p(u\, b_j).$$

A square-root bound $|S_p(u)| \ll \sqrt{p}$ for $u \ne 0$ would give $E(p) \ll p$, hence $Z(p) \ll \sqrt{p}$. This is a *second Fourier layer*: the first (multiplicative Mellin) transform produces the coefficients b_j; the second (additive) transform probes their residual distribution in $\mathbb{F}_p$. Ordinary monodromy of the K3 local system controls the first layer. It does not automatically construct a bounded-conductor object for the second layer. The modular form $f_4 = \eta(2z)^4\eta(4z)^4 \in S_4(\Gamma_0(8))$ controls only the central coefficient $b_{(p-1)/2} \equiv a_p(f_4) \pmod p$ via the quadratic Kummer twist; generic characters t^{-j} with $\mathrm{ord}(\chi_{p,j}) \to \infty$ do not correspond to any fixed modular form. A residual anti-concentration theorem, proving the second-layer bound $E(p) \ll p$ or equivalently the additive exponential-sum bound above, would require genuinely new same-characteristic arithmetic geometry.

12.29 The high-moment criterion

The dispersion hypothesis (Hypothesis 93) is a second-moment statement localized in short windows. An alternative sufficient condition—global in the integer variable, but requiring a single fixed higher moment—is the following.

For a dyadic block of primes $(X, 2X]$ and an integer m, define

$$K_X(m) \;=\; \#\big\{p \in (X,2X] :\; m \bmod p \in Z_p\big\}, \qquad \lambda_X \;=\; \sum_{X<p\le 2X} \frac{Z(p)}{p},$$

so that $\lambda_X \ll X^{2/3}/\log X$ by Proposition 9. Write $(K)_k = K(K-1)\cdots(K-k+1)$ for the falling factorial.

Hypothesis 151 (High-moment bound $(\mathrm{HM})_k$)**.** *For one fixed integer $k \ge 3$, uniformly over dyadic X,*

$$\sum_{0\le m<X^2} \big(K_X(m)\big)_k \;\ll\; X^{2+o(1)}\,\lambda_X^k.$$

Proposition 152 (The case $k = 2$ holds unconditionally)**.** $\sum_{0 \le m < X^2} \big(K_X(m)\big)_2 \;\le\; 5\,X^2\lambda_X^2.$

Proof. Expanding the falling factorial over ordered pairs of distinct primes,

$$\sum_{m<X^2} \big(K_X(m)\big)_2 = \sum_{\substack{p,q \in (X,2X] \\ p \neq q}} \; \sum_{\substack{r \in Z_p \\ s \in Z_q}} \#\big\{m < X^2 : m \equiv r\ (p),\ m \equiv s\ (q)\big\}.$$

By the Chinese remainder theorem the inner count is at most $X^2/(pq) + 1$, and since $pq \le 4X^2$ the additive 1 is at most $4X^2/(pq)$, so the count is at most $5X^2/(pq)$. Summing,

$$\sum_{m<X^2} \big(K_X(m)\big)_2 \le 5X^2 \sum_{\substack{p,q \in (X,2X] \\ p \neq q}} \frac{Z(p)Z(q)}{pq} \le 5X^2\lambda_X^2. \qquad \square$$

Remark 153 (Where the frontier begins). For $k \ge 3$, distinct primes have $p_1 \cdots p_k > X^k > X^2$, so each residue tuple $(r_1, \ldots, r_k) \in Z_{p_1} \times \cdots \times Z_{p_k}$ has *at most one* CRT representative below X^2, and the trivial bound on the k-th moment is the total tuple count $\asymp (X\lambda_X)^k$, which exceeds the conjectured $X^2\lambda_X^k$ by the factor X^{k-2}. Hypothesis $(\mathrm{HM})_k$ thus asserts that only the expected proportion $\asymp X^2/(p_1 \cdots p_k)$ of the actual Apéry zero tuples admit a representative in $[0, X^2)$ — a genuine cross-prime equidistribution statement, strictly beyond Proposition 152, yet far weaker than the Poisson conjecture (Hypothesis 58).

Remark 154 (Computational evidence). Using the actual Apéry zero sets Z_p for all primes $p \in (X, 2X]$, the moment ratios $R_k = \sum_{m<X^2} (K_X(m))_k \,/\, X^2\lambda_X^k$ are bounded and of order one at every tested scale:

X	$\#p$	λ_X	$\max K$	R_2	R_3	R_4
128	23	0.158	3	0.91	0.93	0.00
256	43	0.126	3	0.89	0.46	0.00
512	75	0.093	3	0.97	0.72	0.00
1024	137	0.082	4	0.97	0.83	0.50
2048	255	0.078	4	0.98	0.93	1.27

The $k = 2$ ratio approaches 1 from below (the independence prediction), and R_3, R_4 fluctuate at order one—the expected count $X^2\lambda_X^4$ of 4-clusters is single-digit at these scales, so R_4 is dominated by extreme-value sampling noise. The observed $\max K = 4$ over $X^2 \approx 4 \times 10^6$ integers matches the Poisson extreme-value prediction. Note $\lambda_X \approx 1/\log X$ here (the empirical Poisson mean), far below the worst-case input $X^{2/3}/\log X$ used in Theorem 155.

Theorem 155 ($(\mathrm{HM})_k$ implies a pointwise power saving)**.** *Assume Hypothesis 151 for one fixed $k \ge 3$. Then for* every *n,*

$$\log G_n \;\ll\; n^{2/3+2/k+o(1)}.$$

In particular $k = 8$ gives $\log G_n \ll n^{11/12+o(1)}$, and any fixed $k > 6$ proves the full conjecture $G_n = e^{o(n)}$ for all n.

Proof. Let $M_X = \max_{0 \le m < X^2} K_X(m)$. If $M_X \ge 2k$, then $(M_X)_k \ge (M_X/2)^k$; taking the maximizing m inside the hypothesis,

$$(M_X/2)^k \;\le\; \sum_{m<X^2} \big(K_X(m)\big)_k \;\ll\; X^{2+o(1)}\lambda_X^k.$$

In either case ($M_X < 2k$ or not),

$$M_X \ll X^{2/k+o(1)}\lambda_X + 1 \ll \frac{X^{2/3+2/k+o(1)}}{\log X}.$$

Now fix n. The primes in $(\sqrt{n},\, 2\sqrt{n}]$ contribute at most $\pi(2\sqrt{n})\log(2\sqrt{n}) = O(\sqrt{n})$ trivially. The remaining range $(2\sqrt{n},\, n]$ is covered by dyadic blocks $(X, 2X]$ with $X = 2^j \le n$; every block meeting $(2\sqrt{n},\, n]$ has $2X > 2\sqrt{n}$, hence $X > \sqrt{n}$ and therefore $n < X^2$. Hence $m = n$ is admissible in the maximum, and the block $(X, 2X]$ contains at most M_X lower-digit bad primes, each of weight $\log p \le \log 2X$. Therefore

$$\sum_{\substack{\sqrt{n}<p\le 2n \\ p \mid b_{n \bmod p}}} \log p \ll \sqrt{n} + \sum_{\substack{X=2^j \\ \sqrt{n}<X\le n}} X^{2/3+2/k+o(1)} \ll n^{2/3+2/k+o(1)},$$

the geometric sum being dominated by its top block. Adding the small-prime bound (Proposition 3), the companion-block bound (Proposition 73), and reducing to the lower-digit channel by Lemma 69(iii), the claim follows. For $k > 6$ the exponent satisfies $2/3 + 2/k < 1$, so $\log G_n = o(n)$. □

Remark 156. Among all sufficient conditions collected in this paper (Hypothesis Z, $\bar{Z}$, Poisson, AP–BDH, AMTD), Hypothesis $(\mathrm{HM})_8$ appears to be the weakest that still closes the full conjecture: it requires no pointwise information about any single prime, no short-window localization, and no variance identity—only one fixed factorial moment at the independence scale, compatible with the worst proved cardinality bound $Z(p) \ll p^{2/3}$.

The next two statements make the $(\mathrm{HM})_3$ frontier exact: a sharp reformulation as a positive pair–Palm excess, and an incidence model showing that *all* presently proved primewise and pairwise facts—including the fiber and energy bounds of Section 5.4 and even a uniform $O(1)$ row-codegree—are jointly insufficient for any power saving at $k = 3$. (In this block, Ω_p temporarily denotes the bad set $\{0 \le m < X^2 : m \bmod p \in \mathcal{Z}_p\}$.)

The third factorial moment: exact obstruction

Write

$$\mathcal{P}_X = \{p \text{ prime} : X < p \le 2X\}, \qquad M = X^2,$$

where $X \ge 2$ is integral, and put

$$\Omega_p = \{0 \le m < M : m \bmod p \in \mathcal{Z}_p\}, \qquad f_p(m) = \mathbf{1}_{\Omega_p}(m).$$

Thus $K_X(m) = \sum_{p\in\mathcal{P}_X} f_p(m)$. Set

$$A_p = |\Omega_p|, \qquad \rho_p = \frac{A_p}{M}, \qquad \mu = \sum_{p\in\mathcal{P}_X} \rho_p = \frac{1}{M}\sum_{m<M} K_X(m),$$

and retain

$$\lambda = \lambda_X = \sum_{p\in\mathcal{P}_X} \frac{Z(p)}{p}.$$

For ordered distinct primes $p, q \in \mathcal{P}_X$, define

$$J_{p,q} = |\Omega_p \cap \Omega_q|,$$
$$T_{p,q} = \sum_{m \in \Omega_p \cap \Omega_q} \sum_{\ell \in \mathcal{P}_X \setminus \{p,q\}} f_\ell(m),$$
$$\lambda_{p,q} = \sum_{\ell \in \mathcal{P}_X \setminus \{p,q\}} \frac{Z(\ell)}{\ell}, \qquad \mu_{p,q} = \sum_{\ell \in \mathcal{P}_X \setminus \{p,q\}} \rho_\ell.$$

The two aggregate positive pair–Palm excesses are

$$\mathscr{P}_3(X) = \sum_{p \neq q} \big(T_{p,q} - J_{p,q}\lambda_{p,q}\big)_+, \tag{192}$$
$$\mathscr{P}_3^{\mathrm{fin}}(X) = \sum_{p \neq q} \big(T_{p,q} - J_{p,q}\mu_{p,q}\big)_+. \tag{193}$$

The first uses the complete-period densities $Z(\ell)/\ell$; the second uses the exact marginals on $[0, M)$ and therefore absorbs every one-column boundary error.

Theorem 157 (Sharp pair–Palm characterization of $(\mathrm{HM})_3$)**.** *With all prime tuples ordered, one has the exact identities*

$$\sum_{p \neq q} J_{p,q} = \sum_{m<M} (K_X(m))_2, \tag{194}$$
$$\sum_{p \neq q} T_{p,q} = \sum_{m<M} (K_X(m))_3. \tag{195}$$

Moreover,

$$\mathscr{P}_3(X) \leq \sum_{m<M} (K_X(m))_3 \leq \mathscr{P}_3(X) + 5M\lambda^3, \tag{196}$$
$$\mathscr{P}_3^{\mathrm{fin}}(X) \leq \sum_{m<M} (K_X(m))_3 \leq \mathscr{P}_3^{\mathrm{fin}}(X) + 5(1 + 2/X)M\lambda^3. \tag{197}$$

Consequently $(\mathrm{HM})_3$ *is equivalent to either one of*

$$\mathscr{P}_3(X) \ll X^{2+o(1)}\lambda_X^3, \qquad \mathscr{P}_3^{\mathrm{fin}}(X) \ll X^{2+o(1)}\lambda_X^3.$$

More generally, for every fixed $\theta \geq 0$, the estimate in Goal G2 with exponent θ is equivalent to either positive-excess estimate with right-hand side $X^{2+\theta+o(1)}\lambda_X^3$.

There is also an exact centered formulation. Define

$$V_X = \sum_{m<M} (K_X(m) - \mu)^2, \qquad C_{3,X} = \sum_{m<M} (K_X(m) - \mu)^3$$

and the third factorial-moment defect

$$\mathfrak{D}_3(X) = (C_{3,X} - M\mu) + 3(\mu - 1)(V_X - M\mu). \tag{198}$$

Then

$$\sum_{m<M} (K_X(m))_3 = M\mu^3 + \mathfrak{D}_3(X). \tag{199}$$

Thus $(\mathrm{HM})_3$ *is equivalently the one-sided estimate*

$$\big(\mathfrak{D}_3(X)\big)_+ \ll X^{2+o(1)}\lambda_X^3.$$

The same equivalence holds at every G2 exponent $\theta \geq 0$.

Proof. If $p \neq q$ lie in $\mathcal{P}_X$, then $pq > M$. Hence every residue pair $(a \bmod p, b \bmod q)$ has at most one representative in $[0, M)$. This removes, rather than estimates, the usual CRT boundary term. At an integer with $K_X(m) = k$, there are exactly $(k)_2$ ordered choices of (p, q), and each has $k - 2$ choices of a third prime. Its contribution to $\sum_{p \neq q} T_{p,q}$ is therefore $(k)_2(k-2) = (k)_3$. This proves (194)–(195); in particular, no factor 3! is missing.

For nonnegative A, B,

$$(A - B)_+ \leq A \leq B + (A - B)_+.$$

Apply this with $A = T_{p,q}$ and $B = J_{p,q}\lambda_{p,q}$, sum over ordered pairs, and use $\lambda_{p,q} \leq \lambda$ together with the proved second factorial moment:

$$\sum_{p \neq q} J_{p,q}\lambda_{p,q} \leq \lambda \sum_{m<M} (K_X(m))_2 \leq 5M\lambda^3.$$

This proves (196).

For each p and each residue in $\mathcal{Z}_p$, the number of its representatives below M differs from M/p by less than one. Therefore

$$|A_p - MZ(p)/p| \leq Z(p), \qquad |\mu - \lambda| \leq \frac{1}{M} \sum_p Z(p) \leq \frac{2\lambda}{X},$$

where the final inequality uses $p \leq 2X$. In particular $\mu \leq (1 + 2/X)\lambda$. Applying the same positive-part inequality with $B = J_{p,q}\mu_{p,q}$ gives

$$\sum_{p \neq q} J_{p,q}\mu_{p,q} \leq \mu \sum_{m<M} (K_X(m))_2 \leq 5(1 + 2/X)M\lambda^3,$$

which proves (197). Each equivalence follows in both directions from its sandwich. This is a characterization, not a new estimate hidden in a renamed hypothesis: the unproved arithmetic is exactly the positive conditional overload of third primes on the actual two-prime CRT loci.

Finally write $Y_m = K_X(m) - \mu$, so that $\sum_m Y_m = 0$. Expanding $K^3 - 3K^2 + 2K$ gives

$$\sum_{m<M} (K_X(m))_3 = C_{3,X} + 3(\mu - 1)V_X + M(\mu^3 - 3\mu^2 + 2\mu),$$

which rearranges to (199). Since the main term is nonnegative,

$$(\mathfrak{D}_3(X))_+ \leq \sum_{m<M} (K_X(m))_3 \leq M\mu^3 + (\mathfrak{D}_3(X))_+.$$

Equation (12.29) bounds $M\mu^3$ by a constant multiple of $M\lambda^3$, proving the last equivalences. When $\lambda = 0$ all zero sets, all pair loci, and all displayed quantities vanish. The same is true of the Palm sums when there is no pair locus, so no division by a possibly zero parameter has been suppressed. ☐

Why the available primewise and pairwise tools do not imply G2

Proposition 158 (An anchored rainbow star with bounded codegrees)**.** *For every sufficiently large integral X there is a family $(\mathcal{Z}_p^*)_{p \in \mathcal{P}_X}$ with the following properties. Writing*

$$K_X^*(m) = \sum_{p \in \mathcal{P}_X} \mathbf{1}_{\{m \bmod p \in \mathcal{Z}_p^*\}}, \qquad \lambda_X^* = \sum_{p \in \mathcal{P}_X} \frac{|\mathcal{Z}_p^*|}{p},$$

$$\mu^* = X^{-2} \sum_{m<X^2} K_X^*(m),$$

one has:

1. *Every $\mathcal{Z}_p^*$ has size zero or two. Every nonempty set is reflection-invariant, has no consecutive elements, avoids $0, 1, p-2, p-1$, and satisfies*

$$|\mathcal{Z}_p^* \cap I| \le 2|I|^{2/3}$$

for every nonempty interval $I \subseteq \{0, \dots, p-1\}$.

2. *The universal CRT bound*

$$\sum_{m<X^2} (K_X^*(m))_2 \le 5X^2(\lambda_X^*)^2$$

holds, and even the global centered second moment has the natural scale

$$\sum_{m<X^2} (K_X^*(m) - \mu^*)^2 \ll X^2\lambda_X^*.$$

3. *If $\mathcal{P}^*(m) = \{p : m \bmod p \in \mathcal{Z}_p^*\}$, then for all distinct $0 \le m, n < X^2$,*

$$|\mathcal{P}^*(m) \cap \mathcal{P}^*(n)| \le 7.$$

Thus even a uniform $O(1)$ row-codegree bound does not prevent the model.

4. *Nevertheless,*

$$\lambda_X^* \asymp \frac{1}{\log X}, \qquad \sum_{m<X^2} (K_X^*(m))_3 \gg \frac{X^3}{(\log X)^3}.$$

Consequently the family violates every G2 estimate with fixed $\theta < 1$.

The family can additionally be extended to a reflection-invariant value map whose largest fiber is at most two and whose collision energy is at most $2p$. Hence the numerical consequences of the bordered nonvanishing theorem—the $O(p^{3/4})$ fiber bound and $O(p^{7/4})$ energy bound—also do not rule it out.

If one uses the gap-polynomial reflection identity $N_h(-x-h-1) = (-1)^{h-1}N_h(x)$, the unique pair in every nonempty $\mathcal{Z}_p^*$ also satisfies its corresponding necessary gap certificate modulo p.

Proof. Put $M = X^2$, choose $m_0 = \lfloor 3M/4 \rfloor$, and let $d = 2m_0 + 1$. Delete from $\mathcal{P}_X$ the primes dividing

$$E = (m_0 - 1)m_0(m_0 + 1)(m_0 + 2)d.$$

Only $O(1)$ primes are deleted: $|E| \ll X^{10}$ and every deleted prime exceeds X. For every remaining prime define

$$a_p = m_0 \bmod p, \qquad \mathcal{Z}_p^* = \{a_p, p - 1 - a_p\},$$

and put $\mathcal{Z}_p^* = \varnothing$ for deleted primes. The deletion makes the two displayed residues distinct and excludes the four endpoint residues. Their difference is a nonzero even integer, so they are not consecutive. Reflection is built in, and the interval estimate is immediate because an interval contains at most two marked residues.

Let L be the number of nonempty columns. The prime number theorem and Mertens' theorem give

$$L \sim \frac{X}{\log X}, \qquad \lambda_X^* = 2 \sum_{\substack{X<p\le 2X \\ p \nmid E}} \frac{1}{p} \asymp \frac{1}{\log X}.$$

Every active prime marks m_0, so $K^*_X(m_0) = L$ and

$$\sum_{m<M} (K^*_X(m))_3 \geq (L)_3 \gg \frac{X^3}{(\log X)^3}.$$

Dividing by $X^{2+\theta+o(1)}(\lambda^*_X)^3$ leaves $X^{1-\theta-o(1)}$, which tends to infinity for $\theta < 1$. The second-factorial-moment estimate is universal and follows by the same CRT argument as Proposition 152. Also

$$\begin{aligned}\sum_m (K^*_X(m) - \mu^*)^2 &\leq \sum_m (K^*_X(m))^2 \\ &= \sum_m K^*_X(m) + \sum_m (K^*_X(m))_2 \\ &\ll M\lambda^*_X + M(\lambda^*_X)^2 \ll M\lambda^*_X.\end{aligned}$$

It remains to check the codegree. Fix $0 \leq m < n < M$ and put $h = n - m$. If an active prime marks both integers, their two residues are chosen from a_p and $-1 - a_p$. Equal choices give $p \mid h$; different choices give

$$h \equiv \pm(2a_p + 1) \equiv \pm d \pmod p.$$

Thus every common prime divides the nonzero integer

$$H = h(h - d)(h + d).$$

Here $0 < h < M < d$, and $|H| < 8M^3 = 8X^6$. If there are k common primes, their product exceeds X^k and divides $|H|$, whence

$$k \leq \frac{\log(8X^6)}{\log X} \leq 7$$

for sufficiently large X.

For the value-map assertion, partition the residues into the orbits of $r \mapsto p - 1 - r$. At an active prime, assign zero to the marked orbit and distinct nonzero values to all other orbits. At a deleted prime, assign distinct nonzero values to every orbit; this is possible because there are $(p + 1)/2 \leq p - 1$ orbits. Thus the zero fiber is exactly $\mathcal{Z}^*_p$ in both cases. Every fiber has size at most two, and, since $t^2 \leq 2t$ for $t \in \{0, 1, 2\}$, the fiber energy is at most $2p$.

For completeness, the gap-polynomial reflection identity used here is symbolic. For $h \geq 2$, let $D_h(x)$ be the symmetric $(h - 1) \times (h - 1)$ tridiagonal matrix with diagonal entries $P(x + i)$, $1 \leq i < h$, and off-diagonal entries $(x + i + 1)^3$, $1 \leq i < h - 1$. Expansion in the last row and the defining recurrence for N_h give $\det D_h(x) = N_h(x)$. If R is the matrix that reverses the coordinate order, then $P(-v - 1) = -P(v)$ gives, entry by entry,

$$D_h(-x - h - 1) = -RD_h(x)R.$$

Consequently

$$N_h(-x - h - 1) = (-1)^{h-1} N_h(x). \tag{200}$$

(For $h = 1$ this is immediate from $N_1 = 1$.) This proves the identity for every h, without numerical verification.

Finally let $u = \min(a_p, p - 1 - a_p)$ and $h_p = p - 1 - 2u$. The deleted endpoints ensure $1 \leq u < u + h_p < p$; moreover h_p is positive and even. Since

$$-u - h_p - 1 = u - p \equiv u \pmod p,$$

the gap-polynomial reflection identity gives $N_{h_p}(u) \equiv -N_{h_p}(u) \pmod p$, hence $N_{h_p}(u) \equiv 0 \pmod p$ for odd p. $\square$

Remark 159 (Scope of the obstruction). The preceding family is an incidence model, not the actual zero family of the Apéry recurrence. It proves that reflection, no consecutive zeros, the interval $O(|I|^{2/3})$ bound, the universal CRT second moment, the primewise fiber/energy bounds, the gap-polynomial pair certificates, and even uniform $O(1)$ row codegree do not imply any G2 power saving. The missing arithmetic must constrain, across distinct primes, which of the structurally permitted reflection orbits is selected as the zero fiber. The pair–Palm quantity in Theorem 157 is the minimal aggregate positive overload that measures exactly this selection problem.

Computational findings

The direct CRT enumerator in `hm3_explore.py`, independently cross-checked against an m-scatter, gives the following ordered third factorial moments:

X	$\sum_{m<X^2}(K_X(m))_3$	$X^2\lambda_X^3$	R_3
256	60	131.989926	0.454580146
512	150	209.308570	0.716645284
1024	486	583.330960	0.833146247
2048	1824	1953.964920	0.933486564
4096	7218	7608.237134	0.948708600

At $X = 4096$, the 1203 canonical prime triples are supported on 1161 integers; 1147 have $K_X = 3$ and 14 have $K_X = 4$. The conditional third-prime extension rate per unordered pair hit is 0.073422, close to $\lambda_X = 0.0768283$. Quotient repetitions, reflection-orbit pileups, and residue correlations show no visible clustering. Full histograms and reproduction details are in `hm3_exploration.md`. These data are evidence for $(\mathrm{HM})_3$, not a proof.

Stall report and infrastructure audit

Goals G1 and G2 are not proved here, and no G3 hypothesis is claimed. In particular, a third-order Palm condition should not be advertised as "weaker than AP–BDH": the latter is an L^2 condition and the two statements are not ordered without extra hypotheses. The completed deliverable is G4, Theorem 157, together with the sharpness obstruction above.

Two issues in the supplied route are material to this conclusion. First, Lucas gives only

$$m \bmod p \in \mathcal{Z}_p \quad \Longrightarrow \quad p \mid b_m.$$

The converse can fail through the leading digit. For example, $b_2 = 73$; with $X = 64$, $p = 73$, and $m = 146 < X^2$, Lucas gives $b_{146} \equiv b_2 b_0 \equiv 0 \pmod{73}$, while $m \bmod 73 = 0 \notin \mathcal{Z}_{73}$. Hence $K_X(m) \le \omega_{(X,2X]}(b_m)$, but equality and the corresponding factorial-moment identity asserted in the specification are false.

Second, the written proof of Lemma 76 starts with the unavailable assertion $p > N/2 \ge h$ and then says that the two residues "differ by exactly h". Quotient wrap invalidates that sentence even under the displayed extra inequalities: for $p = 37$ one has $\mathcal{Z}_{37} = \{17, 19\}$, and $m = 93$, $n = 128$, $N = 70$ give $37 > N/2 \ge n-m$ while the residues are 19 and 17. The lemma may be repairable by a separate wrap argument using reflection, but the proof as written cannot be used as an established HM3 input. The anchored-star proposition shows that even a repaired, stronger $O(1)$ row-codegree estimate would still not control a one-row rainbow pileup.

Finally, Proposition 139 in the second-moment section states $\|\rho\|_2^2 \le T_N^2/N$, whereas Cauchy–Schwarz gives the reverse inequality and the proposition's own table reports a ratio near 1.25. No step above uses that proposition.

The trivial tuple estimate used for comparison should also be read as the uniform inequality

$$\sum_{m<X^2} (K_X(m))_3 \le \Big(\sum_{X<p\le 2X} Z(p) \Big)^3 \le 8X^3 \lambda_X^3.$$

It need not be asymptotic to $(2X\lambda_X)^3$ when the zero mass is concentrated on fewer than three primes.

13 Conclusion

The principal unconditional result is the quantitative exceptional-set theorem: for every fixed $\varepsilon > 0$,

$$\#\{n \le X : \log G_n > \varepsilon\, n\} \ \ll_\varepsilon \ (\log X)^2.$$

Consequently $G_n = e^{o(n)}$ on a set of natural density one; the exceptional set has upper Banach density zero and finite harmonic weight. The arithmetic engine is the uniform interval estimate $Z(p) \ll p^{2/3}$, proved by the gap-polynomial and separated-block method.

The all-n theorem reduces to the block second-factorial-moment estimate

$$F_N \ \ll \ T_N + \frac{T_N^2}{N}, \tag{201}$$

which asserts that prime columns in the top block do not systematically align at one integer. The profile decomposition of F_N separates this into the deterministic profile energy $\|\rho\|^2 \sim c_2\, T^2/N$ (with c_2 given exactly by (188)), a cross-term $|\langle \rho, \eta\rangle| \le \rho_{\max} T = O(T^2/N)$, and the residual shot-noise term $\|\eta\|^2$. Since $T^2/N = O(T)$, the estimate (201) is equivalent to the shot-noise bound $\|\eta\|^2 = O(T)$, and computation through $N = 262{,}144$ gives $\|\eta\|^2/T \approx 1.0$.

This gap is genuine. All classical sieves (Selberg, Brun, Rosser–Iwaniec, GPY) are circular for (201): they expand into the same intersection counts $A_{pq}(N)$ that the estimate controls, and their quadratic-form optimization requires the two-prime level of distribution as input (Remark 136). On the geometric side, the K3 lisse sheaf $R^2 f_* \mathbb{Q}_\ell$ has bounded rank and conductor, but the event $p \mid b_j$ is a same-characteristic crystalline condition. The Jacobi counterexample shows that a rank-one sheaf with the same formal properties can have $\gg p$ bad tame characters. The gap-polynomial identities are monochromatic: they control two incidences of the *same* prime and do not constrain the rainbow intersections that determine $\|\eta\|^2$.

Two precise sufficient criteria for Hypothesis Z emerge: a bounded-degree tame Hasse polynomial (giving $Z(p) \le 2D$), or a unique-unit-root theorem for all but $O(1)$ tame twists. The fiberwise Hasse–Witt invariant $A_p(t) = \sum_{n<p} b_n\, t^n$ is explicitly known from Stienstra–Beukers [25], but it controls which *fibers* are nonordinary, not which *Mellin coefficients* vanish; the character-twisted Hasse computation is a fundamentally different problem on the global twisted cohomology $H^1_{\mathrm{rig},c}(U, \mathcal{V}_{\mathrm{tr}} \otimes \mathcal{K}_{\omega^{-j}})$. The raw interpolation $j \mapsto b_j$ has degree $p-2$ already for $p = 11$ and 13; moreover, the Sym^2 factorization forces at least $(p{-}5)/2$ nonzero Fourier coefficients, giving a rigorous lower bound $D \ge (p{-}7)/4$ on any Laurent polynomial of degree D representing the unnormalized Mellin transform. Bounded tame Hasse degree would require a nontrivial Gauss/Jacobi normalization of the first Frobenius block.

Computation through all 78,496 primes $p \le 10^6$ gives $Z(p) \le 12$ and mean ≈ 1; the value-collision energy ratio $E(p)/p \to 3$ (confirmed through $p = 100{,}000$) and the profile constant $c_2 \to 1.262$ (through $N = 262{,}144$) match the palindromic null at every tested scale. These observations support all conjectures but are not used as proofs.

The prime-wise decomposition (Remark 143) provides a structural reduction: $\|\eta\|^2 = D + O$ where $D = \sum_p D_p \approx T$ is an explicit diagonal and $O = \sum_{p\neq q} C_{pq}$ is the cross-prime interaction. Under the independent fixed-cardinality residue model (each Z_p drawn uniformly from $\binom{\mathbb{F}_p}{Z(p)}$, independently across primes), one obtains the exact variance formula

$$\mathrm{Var}(\|\eta\|^2) \;=\; \sum_p \mathrm{Var}(D_p) \;+\; \sum_{p\neq q} \mathrm{Var}(C_{pq}),$$

whose first sum is $O(T)$ (fluctuation of the active tail mass Z_p^{act}) and whose second is $O(T^2/N)$ (CRT averaging in intervals of length $\sim N$ within period $pq \sim N^2$). The null-model conclusion $\|\eta\|^2/T \to 1$ in L^2 is therefore unconditional within the model.

However, CRT does not imply independence for *deterministic* zero-sets: distinct residue fields provide separate coordinates, not a product measure. The adversarial construction $Z_p = \{2N \bmod p\}$ for all p places every shifted zero at $m = 2N$, giving $O = \Theta(T^2)$; this is a single explicit moving-target choice with $|Z_p| = 1$ for all p, so the two-prime dispersion statement is *false* for general $O(1)$-size zero-sets. For the Apéry sequence, this pattern is impossible: the fixed-column $Z_p = \{c - p \bmod p\}$ forces $p \mid b_c$ for all p, contradicting $b_c \in \mathbb{Z} \setminus \{0\}$. More generally, the palindromic structure (Z_p consisting of reflected pairs $\{j_p, p{-}1{-}j_p\}$) and the Galois-group structure of the gap polynomials prevent systematic alignment of zero positions across primes. Nonetheless, the quantitative control—that the residual alignment is $o(T)$, not merely $o(T^2)$—is a stronger statement, and cannot be established by any purely black-box argument: Bombieri–Vinogradov, the large sieve, and Barban–Davenport–Halberstam address single-modulus discrepancies, not cross-prime collisions, and the spectral bilinear framework of Duke–Friedlander–Iwaniec (“bilinear forms with Kloosterman fractions”) requires a fixed numerator, which the moving-target zero-sets do not provide.

Transfer of the null-model theorem to the actual Apéry zeros is captured by the following precise conjecture.

Conjecture 160 (Two-prime dispersion)**.** *For $N \to \infty$ and primes $p, q \in (N, 2N]$ with $p \neq q$, define $C_{pq} = \sum_m X_{m,p} X_{m,q}$. Then*

$$\sum_{\substack{p,q\in(N,2N] \\ p\neq q}} |C_{pq}| \;=\; o(T_N).$$

This is equivalent to the BFM2 bound (201) and hence to the all-n conjecture $G_n = e^{o(n)}$.

Remark 161 (Anatomy-of-integers formulation)*.* For each $j \leq N$, define $\omega(j) = \#\{p \in (N, 2N] : p \mid b_j\}$. Then $\sum_j \omega(j) = T + F$ (counting primes that divide some b_j in the top block), and

$$\sum_{p\neq q}\sum_{j\leq N} \mathbf{1}[pq \mid b_j] \;=\; \sum_{j\leq N} \binom{\omega(j)}{2}.$$

The two-prime dispersion is equivalent to $\sum_j \omega(j)(\omega(j){-}1) = o(T)$: at most $o(T)$ indices j are divisible by two or more primes from the same dyadic window. Since $\log b_N \sim N \log(17 + 12\sqrt{2}) \approx 3.53\,N$ while $\sum_{p\in(N,2N]} \log p \sim N$, each b_j has room in principle for every prime in the window as a factor. Pure size/counting arguments cannot exclude this conspiracy; the Poisson prediction relies on the equidistribution of the p-adic digits of b_j across independent residue fields.

Table 8 verifies the anatomy-of-integers formulation directly: at every tested scale, the distribution of $\omega(j)$ matches Poisson(λ) with $\lambda = T/(N+1)$ to within sampling noise. In particular, $\max_j \omega(j) \le 3$ through $N = 131{,}072$, consistent with the Poisson prediction $N \cdot \lambda^3/6 \approx 5$ at this scale. The second factorial moment satisfies $\sum_j \omega(j)(\omega(j)-1)/T \approx \lambda$, which is exactly the collision rate $T/N \approx 1/\ln N$ predicted by Mertens—the same rate appearing in the two-prime scaling law below.

Table 8: Distribution of $\omega(j) = \#\{p \in (N, 2N] : p \mid b_j\}$. $\lambda = T/(N+1)$ is the mean; the Poisson column gives $e^{-\lambda}\lambda^k/k!$.

N	T	λ	$\max\omega$	$\#\{\omega\ge 2\}$	$\sum \omega(\omega-1)/T$	Poisson fit
8192	608	0.074	2	22	0.072	$< 0.1\%$
32768	2048	0.063	3	53	0.056	$< 0.1\%$
65536	3944	0.060	3	108	0.057	$< 0.1\%$
131072	7584	0.058	3	227	0.063	$< 0.1\%$
262144	14104	0.054	3	351	0.051	$< 0.1\%$

Computation through $N = 262{,}144$ gives $|O|/T < 2 \times 10^{-3}$ (Table 7), far stronger than the conjecture requires.

Table 9: Two-prime dispersion scaling. $\Sigma_{\text{abs}} = \sum_{p \ne q} |C_{pq}|$ measures the total unsigned cross-prime interaction. The ratio Σ_{abs}/T decreases as $\approx c/\ln N$ with $c \approx 1.08$.

N	T	Σ_{abs}	Σ_{abs}/T	$\ln N \cdot \Sigma_{\text{abs}}/T$	joint
8192	354	43.1	0.122	1.10	10
16384	664	75.3	0.113	1.10	17
32768	1182	123.2	0.104	1.08	29
65536	2237	214.1	0.096	1.06	47
131072	4317	391.1	0.091	1.07	87
262144	7940	682.3	0.086	1.07	156

Table 9 decomposes the cross-prime sum into its unsigned magnitude. The product $(\ln N) \cdot \Sigma_{\text{abs}}/T$ remains approximately constant at ≈ 1.08, giving the empirical scaling law

$$\frac{\sum_{p \ne q} |C_{pq}|}{T_N} \approx \frac{1.08}{\ln N}.$$

This is explained by a counting argument. Since $L \le p$ for primes $p > N$, the map $n \mapsto n \bmod p$ restricted to $[1, L]$ is injective; CRT lifts at coprime moduli hit the interval at most once, giving the sharp pointwise bound $|C_{pq}| \le \min(Z(p), Z(q))$. In practice, most pairs have no joint CRT hit (the "joint" column of Table 9 counts pairs with at least one), so $|C_{pq}| \approx Z(p)\, Z(q)\, N/(pq) = O(1/N)$ for the bulk. With $O(N^2/(\ln N)^2)$ pairs and $T = \Theta(N/\ln N)$, the sum-to-T ratio is $O(1/\ln N)$. The unsigned ratio $\Sigma_{\text{abs}}/T \sim 1/\ln N$ is therefore essentially *forced*—it measures the expected collision count $\sum_{p \ne q} E_{pq} \approx T^2/N$, which is $\Theta(T/\ln N)$ by Mertens. The genuine arithmetic content is in the *signed* sum: $|O|/T < 2 \times 10^{-3}$ says that the actual collision count $\sum_m r(m)(r(m)-1)$ matches its Poisson prediction to within a few percent.

For non-CM motives over $\mathbb{Q}$, independence of Frobenius traces at distinct primes is a consequence of the generalized Sato–Tate conjecture (known for GL_2 by the work of Barnet-Lamb–Geraghty–Harris–Taylor, extended to the symmetric square L-function). However, the single-prime $Z(p)$ statistics are a *vertical* problem—equidistribution of Mellin-transform values as the character varies within one $\mathbb{F}_p$—for which Katz's finite-field Mellin equidistribution theory [15] is the right tool. The two-prime dispersion is a *horizontal* problem: correlating arithmetic events at distinct primes for a fixed motive, which is of Lang–Trotter type. No equidistribution theorem over a single finite field can see cross-prime structure, and no analog of the joint Sato–Tate theorem across two distinct primes exists in the literature. The two-prime dispersion asks for a compatible statement at the level of zero-counts $Z(p)$, which lives in the Kummer-twisted cohomology rather than the fiber-variable cohomology. The adversarial construction $\mathcal{Z}_p = \{2N \bmod p\}$ shows that any proof must use Apéry-specific structure (cf. Friedlander–Granville [10] for the general impossibility of target-agnostic equidistribution beyond the large-sieve range).

The strongest existing results for structured moving targets in sieve theory follow the Duke–Friedlander–Iwaniec template [8], which proves equidistribution of roots of a quadratic congruence to moduli beyond $\sqrt{x}$ using an integral formula over automorphic forms on GL_3. Their method requires the target set $\mathcal{Z}_p$ to arise as roots of a *fixed* polynomial with a global algebraic-variety structure—precisely the ingredient the Apéry recurrence lacks (the degree grows linearly with p, and no single variety parametrizes the zero set). The only result achieving the moving-target beyond-$\sqrt{x}$ barrier without a fixed polynomial degree is Maynard's breakthrough [17], which handles targets of size $O(p^{1/3})$ in arithmetic progressions using new estimates for trilinear Kloosterman sums—still requiring a multiplicative character structure absent from the Apéry setting. In summary, the two-prime dispersion for Apéry numbers falls in the gap between what sieve methods can reach (targets with global algebraic structure) and what the Apéry recurrence provides (a local, p-dependent set of size $O(1)$).

The Diophantine-approximation approach to GCDs of recurrence terms—the Subspace-Theorem framework of Bugeaud–Corvaja–Zannier [6], Levin [16], and Xiao [26]—requires the sequence to admit a finite exponential-polynomial representation (a constant-coefficient linear recurrence). The Apéry sequence satisfies a recurrence with polynomial coefficients and obeys $b_n \sim C\alpha^n n^{-3/2}$, where the fractional exponent $-3/2$ is incompatible with constant-coefficient recurrences. Moreover, even in the exponential-polynomial setting, same-sequence GCD bounds fail: $\gcd(2^m-1, 2^n-1) = 2^{\gcd(m,n)}-1$ is exponentially large when $m \mid n$. Therefore no existing Subspace-Theorem argument yields $G_n = e^{o(n)}$ for all n.

The most direct computational path is an explicit Hasse–Witt calculation for the rank-three transcendental crystal, determining the slope-zero multiplicity of the Kummer-twisted compactly supported cohomology as a function of the character parameter. The Galois-group analysis of reductions of Apéry-like series by Caruso–Fürnsinn–Vargas-Montoya–Zudilin [7] may provide algebraic input, though their results concern the fiber parameter and do not directly control the character-twisted Mellin structure or the two-prime dispersion.

On the single-prime side, the bordered-certificate analysis (Section 5.4) settles the value-distribution question at the 3/4 exponent: $\max_a N_p(a) = O(p^{3/4})$ and $E(p) = O(p^{7/4})$ unconditionally, with the certificate nonvanishing (Theorem 32) sharp at $k < p$. By the impossibility construction discussed in Section 12, no package of single-prime distributional facts—including these—can close the pointwise conjecture; their role is in the averaged theorems and as input to any future cross-prime argument.

Finally, the open problem now has a sharply localized form. The quotient reduction (Proposition 74) shows that prime counting alone disposes of every prime below $n/(f(n)\log n)$, so the conjecture is equivalent to controlling the lower-digit channel uniformly over the bounded quotient classes $q = \lfloor n/p \rfloor < f(n)\log n$; and among all sufficient conditions collected here, the single fixed factorial moment $(\mathrm{HM})_8$ (Hypothesis 151), proved unconditionally for $k = 2$ (Proposition 152), appears to be the weakest cross-prime statement that closes the full conjecture (Theorem 155).

14 Structure of the marked scalar, and the exact location of the obstruction

Throughout, $\Lambda(x,y,z) = \dfrac{(1+x)(1+y)(1+z)\big((1+y)(1+z)+xyz\big)}{xyz} = \sum_{\kappa\in P}\lambda_\kappa X^\kappa$ is the Apéry Laurent polynomial, $b_r = \mathrm{CT}\,\Lambda^r$, and $C_M(d)$ denotes the d-section of the coefficient array of Λ^M.

14.1 The marked scalar is a moment of point counts

Proposition 162. *For every prime $p \geq 5$ and every M,*

$$C_M(p-1) \;\equiv\; -\sum_{x,y,z\in\mathbb{F}_p^\times} \Lambda(x,y,z)^M \;\equiv\; -\sum_{t\in\mathbb{F}_p^\times} t^r N_p(t) \pmod{p}, \qquad r = M \bmod (p-1),$$

where $N_p(t) = \#\{(x,y,z) \in (\mathbb{F}_p^\times)^3 : \Lambda(x,y,z) = t\}$. Consequently, writing $n = p + r$, the target condition $p \mid b_n$ is exactly the vanishing modulo p of the r-th moment of the point-count function of the Apéry family.

Audit: `q32_marked_scalar_character_sum.py` — 95 shell-versus-exponential-sum checks, 95 shell-versus-moment checks, 44 shell-versus-Apéry checks.

14.2 The reflection law, and the parity of $|Z_p|$

Proposition 162 converts a geometric symmetry into an arithmetic one.

Proposition 163. *$N_p(t) = N_p(t^{-1})$ for every $t \in \mathbb{F}_p^\times$; hence*

$$b_{p-1-r} \equiv b_r \pmod{p} \qquad (0 \leq r < p-1),$$

so $Z_p = \{r < p : \; p \mid b_r\}$ is stable under the involution $r \mapsto p-1-r$, and $|Z_p|$ is even unless the fixed point $\frac{p-1}{2}$ lies in Z_p.

Inversion-invariance is not new to the project: it is the hypothesis defining the space V_p^+ in the random model recorded in `DOCTRINE.md` (Q5387), which yields $K \sim \mathrm{Bin}((p-3)/2, 1/p) \to \mathrm{Poisson}(1/2)$. What Proposition 163 adds is that the invariance is a *theorem* for the actual sequence, derived from the geometry through Proposition 162, together with the exact classification of the exceptions: the Poisson law is obeyed by the *pair* count $|Z_p|/2$ because the involution forces the pairing, and the "central exceptions" to evenness are exactly the fixed-point primes. For $p < 4000$ these are $p = 11$ with $Z_{11} = \{5\}$ and $p = 3137$ with $Z_{3137} = \{1568\}$, and in both cases the unique zero equals $(p-1)/2$.
Audit: same script — 420 inversion checks, 420 palindromy checks (the ratio b_{p-1-r}/b_r is identically 1, not merely ± 1), and the involution verified on all 548 primes $5 \leq p < 4000$ without exception.

14.3 Parity of $|Z_p|$ is governed by ordinarity

Combining Proposition 163 with the Beukers/Ahlgren–Ono supercongruence $b_{(p-1)/2} \equiv a_p \pmod{p^2}$ for the newform 8.4.*a*.*a* gives

Proposition 164. *$|Z_p|$ is odd if and only if p is non-ordinary for* 8.4.*a*.*a, i.e. $p \mid a_p$.*

Proof. By Proposition 163, Z_p is stable under $r \mapsto p-1-r$, whose unique fixed point is $(p-1)/2$; hence $|Z_p|$ is odd exactly when $(p-1)/2 \in Z_p$, i.e. when $p \mid b_{(p-1)/2}$, which by the supercongruence is equivalent to $p \mid a_p$. $\square$

Verified over all 2260 primes $5 \le p \le 20\,000$: $|Z_p|$ is odd exactly at $p = 11$ and $p = 3137$, and these are exactly the primes with $p \mid a_p$ — no mismatches. Since non-ordinary primes have density zero, $|Z_p|$ is even for almost all p; this is the structural reason the Poisson law of the ledger is stated for the pair count $|Z_p|/2$.

14.4 Where G_n actually lives

The factorisation below is already recorded in `DOCTRINE.md` (Q5386); what is added here is the direct computation of both factors.

Proposition 165. *Write $a_n = A_n/D_n$ in lowest terms, so $D_n \mid d_n$. Then $G_n = \gcd(d_n a_n, d_n b_n) = (d_n/D_n) \cdot \gcd(A_n, b_n)$, and for every $n \le 330$:*

$$\gcd(A_n, b_n) = 1,$$

so $G_n = d_n/D_n$ exactly. Moreover the large prime factors of G_n are precisely the targets $p \in (n/2, n]$ with $p \mid b_n$; e.g.

$$G_{200} = 2 \cdot 3^3 \cdot 5 \cdot 17 \cdot 19 \cdot \mathbf{139} \cdot \mathbf{181}, \qquad G_{321} = 2 \cdot 3 \cdot 5 \cdot 7^2 \cdot \mathbf{179} \cdot \mathbf{193} \cdot \mathbf{211},$$

and G_n never exceeds 34 bits on that range.

So the conjecture is equivalent to the statement that Apéry's classical denominator $d_n = \mathrm{lcm}(1, \dots, n)^3$ is optimal up to $e^{o(n)}$, the only losses coming from top-window primes dividing b_n.
Audit: `q32_actual_Gn_audit.py`.

14.5 The target count

Proposition 166. *Let $K(n) = \#\{p \in (n/2, n] : p \mid b_n\}$, so that the quantity to be bounded is $T(n) = \sum_p \log p \le K(n) \log n$. Then*

$$K(n) \le 3 \qquad \text{for every } n \le 200\,000,$$

with 412 indices having $K \ge 2$, exactly 12 having $K = 3$, and none $K \ge 4$; the running maximum increases only at $n = 6, 200, 321$. The mean of K is ≈ 0.073, matching the Mertens prediction $\log 2/\log n$. The same computation gives $\sum_{p \le 200\,000} |Z_p| = 18\,126$ (mean 1.016) and $\max |Z_p| = 12$ at $p = 159\,977$.

Audit: `q32_k_scan.c`, validated against the independent Python implementation `q32_top_window_target_counts.` on $n \le 20\,000$.

14.6 The denominator-defect law, and what it discharges unconditionally

Write $G_n = \prod_p p^{e_p(n)}$ with $e_p(n) = v_p(d_n) - v_p(D_n) \ge 0$; since $D_n \mid d_n$ we always have

$$0 \le e_p(n) \le v_p(d_n) = 3\lfloor \log_p n \rfloor. \tag{$*$}$$

The law below is in fact a consequence of an exact congruence for the *numerator*, which removes the empirical step from the reduction.

Proposition 167 (numerator congruence)**.** *For every prime $p \ge 7$ and every n with $n/2 < p \le n$,*

$$\boxed{p^3 a_n \equiv \tfrac{6}{5}\, b_n \pmod{p}.}$$

Consequently, if $p \nmid b_n$ then $p^3 a_n$ is a p-unit, so $v_p(D_n) = 3$ and $e_p(n) = 0$: only primes dividing b_n can contribute to G_n.

Writing $n = p + r$ with $0 \le r < p$, the statement takes its natural form as a *vector* Lucas property:

$$\boxed{\big(p^3 a_{p+r},\ b_{p+r}\big) \ \equiv\ b_r \cdot (a_1, b_1)\ =\ b_r \cdot (6,5) \pmod{p}.}$$

The second coordinate is the classical Apéry–Lucas congruence $b_{p+r} \equiv 5\, b_r$; the first, $p^3 a_{p+r} \equiv 6\, b_r$, is its numerator counterpart, the factor p^3 appearing precisely because the leading coefficient $(n+1)^3$ of the recurrence vanishes at $n = p-1$. So modulo p the solution vector collapses onto the line spanned by the initial vector (a_1, b_1), which is exactly why a prime can contribute to G_n only through b_n.

Proof. Write $n = p + r$ with $0 \le r < p$, and recall Apéry's formula $a_n = \sum_k \beta_{n,k}\big(H_n^{(3)} + c_{n,k}\big)$ with $\beta_{n,k} = \binom{n}{k}^2\binom{n+k}{k}^2$ and $c_{n,k} = \sum_{m\le k} \frac{(-1)^{m-1}}{2m^3\binom{n}{m}\binom{n+m}{m}}$.

Step 1. Since $n < 2p$, the only multiple of p in $[1,n]$ is p itself, so $H_n^{(3)} = p^{-3} + h$ with h p-integral, whence $p^3 H_n^{(3)} \equiv 1 \pmod{p}$.

Step 2. In $c_{n,k}$ the denominator $2m^3\binom{n}{m}\binom{n+m}{m}$ has $v_p = 3$ only for $m = p$. Indeed $v_p(m^3) = 3$ forces $m = p$ because $m \le n < 2p$; and $v_p\big(\binom{n}{m}\binom{n+m}{m}\big) \le 2$, since $n, n+m < p^2$ gives $v_p \le 1$ for each binomial by Kummer. Every term with $v_p \le 2$ contributes $p^{3-v_p} \equiv 0 \pmod{p}$ after multiplication by p^3. The surviving $m = p$ term occurs exactly for $k \ge p$.

Step 3. By Lucas, $\binom{n}{p} = \binom{p+r}{p} \equiv 1$ and $\binom{n+p}{p} = \binom{2p+r}{p} \equiv 2 \pmod{p}$, so the $m = p$ term contributes $p^3 \cdot \frac{(-1)^{p-1}}{2p^3\binom{n}{p}\binom{n+p}{p}} \equiv \frac{1}{4} \pmod{p}$.

Step $3\frac{1}{2}$ (why only edge terms survive). For $p \in (n/2, n]$ Kummer's theorem gives

$$v_p\left(\binom{n}{k}\right) = \begin{cases} 1, & n-p < k < p,\\ 0, & \text{otherwise,}\end{cases}$$

since adding k and $n-k$ in base p produces a carry exactly in that range (checked on 3267 triples (n,p,k) with no mismatch). Hence modulo p only the *edge* terms $k \le n-p$ and $k \ge p$ of $b_n = \sum_k \binom{n}{k}^2\binom{n+k}{k}^2$ survive, which is the split used next.

Step 4. Again by Lucas, for $k < p$ one has $\binom{n}{k} \equiv \binom{r}{k}$ and $\binom{n+k}{k} \equiv \binom{r+k}{k}$, while for $k = p + k_0$ with $k_0 \le r$ one has $\binom{n}{k} \equiv \binom{r}{k_0}$ and $\binom{n+k}{k} \equiv 2\binom{r+k_0}{k_0}$. Hence

$$\sum_{k<p} \beta_{n,k} \equiv b_r, \qquad \sum_{k \ge p} \beta_{n,k} \equiv 4 b_r \pmod{p},$$

and adding gives the classical $b_n \equiv 5 b_r$.

Step 5. Combining,

$$p^3 a_n = \sum_k \beta_{n,k}\big(p^3 H_n^{(3)} + p^3 c_{n,k}\big) \equiv \sum_k \beta_{n,k} + \tfrac{1}{4}\sum_{k\ge p} \beta_{n,k} \equiv 5b_r + \tfrac{1}{4}\cdot 4b_r = 6b_r \pmod{p}. \qquad \square$$

General form. The statement extends to all digits:

$$\boxed{\big(p^3 a_{qp+r},\ b_{qp+r}\big) \ \equiv\ b_r \cdot (a_q, b_q) \pmod p, \qquad 1 \le q < p,\ 0 \le r < p.}$$

Verified on 9837 checks per coordinate over all $7 \le p < 80$, $1 \le q < p$, $0 \le r < p$ with $qp + r < 700$: no violations.

The proof for $q \ge 2$ does not extend the $q = 1$ argument verbatim (Step 2 there used $m \le n < 2p$, whereas for $q \ge 2$ every $m = tp$ with $t \le q$ has $v_p(m^3) = 3$). It splits instead into a boundary computation and a propagation.

Proof of the general form. (A) Boundary, $n = qp$. The multiples of p in $[1, qp]$ are $p, 2p, \ldots, qp$, so $H^{(3)}_{qp} = p^{-3} H^{(3)}_q + (p\text{-integral})$ and $p^3 H^{(3)}_{qp} \equiv H^{(3)}_q$. By Kummer, $v_p\binom{qp}{k} = 0$ exactly when $p \mid k$: writing $k = k_1 p + k_0$, a nonzero k_0 forces a units-digit carry, while for $k_0 = 0$ the top digits add to $q < p$ without carry. Hence $v_p(\beta_{qp,k}) \ge 2$ for $p \nmid k$, and for $k = k_1 p$ one has $\beta_{qp,k_1 p} \equiv \beta_{q,k_1}$. The last step is cleanest in the relative form: for $A \ge B \ge 0$ put $U_p(A, B) = \binom{Ap}{Bp}/\binom{A}{B}$; deleting the multiples of p from the three factorials shows $U_p(A, B)$ is a p-adic unit, and each remaining block $\prod_{r=1}^{p-1}(ip+r) \equiv (p-1)!$ gives $U_p(A, B) \equiv 1 \pmod p$. Then $\beta_{qp,jp} = \beta_{q,j}\, U_p(q, j)^2 U_p(q+j, j)^2 \equiv \beta_{q,j}$, which stays valid even when $\binom{q}{j}$ is itself divisible by p (both sides then vanish). Only p odd and $1 \le q < p$ are needed. In $c_{qp,k}$, a term with $p \nmid m$ has $v_p(m^3) = 0$ and $v_p\big(\binom{qp}{m}\binom{qp+m}{m}\big) \le 2$, so its denominator has $v_p \le 2$ and it dies after multiplication by p^3. The bound is exact, by Kummer: writing $m = m_1 p + m_0$ with $m_0 \ne 0$,

- adding m and $qp - m$ in base p, the units digits are m_0 and $p - m_0$, which sum to exactly p — so there is always a carry at position 0; and no carry can leave position 1 because $qp < p^2$. Hence $v_p\binom{qp}{m} = 1$ exactly.

- adding m and qp, the units digit is $m_0 + 0 < p$, so the only possible carry is at position 1, occurring iff $m_1 + q \ge p$. Hence $v_p\binom{qp+m}{m} \le 1$.

(Both statements verified on 34 578 pairs with no violation.) A term with $m = tp$ has $v_p(m^3) = 3$ and $\binom{qp}{tp}\binom{qp+tp}{tp} \equiv \binom{q}{t}\binom{q+t}{t}$ by Lucas, while $(-1)^{tp-1} = (-1)^{t-1}$ for odd p.

The c-summands must be weighed *together with* the outer factor, not termwise. Write $k = \ell p + s$ and $m = tp + u$ with $0 \le s, u < p$; the exact Kummer formulas are

$$v_p\binom{qp}{m} = \mathbf{1}_{u>0}, \qquad v_p\binom{qp+m}{m} = \mathbf{1}_{q+t\ge p}, \qquad v_p(\beta_{qp,k}) = 2 \cdot \mathbf{1}_{s>0} + 2 \cdot \mathbf{1}_{q+\ell \ge p}$$

(for the second, use $\binom{qp+m}{m} = \binom{qp+m}{qp}$: adding $(t, u)_p$ to $(q, 0)_p$ has no units carry and at most the single tens carry $q + t \ge p$). Hence for $p \nmid m$ the denominator valuation is $1 + \mathbf{1}_{q+t\ge p} \le 2$ and valuation 3 is impossible, so after multiplying by $p^3 \beta_{qp,k}$ such a summand still has valuation ≥ 1 and vanishes. For $u = 0$ the summand's total valuation is $v_p(\beta_{qp,k}) - \mathbf{1}_{q+t\ge p}$, so a nonzero contribution requires $s = 0$ *and* $q + \ell < p$; then $t \le \ell$ forces $q + t < p$, every relevant binomial is a unit, and Lucas returns exactly the (q, ℓ, t) summand of a_q. If $q + \ell \ge p$ then $v_p(\beta_{qp,k}) = 2$ while the bracket has valuation ≥ -1, so the term vanishes on both sides. Summing,

$$p^3 a_{qp} \equiv \sum_{\ell} \beta_{q,\ell}\big(H^{(3)}_q + c_{q,\ell}\big) = a_q \pmod p.$$

(B) Propagation, $r \ge 1$. For $n = qp + r$ one has $n \equiv r \pmod p$, so modulo p the recurrence reads $(r+1)^3 u_{n+1} \equiv P(r) u_n - r^3 u_{n-1}$ — *exactly* the b-recurrence in the variable r, and $(r+1)^3$

is invertible since $r+1 \le p-1$. At the left edge, $P(0) = 5$ and $(qp)^3 \equiv 0$ give $u_{qp+1} \equiv 5\, u_{qp}$, matching $b_1 = 5$. So both coordinates satisfy the same recurrence with initial data proportional to $(1, 5)$, whence $u_{qp+r} \equiv u_{qp}\, b_r$; here $r = 0$ is the base value, the recurrence at $n = qp$ supplies $r = 1$, and the induction runs for $1 \le r \le p-2$, so one never divides by p^3 at $r = p-1$. Note $p^3 a_j$ is p-integral because $j < p^2$: the hypothesis $q < p$ is essential to the normalisation, not cosmetic.

Applying (B) to each coordinate and inserting (A) and $b_{qp} \equiv b_q$ gives the theorem. □

The two ingredients were also verified numerically: the boundary congruence in 110 cases and the propagation in 2339 cases, with no violations.

On terminology. Calling this a "rank-one collapse" is inaccurate: when $b_r \neq 0$ the displayed action is $b_r \cdot I$, which has rank two. It is better described as a *Cartier/Frobenius block law*: the pair at index $qp + r$ is a scalar multiple of the pair at index q, the scalar being the low digit's b-value.

Novelty. A literature check against Gessel (1982), Beukers (1985, 1987), Coster, Malik–Straub (2016), Delaygue (2018), Rowland–Yassawi–Krattenthaler (2021) and Straub (2024) found no companion/numerator statement: those works give Lucas and Dwork congruences for the distinguished *integral* sequence, not for the rational second solution after pole-clearing by p^3. The defensible claim is therefore "apparently new in this normalisation", not absolute novelty.

A property of Apéry-like pairs, not of $\zeta(3)$ (empirically). The same statement holds for the $\zeta(2)$ pair, with the exponent tracking the order of the leading coefficient $(n+1)^\kappa$ of the recurrence. For $(n+1)^2 u_{n+1} = (11n^2 + 11n + 3)u_n + n^2 u_{n-1}$ with $(a_1, b_1) = (5, 3)$,

$$\big(p^2 a_{qp+r},\ b_{qp+r}\big) \equiv b_r \cdot (a_q, b_q) \pmod p,$$

verified on 3198 checks per coordinate with no violations. The r-propagation is formal for a weight-κ recurrence once $p^\kappa A_n$ is p-integral, but the boundary identity $p^\kappa A_{qp} \equiv A_q$ is separate arithmetic input for each family — Lucas for the integral solution does not imply it — so each sporadic pair must be checked individually. For the two Apéry pairs it is confirmed numerically here.

This unifies the trichotomy used in the draft. Modulo p the pair at index n is b_r times the pair at index q, so the right-hand side can vanish in exactly three ways — $p \mid b_r$ (lower-digit bad), $p \mid b_q$ (leading-digit), and $p \mid$ numerator of a_q (companion-block) — and for the top window $q = 1$, where $a_1 = 6$ and $b_1 = 5$, only the first survives for $p \ge 7$. That is the structural reason the top-window analysis is clean.

All five steps were also checked numerically with no exceptions: the three Lucas evaluations of Step 3–4 over every (p, r) with $7 \le p < 60$, $r < p$, $p + r \le 150$; and the conclusion on all 3305 pairs (n, p) with $7 \le p$, $n/2 < p \le n$, $20 \le n < 260$ — including all 28 pairs with $p \mid b_n$ — and on 3320 checks per coordinate in the (p, r) form (`q32_numerator_congruence_audit.py`).

Combined with the universal bound $e_p(n) \le 3$, Proposition 167 makes the top-window step of the reduction *exact*: the contribution of the window to $\log G_n$ is at most $3 \log \operatorname{rad}_{(n/2,n]}(b_n)$, with no unproved input.

A sharper valuation law (verified, proof of the exact valuation still open). The mod-p congruence gives $e_p(n) \ge 1$ when $p \mid b_n$; the data say much more. For every $p \ge 7$ in the top window,

$$\boxed{v_p(a_n) = -3 + v_p(b_n), \qquad \text{equivalently} \qquad e_p(n) = v_p(b_n).}$$

Verified on all 4340 pairs (n, p) with $7 \le p$, $n/2 < p \le n$, $20 \le n < 300$: no violations. Two consequences, both sharper than the bound above:

- the top-window part of G_n *equals* the top-window part of b_n exactly, so the conjecture is precisely the statement that the $(n/2, n]$-part of b_n is $e^{o(n)}$ — no factor 3, and multiplicities rather than radicals;

- $\gcd(A_n, b_n)$ is coprime at such p *iff* $v_p(b_n) \le 3$ (checked: no violations, and the largest multiplicity observed in the window is 2). So the primitive coprimality conjecture reduces, in this range, to a multiplicity bound.

Proof of the valuation law when $v_p(b_n) \le 2$. Let $p \ge 7$ lie in the top window and put $t = v_p(b_n)$, $\alpha = v_p(a_n)$.

If $t = 0$ there is nothing to do: writing $n = p + r$, the block law gives $p^3 a_n \equiv 6 b_r$ with b_r a unit, so $v_p(a_n) = -3$ as claimed. (The no-two-consecutive-zeros lemma is silent when $t = 0$, so this case must be handled separately rather than through the Casoratian.)

Now let $t \in \{1, 2\}$ and suppose first $p \nmid n + 1$. The exact Casoratian $a_n b_{n+1} - a_{n+1} b_n = -6/(n+1)^3$ has $v_p = 0$, since $p \ge 7$ gives $p \nmid 6$. Here $b_r \equiv 0$ and $r + 1 \le p - 1$, so there is no wraparound and the no-two-consecutive-zeros lemma does apply, giving $b_{r+1} \not\equiv 0$; the block law at $n+1 = p+(r+1)$ then yields $v_p(b_{n+1}) = 0$ and $v_p(a_{n+1}) = -3$ exactly. The two products therefore have valuations

$$v_p\big(a_n b_{n+1}\big) = \alpha, \qquad v_p\big(a_{n+1} b_n\big) = -3 + t.$$

If $\alpha \ne -3 + t$ the difference would have valuation $\min(\alpha, -3 + t) = 0$, forcing $-3 + t \ge 0$; this is impossible for $t \le 2$. Hence $\alpha = -3 + t$, i.e. $e_p(n) = v_p(b_n)$.

The excluded case $p \mid n + 1$ means $n + 1 = 2p$, i.e. $n = 2p - 1$ and $r = p - 1$. There palindromy gives $b_{p-1} \equiv b_0 = 1 \pmod p$, so $p \nmid b_n$, $t = 0$, and the law holds trivially (verified: no prime $p < 400$ has $p \mid b_{p-1}$). $\square$

The same argument in fact covers every multiplicity, giving the saturated law

$$\alpha = t - 3 \ (t < 3), \qquad \alpha \ge 0 \ (t = 3), \qquad \alpha = 0 \ (t > 3),$$

and hence, for *all* t,

$$\boxed{v_p(D_n) = 3 - \min(t, 3), \qquad e_p(n) = \min\big(v_p(b_n), 3\big).}$$

The cap is simply $v_p(d_n) = 3$.

Two bookkeeping points. First, $v_p(G_n) = e_p(n) + \min\big(v_p(A_n), v_p(b_n)\big)$, so $G_n = \prod_p p^{e_p(n)}$ presumes coprimality; in the top window with $t \le 2$ the two agree, since $v_p(a_n) = t - 3 < 0$ forces $v_p(A_n) = 0$, but that is an argument rather than a definition, and at $t = 3$ it is exactly the open point. Second, the companion channel in the middle range is $p \mid A_q$, not $p \mid b_q$: for $p > q$ the denominator D_q is a p-unit, so the block law gives $p \mid G_n \iff p \mid A_q b_r$. Since $\log |A_q| = O(q)$, the combined carrier still satisfies $\sum_{q \le \log n} \log \operatorname{rad}(A_q b_q) = O(\log^2 n)$.

So the denominator-defect law is completely proved, and the top-window contribution to $\log G_n$ equals $\sum_p \min(v_p(b_n), 3) \log p$ exactly. *But the top window alone is not the whole story*: the justified decomposition is

$$\log G_n = \sum_{n/2 < p \le n} \min\big(v_p(b_n), 3\big) \log p + M_n + O\!\left(\frac{n}{\log n} + \log^2 n\right), \qquad M_n = \sum_{\substack{n/\log n < p \le n/2 \\ p \,|\, b_{n \bmod p}}} v_p(G_n) \log p,$$

and from $p \mid b_n$ alone the middle range admits only the bound $O(\log b_n) = O(n)$. Hence the conjecture is equivalent to the *full* large-prime condition $\sum_{n/\log n < p \le n} \min(v_p(b_n), 3) \log p = o(n)$;

the top-window sum alone suffices only once $M_n = o(n)$ is proved separately. This is not a formality: the two ranges contribute comparably in practice. For instance the top-window and middle-range sums are 5.60 and 5.19 at $n = 500$, and 0 and 12.23 at $n = 2500$ — the middle range carries roughly half the mass and sometimes all of it.

One structural remark in its favour: the occupancy-one barrier is specific to $p \asymp n$. For $p \asymp n/\log n$ an interval of length n contains $\asymp \log n$ representatives of each residue class, so the large-sieve obstruction is weaker there — though for a *single* n this still only helps on average. What is not settled is the value $\alpha = 0$ at $t = 3$ — equivalently, excluding a common p-factor of the reduced numerator and b_n — which bears on the primitive coprimality conjecture but not on G_n; no $t \geq 3$ occurs in the tested range: computing $v_p(b_n)$ *directly* from the exact integer b_n for every bad pair with $n < 300$ gives maximum multiplicity 2.

A caution. One cannot substitute $v_p(b_r)$ for $v_p(b_n)$ here. The first-block expansion reads $b_{p+r} \equiv 5\big(b_r + pG_r + p^2H_r\big) \pmod{p^3}$, so $p^3 \mid b_{p+r}$ is a condition on that whole combination, not on b_r alone — higher block digits can cancel the vertical digits of b_r. Concretely $b_3 = 5 \cdot 17^2$ has $v_{17}(b_3) = 2$, yet the corrected first quotient digit is a unit, so $v_{17}(b_{20}) = 2$ rather than 3. A survey of $v_p(b_r)$ over the zero pairs (multiplicity 1 in 2072 of 2074 cases, 2 in the remaining two) is therefore evidence about b_r, not about b_n, and does not extend the multiplicity claim. (The other neighbour gives no more: the Casoratian $a_{n-1}b_n - a_nb_{n-1} = -6/n^3$ also has $v_p = 0$, and $p \nmid b_{n-1}$, so it yields the same constraint $\min(\alpha, t-3) = 0$ and likewise stalls at $t \geq 3$.)

The three ingredients were checked independently: the Casoratian has $v_p = 0$ in all 4282 applicable pairs, no two consecutive zeros occur in any of them, and $v_p(a_{n+1}) = -3$ in all 4251 pairs where it applies — no violations.

The mod-p^2 route, by contrast, does not help: the corrections vanish at $r = 0$ (Beukers' $b_p \equiv 5 \bmod p^3$) but are irregular for $r \geq 1$.

Proposition 168 (verified law)**.** *For primes $p \geq 7$ with $n/2 < p \leq n$,*

$$e_p(n) > 0 \iff p \mid b_n,$$

and then $e_p(n) = 1$. Verified for all such pairs with $n \leq 250$: 3121 of 3124 cases, the three exceptions being $p = 5$ at $n = 5, 7, 9$ — exactly the prime dividing $b_1 = 5$, which is the same exception carried by the Apéry–Lucas step. For $\sqrt{n} < p \leq n/2$ the defect takes values in $\{1, 2, 3\}$ and occurs precisely when a base-p digit of n lies in Z_p; e.g. the defect residues are $\{5\}$ for $p = 11$, $\{3, 13\}$ for $p = 17$, $\{8, 10\}$ for $p = 19$, $\{8, 22\}$ for $p = 31$, $\{17, 19\}$ for $p = 37$, $\{10, 30\}$ for $p = 41$, $\{9, 49\}$ for $p = 59$ — in each case exactly Z_p, and each pair summing to $p - 1$ as Proposition 163 requires.

Two consequences, the first of which needs no hypothesis at all.

Lemma 169 (unconditional)**.** $\displaystyle\sum_{p \leq n/\log n} e_p(n)\log p \;\leq\; 3\,\pi\!\left(\frac{n}{\log n}\right)\log n \;=\; O\!\left(\frac{n}{\log n}\right) \;=\; o(n).$

Proof. By $(*)$, $e_p(n)\log p \leq 3\lfloor \log_p n\rfloor \log p \leq 3\log n$; sum over the $\pi(n/\log n) \sim n/\log^2 n$ primes involved. ☐

So *every prime below $n/\log n$ is already harmless*, whatever the arithmetic of b_n. For $p > n/\log n$ (hence $p > \sqrt{n}$ for large n) the base-p expansion of n has exactly two digits, $n = qp + k$ with $1 \leq q < \log n$, and Apéry–Lucas gives $b_n \equiv b_q b_k \pmod{p}$. The digit q can only contribute primes dividing one of $b_1, \ldots, b_{\lfloor \log n\rfloor}$, and

$$\sum_{j \leq \log n} \log \operatorname{rad}(b_j) \leq \sum_{j \leq \log n} \log b_j = O(\log^2 n),$$

since $\log b_j \sim 3.4j$ (checked: $\log \operatorname{rad}(b_j)$ equals $\log b_j$ to within a few percent for $j \le 14$). The block law shows this covers the *companion* primes too: $p^3 a_n \equiv a_q b_r$, so $e_p(n) > 0$ forces $p \mid b_r$ (hence $p \mid b_n$), or $p \mid b_q$, or p divides the numerator of a_q; and since $q \le \log n$, the last two together contribute at most $\sum_{q \le \log n} \big(\log b_q + \log |\mathrm{num}\, a_q|\big) = O(\log^2 n)$. Hence

$$\boxed{\log G_n \;=\; O\!\left(\frac{n}{\log n}\right) + O(\log^2 n) + 3 \sum_{\substack{n/\log n < p \le n \\ (n \bmod p) \in Z_p}} \log p}$$

and the conjecture is *equivalent* to the last sum being $o(n)$. This is a sharper localisation than the top-half reduction: the entire range $p \le n/\log n$ and the whole q-digit contribution are discharged with Chebyshev's bound alone.

14.7 How large the remaining sum actually is

Using the Lucas criterion ($p \mid b_n$ iff some base-p digit of n lies in Z_p) one can compute the full quantity $R(n) = \log \operatorname{rad}_{p \le n}(b_n)$ for every n in a range. For $n \le 20\,000$:

range	$\max R(n)/n$
$100 \le n \le 1000$	0.1091 ($n = 200$)
$1000 \le n \le 10\,000$	0.0251 ($n = 1041$)
$10\,000 \le n \le 100\,000$	0.0041 ($n = 11\,181$)

$$\max_{n \le 10^5} \#\{p \le n : p \mid b_n\} = 14.$$

Thus $R(n)$ itself grows extremely slowly. Quantitatively, $R(200) \approx 21.8$ and $R(11\,181) \approx 45.5$: the value not quite doubles while n grows by a factor 56. The heuristic prediction from $\Pr[p \mid b_n] \approx |Z_p| \log_p n/p$ is $R(n) \asymp \log n \log\log n$, which predicts a ratio 2.4 over that range against the observed 2.1. So the truth appears to be

$$R(n) = O(\log n \log\log n),$$

while the conjecture asks only for $o(n)$ — the gap between what is true and what must be proved is enormous, and the entire difficulty is that no unconditional argument controls the prime factors of one specific value b_n.

A complementary datum explains *why* the count is so small: b_n has very few prime factors, and they are huge. Complete factorisations give

n	5	8	11	14	17	20	23	26
$\omega(b_n)$	3	4	3	3	4	4	7	3
$\Omega(b_n)$	3	5	6	3	4	4	7	4
bits of b_n	20	34	49	64	79	94	108	123
bits of largest $p \mid b_n$	14	21	22	42	63	48	23	60

so $\omega(b_n)$ stays near $\log n$ while $\log b_n \sim 3.53n$, and a single prime factor carries a constant fraction of the size. Since $H(n) \le \omega(b_n)$, the conjecture would follow from $\omega(b_n) = o(n/\log n)$ — much weaker than the observed $\omega(b_n) \approx \log n$ — but that is not provable either: splitting at any Y gives only $\omega(b_n) \le \pi(Y) + \log b_n/\log Y$, which is $\gg n/\log n$ for every choice of Y.

14.8 What is load-bearing: the conjecture is FALSE inside the same family

It is natural to hope that the conjecture follows from soft structure — D-finiteness, modularity, Lucas congruences, or reflexivity of the Newton polytope. It does not, and the reason is a counterexample inside the sporadic Apéry-like family itself.

Malik–Straub (*Divisibility properties of sporadic Apéry-like numbers*, Res. Number Theory 2 (2016), Thm. 6.6) prove that Cooper's sporadic sequences satisfy

$$s_7(p-j) \equiv 0\ (p) \quad 1 \le j \le \tfrac{p+1}{3}, \qquad s_{10}(p-j),\, s_{18}(p-j) \equiv 0\ (p) \quad 1 \le j \le \tfrac{p+2}{4}.$$

That is, their zero sets Z_p contain a *macroscopic interval*. Lucas then propagates the interval: for $n/2 < p \le (3n+1)/5$ one gets $p \mid s_7(n)$, so by the prime number theorem

$$\log \operatorname{rad}_{p\le n}\big(s_7(n)\big) \;\ge\; \big(\tfrac{1}{10}+o(1)\big)n, \qquad \log \operatorname{rad}_{p\le n}\big(s_{10}(n)\big),\; \log \operatorname{rad}_{p\le n}\big(s_{18}(n)\big) \;\ge\; \big(\tfrac{1}{14}+o(1)\big)n.$$

More generally, a terminal interval of length αp in Z_p forces $\log \operatorname{rad}_{p\le n}(u_n) \ge \big(\frac{\alpha}{2(2-\alpha)} + o(1)\big)n$.

So three bona fide modular Apéry-like sequences *violate* the analogue of the conjecture. They are D-finite, they satisfy Lucas congruences, they are modular, and (by Gorodetsky's constant-term representations) 14 of the 15 sporadics — including all three counterexamples — have the origin as the unique interior lattice point of the Newton polytope, exactly like Λ. Even $\binom{2n}{n} = \mathrm{CT}(2+x+x^{-1})^n$ shares that property.

Consequence: no proof for b_n can proceed from D-finiteness, modularity, Lucas congruences or polytope reflexivity alone, since sequences with all four properties violate the conclusion. Any proof must use the size and geometry of Z_p.

This also qualifies the rank-one discussion below: the Beukers–Vlasenko rank-one statement explains why every *local elimination* aliases, but it does not separate the true cases from the false ones, since the counterexamples have the same polytope invariant.

14.9 Numerical separation, and a check of the sharp factorial-ratio constant

Computing $R(n) = \log \operatorname{rad}_{p\le n}(u_n)$ for $n \in [500, 2000]$ across the family (script `q32_family_compare.py`; Z_p from the division-free scaled recurrence, then the Lucas digit criterion) gives

sequence	$\max R(n)/n$	$\max \#\{p \le n : p \mid u_n\}$
Apéry $\zeta(3)$	0.042	7
Apéry $\zeta(2)$	0.042	8
Franel	0.036	9
Domb	0.036	8
Almkvist–Zudilin	0.031	4
Cooper s_7	0.301	70
Cooper s_{10}	0.227	53
Cooper s_{18}	0.225	52
$\binom{2n}{n}$ (control)	0.420	101

a sharp dichotomy: the first five decay with n, the last four do not. The three Cooper sequences were checked directly here — their recurrences reproduce $s_{10}(n) = \sum_k \binom{n}{k}^4$ exactly, and their zero sets have $|Z_p| \approx 26\%$–35% of p (against $|Z_p| \approx 1$ for Apéry), with the Malik–Straub terminal intervals verified to be contained in Z_p at $p = 23, 31, 43$. For factorial ratios there is a sharp theorem: with $\Delta(x) = \sum_i \lfloor a_i x \rfloor - \sum_j \lfloor b_j x \rfloor$,

$$\log \operatorname{rad}_{p\le n}(F_n) = \kappa_\Delta n + o(n), \qquad \kappa_\Delta = \int_1^\infty \mathbf{1}_{\Delta(t)>0}\,\frac{dt}{t^2},$$

which for $\binom{2n}{n}$ gives $\kappa = 2\log 2 - 1 = 0.386294\ldots$. Direct computation confirms this and validates the scanner:

n	10^3	10^4	10^5	10^6
$R(n)/n$	0.38536	0.38780	0.38614	0.38573

14.10 An invalid step in the conditional theorem, and its repair

Correction of a status claim, not of the paper. Theorem 80 of `proof.tex` states its conclusion for a set of n of natural density 1, and its Markov argument is correct. It was `STATUS.md` (and the handoff prose derived from it) that recorded the conclusion as holding "for ALL n". That upgrade is **invalid**, and cannot be repaired from Hypothesis $\bar{Z}$ ($\sum_{p\le x} Z(p) \ll \pi(x)$) together with the structure of Z_p proved so far.

Double counting gives

$$\sum_{m\le N}\sum_{p\in P}\mathbf{1}_{m \bmod p\in Z_p} = \sum_{p\in P} Z(p)\Big(\frac{N}{p}+O(1)\Big),$$

so $\bar{Z}$ controls the *mean* diagonal incidence over m, and Markov yields a density-one conclusion. Passing from that mean to a bound for every individual n is the invalid step.

Proposition 170 (the structural properties do not suffice)**.** *There exist sets $S_p \subseteq [0,p)$ with $|S_p| \le 2$, stable under $r \mapsto p-1-r$, with no two consecutive elements and with $\sum_{p\le x}|S_p| \ll \pi(x)$, for which $T(N) \asymp N$.*

Proof. Fix N and set $S_p = \{N-p,\ p-1-(N-p)\}$ for $p \in (N/2, N]$ and $S_p = \varnothing$ otherwise. Then $N \bmod p = N-p \in S_p$ for every p in the window. ☐

Verified numerically at $N = 20\,000$: $\max_p |S_p| = 2$, no reflection violation, no consecutive pair, $\sum_p |S_p|/\pi(N) = 0.913$ — so $\bar{Z}$ holds with constant below 1 — and yet $T(N)/N = 0.4955$. Hence *every* structural property proved so far about Z_p (size, reflection symmetry, no consecutive zeros) is consistent with total alignment at a single n. The obstruction is alignment, not size.

So the density-one theorem is exactly what the present hypotheses give. To reach every n one needs a uniform diagonal-discrepancy hypothesis:

$$\textbf{(DA)} \qquad \sup_n \Big|\sum_{n/2<p\le n}\Big(\mathbf{1}_{n-p\in Z_p} - \frac{Z(p)}{p}\Big)\Big| \ll \frac{\sqrt{n}}{\log n}.$$

By partial summation $\bar{Z}$ gives $\sum_{n/2<p\le n} Z(p)/p \ll 1/\log n$, so $\bar{Z}$ together with (DA) yields $T(n) = O(\sqrt{n})$ and closes the conjecture. The all-n theorem should therefore be stated as "$\bar{Z}$ and (DA) imply $\log G_n = O(\sqrt{n})$ for all n"; $\bar{Z}$ alone gives the density-one theorem, which is what `proof.tex` correctly proves.

Note also that Hypothesis Z in its pointwise form ($Z(p) \le C$) is expected to be *false*: extreme-value theory for the $2\cdot\mathrm{Poisson}(1/2)$ law predicts $\max_{p\le N} Z(p) \sim 2\log\log N$, matching the observed record progression $2, 4, 6, 8, 10, 12$ with $Z(p) = 12$ first at $p = 159\,977$.

14.11 A p-independent criterion and an explicit shift operator

Two exact facts, both audited in `q32_shift_operator_audit.py` (12 criterion checks, 96 operator checks).

The 27-term criterion. Since $p \equiv 1 \pmod{p-1}$ we have $n-p \equiv n-1 \pmod{p-1}$, so $t^{n-p} = t^{n-1}$ on $\mathbb{F}_p^\times$ and the exponent in the criterion is *independent of* p. With $m = n-1$ and $p \in (n/2, n]$,

$$p \mid b_n \iff \sum_{x,y,z\in\mathbb{F}_p^\times} \Lambda(x,y,z)^m \equiv 0\ (p) \iff p \mid V(n, p-1), \qquad V(n,s) = \sum_{\varepsilon\in\{-1,0,1\}^3} c_m(\varepsilon_1 s, \varepsilon_2 s, \varepsilon_3 s),$$

the 27 terms being the only lattice points of mP in the sublattice $s\mathbb{Z}^3$ once $s > m/2$. Verified at all four known bad primes ($n = 200$: $p = 139, 181$; $n = 321$: $p = 179, 193, 211$) and at non-bad primes.

The shift operator. The coefficient family $C(s) = c_m(s,0,0)$ is P-recursive in the *shift* variable s, of order 2 and degree 3:

$$\boxed{q_2(s)C(s+2) + q_1(s)C(s+1) + q_0(s)C(s) = 0,}$$

$$\begin{aligned}
q_0(s) &= (s-m)^3,\\
q_1(s) &= 2s^3 - 4(m-1)s^2 + (m^2 - 7m + 3)s + (2m^3 + 4m^2 - 2m + 1),\\
q_2(s) &= s^3 - (m-4)s^2 - (m+5)(m-1)s + (m-1)^2(m+2).
\end{aligned}$$

The coefficients are polynomial in m as well, and q_0 vanishes to order 3 at $s = m$, the edge of the support ($C(s) = 0$ for $s > m$).

Why this is the natural home for a certificate. The criterion evaluates C at $s = p-1 \equiv -1 \pmod{p}$. A certificate — an integer $C_n \neq 0$ divisible by every bad prime with $\log|C_n| = o(n)$ — would close the conjecture, since then $\#\text{bad} \leq \log|C_n|/\log(n/2)$. Two routes are now concrete rather than speculative:

1. If $V(n,\cdot)$ agreed with a polynomial $Q_n \in \mathbb{Z}[s]$ of degree D and height H on the relevant range, then $V(n, p-1) \equiv Q_n(-1)$ and $C_n = Q_n(-1)$ would work. *This fails*: eleven finite differences of $V(120, s)$ reduce the size only from 611 to 593 bits, so V is not polynomial of any low degree.

2. The transfer-matrix product $\prod_s M(s) \bmod p$ of the operator above over a full period is exactly the p-curvature object of a Picard–Fuchs-type operator, which is *nilpotent* for globally nilpotent operators (Katz). This is the remaining concrete mechanism by which $C(p-1) \bmod p$ could be expressed through data of size $O(\log n)$.

14.12 A second-moment target with large slack

The pointwise statement follows from a *second moment*, and with an enormous margin. Let

$$I = \{(r,p) : p \text{ prime},\ r < p,\ p \mid b_r\}, \qquad H(n) = \#\{(r,p) \in I :\ r + p = n\},$$

so $H(n)$ is exactly the number of bad primes at level n (Apéry–Lucas). Restrict to $N < n \leq 2N$ and set

$$E = \sum_{N<n\leq 2N} H(n)^2 = \#\big\{\big((r,p),(r',p')\big) \in I^2 :\ r + p = r' + p'\big\},$$

the additive energy of I along the diagonal. Since $\max_n H(n)^2 \le E$,

$$\boxed{E = o\left(\frac{N^2}{\log^2 N}\right) \implies \max_n H(n) = o\left(\frac{N}{\log N}\right)},$$

which closes the conjecture. *This is an equivalence, not a weakening*: since $E \le (\max_n H) \sum_n H = |I| \max_n H \ll (N/\log N) \max_n H$ under $\bar{Z}$, one also has $\max_n H = o(N/\log N) \Rightarrow E = o(N^2/\log^2 N)$. In particular a constant-factor saving in E buys only a constant-factor saving in $\max_n H$. What the reformulation provides is a different *shape* — an additive-energy statement about the incidence set, for which the available tools (Cauchy–Schwarz, sieve upper bounds for prime pairs, CRT independence) are at least standard — not a lower bar.

How much slack is there? Measured at $N = 60\,000$ (from the exact Z_p data):

$$|I| = \sum_n H(n) = 4398, \qquad E = \sum_n H(n)^2 = 4714, \qquad E/|I| = 1.072,$$

against a requirement of $E = o(2.97 \cdot 10^7)$ — the truth is smaller by a factor 6309, and the gap grows with N. Moreover the random model predicts $|I| + |I|^2/N = 4720$ against the observed 4714: agreement to 0.1%, i.e. the energy is entirely diagonal, with the off-diagonal contributing the predicted $|I|^2/N$ and nothing more.

The companion discrepancy target of §14.10 is equally comfortable:

$$\max_{1000 \le n \le 60\,000} \left| \sum_{n/2 < p \le n} \left(\mathbf{1}_{n-p \in Z_p} - \frac{Z(p)}{p} \right) \right| = 2.933 \quad (n = 47\,859),$$

against the $\sqrt{n}/\log n = 20.3$ that (DA) demands there; the observed discrepancy looks bounded rather than growing.

One further structural point, which explains why the energy formulation meets the same wall. For p in the top window the quotient $\lfloor n/p \rfloor$ equals 1, so the bad condition is a *pure congruence* $n \bmod p \in Z_p$. Hence for distinct p, p' the joint condition is a congruence modulo pp', and by CRT its density is exactly $|Z_p||Z_{p'}|/(pp')$ — which correctly predicts the observed off-diagonal $|I|^2/N$. But $pp' \asymp N^2$ exceeds the length N of the range of n, so each pair contributes 0 or 1 and the CRT main term is swamped by the error. This is precisely the large-sieve Q^2 barrier in another guise: the problem carries one degree of freedom per prime.

So the remaining gap is not quantitative delicacy — both sufficient hypotheses hold in the data by factors of 10^1–10^4 — but the absence of any method that controls the *alignment* of the sets Z_p along a single diagonal $r + p = n$. In the energy formulation this is the statement that the incidence set I, of size $\approx N/\log N$ inside a box with N diagonals, does not concentrate a positive proportion of its points on one diagonal.

14.13 Why only the first moment is accessible

The k-th moment of the target count factors as

$$\sum_{n \le N} H(n)^k = \sum_{p_1, \dots, p_k} \#\{n \le N : \ n \bmod p_i \in Z_{p_i} \ (1 \le i \le k)\},$$

because for p in the top window the quotient $\lfloor n/p \rfloor$ is 1 and the bad condition is a pure congruence modulo p. The inner count is governed by the modulus $p_1 \cdots p_k \asymp N^k$:

- $k = 1$: the modulus is $p \asymp N$, the range $n \le N$ contains $N/p + O(1) = O(1)$ complete residue classes, and one gets the honest main term $\sum_p Z(p)(N/p + O(1))$. This is exactly the input for the density-one theorem.
- $k \ge 2$: the modulus N^k exceeds the length of the range, so each tuple contributes 0 or 1 and no CRT/equidistribution gain is available.

So the first moment is the only one the range can support, and Markov applied to it yields a density-one statement and nothing stronger. This is the same Q^2 obstruction as in the large sieve and in the energy formulation of §14.12 — three faces of "one degree of freedom per prime". It explains, rather than merely reports, why the exceptional set of the conditional theorem stalls at a power of $\log N$.

The data are consistent with all moments being of the same size: at $N = 60\,000$,

k	1	2	3	4
$\sum_{n \le N} H(n)^k$	4398	4714	5388	6862

$N/\log N = 5454$,

i.e. every moment is $\asymp N/\log N$, as a Poisson law with mean $\asymp 1/\log N$ predicts. A proof of even the second moment at that strength would immediately give $\max_n H(n) \ll \sqrt{N/\log N} = o(N/\log N)$ and close the conjecture.

14.14 No second relation: the zeros are statistically generic

Beating the trivial constant 1/2 would follow from a "non-forced second relation" excluding a positive proportion of the candidate primes. No such relation exists at the level of congruences or quadratic characters. Over the 950 pairs (p, z) with $z \in Z_p$, $5 \le p < 8000$:

$z \bmod 2$	47.5%, 52.5%
$z \bmod 3$	31.3%, 33.5%, 35.3%
$z \bmod 4$	23.6%, 26.7%, 23.9%, 25.8%
$z \bmod 8$	all eight classes 11.2%–13.9%
$\left(\frac{z}{p}\right)$	$+1$: 50.3%, -1 : 49.7%

and the normalised zeros z/p are uniform in $[0, 1]$ (ten bins, 69–81 against 72.4 expected). Note that reflection preserves parity, since $p - 1$ is even, so a parity bias was *a priori* possible; there is none.

Two structural remarks close the remaining elementary routes. First, the reflection partner of a hit at level n is a hit at $n^* = 3p - 1 - n$, a *different* level, so reflection injects hits into other diagonals and yields no pointwise saving. Second, the gap continuant N_h has the forced factor $2X + h + 1$; at $X = z$ and $h = p - 1 - 2z$ this equals p, so $N_h(z) \equiv 0 \pmod p$ is tautological rather than a second condition. Finally, the factor-count bound $H_R(n) \le (\lambda/2 + o(1))R^2/\log n$, $\lambda = \log(17 + 12\sqrt{2})$, beats prime counting only for $R < 2/\lambda = 0.567$, i.e. never for a growing range; the genuinely free prefix $r \le \lambda^{-1} \log n$ (where $b_r < n/2$, so no hit is possible) removes only $O(\log n)$ residues.

14.15 The certificate exists for codegree, not for degree

It is worth recording precisely which certificate the paper already has, since it delimits the gap. Lemma 76 proves that for $m, n \in (N, 2N]$ with $h = n - m$, the number of primes $p > P_0$ simultaneously lower-digit bad for m and for n is $O\big(h \log N/\log P_0\big)$, hence $O(h)$ for $P_0 = N^\delta$. Its proof is exactly a certificate argument:

- *no wrap*: the two zeros lie at distance h, so $p \mid N_h(m \bmod p) \equiv N_h(m)$, and every such prime divides the *fixed integer* $N_h(m)$, of height $\le 3h\log(CN)$;
- *wrap*: $p \mid \prod_{j\le h}(m+j)$, of height $h\log 3N$.

So a low-height certificate *does* exist for a *pair* of indices, and it is what powers the polylogarithmic exceptional set via a Kővári–Sós–Turán argument — the exceptional-set theorem is not a Markov bound on a first moment.

What is missing is the same statement for a *single* index: a certificate divisible by all primes bad for one n. The codegree bound cannot be promoted to a degree bound. Summing $\operatorname{codeg}(m, n_0) \le C|n_0 - m|$ over all $m \in (N, 2N]$ gives only $H(n_0) \ll N^2$; and within a window of length $H_0 < p$ the residues $m \bmod p$ are distinct, so each bad prime for n_0 is bad for at most $O(1)$ nearby m, and no contradiction with a large $H(n_0)$ arises. In graph terms: bounded codegree limits how many vertices can have large degree — which is exactly the exceptional-set theorem — but never excludes a single high-degree vertex.

The incidence structure is entirely explicit: a prime p is bad for exactly $|Z_p|$ levels, namely $n = p+z$ with $z \in Z_p$, and reflection pairs these into $\{p+z,\ p+(p-1-z)\}$, two levels summing to $3p-1$ — so the levels served by p are symmetric about $(3p-1)/2$. Triples of levels sharing a bad prime therefore occur exactly when $|Z_p| \ge 3$; the smallest instance in range is $Z_{181} = \{19, 47, 133, 161\}$ (two reflection pairs), which makes 181 bad for the four levels $200, 228, 314, 342$. Consequently the summed triple codegree is $\sum_p \binom{|Z_p|}{3}$, which the Poisson law makes $O(\pi(N))$ — the right shape for a hypergraph refinement of the Kővári–Sós–Turán argument, though the pointwise obstruction is untouched by it.

The certificate for triples is available in principle. For three nonwrapping zeros at $z,\ z+h_1,\ z+H$ ($H = h_1 + h_2$) the correct pair of detectors is $F(X) = N_{h_1}(X)$ and $G(X) = N_{h_2}(X + h_1)$ — *not* N_H, since continuant addition gives

$$N_H(X) = N_{h_1}(X)N_{h_2+1}(X + h_1 - 1) - (X + h_1)^6 N_{h_1-1}(X)N_{h_2}(X + h_1),$$

so the long return supplies no independent equation. Writing $\widehat{R}_{h_1,h_2}$ for the resultant of F and G after removing primitive contents, their characteristic-zero gcd and the forced reflection factors, one has $\deg N_h \le 3(h-1)$ and $\log H(N_h) = O(h\log h)$, so Sylvester–Hadamard gives $\log|\widehat{R}_{h_1,h_2}| = O(h_1h_2\log H)$. What must be proved is $\widehat{R}_{h_1,h_2} \ne 0$; no residual characteristic-zero collision was found in the tested range. That would give a triple-codegree bound and, through a hypergraph KST argument, an improvement of the polylogarithmic exceptional set — a genuine secondary target, though not the pointwise theorem. Indeed the incidence graph is $K_{2,2}$-free on the prime side (two primes p, p' share a common level only if n solves a CRT system modulo $pp' \asymp N^2 > N$, so at most one), and a star is $K_{2,2}$-free; no purely combinatorial argument can therefore exclude one bad n.

The codegree lemma itself was audited numerically here. Over all pairs $(m, m+h)$ with $h \le 8$ and $m \le 200\,000$ (top window), the maximum number of common bad primes is

h	1	2	3	4	5	6	7	8
max codeg	0	1	0	1	1	0	1	1

comfortably inside the proved $O(h)$, and the value 0 at $h = 1$ is the no-two-consecutive-zeros theorem appearing independently.

14.16 Provenance, and a separate coprimality conjecture

A literature search located no pre-2026 statement of this problem. Its provenance is the challenge itself: M. Shalyt et al., *The Ramanujan Challenge for AI*, arXiv:2607.09721v1 (27 June 2026), §3.2, contributed by Shachar Weinbaum, whose §3 states that its open problems are "presented here for the first time". The obvious predecessor chain — Apéry (1979); Beukers (1979); Hata (1990, 2000); Rhin–Viola (2001); Fischler (Astérisque 294, 2004); Zudilin (2004); Krattenthaler–Rivoal (Mem. AMS 186, 2007) — does not contain it; in particular the Krattenthaler–Rivoal "denominator conjecture" concerns common denominators of engineered multi-zeta forms, not the exact content of Apéry's own pair.

The title is explained by the following computation. With $\alpha = 17 + 12\sqrt{2}$, $Q = 3 + \log\alpha$ and $E = \log\alpha - 3$, Apéry's own pair yields the irrationality-measure bound $1 + Q/E = 13.4178\ldots$; if $\log G_n = cn + o(n)$ then primitive reduction improves this to $1 + (Q - c)/(E + c)$. Since $\mu(\zeta(3)) \geq 2$ always, $c \leq (Q - E)/2 = 3$ unconditionally — so the conjecture asserts that Apéry's bound is exactly what his construction gives, with no cancellation to exploit. (This also shows the irrationality-measure lever is weak here: it yields only $c \leq 3$, whereas prime counting already gives $\log G_n \leq (3/2 + o(1))n$.) The Casoratian pair bound is likewise not binding: $G_n G_{n+1} \mid 6 d_n d_{n+1}/(n+1)^3$ gives $\log G_n + \log G_{n+1} \leq 6n + o(n)$, while prime counting already forces $\log G_n + \log G_{n+1} \leq 3n + o(n)$. So both classical levers are dominated by the trivial prime count, and neither can contribute.

Separately, our computation $\gcd(A_n, b_n) = 1$ for all $n \leq 330$ is, as far as could be determined, unrecorded. Since $G_n = (d_n/D_n)\gcd(A_n, b_n)$, coprimality is *not* needed for the subexponential-content conjecture; it is a strictly stronger statement, and deserves to be stated separately as a *primitive Apéry coprimality conjecture.*

14.17 Is the conjecture safe?

No positive heuristic or structural mechanism for failure could be identified. Three specific worries can be dismissed.

Arithmetic of n. For a top prime, $p \mid b_n$ is the single-block condition $n - p \in Z_p$; divisibility of n by many small integers does not enter. Likewise a "supersingular parameter" intuition is misleading: $r = n - p$ is a coefficient index that changes with p, not one geometric parameter reduced modulo many primes.

Short progressions inside one Z_p. Harmless: each zero z contributes to a *different* level $p + z$, so an arithmetic progression of zeros merely spreads one prime's few contributions across several n. The Cooper-type failure needs a macroscopic, systematically positioned block, which Apéry excludes by reflection, nonzero endpoints, and no consecutive zeros.

The record $|Z_p| = 12$ at $p = 159\,977$. That is six reflection pairs. With Poisson mean $\lambda \approx 0.504$ over the $\approx 17\,982$ primes below $2 \cdot 10^5$, the expected number with at least six pairs is ≈ 0.27, so observing one has probability $\approx 23\%$: entirely ordinary, not a heavy tail.

What remains is a purely logical vulnerability: one can adversarially plant a diagonal while preserving Poisson cardinalities and near-uniform positions for every individual prime. So the data do not prove the conjecture — but a counterexample with $H(n) \gg n^{\varepsilon}$ would need a cross-characteristic resonance for which there is at present neither an algebraic mechanism nor a numerical hint.

14.18 The strongest currently provable statement is localized

The exceptional-set theorem can be upgraded, with no new input, from a global count to a *local* one. The codegree argument in fact yields $O_{\varepsilon}(1)$ exceptional levels in every interval of length $c_{\varepsilon} N/\log N$

(one may take $c_\varepsilon \asymp \varepsilon^2$), whence:

Proposition 171 (localized exceptional set)**.** *For every $\varepsilon > 0$ there is C_ε such that, for all large N and* every *interval $I \subseteq (N, 2N]$,*

$$\#\{m \in I :\ \log G_m > \varepsilon m\} \ \le\ C_\varepsilon\Big(1 + \frac{|I| \log N}{N}\Big).$$

Proof. Let K be the codegree constant and A absolute with $\pi(2N) \le AN/\log N$. Put $d := \varepsilon N/(4 \log 2N) \ge \varepsilon N/(8 \log N)$, so every exceptional m has $|P(m)| \ge d$, and let $Q \le \pi(2N) \le AN/\log N$ be the number of available primes. For an interval I of length L set $E = \{m \in I : |P(m)| \ge d\}$, $M = |E|$, and $\nu_p = \#\{m \in E : p \in P(m)\}$, $S = \sum_p \nu_p$. Then $S \ge Md$ and, by Cauchy–Schwarz and the codegree bound $|P(m) \cap P(n)| \le K|m - n| \le KL$,

$$(Md)^2 \le S^2 \le Q \sum_p \nu_p^2 = Q\Big(S + 2 \sum_{\substack{m<n \\ m,n \in E}} |P(m) \cap P(n)|\Big) \le Q\big(MQ + KLM^2\big),$$

whence $M\big(d^2 - KQL\big) \le Q^2$. Choosing $L_0 := c_\varepsilon N/\log N$ with $c_\varepsilon := \varepsilon^2/(128AK)$ gives $KQL \le \varepsilon^2 N^2/(128 \log^2 N) \le d^2/2$ for $L \le L_0$, hence

$$M \le 2\Big(\frac{Q}{d}\Big)^2 \le \frac{128A^2}{\varepsilon^2} = O_\varepsilon(1).$$

The admissible length is $d^2/(KQ) \asymp \varepsilon^2 N/\log N$, which is where the ε^2 comes from. For arbitrary I, partition it into at most $1 + L/L_0$ subintervals and sum. ☐

Summing over a shell gives $O_\varepsilon(\log N)$ and cumulatively $O_\varepsilon((\log X)^2)$, recovering the stated theorem; but the local form is strictly stronger, since it forbids exceptional levels from clustering anywhere.

This is also the natural stopping point for the pairwise method: a configuration with one dense exceptional row every $\asymp N/\log N$ is compatible with the codegree bound $|P(m) \cap P(n)| \ll |m - n|$, so no more efficient pairwise Kővári–Sós–Turán argument can improve the exponent. Only a triple-codegree bound $\Delta_3(N)$ — for which §14.15 supplies the resultant certificate — can go further.

14.19 The obstruction, stated precisely

The results above localise the difficulty completely: G_n is a handful of small primes times the $K(n) \le 3$ targets, and every route we have tried terminates in the same place — bounding the number of prime factors of the *single* integer b_n inside the short window $(n/2, n]$. The correct trivial bound here is the ambient prime count,

$$H(n) \le \pi(n) - \pi(n/2) = \Big(\tfrac{1}{2} + o(1)\Big)\frac{n}{\log n},$$

which is stronger than the height bound $\log b_n/\log(n/2) \sim 2L$ with $L = \log(17 + 12\sqrt{2}) = 3.5255$. **The first Apéry-specific unconditional progress would therefore be any constant below** 1/2, or any $o(n/\log n)$. Known divisors do not help: Delaygue (Compos. Math. 154 (2018)) proves $5^{\alpha_5(n)} 11^{\alpha_{11}(n)} \mid b_n$ with α_5 counting base-5 digits in $\{1, 3\}$ and α_{11} counting base-11 digits equal to 5, but both are $O(\log n)$; and $2, 3, 7 \nmid b_n$ for every n, so no lcm-type divisor exists. Three separate barriers are now identified, each with a reason.

1. **Rank one is forced, not accidental.** The Newton polytope of Λ has a unique interior lattice point (the origin), so by Beukers–Vlasenko *Dwork Crystals I* (IMRN 2021, Thm. 11) the Cartier-stable quotient has rank 1; equivalently, for the K3 fibre $h^{0,2} = 1$, so the Hasse–Witt operator is 1×1 (Katz, *Une formule de congruence pour la fonction zêta*, SGA 7 II, Exp. XXII). Hence *every* mod-p observable built from the coefficient array is a multiple of the single marked scalar of Proposition 162, and any elimination of it reacquires the candidate primorial, of logarithmic height $\sim n/2$. This is the structural reason behind every rank-one alias encountered in the terminal, Turán, Hankel, Newton-jet and cross-index families.

2. **The second p-adic digit is not a second selective condition.** Modulo p^2 one does obtain a genuinely new coordinate, namely $D_{p,r} = R_{p-1}(\Lambda^{r-1}\Phi_p)$ with $\Phi_p = (\Lambda^p - \Lambda(X^p))/p$; but it is not target-selective. Direct computation gives $D_{19,10} = D_{31,8} = 0$ at genuine targets, and also $D_{19,9} = 0$ at a non-target. No codimension-two statement is available.

3. **The vertical average is equally out of reach.** The pointwise problem would follow from Hypothesis Z, $\sum_{p \le P} |Z_p| = O(P/\log P)$. The best unconditional aggregate implied by $|Z_p| = O(p^{2/3})$ is $O(P^{5/3}/\log P)$, and no improvement is available — not to $P^{1+\varepsilon}/\log P$, and not even $|Z_p| = O(1)$ on a positive-density set of primes. Katz's finite-field Mellin equidistribution (*Convolution and Equidistribution*, Ann. of Math. Studies 180, 2012) controls archimedean distributions of normalised traces, not exact vanishing modulo p; and Serre's bound $\#\{p \le X : a_p(E) = 0\} \ll X/(\log X)^{3/2-\varepsilon}$ (Publ. Math. IHES 54 (1981)) is one trace test per prime, whereas $|Z_p|$ involves p character-indexed tests inside each prime.

Naming the obstruction. It is worth separating our barrier from the one that modern sieve technology does break. Bombieri–Friedlander–Iwaniec, Fouvry–Iwaniec and Zhang cross the *level-of-distribution* barrier $Q = x^{1/2}$: at $q = x^{1/2+\delta}$ each progression still contains $x^{1/2-\delta}$ candidates, and dispersion plus a factorisation $q = q_1 q_2$ creates genuine bilinear boxes with Kloosterman input. Our barrier is the *occupancy-one* barrier $Q = x$: with $p \asymp N$ and n ranging over an interval of length N, each modulus contains exactly one candidate, and p is simultaneously the prime variable and the modulus, so it admits no balanced factorisation.

Varying residues are *not* the obstacle — Zhang handles all roots of a fixed f modulo smooth q, Maynard's uniform-residue theorem takes a supremum over arbitrary a_q when q has a conveniently sized factor, and Duke–Friedlander–Iwaniec handle roots of a fixed quadratic congruence as the prime modulus varies. What those all require is a factorable modulus or a fixed low-complexity algebraic kernel; ours is a prime modulus with a holonomic zero indicator, and we verified in §14.14 that the residues are statistically generic.

There is an equivalent and perhaps cleaner way to say what is missing. Identify $\mathbb{Z}/M$, $M = \prod_{p \in P} p$, with $\prod_{p \in P} \mathbb{F}_p$; then B_p is the cylinder $x_p \in Z_p$, and

$$\bigcap_{p \in P} B_p \;=\; \prod_{p \in P} Z_p$$

is *automatically nonempty*: it consists of $\prod_p |Z_p|$ residue classes modulo M, each of multiplicity $|P|$. So a totally aligned n always exists modulo M; the entire question is whether one of those CRT representatives happens to lie in the short interval $(N, 2N]$, where $N \ll M$. Covering-system technology (Filaseta–Ford–Konyagin–Pomerance–Yu, JAMS 20 (2007); Hough, Ann. of Math. 181 (2015); Balister–Bollobás–Morris–Sahasrabudhe–Tiba, Invent. Math. 228 (2022)) controls the *measure* of a union and deliberately tolerates microscopic hot spots, so it says nothing here; a natural descriptive name is *short-interval CRT discrepancy*, or microscopic non-clustering of CRT representatives.

The precise statement needed may therefore be named *microscopic moving-target equidistribution for Apéry zero residues*. It is strictly stronger than ordinary prime-modulus root equidistribution (Hooley; Duke–Friedlander–Iwaniec), because the target window has length 1 — relative scale $1/p$ — rather than fixed positive relative length. For calibration, even the cubic analogue $x^3 \equiv 2 \pmod{p}$ lacks comparable prime-modulus machinery, so a holonomic zero set at scale $1/p$ is well beyond current reach.

Size arguments are vacuous here, quantitatively. If $H(n) = cn/\log n$ then the product of the bad primes has logarithm $\sim cn$ and divides b_n, giving $c \le \log\alpha = 3.5255$ with $\alpha = 17 + 12\sqrt{2}$. But prime counting already gives $c \le 1/2$, since $\theta(n) - \theta(n/2) \sim n/2$. So the size bound is worse than the trivial one by a factor seven, and no renormalisation helps: for a size argument to yield $o(n)$ one would need $\log b_n = o(n)$, i.e. b_n subexponential, or a known divisor capturing all but $o(n)$ of $\log b_n$ — impossible since b_n is rough (only 0–15% of $\log b_n$ comes from primes $\le n^2$).

What the multiplicative case has and we do not. For $u_n = a^n - 1$ a nontrivial pointwise bound *is* available: partition the primes $p \mid a^n - 1$ by $d = \mathrm{ord}_p(a)$, so that $d \mid n$, $p \equiv 1 \pmod{d}$ and $p \mid \Phi_d(a)$; then the count T_d of such p in $(n/2, n]$ obeys both $T_d \ll n/d$ (candidates $p = 1 + kd$) and $T_d \le \varphi(d)\log(|a|+1)/\log(n/2)$ (size of $\Phi_d(a)$), and splitting the divisor sum at $d = \sqrt{n\log n}$ gives a genuine saving. The rigid invariant there is the multiplicative order. Our zero set Z_p carries no group structure, and there is no analogue.

Conversely, D-finiteness by itself supplies nothing: for fixed $K \ge 2$ the factorial ratios $U_n^{(K)}$ are hypergeometric, hence P-recursive, with $\log U_n^{(K)} \asymp n$, and Legendre's formula telescopes to give $p \mid U_n^{(K)}$ exactly for $p \le Kn/(K+1)$ — a positive proportion of the window. So no theorem of the form "P-recursive $\Rightarrow$ few prime factors in a short window" can exist, and the Apéry case must be decided by the arithmetic of Z_p itself.

Accordingly the unconditional pointwise statement appears to require genuinely new technology rather than a refinement of the arguments used here.

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